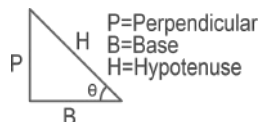


Trigonometry

It is that branch of Mathematics which deals with the measurements of the sides and angles of a triangle and also the problems associated with angles.

Important Formulae

In a right angled triangle, as shown below



$$\begin{aligned}\sin \theta &= \frac{P}{H} & \operatorname{cosec} \theta &= \frac{H}{P} \\ \cos \theta &= \frac{B}{H} & \sec \theta &= \frac{H}{B} \\ \tan \theta &= \frac{P}{B} & \cot \theta &= \frac{B}{P}\end{aligned}$$

Note that:

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}, \therefore \operatorname{cosec} \theta \times \sin \theta = 1$$

$$\text{Similarly, } \sec \theta = \frac{1}{\cos \theta}$$

$$\therefore \sec \theta \times \cos \theta = 1$$

$$\text{And, } \cot \theta = \frac{1}{\tan \theta}$$

$$\therefore \cot \theta \times \tan \theta = 1$$

$$\text{Also, } \tan \theta = \frac{\sin \theta}{\cos \theta}$$

- $\sin \theta = \cos (90^\circ - \theta)$
- $\operatorname{cosec} \theta = \sec (90^\circ - \theta)$
- $\tan \theta = \cot (90^\circ - \theta)$

Conversion of trigonometric ratios

- $\sin (-x) = -\sin x$
- $\operatorname{cosec} (-x) = -\operatorname{cosec} x$
- $\cos (-x) = \cos x$
- $\sec (-x) = \sec x$
- $\tan (-x) = -\tan x$
- $\cot (-x) = -\cot x$

Signs of the trigonometric ratios in different quadrants :

2nd quadrant Sine and cosecant are positive	1st quadrant All positive
3rd quadrant tangent and cotangent are positive	4th quadrant cosine and secant are positive

- x axis $\pm \theta \rightarrow$ no change
- y axis $\pm \theta \rightarrow$ changes like
- $\sin \theta \leftrightarrow \cos \theta$
- $\tan \theta \leftrightarrow \cot \theta$
- $\operatorname{cosec} \theta \leftrightarrow \sec \theta$

Important Identities

- $\sin^2 \theta + \cos^2 \theta = 1$
- $\sec^2 \theta - \tan^2 \theta = 1$
- $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$

We can derive all other identities from

these three basic identities.

Values of Trigonometric functions for different values of θ .

	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
$\cot \theta$	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
$\operatorname{cosec} \theta$	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

The value of $\sin \theta$ or $\cos \theta$ never exceeds 1, whereas the value of $\sec \theta$ or $\operatorname{cosec} \theta$ is always greater than or equal to 1.

Range of all trigonometric functions

	Minimum	Maximum
$\sin x, \cos x$ (odd powers)	-1	+1
$\sin x, \cos x$ (even powers)	0	+1
$\tan x, \cot x$ (odd powers)	$-\infty$	$+\infty$
$\tan x, \cot x$ (even powers)	0	$+\infty$
$\sec x, \operatorname{cosec} x$ (odd powers)	$-\infty$	$+\infty$
$\sec x, \operatorname{cosec} x$ (even powers)	+1	$+\infty$

Maximum and Minimum value of the trigonometric expressions of the different form :

$$\bullet a \sin x \pm b \cos x$$

$$\text{Maximum value} = \sqrt{a^2 + b^2}$$

$$\text{Minimum value} = -\sqrt{a^2 + b^2}$$

$$\bullet a \sin^2 x + b \cos^2 x$$

$$\text{If, } a > b \text{ Maximum value} = a$$

$$\text{Minimum value} = b$$

$$\text{If } a < b \text{ Maximum value} = b$$

$$\text{Minimum value} = a$$

$$\bullet a \operatorname{cosec}^2 x + b \sec^2 x$$

$$\text{Maximum value} = \infty$$

$$\text{Minimum value} = (\sqrt{a} + \sqrt{b})^2$$

$$\bullet a \sin^2 x + b \operatorname{cosec}^2 x$$

$$\bullet a \tan^2 x + b \cot^2 x$$

$$\bullet a \cos^2 x + b \sec^2 x$$

Maximum and Minimum value of these three trigonometric expression is ∞ and $2\sqrt{ab}$ respectively.

$$\bullet \sin^{2n} x + \cos^{2m} x$$

$$\text{Maximum value} = 1$$

$$\text{Minimum value} = (\text{use } x = 45^\circ)$$

Example : Find the minimum and maximum value of $\sin^2 x + \cos^4 x$?

$$\text{Maximum value of } \sin^2 x + \cos^4 x = 1$$

$$\text{Minimum value} = \sin^2 45^\circ + \cos^4 45^\circ =$$

$$\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$\bullet (\sin x \cos x)^n \text{ (when } n = \text{even)}$$

$$\text{Max. value} = +\frac{1}{2^n} \text{ and Min. value} = 0$$

When (n = odd)

$$\text{Max. value} = +\frac{1}{2^n} \text{ and Min. value} = -\frac{1}{2^n}$$

Compound Formulae

- $\sin (A + B)$
 $= \sin A \cos B + \cos A \sin B$
- $\sin (A - B)$
 $= \sin A \cos B - \cos A \sin B$
- $\cos (A + B)$
 $= \cos A \cos B - \sin A \sin B$
- $\cos (A - B)$
 $= \cos A \cos B + \sin A \sin B$
- $\sin 2A = 2 \sin A \cos A$
 $= \frac{2 \tan A}{1 + \tan^2 A}$
- $\cos 2A = \cos^2 A - \sin^2 A$
 $= 1 - 2 \sin^2 A$
- $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$
 $= 2 \cos^2 A - 1$
- $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
- $\sin 3A = 3 \sin A - 4 \sin^3 A$
- $\cos 3A = 4 \cos^3 A - 3 \cos A$
- $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$
- $\sin \theta \cdot \sin 2\theta \cdot \sin 4\theta$
 $= \frac{1}{4} \sin 3\theta$
- $\cos \theta \cdot \cos 2\theta \cdot \cos 4\theta$
 $= \frac{1}{4} \cos 3\theta$
- $\tan \theta \cdot \tan 2\theta \cdot \tan 4\theta = \tan 3\theta$
- $\tan (A + B) = \frac{\tan A + \tan B}{1 - (\tan A \times \tan B)}$
- $\tan (A - B) = \frac{\tan A - \tan B}{1 + (\tan A \times \tan B)}$
- $\cot (A + B) = \frac{\cot B \cot A - 1}{\cot B + \cot A}$
- $\cot (A - B) = \frac{\cot B \cot A + 1}{\cot B - \cot A}$

Transformations of sums or differences into products

- $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$
- $\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$
- $\cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$
- $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$

Concepts

$$\bullet a \sin X + b \cos X = m$$

$$a \cos X - b \sin X = n$$

$$\text{Then, } a^2 + b^2 = m^2 + n^2$$

Example : if $3 \sin \theta + 5 \cos \theta = 5$. Find the value of $3 \cos \theta - 5 \sin \theta$?

$$3 \cos \theta - 5 \sin \theta = \sqrt{5^2 + 3^2 - 5^2} = \pm 3$$

$$\bullet a \sec X + b \tan X = m$$

$$a \tan X - b \sec X = n$$

$$\text{Then, } a^2 - b^2 = m^2 - n^2$$

If $\sec x$ and $\tan x$ are replaced by $\operatorname{cosec} x$ and $\cot x$ respectively. Then the outcome will be the same.

Example : if $29 \sec \theta - 21 \tan \theta = 20$ Find the value of $29 \tan \theta - 21 \sec \theta$?

$$\text{Sol:- } = 29 \tan \theta - 21 \sec \theta$$

$$= \sqrt{20^2 - 29^2 + 21^2} = 0$$

$$\bullet \sec x + \tan x = \frac{1}{\sec x - \tan x}$$

$$\bullet \operatorname{cosec} x + \cot x = \frac{1}{\operatorname{cosec} x - \cot x}$$

Some Important results

(i). If $A + B + C = 180^\circ$. then,

$$\bullet \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$\bullet \cot A \cot B + \cot B \cot C + \cot C \cot A = 1$$

(ii). If $A + B + C = 90^\circ$. then,

$$\bullet \tan A \tan B + \tan B \tan C + \tan C \tan A = 1$$

$$\bullet \cot A + \cot B + \cot C = \cot A \cot B \cot C$$

(iii). If $A + B = 45^\circ$. then

$$\bullet (1 + \tan A)(1 + \tan B) = 2$$

$$\bullet (1 - \cot A)(1 - \cot B) = 2$$

Variety Questions

Q.1. If $\sin(P) = \cos(Q)$, $\cos(P) = \sin(R)$, and $P + Q + R = 90^\circ$, then what is the value of angle P?

SSC CGL 21/09/2025 (Shift 3)

- (a) 90° (b) 60° (c) 30° (d) 15°

Q.2. If $\cot A + \operatorname{cosec} A = y$, express $\operatorname{cosec} A$ in terms of y .

Graduate Level 29/07/2025 (Shift - 3)

(a) $\frac{(y^2 + 1)}{2y}$ (b) $y^2 + \frac{1}{y}$

(c) $2y$ (d) $\frac{(y^2 - 1)}{2y}$

Q.3. If $\cot(2y) = \frac{3}{4}$, where y lies in the first quadrant, find the exact value of $\cot(6y)$

Higher Secondary 30/07/2025 (Shift - 2)

(a) $\frac{112}{55}$ (b) $-\frac{117}{44}$ (c) $\frac{130}{51}$ (d) $-\frac{110}{77}$

Q.4. A Radar System is aligned so that it forms an angle x with the horizontal. It is

known that $\cos x = -\frac{12}{13}$ and x lies in the third quadrant. What is the sin of half the elevation angle $(\frac{x}{2})$, assuming precision is necessary for signal locking?

Higher Secondary 26/07/2025 (Shift - 2)

(a) $\frac{2}{\sqrt{26}}$ (b) $\frac{3}{\sqrt{26}}$ (c) $\frac{5}{\sqrt{26}}$ (d) $-\frac{5}{\sqrt{26}}$

Q.5. Find the value of $\tan 72^\circ - \tan 27^\circ - \tan 72^\circ \tan 27^\circ$.

- (a) -1 (b) 1 (c) -2 (d) 0

Q.6. Simplify $\frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta}$.

(a) $\cos \frac{\theta}{2}$ (b) $\sin \frac{\theta}{2}$ (c) $\cot \frac{\theta}{2}$ (d) $\tan \frac{\theta}{2}$

Q.7. The value of the expression

$$\frac{1 - \sin(2t)}{1 + \sin(2t)} \times \frac{\cos(t) + \sin(t)}{\cos(t) - \sin(t)} \text{ is}$$

(a) $\frac{1 - 2 \tan(t)}{1 + 2 \tan(t)}$ (b) $\frac{1 - \tan(t)}{1 + \tan(t)}$

(c) $\frac{1 + 2 \tan(t)}{1 - 2 \tan(t)}$ (d) $\frac{1 + \tan(t)}{1 - \tan(t)}$

Q.8. $\sin A \sin B =$ _____.

(a) $\frac{1}{2} \{(\sin(A + B) + \sin(A - B))\}$

(b) $\frac{1}{2} \{(\sin(A + B) - \sin(A - B))\}$

(c) $\frac{1}{2} \{(\cos(A + B) - \cos(A - B))\}$

(d) $\frac{1}{2} \{(\cos(A - B) - \cos(A + B))\}$

Q.9. If $a \sin^3 x + b \cos^3 x = \sin x \cos x$ and $a \sin x = b \cos x$, then find the value of $a^2 + b^2$, provided that x is neither 0° nor 90° .

- (a) 0 (b) 1 (c) $a^2 - b^2$ (d) $a^2 + b^2$

Q.10. If $p \cos A = 2q \sin A$ and $2p \operatorname{cosec} A - q \sec A = 3$, then the value of $p^2 + 4q^2$ is:

- (a) 2 (b) 4 (c) 1 (d) 3

Q.11. If $\tan A + \sin A = x$, $\tan A - \sin A = y$, what is the value of $x^2 - y^2$?

(a) $4\sqrt{xy}$ (b) $2\sqrt{\frac{x}{y}}$ (c) $\sqrt{xy}$ (d) $4\sqrt{\frac{x}{y}}$

Q.12. What is the value of $\cos^{-1}(\frac{7}{25}) - \cos^{-1}(\frac{15}{17})$?

(a) $\cos^{-1}(\frac{304}{425})$ (b) $\cot^{-1}(\frac{297}{425})$

(c) $\tan^{-1}(\frac{297}{425})$ (d) $\sin^{-1}(\frac{304}{425})$

Q.13. If $f(p) = \sin^{p+2} x + \cos^{p+2} x$, then the value of $6f(2) - 4f(4) + 10f(0)$ is:

- (a) 12 (b) 10 (c) 11 (d) 14

Q.14. If $\tan A = a \tan B$ and $\sin A = b \sin B$, then the value of $\cos^2 A$ is:

(a) $\frac{b^2 + 1}{a^2 + 1}$ (b) $\frac{b^2 - 1}{a^2 - 1}$

(c) $\frac{a^2 + 1}{b^2 + 1}$ (d) $\frac{a^2 - 1}{b^2 - 1}$

Q.15. If $\sin(X - Y) = \sin X \cos Y - \cos X \sin Y$, then the value of $\frac{\sin(A - B)}{\cos A \cos B} + \frac{\sin(B - C)}{\cos B \cos C}$ is:

(a) $\frac{\cos(A + C)}{\cos A \cos C}$ (b) $\frac{\sin(C - A)}{\cos A \cos C}$

(c) $\frac{\cos(A + C)}{\sin A \sin C}$ (d) $\frac{\sin(A - C)}{\cos A \cos C}$

Q.16. $\frac{3 \cos^3 \theta - 2 \cos \theta}{\sin \theta - 3 \sin^3 \theta}$ is equal to:

- (a) $\tan^2 \theta$ (b) $\cot^2 \theta$ (c) $\tan \theta$ (d) $\cot \theta$

Q.17. If $\cot A = 7$, then the value of

$$\frac{5 \cos A + 4 \sin A}{\cos^3 A + 7 \sin^3 A + 6 \sin A}$$
 is:

- (a) 2 (b) 0 (c) 3 (d) 1

Q.18. What is the value of $\frac{[(\sin x + \sin y)(\sin y - \sin x)]}{[(\cos x + \cos y)(\cos y - \cos x)]}$

- (a) 0 (b) 1 (c) -1 (d) 2

Q.19. Simplify $\frac{\cos 13^\circ + \sin 13^\circ}{\cos 13^\circ - \sin 13^\circ}$

(a) $\cot 58^\circ$ (b) $\tan 32^\circ$

(c) $\tan 58^\circ$ (d) $\cos 26^\circ$

Q.20. If $(\sin x + \sin y) = a$ and $(\cos x + \cos y) = b$, then find the value of $(\sin x \sin y + \cos x \cos y)$.

(a) $\frac{a^2 + b^2 - 1}{2}$ (b) $\frac{a^2 - b^2 - 2}{2}$

(c) $\frac{a^2 + b^2 - 2}{2}$ (d) $\frac{a^2 + b^2 + 2}{2}$

Q.21. If $\tan(t) + \cot(t) = 1$, then one of the values of the expression $\frac{1}{\sin(t) + \cos(t)}$ is _____.

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{2}{\sqrt{3}}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{3}$

Q.22. If $3 \sec^4 \theta + 8 = 10 \sec^2 \theta$, then the value of $\tan \theta$ can be:

(a) $2\sqrt{3}$ (b) $1\sqrt{3}$ (c) $2\frac{1}{\sqrt{3}}$ (d) $1\frac{1}{\sqrt{3}}$

Q.23. If $\cot^4 A + \cot^2 A = 1$, then find the value of $\cos^4 A - 3 \cos^2 A$.

- (a) -1 (b) -2 (c) 2 (d) 1

Q.24. If $\cot^2 \theta = 1 - e^2$, then the value of $\operatorname{cosec} \theta + \cot^3 \theta \sec \theta$ is:

(a) $(2 - e^2)^{\frac{3}{2}}$ (b) $(1 - e^2)$

(c) $(1 - e^2)^{\frac{3}{2}}$ (d) $(2 - e^2)^{\frac{1}{2}}$

Q.25. Find the least value of $16 \operatorname{cosec}^2 \theta + 25 \sin^2 \theta$.

- (a) 35 (b) 38 (c) 42 (d) 40

Q.26. If $\operatorname{cosec} A + \cot A = a\sqrt{b}$, then find

the value of $\frac{(a^2 b - 1)}{(a^2 b + 1)}$.

- (a) $\cos A$ (b) $\tan A$ (c) $\frac{1}{\sin A}$ (d) $\frac{1}{\cot A}$

Q.27. If $\sin \theta + \tan \theta = p$ and $\tan \theta - \sin \theta = q$, then which of the relation satisfies for given values of p and q ?

- (a) $p^2 - q^2 = 4\sqrt{pq}$ (b) $p^2 - q^2 = 8\sqrt{pq}$
(c) $p^2 + q^2 = 4\sqrt{pq}$ (d) $p^2 + q^2 = 4pq$

Q.28. If θ is an acute angle and $\sin \theta \cos \theta = 2 \cos^3 \theta - \frac{1}{4} \cos \theta$, then the value of $\sin \theta$ is:

- (a) $\frac{\sqrt{15}-1}{8}$ (b) $\frac{\sqrt{15}-1}{4}$
(c) $\frac{\sqrt{15}+1}{4}$ (d) $\frac{\sqrt{15}-1}{2}$

Q.29. Simplify the following: $7(\cot 11^\circ \cot 13^\circ \cot 19^\circ \cot 29^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ)$

- (a) 7 (b) -7 (c) -1 (d) 1

Q.30. The given expression is equal to :

$$\frac{\sin^4 A + \cos^4 A}{1 - 2\sin^2 A \cos^2 A}$$

- (a) 1 (b) -1 (c) 0 (d) 2

Q.31. If $\cot A + \cos A = p$ and $\cot A - \cos A = q$, then which of the following relation is correct?

- (a) $\sqrt{\frac{1}{16pq}} = p^2 + q^2$ (b) $\frac{1}{4pq} = p^2 + q^2$
(c) $\sqrt{16pq} = p^2 - q^2$ (d) $4pq = p^2 + q^2$

Q.32. The value of the expression $4(\sin^6 A + \cos^6 A) - 6(\sin^4 A + \cos^4 A) + 8$ is:

- (a) 4 (b) 8 (c) 7 (d) 6

Q.33. Evaluate the following:

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + 2\cos 8\theta}}}$$

- (a) $2\cos \theta$ (b) $2\cos 2\theta$ (c) $\sin 2\theta$ (d) $\cos 2\theta$

Q.34. Find the value of $\sqrt{\frac{1 - \sin 3\theta}{1 + \sin 3\theta}}$

- (a) $\sec 3\theta - \tan 3\theta$ (b) $(\sec 3\theta - \tan 3\theta)^3$
(c) $(\sec 3\theta - \tan 3\theta)^2$ (d) $\sec 3\theta + \tan 3\theta$

Q.35. The solution of the equation $\cos^2 \theta + \sin \theta + 1 = 0$ lies between :

- (a) $(\frac{3\pi}{4}, \frac{5\pi}{4})$ (b) $(\frac{5\pi}{4}, \frac{7\pi}{4})$
(c) $(\frac{\pi}{4}, \frac{3\pi}{4})$ (d) $(-\frac{\pi}{4}, \frac{\pi}{4})$

Q.36. What is the value of $(1 + \cot A + \tan A)(\sin A - \cos A)(\frac{\sin A \cos A}{\sin^3 A - \cos^3 A})$?

- (a) 1 (b) -1 (c) 0 (d) 2

Q.37. If $\sin m + \sin n = p$, $\cos m + \cos n = q$, then find the value of $\sin m \times \sin n + \cos m \times \cos n$.

- (a) $p^2 + q^2 - 2$ (b) $\frac{(p^2 + q^2 - 2)}{2}$
(c) $p + q - pq$ (d) $p + q + pq$

Q.38. If $a \cot \theta + b \operatorname{cosec} \theta = p$ and $b \cot \theta + a \operatorname{cosec} \theta = q$ then $p^2 - q^2$ is equal to _____.

- (a) $b^2 - a^2$ (b) $a^2 - b^2$ (c) $b - a$ (d) $a^2 + b^2$

Q.39. If A is an acute angle, the simplified form of

$$\frac{\cos(\pi - A) \cdot \cot(\frac{\pi}{2} + A) \cos(-A)}{\tan(\pi + A) \tan(\frac{3\pi}{2} + A) \sin(2\pi - A)}$$

- (a) $\cos^2 A$ (b) $\sin A$ (c) $\sin^2 A$ (d) $\cos A$

Q.40. What is the value of the expression $\cos 2A \cos 2B + \sin^2(A - B) - \sin^2(A + B)$?

- (a) $\sin(2A - 2B)$ (b) $\sin(2A + 2B)$
(c) $\cos(2A + 2B)$ (d) $\cos(2A - 2B)$

Q.41. If $\frac{k - k \cot^2 30^\circ}{1 + \cot^2 30^\circ} = \sin^2 60^\circ + 4 \tan^2 45^\circ - \operatorname{cosec}^2 60^\circ$, then the value of k (correct to two decimal places) is:

- (a) 5.55 (b) -6.83 (c) -5.58 (d) 6.83

Q.42. If $\cos \theta + \sin \theta = \sqrt{2} \cos \theta$, find the value of $(\cos \theta - \sin \theta)$.

- (a) $\sqrt{2} \sin \theta$ (b) $\sqrt{2} \cos \theta$
(c) $\frac{1}{\sqrt{2}} \sin \theta$ (d) $\frac{1}{2} \cos \theta$

Q.43. The value of $\sin 73^\circ + \cos 137^\circ$ is:

- (a) $\sin 13^\circ$ (b) $\cos 13^\circ$
(c) $\cos 18^\circ$ (d) $\sin 18^\circ$

Q.44. The value of $\sin 18^\circ$ is given as $\frac{\sqrt{5}-1}{4}$. Using this value, find the value of $\cos 18^\circ$.

- (a) $\frac{\sqrt{10+2\sqrt{5}}}{4}$ (b) $\frac{\sqrt{5}+1}{4}$
(c) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (d) $\frac{\sqrt{10-2\sqrt{5}}}{4}$

Q.45. The value of $\frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$ is:

- (a) $\tan 3\theta$ (b) $\cot 3\theta$ (c) $\tan \theta$ (d) $\cot \theta$

Q.46. The value of $\cot^2 46^\circ - \sec^2 44^\circ + (\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ)$

- (a) 23.5 (b) 20.5 (c) 22.5 (d) 21.5

Q.47. If A lies between 45° and 540° , and $\sin A = 0.5$, what is the value of $\frac{A}{3}$ in degree?

- (a) 170° (b) 175° (c) 165° (d) 160°

Q.48. If $\cos 27^\circ = x$, then the value of $\tan 63^\circ$ is:

- (a) $\frac{\sqrt{1+x^2}}{x}$ (b) $\frac{x}{\sqrt{1+x^2}}$
(c) $\frac{\sqrt{1-x^2}}{x}$ (d) $\frac{x}{\sqrt{1-x^2}}$

Q.49. If $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ and $x \sin \theta = y \cos \theta$, then the value of $x^2 + y^2$ is:

- (a) 0 (b) 4 (c) 1 (d) 2

Q.50. If $\cos^2 \theta + \cos^4 \theta = 1$, then the value of $\sin \theta + \sin^2 \theta$ is:

- (a) 0 (b) $\frac{1}{2}$ (c) 2 (d) 1

Q.51. If $\sin \theta - \cos \theta = \frac{7}{17}$, then find the value of $\sin \theta + \cos \theta$.

- (a) $\frac{8}{17}$ (b) $\frac{23}{13}$ (c) $\frac{23}{17}$ (d) $\frac{8}{13}$

Q.52. If $\sec \theta$ and $\sin \theta$ ($0^\circ < \theta < 90^\circ$) are the roots of the equation $\sqrt{6}x^2 - kx + \sqrt{6} = 0$, then the value of k is:

- (a) $\sqrt{3}$ (b) $3\sqrt{2}$ (c) $2\sqrt{3}$ (d) $3\sqrt{3}$

Q.53. If $\cos A$, $\sin A$, $\cot A$ are in geometric progression, then the value of $\tan^6 A - \tan^2 A$ is:

- (a) $\frac{1}{2}$ (b) 3 (c) $\frac{1}{3}$ (d) 1

Q.54. Find the value of $\cos 225^\circ$

- (a) -0.7071 (b) -0.866
(c) 0.7071 (d) 0.866

Q.55. If $0^\circ < A, B < 45^\circ$, $\cos(A+B) = \frac{24}{25}$

and $\sin(A-B) = \frac{15}{17}$, then $\tan 2A = ?$

- (a) 0 (b) 1 (c) $\frac{416}{87}$ (d) $\frac{213}{4}$

Q.56. If $\cos^2 \theta - \sin^2 \theta = \tan^2 \theta$, then which of the following is true?

- (a) $\cos \theta \cos \theta = 1$
(b) $\cos \theta \cos \theta = \sqrt{2}$
(c) $\cos^2 \theta - \sin^2 \theta = \cot^2 \theta$
(d) $\cos^2 \theta - \sin^2 \theta = \tan^2 \theta$

Q.57. everything between 0° and 90° For α'_i 's, ($i = 1, 2, 3, \dots, 20$) it is given that $\sin \alpha_1 + \sin \alpha_2 + \sin \alpha_3 + \dots + \sin \alpha_{20} = 20$. Find the value (in degrees) of $(\alpha_1 + \alpha_2 + \alpha_3 + \dots + \alpha_{20})$.

- (a) 1800° (b) 900° (c) 0° (d) 20°

Q.58. For α and β both being acute angles, it is given that $\sin(\alpha + \beta) = 1$, $\cos(\alpha - \beta) = \frac{1}{2}$. The values of α and β are:

- (a) $75^\circ, 15^\circ$ (b) $45^\circ, 15^\circ$
(c) $75^\circ, 45^\circ$ (d) $60^\circ, 30^\circ$

Q.59. If $\theta = 9^\circ$, then what is the value of $\cot \theta \cdot \cot 2\theta \cdot \cot 3\theta \cdot \cot 4\theta \cdot \cot 5\theta \cdot \cot 6\theta \cdot \cot 7\theta \cdot \cot 8\theta \cdot \cot 9\theta$?

- (a) $\sqrt{3} - 1$ (b) 1 (c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$

Q.60. Which among the following is an irrational quantity ?

- (a) $\tan 30^\circ \cdot \tan 60^\circ$ (b) $\sin 30^\circ$
(c) $\tan 45^\circ$ (d) $\cos 30^\circ$

Q.61. Which among the following increases continuously in the range $0^\circ < \theta < 90^\circ$?

- (a) $\cot \theta$ (b) $\operatorname{cosec} \theta$ (c) $\tan \theta$ (d) $\cos \theta$

Q.62. If $\sec \theta = 4x$ and $\tan \theta = \frac{4}{x}$, ($x \neq 0$) then the value of $8(x^2 - \frac{1}{x^2})$ is :

- (a) $\frac{1}{16}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{8}$

Q.63. If $\sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \times \sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} = \frac{1-r}{1+r}$, then the value of r is:

- (a) $\sin \theta$ (b) $\operatorname{Cosec} \theta$ (c) $\sec \theta$ (d) $\cos \theta$

Q.64. If $\frac{(\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)}{\tan^2 \theta - \sin^2 \theta} = r^3$, then $r = ?$

- (a) $\sin \theta \cos \theta$ (b) $\tan \theta$
(c) $\cot \theta$ (d) $\operatorname{Cosec} \theta \sec \theta$

Q.65. If $3 - 2\sin^2 \theta - 3\cos \theta = 0$, $0^\circ < \theta < 90^\circ$, then what is the value of $(2\operatorname{Cosec} \theta + \tan \theta)$?

- (a) $7\sqrt{3}$ (b) $5\sqrt{3}$ (c) $\frac{5\sqrt{3}}{3}$ (d) $\frac{7\sqrt{3}}{3}$

Q.66. A simplified of

$$\left(\frac{\sin \theta}{1+\cos \theta} + \frac{1+\cos \theta}{\sin \theta}\right) \left(\frac{1}{\tan \theta + \cot \theta}\right) \text{ is :}$$

- (a) $\cos \theta$ (b) $2\sin \theta$ (c) $\sin \theta$ (d) $2\cos \theta$

Q.67. The value of

$$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\sin 56^\circ \cdot \sec 34^\circ + \cos 25^\circ \cdot \operatorname{cosec} 65^\circ} \text{ is:}$$

- (a) $\frac{1}{2}$ (b) 4 (c) 2 (d) 1

Q.68. $\frac{(\sec \theta + \tan \theta)(1 - \sin \theta)}{\operatorname{cosec} \theta(1 + \cos \theta)(\operatorname{cosec} \theta - \cot \theta)}$ is equal to:

- (a) $\sin \theta$ (b) $\sec \theta$ (c) $\cos \theta$ (d) $\operatorname{Cosec} \theta$

Q.69. If $3\cos^2 A + 7\sin^2 A = 4$, then what is the value of $\cot A$, given that A is an acute angle ?

- (a) 1 (b) $\sqrt{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{\sqrt{3}}$

Q.70. If $\operatorname{Cosec} \theta = \frac{13}{12}$, then $\sin \theta + \cos \theta - \tan \theta$ is equal to :

- (a) $\frac{91}{65}$ (b) $\frac{139}{65}$ (c) $\frac{71}{65}$ (d) $-\frac{71}{65}$

Q.71. If $\sec 4\theta = \operatorname{Cosec}(\theta + 20^\circ)$, then θ is equal to:

- (a) 22° (b) 18° (c) 14° (d) 20°

Q.72. If $3\sin \theta = 2\cos^2 \theta$, $0^\circ < \theta < 90^\circ$, then the value of $(\tan^2 \theta + \sec^2 \theta - \operatorname{cosec}^2 \theta)$ is :

- (a) -2 (b) $-\frac{7}{3}$ (c) $\frac{7}{3}$ (d) 2

Q.73. If $\frac{\tan \theta + \sin \theta}{\tan \theta - \sin \theta} = \frac{k+1}{k-1}$, then $k = ?$

- (a) $\operatorname{cosec} \theta$ (b) $\sec \theta$ (c) $\cos \theta$ (d) $\sin \theta$

Q.74. If $\frac{1}{\operatorname{cosec} \theta - 1} + \frac{1}{\operatorname{cosec} \theta + 1} = 2 \sec \theta$, $0^\circ < \theta < 90^\circ$, then the value of $(\cot \theta + \cos \theta)$ is:

- (a) $\frac{1+\sqrt{2}}{2}$ (b) $\frac{2+\sqrt{2}}{2}$
(c) $\frac{2+\sqrt{3}}{\sqrt{2}}$ (d) $1 + \sqrt{2}$

Q.75. If $\sin \theta = \frac{a}{\sqrt{a^2 + b^2}}$, $0^\circ < \theta < 90^\circ$,

then the value of $\sec \theta + \tan \theta$ is :

- (a) $\frac{\sqrt{a^2 + b^2} + a}{b}$ (b) $\frac{\sqrt{a^2 + b^2} + b}{2a}$
(c) $\frac{\sqrt{a^2 + b^2} + a}{2b}$ (d) $\frac{\sqrt{a^2 + b^2} + b}{a}$

Q.76. If $\cos^2 \theta - 3\cos \theta + 2 = \sin^2 \theta$, $0^\circ < \theta < 90^\circ$, then the value of $2\operatorname{cosec} \theta + 4\cot \theta$ is :

- (a) $\frac{8\sqrt{3}}{3}$ (b) $\frac{4\sqrt{3}}{4}$ (c) $2\sqrt{3}$ (d) $4\sqrt{3}$

Q.77. The value of $\frac{1}{\sec x - \tan x} - \frac{1}{\cos x}$, $0^\circ < x < 90^\circ$, is equal to :

- (a) $2\sec x$ (b) $\tan x$ (c) $2\cos x$ (d) $\cot x$

Q.78. If $\tan^2 \theta - 3\sec \theta + 3 = 0$, $0^\circ < x < 90^\circ$, then the value of $\sin \theta + \cot \theta$ is :

- (a) $\frac{5\sqrt{3}}{6}$ (b) $2\sqrt{3}$ (c) $\frac{5\sqrt{3}}{3}$ (d) $3\sqrt{3}$

Q.79. If $(1 + \tan^2 \theta) + (1 + (\tan^2 \theta)^{-1}) = k$, then $\sqrt{k} = ?$

- (a) $\operatorname{cosec} \theta \cdot \sec \theta$ (b) $\operatorname{cosec} \theta \cdot \cos \theta$
(c) $\sin \theta \cdot \cos \theta$ (d) $\sin \theta \cdot \sec \theta$

Q.80. The value of $\frac{1}{\sin \theta} - \frac{\cot^2 \theta}{1 + \operatorname{cosec} \theta}$ is

- (a) 0 (b) 1 (c) 2 (d) -1

Q.81. $\left(\frac{1}{1 + \sin^2 \theta} + \frac{1}{1 + \operatorname{cosec}^2 \theta}\right) = ?$

- (a) $\sin^2 \theta$ (b) 1 (c) $\operatorname{cosec}^2 \theta$ (d) 2

Q.82. If $\frac{\cos \theta}{1 - \sin \theta} + \frac{\cos \theta}{1 + \sin \theta} = 4$,

$0^\circ < \theta < 90^\circ$, then the value of $(\tan \theta + \operatorname{cosec} \theta)$ is :

- (a) $\frac{5\sqrt{2}}{2}$ (b) $\frac{5\sqrt{3}}{3}$ (c) $\frac{4\sqrt{3}}{3}$ (d) $\frac{5\sqrt{2}}{3}$

Q.83. $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = ?$

- (a) $\sec \theta \sin \theta$ (b) $\sec \theta \tan \theta$
(c) $\sec \theta + \tan \theta$ (d) $\sec \theta - \tan \theta$

Q.84. The value of

$$\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta} \times \sqrt{\tan^2 \theta - \sin^2 \theta} \text{ is equal to :}$$

- (a) $\operatorname{cosec} \theta \sec^2 \theta$ (b) $\sin \theta \sec^2 \theta$
(c) $\sin \theta \cos^2 \theta$ (d) $\operatorname{cosec} \theta \cos^2 \theta$

Q.85. If $\sec \theta - \tan \theta = P$, then $\operatorname{cosec} \theta = ?$

- (a) $\frac{2P}{1 - P^2}$ (b) $\frac{1 - P^2}{1 + P^2}$
(c) $\frac{P^2 + 1}{1 - P^2}$ (d) $\frac{2P}{1 + P^2}$

Q.86. If $0^\circ < \theta < 90^\circ$ and $\cos^2 \theta$

$= 3(\cot^2 \theta - \cos^2 \theta)$ then the value of $\left(\frac{1}{\sec \theta + \sin \theta}\right)^{-1}$ is :

- (a) $\sqrt{3} + 2$ (b) $2(2 - \sqrt{3})$
(c) $2(\sqrt{3} - 1)$ (d) $\sqrt{3} + 1$

Q.87. $\left(\frac{\sin \theta - 2\sin^3 \theta}{2\cos^3 \theta - \cos \theta}\right)^2 + 1$, $\theta \neq 45^\circ$ is equal to.

- (a) $\operatorname{cosec}^2 \theta$ (b) $\sec^2 \theta$ (c) $\cot^2 \theta$ (d) $2\tan^2 \theta$

Q.88. $(\operatorname{cosec} A - \sin A)^2 + (\sec A - \cos A)^2 - (\cot A - \tan A)^2$ is equal to:

- (a) 2 (b) 0 (c) 1 (d) -1

Q.89. Find the value of $1 + \frac{\tan^2 A}{1 + \sec A}$

- (a) $\operatorname{cosec} A$ (b) $\cos A$ (c) $\sec A$ (d) $\sin A$

Examination wise Questions

SSC CGL 2025 Tier - I

Q.90. If $\tan(90^\circ - \theta) = \sqrt{3}$, what is $\cos \theta$?
SSC CGL 12/09/2025 (Shift 2)

- (a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{\sqrt{2}}{2}$ (d) $\frac{3}{4}$

Q.91. If $\sec A = \frac{13}{5}$ and A is acute, find $\tan A$.
SSC CGL 12/09/2025 (Shift 2)

- (a) $\frac{13}{5}$ (b) $\frac{12}{5}$ (c) $\frac{13}{12}$ (d) $\frac{12}{13}$

Q.92. If $\cos(90^\circ - x) = \sin(2x)$, then what is x ?
SSC CGL 13/09/2025 (Shift 1)

- (a) 0° (b) 30° (c) 25° (d) 15°

Q.93. Given, A and B are complementary angles, and $\cos A = \frac{4}{5}$, what is $\tan B$?
SSC CGL 13/09/2025 (Shift 1)

- (a) $\frac{4}{3}$ (b) $\frac{7}{3}$ (c) $\frac{2}{5}$ (d) $\frac{5}{3}$



Q.94. If $\sin^2 A - \cos^2 A = \frac{1}{4}$, find the value

of $\frac{1}{\sec^2 A}$.

SSC CGL 13/09/2025 (Shift 1)

- (a) $\frac{3}{7}$ (b) $\frac{3}{8}$ (c) $\frac{1}{8}$ (d) $\frac{1}{3}$

Q.95. If $\sin A = \cos(2A - 30^\circ)$, then what is the value of A?

SSC CGL 14/09/2025 (Shift 3)

- (a) 40° (b) 25° (c) 18° (d) 30°

Q.96. If $\sin A + \cos A = \sqrt{2} \cos A$, then what is the value of $\cot A$?

SSC CGL 14/09/2025 (Shift 3)

- (a) $\sqrt{2}$ (b) 1 (c) $\sqrt{2} + 1$ (d) $\sqrt{2} - 1$

Q.97. If $\sin \theta = \frac{15}{17}$, and $\theta \in (0, \frac{\pi}{2})$, then what is the value of $\tan \theta$?

SSC CGL 17/09/2025 (Shift 1)

- (a) $\frac{9}{16}$ (b) $\frac{25}{9}$ (c) $\frac{15}{8}$ (d) $\frac{16}{9}$

Q.98. What is the value of $\tan X + \cot X$ if $\tan X = 3$?

SSC CGL 17/09/2025 (Shift 1)

- (a) $\frac{10}{3}$ (b) $\frac{5}{3}$ (c) $\frac{1}{3}$ (d) $\frac{7}{3}$

Q.99. If $\sin B + \cos B = \frac{13}{12}$, find $\sin 2B$.

SSC CGL 17/09/2025 (Shift 1)

- (a) $\frac{169}{144}$ (b) $\frac{5}{148}$ (c) $\frac{7}{8}$ (d) $\frac{25}{144}$

Q.100. If $\sin A = \frac{5}{13}$ and A lies in the second quadrant, find the value of $(\sin A + \cos A)^2$.

SSC CGL 17/09/2025 (Shift 3)

- (a) 0 (b) 1 (c) $\frac{49}{169}$ (d) $\frac{39}{169}$

Q.101. If $\sin A = \frac{p}{q}$, then what is the value of $(1 + \tan^2 A)$?

SSC CGL 17/09/2025 (Shift 3)

- (a) $\frac{q^2}{q^2 - p^2}$ (b) $\frac{2q^2}{q^2 - p^2}$
(c) $\frac{q^2}{2(q^2 - p^2)}$ (d) $\frac{5q^2}{q^2 - p^2}$

Q.102. If $\tan A = x + \frac{1}{x}$, find $\sec^2 A$.

SSC CGL 17/09/2025 (Shift 3)

- (a) $x^2 + \frac{1}{x^2} + 5$ (b) $x^2 + \frac{1}{x^2} + 2$
(c) $x^2 + \frac{1}{x^2} + 3$ (d) $x^2 + \frac{1}{x^2} + 4$

Q.103. If $\sin^4 A - \cos^4 A = \frac{1}{2}$, find the value of $\cos^2 A$.

SSC CGL 18/09/2025 (Shift 1)

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{2}{3}$

Q.104. If $\cot A = x + \frac{1}{x}$, find $\sec^2 A$.

SSC CGL 18/09/2025 (Shift 1)

- (a) $1 + (1 + \frac{1}{x})^{-2}$ (b) $2 + (x + \frac{1}{x})^{-2}$
(c) $1 + (x + \frac{1}{x})^{-2}$ (d) $2 + (x + \frac{1}{x})^{-2}$

Q.105. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, A

and $B \in (0, \frac{\pi}{2})$, find $\tan(A + B)$.

SSC CGL 18/09/2025 (Shift 2)

- (a) 3 (b) 1 (c) 7 (d) 9

Q.106. If $\sec A = 2$, then $\tan A - \operatorname{cosec} A = ?$

SSC CGL 18/09/2025 (Shift 2)

- (a) $\frac{2}{\sqrt{6}}$ (b) $\frac{1}{\sqrt{3}}$ (c) 0 (d) $\frac{1}{3}$

Q.107. If $\tan B + \cot B = 4$ and $\tan B \cdot \cot B = 1$, find $\tan^2 B + \sec^2 B$.

SSC CGL 18/09/2025 (Shift 3)

- (a) $15 + 6\sqrt{3}$ (b) $15 + 8\sqrt{3}$
(c) $16 + 8\sqrt{3}$ (d) $16 + 6\sqrt{3}$

Q.108. If $\sin A = \frac{2x}{1+x^2}$, What is $\tan A$?

SSC CGL 18/09/2025 (Shift 3)

- (a) $\frac{1+x^2}{1-x^2}$ (b) $\frac{1-x^2}{1+x^2}$
(c) $\frac{2x}{1-x^2}$ (d) $\frac{1-x^2}{2x}$

Q.109. If $\sin A + \cos A = \frac{3}{2}$ and $A \in (0, \frac{\pi}{2})$, then what is $\sin A \cos A$?

SSC CGL 19/09/2025 (Shift 1)

- (a) 1 (b) 2
(c) $\frac{5}{8}$ (d) Cannot be determined

Q.110. What is the value of $\sin 40^\circ \cos 20^\circ + \cos 40^\circ \sin 20^\circ$?

SSC CGL 19/09/2025 (Shift 1)

- (a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{4}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{7}{8}$

Q.111. If $\sin x = \frac{7}{25}$ and $x \in (0, \frac{\pi}{2})$, then

find $\frac{1 + \tan x}{1 - \tan x}$

SSC CGL 19/09/2025 (Shift 1)

- (a) 6 (b) $\frac{31}{17}$ (c) $\frac{24}{17}$ (d) 9

Q.112. If $\sin A + \cos A = Y$, then find the value of $\sin^2 A + \cos^2 A + 2\sin A \cos A$.

SSC CGL 19/09/2025 (Shift 2)

- (a) Y (b) Y^2 (c) 1 (d) $Y + 1$

Q.113. If $\sin x = \sqrt{3} \cos x$, find $\sin^4 x + \cos^4 x$.

SSC CGL 19/09/2025 (Shift 2)

- (a) $\frac{3}{8}$ (b) $\frac{5}{8}$ (c) $\frac{1}{8}$ (d) $\frac{1}{6}$

Q.114. If $\tan A = \frac{1}{\sqrt{3}}$, what is the value of

$(1 + \sec A)(1 + \operatorname{cosec} A)$?

SSC CGL 19/09/2025 (Shift 3)

- (a) $\frac{3\sqrt{3} + 6}{2\sqrt{3}}$ (b) $\frac{3\sqrt{3} + 4}{\sqrt{3}}$
(c) $\frac{\sqrt{3} + 6}{\sqrt{3}}$ (d) $\frac{3\sqrt{3} + 6}{\sqrt{3}}$

Q.115. If $\sec x + \tan x = \sqrt{2}$, what is the value of $\sec x - \tan x$?

SSC CGL 19/09/2025 (Shift 3)

- (a) $\frac{2}{\sqrt{2}}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{2\sqrt{2}}$ (d) $\frac{3}{\sqrt{2}}$

Q.116. If $\operatorname{Cosec} A = y$, then what is $\cot^2 A$ in terms of y?

SSC CGL 19/09/2025 (Shift 3)

- (a) $y^2 - 3$ (b) $y^2 - 2$ (c) $y^2 - 1$ (d) $2y^2 - 1$

Q.117. If $\cot A = \frac{11}{60}$, then what is $\operatorname{cosec} A$?

SSC CGL 20/09/2025 (Shift 1)

- (a) $\frac{61}{60}$ (b) $\frac{60}{61}$ (c) $\frac{61}{11}$ (d) $\frac{61}{59}$

Q.118. If $\sin x = 0.8$ and $x \in (0, \frac{\pi}{2})$, find $\sin 2x + \cos 2x$.

SSC CGL 20/09/2025 (Shift 1)

- (a) 0.68 (b) 1.52 (c) 1.68 (d) 1.04

Q.119. If $\tan \theta = \cot \theta$, what is the value of θ (in degrees)?

SSC CGL 20/09/2025 (Shift 2)

- (a) 45° (b) 60° (c) 90° (d) 55°

Q.120. If $\cos B = 1 - 2\sin^2 B$, then what is the value of $\cos 2B$?

SSC CGL 20/09/2025 (Shift 2)

- (a) $\sin B$ (b) $1 - 2\sin^2 B$
(c) $\sec B$ (d) Cannot be determined

Q.121. If $\sin(2x + 30^\circ) = \cos(5x - 3^\circ)$, find x.

SSC CGL 20/09/2025 (Shift 2)

- (a) 27° (b) 30° (c) 12° (d) 9°

Q.122. If $\sin A = \frac{5}{13}$ and A is acute, then what is the value of $\cos A$?

SSC CGL 20/09/2025 (Shift 3)

- (a) $\frac{12}{13}$ (b) $\frac{3}{5}$ (c) $\frac{\sqrt{7}}{2}$ (d) $\frac{1}{8}$

Q.123. If $\sin A = \frac{3}{5}$, find the value of $\sin^4 A + \cos^4 A$.

SSC CGL 20/09/2025 (Shift 3)

- (a) $\frac{337}{625}$ (b) $\frac{125}{196}$ (c) $\frac{361}{861}$ (d) $\frac{256}{289}$

Q.124. If $\sin A + \cos A = \sqrt{2}$, then find the value of $\sin 2A$.

SSC CGL 21/09/2025 (Shift 1)

- (a) 1 (b) 0 (c) -1 (d) $\frac{1}{2}$

Q.125. If $\sin A = \frac{3}{5}$ and $\cos A = \frac{4}{5}$, what

is $\tan A$?

SSC CGL 21/09/2025 (Shift 1)

- (a) $\frac{3}{4}$ (b) $\frac{4}{3}$ (c) $\frac{5}{4}$ (d) $\frac{2}{3}$

Q.126. If $\cos A = \frac{3}{5}$ and A is an acute angle, find $\tan(90^\circ - A)$.

SSC CGL 21/09/2025 (Shift 1)

- (a) $\frac{3}{4}$ (b) $\frac{4}{3}$ (c) $\frac{5}{4}$ (d) $\frac{2}{3}$

Q.127. If $\tan A = x + \frac{1}{x}$, then what is $\sec^2 A$?

SSC CGL 21/09/2025 (Shift 2)

- (a) $x^2 + \frac{1}{x^2} + 3$ (b) $x^2 + \frac{1}{x^2} - 1$
(c) $x^2 + \frac{1}{x^2} + 1$ (d) $x^2 + \frac{1}{x^2} - 2$

Q.128. If $\sec A + \tan A = \frac{3}{2}$, then what is $\sec A - \tan A$?

SSC CGL 21/09/2025 (Shift 2)

- (a) $\frac{2}{3}$ (b) $\frac{1}{2}$ (c) $\frac{3}{2}$ (d) $\frac{2}{5}$

Q.129. If $\sin A = \frac{1}{\sqrt{17}}$, then find the value of $\tan A + \sec A$.

SSC CGL 21/09/2025 (Shift 3)

- (a) $\frac{(1+\sqrt{17})}{4}$ (b) $\frac{(5+\sqrt{17})}{5}$
(c) $\frac{(6+\sqrt{17})}{3}$ (d) $\frac{(1+\sqrt{17})}{5}$

Q.130. Find the value of $\cos^2 90^\circ + \sin^2 60^\circ$.

SSC CGL 21/09/2025 (Shift 3)

- (a) $\frac{3}{4}$ (b) $\frac{11}{5}$ (c) $\frac{6}{5}$ (d) $\frac{7}{4}$

Q.131. If $\sin A = \frac{9}{41}$ and A lies in 2nd quadrant, then find the value of $\sin A \times \sec A$.

SSC CGL 22/09/2025 (Shift 1)

- (a) $\frac{9}{40}$ (b) $-\frac{9}{40}$ (c) $\frac{41}{40}$ (d) $-\frac{41}{40}$

Q.132. If $\tan A = \frac{x}{y}$ and $A \in (0, \frac{\pi}{2})$, then what is $\sin A$ in terms of x and y?

SSC CGL 22/09/2025 (Shift 1)

- (a) $\frac{\sqrt{x^2 - y^2}}{x}$ (b) $\frac{y}{\sqrt{x^2 + y^2}}$
(c) $\frac{x}{\sqrt{x^2 + y^2}}$ (d) $\frac{\sqrt{x^2 + y^2}}{y}$

Q.133. If $\sin A = \frac{3}{5}$, find the value of $(15\sin A + 20\cos A)^2 + (5\sin A + 10\cos A)^2$.

SSC CGL 22/09/2025 (Shift 2)

- (a) 746 (b) 625 (c) 713 (d) 734

Q.134. If $\sin \beta = \cos \beta$, then find $\sin^3 \beta + \cos^3 \beta$.

SSC CGL 22/09/2025 (Shift 2)

- (a) $\frac{\sqrt{2}}{2}$ (b) $\frac{\sqrt{3}}{5}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{5}}{3}$

Q.135. If $\tan 2\theta = \cot(3\theta + 10^\circ)$, then what is the value of θ ?

SSC CGL 22/09/2025 (Shift 2)

- (a) 30° (b) 25° (c) 16° (d) 18°

Q.136. If $\sin \theta + \cos \theta = \frac{\sqrt{3}}{2}$, find the value of $\sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta$.

SSC CGL 22/09/2025 (Shift 3)

- (a) 1 (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{3}{4}$ (d) $\frac{1}{2}$

Q.137. If $\sin C = \frac{1}{\sqrt{10}}$, then find $\cot C$ in simplest form.

SSC CGL 22/09/2025 (Shift 3)

- (a) $\sqrt{10}$ (b) $\frac{1}{\sqrt{10}}$ (c) 3 (d) $\frac{3}{\sqrt{10}}$

Q.138. If $\sin A = 0.8$, what is the value of $\tan A$?

SSC CGL 23/09/2025 (Shift 1)

- (a) 1.50 (b) 1.33 (c) 1.25 (d) 2.56

Q.139. If $\operatorname{cosec} A + \cot A = r$, then what is $\operatorname{cosec} A$ in terms of r?

SSC CGL 23/09/2025 (Shift 1)

- (a) $\frac{r^2+2}{2r}$ (b) $\frac{r^2-1}{2r}$ (c) $\frac{r^2+1}{2r}$ (d) $\frac{2r^2+1}{r}$

Q.140. Which of the following is true for all acute angles A ?

SSC CGL 23/09/2025 (Shift 1)

- (a) $\tan(90^\circ - A) = \sin A$
(b) $\cos(90^\circ - A) = \sin A$
(c) $\sin(90^\circ - A) = \tan A$
(d) $\cot(90^\circ - A) = \cos A$

Q.141. If $\cos \theta = \frac{5}{13}$ and θ lies in 2nd quadrant, find the value of $(1 + \sin \theta)(1 - \sin \theta)$.

SSC CGL 23/09/2025 (Shift 2)

- (a) $-\frac{19}{125}$ (b) $\frac{26}{163}$ (c) $\frac{21}{159}$ (d) $\frac{25}{169}$

Q.142. If $\cos x = \sin(3x - 10^\circ)$, then what is the value of x?

SSC CGL 23/09/2025 (Shift 2)

- (a) 25° (b) 35° (c) 28° (d) 10°

Q.143. If $\cos A = x$, then what is the value of $\cot A$ in terms of x ?

SSC CGL 23/09/2025 (Shift 3)

- (a) $\frac{x}{\sqrt{1-x^2}}$ (b) $\frac{\sqrt{1-x^2}}{x^2}$ (c) $\frac{2}{x}$ (d) $2x^2$

Q.144. If $\cos A = \frac{3}{5}$ and A lies in the 1st quadrant, what is the value of $\cot A$?

SSC CGL 23/09/2025 (Shift 3)

- (a) $-\frac{3}{4}$ (b) $\frac{3}{4}$ (c) $\frac{4}{5}$ (d) $\frac{7}{3}$

Q.145. If $\tan x = \cot(2x - 10^\circ)$, find the value of x.

SSC CGL 23/09/2025 (Shift 3)

- (a) 50° (b) 25.52° (c) 40° (d) 33.33°

Q.146. If $\sin A = \frac{35}{37}$ and A is acute, find $\cot A$.

SSC CGL 24/09/2025 (Shift 1)

- (a) $\frac{5}{37}$ (b) $\frac{12}{35}$ (c) $\frac{5}{12}$ (d) $\frac{37}{35}$

Q.147. If $A + B = 90^\circ$ and $\sin A = \frac{3}{5}$, determine the value of $\sin A \times \sin B$.

SSC CGL 24/09/2025 (Shift 1)

- (a) $\frac{16}{25}$ (b) $\frac{25}{12}$ (c) $\frac{12}{25}$ (d) $\frac{4}{5}$

Q.148. If $\cot A = \frac{4}{3}$, then what is the value of $\operatorname{cosec} A$?

SSC CGL 24/09/2025 (Shift 2)

- (a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{5}{3}$ (d) 1

Q.149. If $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$, then what is the value of $3\operatorname{cosec}^2 30^\circ - 3\cot^2 30^\circ$?

SSC CGL 24/09/2025 (Shift 2)

- (a) 5 (b) 1 (c) 3 (d) 6

Q.150. If $\sec(90^\circ - \theta) = y$ and θ is an acute angle, then what is the value of $\sin \theta$ in terms of y ?

SSC CGL 24/09/2025 (Shift 2)

- (a) y (b) $\sqrt{1 - \frac{1}{y^2}}$ (c) $\frac{1}{y}$ (d) Undefined

Q.151. If $\sec A - \tan A = \frac{1}{4}$, then what is the value of $\sec A + \tan A$?

SSC CGL 24/09/2025 (Shift 3)

- (a) 8 (b) 6 (c) 4 (d) 2

Q.152. If $\sin(2x) = \cos(3x - 60^\circ)$, what is the value of x ?

SSC CGL 24/09/2025 (Shift 3)

- (a) 30° (b) 25° (c) 18° (d) 20°

Q.153. If $\tan A + \cot A = \frac{4}{5}$, find $\tan^2 A + \cot^2 A + 2\tan A \cot A$.

SSC CGL 25/09/2025 (Shift 1)

- (a) $\sqrt{\frac{16}{25}}$ (b) $\frac{4}{5}$ (c) $\frac{17}{16}$ (d) $\frac{16}{25}$

Q.154. If $\cos 2A = 3$, then what is the value of $\cos A$?

SSC CGL 25/09/2025 (Shift 1)

- (a) 2 (b) 4 (c) $\sqrt{2}$ (d) $\sqrt{3}$

Q.155. If $\operatorname{cosec} x = \sec(4x - 40^\circ)$, find x.

SSC CGL 25/09/2025 (Shift 1)

- (a) 45° (b) 26° (c) 30° (d) 60°



Q.156. If $\cot A = 3$ and A is acute, then find $\cos A$.

SSC CGL 25/09/2025 (Shift 2)

- (a) $\frac{5}{\sqrt{10}}$ (b) $\frac{\sqrt{10}}{5}$ (c) $\frac{1}{\sqrt{10}}$ (d) $\frac{3}{\sqrt{10}}$

Q.157. If $\tan P = \cot Q$ and both P and Q are acute, what is $P + Q$?

SSC CGL 25/09/2025 (Shift 2)

- (a) 60° (b) 90° (c) 120° (d) 180°

Q.158. If $\cos A = \frac{4}{5}$, then what is $\sin^2 A + \cos^2 A - 2\sin A \cos A$?

SSC CGL 25/09/2025 (Shift 2)

- (a) 1 (b) 0 (c) $\frac{1}{25}$ (d) $\frac{16}{25}$

Q.159. If $\sec A - \cos A = 2$, then find the value of $\cos A$.

SSC CGL 25/09/2025 (Shift 3)

- (a) $-1 + \sqrt{2}$ (b) $3 + \sqrt{2}$
(c) $\sqrt{3} + 2$ (d) No real value

Q.160. If $\cos \theta = \frac{8}{17}$, and θ lies in the 3rd quadrant, find $\sin \theta$.

SSC CGL 25/09/2025 (Shift 3)

- (a) $-\frac{15}{17}$ (b) $\frac{12}{17}$ (c) $\frac{14}{17}$ (d) $-\frac{12}{17}$

Q.161. If $\cos \theta = \frac{3}{5}$ and θ is acute, then find the value of $\cot \theta \times \tan(90^\circ - \theta)$.

SSC CGL 25/09/2025 (Shift 3)

- (a) 1 (b) $\frac{9}{16}$ (c) $\frac{16}{25}$ (d) $\frac{25}{16}$

Q.162. If $4\tan A = 3$, then find the value of $(3\sin A - 4\cos A)$.

SSC CGL 26/09/2025 (Shift 1)

- (a) $-\frac{7}{5}$ (b) $-\frac{4}{5}$ (c) $\frac{3}{5}$ (d) $\frac{7}{5}$

Q.163. If $\sin 2x = \cos(3x - 20^\circ)$, what is the value of x ?

SSC CGL 26/09/2025 (Shift 1)

- (a) 35° (b) 37.5° (c) 22° (d) 25.8°

Q.164. If $\tan A + \sec A = 5$, find $\tan A$.

SSC CGL 26/09/2025 (Shift 1)

- (a) $\frac{13}{5}$ (b) $\frac{4}{3}$ (c) $\frac{12}{5}$ (d) $\frac{7}{5}$

Q.165. If $\sin A = \frac{2x}{(1+x^2)}$, then what is $\cot A$ in terms of x ?

SSC CGL 26/09/2025 (Shift 2)

- (a) $\frac{2x}{(1-x^2)}$ (b) x (c) $\frac{1}{x}$ (d) $\frac{(1-x^2)}{2x}$

Q.166. What is the radian measure of 160° ?

SSC CGL 26/09/2025 (Shift 2)

- (a) $\frac{8\pi}{9}$ (b) $\frac{\pi}{9}$ (c) $\frac{8\pi}{3}$ (d) $\frac{5\pi}{9}$

Q.167. If $\tan x = \cot y$, and $\cot x = \tan y$, then $x + y$ equals:

SSC CGL 26/09/2025 (Shift 2)

- (a) 45° (b) 60° (c) 90° (d) 180°

Q.168. In $\triangle ABC$, right-angled at C , if $\sin A = \frac{3}{5}$, then $\cos B =$?

SSC CGL 26/09/2025 (Shift 3)

- (a) $\frac{2}{13}$ (b) $\frac{21}{13}$ (c) $\frac{3}{5}$ (d) 1

Q.169. If $A + B = 90^\circ$, and $\sin A = \frac{35}{37}$, then what is $\cos B$?

SSC CGL 26/09/2025 (Shift 3)

- (a) $\frac{65}{63}$ (b) $\frac{53}{45}$ (c) $\frac{35}{37}$ (d) 0

Q.170. If $\tan A = \frac{7}{24}$, find $\sin A$.

SSC CGL 26/09/2025 (Shift 3)

- (a) $\frac{7}{25}$ (b) $\frac{4}{25}$ (c) $\frac{5}{3}$ (d) $\frac{15}{14}$

Q.171. If $\sec x - \tan x = \frac{1}{3}$, then what is the value of $\sec x + \tan x$?

SSC CGL 14/10/2025 (Shift 1)

- (a) 9 (b) 3 (c) 4 (d) 1

Q.172. If $\tan \theta = \frac{5}{12}$ What is $\sec \theta$?

SSC CGL 14/10/2025 (Shift 2)

- (a) $\frac{12}{5}$ (b) $\frac{11}{5}$ (c) $\frac{6}{5}$ (d) $\frac{13}{12}$

Q.173. If $\cos(2x + 15^\circ) = \sin(x - 5^\circ)$, what is the value of x ?

SSC CGL 14/10/2025 (Shift 2)

- (a) 45.57° (b) 60° (c) 30° (d) 26.67°

Q.174. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, A

and $B \in (0, \frac{\pi}{2})$, find $\tan(A - B)$.

SSC CGL 14/10/2025 (Shift 3)

- (a) $\frac{2}{7}$ (b) $\frac{1}{7}$ (c) $\frac{3}{7}$ (d) $\frac{4}{7}$

SSC Selection Post (XIII)

Q.175. What is the value of $\cos^4 \theta$ in terms of $\cos 2\theta$?

Higher Secondary 24/07/2025 (Shift - 2)

- (a) $\frac{(1 - \cos 2\theta)^2}{4}$
(b) $\frac{(1 + \cos 2\theta)^2}{4}$
(c) $1 + 2\cos 2\theta + \frac{\cos^2 2\theta}{4}$
(d) $\frac{(1 + \cos^2 2\theta)}{2}$

Q.176. If $\cos A = \frac{4}{5}$, and A is acute, then $\sin A + \cot A$ is:

Higher Secondary 25/07/2025 (Shift - 2)

- (a) $\frac{31}{20}$ (b) $\frac{29}{15}$ (c) $\frac{7}{5}$ (d) $\frac{22}{15}$

Q.177. If $\sin A = \frac{5}{13}$, then what is the value of $\cot A$?

Higher Secondary 25/07/2025 (Shift - 2)

- (a) $\frac{13}{5}$ (b) $\frac{12}{5}$ (c) $\frac{5}{12}$ (d) $\frac{13}{12}$

Q.178. If $\tan P = \cot Q$ and both P and Q are acute angles, then the value of $P + Q$ is:

Higher Secondary 25/07/2025 (Shift - 2)

- (a) 45° (b) 60° (c) 90° (d) 180°

Q.179. The value of $\sec^4 \theta + \tan^4 \theta$ is equal to:

Higher Secondary 26/07/2025 (Shift - 2)

- (a) 1 (b) $1 - 2\sec^2 \theta \tan^2 \theta$
(c) $\sec^2 \theta + \tan^2 \theta$ (d) $1 + 2\sec^2 \theta \tan^2 \theta$

Q.180. Let x and y be angles such that $\sin(x - y) = 1$ and $\cos(x + y) = 0$. What is the value of the expression:

$(\cos x + \cos y)^2 + (\sin x + \sin y)^2$?

Higher Secondary 28/07/2025 (Shift - 2)

- (a) 4 (b) 2 (c) 0 (d) 1

Q.181. If $\tan \theta - \cot \theta = k$, then $\tan^2 \theta + \cot^2 \theta$ is equal to:

Higher Secondary 28/07/2025 (Shift - 2)

- (a) $k^2 + 2$ (b) $k^2 - 2$ (c) $k + 2$ (d) $k - 2$

Q.182. If $\sec \theta + \tan \theta = p$, what is the value of $\sin \theta$ in terms of p ?

Higher Secondary 29/07/2025 (Shift - 2)

- (a) $\frac{(p^2 + 1)}{(p^2 - 1)}$ (b) $\frac{(p^2 - 1)}{2p}$
(c) $\frac{(p^2 - 1)}{(p^2 + 1)}$ (d) $\frac{(p - 1)}{(p + 1)}$

Q.183. If $5\cos^2 \theta + 4\sin^2 \theta = \frac{17}{4}$, then $\cos^2 \theta$ is:

Higher Secondary 30/07/2025 (Shift - 2)

- (a) $\frac{3}{4}$ (b) $\frac{2}{3}$ (c) $\frac{1}{4}$ (d) $\frac{5}{7}$

Q.184. If $\sin \theta + \cos \theta = \frac{7}{5}$, and θ is acute, what is the value of $\sin^4 \theta + \cos^4 \theta$?

Higher Secondary 31/07/2025 (Shift - 2)

- (a) $\frac{361}{25}$ (b) $\frac{144}{125}$ (c) $\frac{529}{50}$ (d) $\frac{337}{625}$

Q.185. If $\sin(90^\circ - \theta) = \frac{3}{4}$, what is $\cos \theta$?

Higher Secondary 31/07/2025 (Shift - 2)

- (a) $\frac{5}{4}$ (b) $\frac{4}{3}$ (c) 1 (d) $\frac{3}{4}$

Q.186. If $\sec(90^\circ - x) = \operatorname{cosec}(2x - 30^\circ)$, and x is acute, find the value of x .

Higher Secondary 31/07/2025 (Shift - 2)

- (a) 90° (b) 60° (c) 30° (d) 45°



Q.187. For an angle measuring precisely 45 degrees, determine the numeric value of its sine multiplied by its cosine.

Higher Secondary 02/08/2025 (Shift - 2)

- (a) $\frac{1}{3}$ (b) 0 (c) $\frac{1}{2}$ (d) $\frac{1}{\sqrt{2}}$

Q.188. If $\sin\theta = \frac{8}{17}$ and θ lies in the first quadrant, find the value of $\cos\theta + \sin\theta$.

Higher Secondary 29/08/2025 (Shift - 2)

- (a) $-\frac{8}{17}$ (b) $\frac{8}{15}$ (c) $\frac{23}{17}$ (d) $-\frac{23}{17}$

Q.189. If $\cos\theta = \sin 2\theta$ if θ is an acute angle, find the value of $\sin\theta$

Higher Secondary 29/08/2025 (Shift - 2)

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{\sqrt{3}}{2}$

Q.190. If $\sin\theta - \cos\theta = \frac{1}{5}$ and $\theta \in (0, \frac{\pi}{2})$, then what is $\sin 2\theta$?

Graduate Level 24/07/2025 (Shift - 3)

- (a) $\frac{12}{25}$ (b) $\frac{23}{25}$ (c) $\frac{24}{25}$ (d) $\frac{1}{25}$

Q.191. If $\sin A = \frac{24}{25}$, and A is acute, find $\cot A$.

Graduate Level 25/07/2025 (Shift - 3)

- (a) $\frac{24}{7}$ (b) $\frac{25}{7}$ (c) $\frac{24}{25}$ (d) $\frac{7}{24}$

Q.192. If $\sec A + \tan A = 8$ and A is acute, find the value of $\sec A - \tan A$.

Graduate Level 26/07/2025 (Shift - 3)

- (a) 0.125 (b) 0.215 (c) 1.125 (d) 0.135

Q.193. If $\cot\theta = \frac{40}{41}$, and θ is acute, find the value of $(\sec^2\theta - 1)$.

Graduate Level 28/07/2025 (Shift - 3)

- (a) $\frac{81}{1600}$ (b) $\frac{1681}{1600}$ (c) $\frac{1681}{81}$ (d) $\frac{16}{81}$

Q.194. If $\operatorname{cosec}\theta = \frac{3}{4}$, find the value of $1 + \cot^2\theta$.

Graduate Level 31/07/2025 (Shift - 3)

- (a) $\frac{25}{9}$ (b) 1 (c) $\frac{16}{9}$ (d) $\frac{9}{16}$

Q.195. If $\operatorname{cosec} A + \cot A = 5$ and A is acute, then find the value of $\operatorname{cosec} A$.

Graduate Level 02/08/2025 (Shift - 3)

- (a) 3.4 (b) 4.5 (c) 1.4 (d) 2.6

eduquity-based pattern (ebp) Questions

Q.196. Find the value of $(\sin^6 A + \cos^6 A + 3\sin^2 A \cos^2 A) - (\sin^4 A - \cos^4 A + 1) \operatorname{cosec}^2 A$.

- (a) 3 (b) -3 (c) 0 (d) -1

Q.197. Find the value of $3(\sec^2 A - 1)(\operatorname{cosec} A - 1)(\operatorname{cosec} A + 1) + 4(\operatorname{cosec}^2 A - \cot^2 A)$.

- (a) 12 (b) 3 (c) 10 (d) 7

Q.198. The value of $\sin^2 30^\circ - \sin^2 40^\circ + \sin^2 45^\circ - \sin^2 55^\circ - \sin^2 35^\circ + \sin^2 45^\circ - \sin^2 50^\circ + \sin^2 60^\circ$ is:

- (a) 1 (b) 2 (c) 4 (d) 0

Q.199. The value of $\cos^2 29^\circ + \cos^2 61^\circ$ is:

- (a) $\frac{\sqrt{3}}{2}$ (b) 2 (c) 1 (d) 0

Q.200. If $\operatorname{cosec} \theta = \frac{5}{3}$, then evaluate

$$(\sec^2 \theta - 1) \times \cot^2 \theta \times (1 + \cot^2 \theta)$$

- (a) $\frac{25}{16}$ (b) $\frac{9}{16}$ (c) $\frac{25}{9}$ (d) $\frac{4}{5}$

Q.201. Simplify the following. $\frac{\sin^3 A - \cos^3 A}{\sin A - \cos A}$ where A is an acute angle.

- (a) $1 + \sin A \cos A$ (b) $1 - 3 \sin A$
(c) $3 \cos A - 1$ (d) $\sin A + \cos A$

Q.202. If $\frac{\sec \alpha + \tan \alpha}{\sec \alpha - \tan \alpha} = \frac{7}{4}$, then find the value of $\operatorname{cosec} \alpha$.

- (a) $\frac{1}{3}$ (b) $\frac{1}{11}$ (c) $\frac{11}{3}$ (d) $\frac{3}{11}$

Q.203. If A is an acute angle, which of the following is equal to $\frac{\sin A}{1 + \cos A}$?

- (a) $\frac{1 - \cos A}{\sin A}$ (b) $\frac{1 + \cos A}{\sin A}$
(c) $\frac{1 - \sin A}{\cos A}$ (d) $\frac{1 + \sin A}{\cos A}$

Q.204. If $\frac{7\sin\theta + 4\cos\theta}{9\sin\theta - 2\cos\theta} = \frac{5}{4}$, then the value of $\frac{\tan^2\theta + 5}{\tan^2\theta - 5}$ is:

- (a) $\frac{2121}{769}$ (b) $-\frac{2121}{769}$ (c) $\frac{769}{2121}$ (d) $-\frac{769}{2121}$

Q.205. $\frac{2 + \tan^2 2A + \cot^2 2A}{\sec 2A \operatorname{cosec} 2A}$ is equal to.

- (a) $\sec 2A + \tan 2A$ (b) $\sec 2A + \operatorname{cosec} 2A$
(c) $\cot 2A - \operatorname{cosec} 2A$ (d) $\sec 2A \operatorname{cosec} 2A$

Q.206. Let $0^\circ < t < 90^\circ$. Then which of the following is true?

- (a) $\sin(t) \neq \cos(t)$ when $t = 45^\circ$
(b) $\sin(t) > \cos(t)$ when $t < 45^\circ$
(c) $\sin(t) < \cos(t)$ when $t < 45^\circ$
(d) $\sin(t) < \cos(t)$ when $t > 45^\circ$

Q.207. If $2 \tan \theta = 3$, then $\frac{3 \sin \theta - 2 \cos \theta}{3 \sin \theta + 2 \cos \theta}$ is equal to:

- (a) $\frac{6}{13}$ (b) $\frac{6}{7}$ (c) $\frac{5}{7}$ (d) $\frac{5}{13}$

Q.208. If $\cot \theta = \frac{4}{3}$, then evaluate $\frac{(\sec \theta (1 + \cot^2 \theta) (\operatorname{cosec}^2 \theta - \cot^2 \theta))}{\operatorname{cosec}^3 \theta}$.

- (a) $\frac{3}{4}$ (b) $\frac{4}{5}$ (c) $\frac{4}{3}$ (d) $\frac{3}{5}$

Q.209. $\frac{3 \sin 58^\circ}{5 \cos 32^\circ} + \frac{2 \cos 62^\circ}{5 \sin 28^\circ}$ is equal to:

- (a) 0 (b) 2 (c) 3 (d) 1

Q.210. If $7 \cos^2 \theta + 5 \sin^2 \theta - 6 = 0$, ($0^\circ < \theta < 90^\circ$), then what is the value of $1 + \sqrt{\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta}}$?

- (a) $\sqrt{2} + 2$ (b) $\sqrt{2} + 1$ (c) $\sqrt{2} - 1$ (d) $\sqrt{2} - 2$

Q.211. Find the value of

$$\frac{\sec 35^\circ}{\operatorname{cosec} 55^\circ} - \frac{\cos 25^\circ}{\sin 65^\circ}$$

- (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) 2

Q.212. Evaluate the value of $(\operatorname{cosec} 56^\circ \cos 34^\circ - \cos 59^\circ \operatorname{cosec} 31^\circ)$.

- (a) 1 (b) 2 (c) -1 (d) 0

Q.213. Find the value of: $[\frac{4}{3} \tan^2 60^\circ + 3$

$$\cos^2 30^\circ - 2 \sec^2 30^\circ - \frac{3}{4} \cot^2 60^\circ] \div [\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ]$$

- (a) $6\frac{2}{3}$ (b) $3\frac{1}{2}$ (c) $4\frac{1}{3}$ (d) $6\frac{3}{2}$

Q.214. Express $\sin 74^\circ + \tan 74^\circ$ in terms of trigonometric ratios of angles between 0° and 45° .

- (a) $\sec 16^\circ + \cot 16^\circ$
(b) $\operatorname{cosec} 16^\circ + \sec 16^\circ$
(c) $\operatorname{cosec} 16^\circ + \cot 16^\circ$
(d) $\cos 16^\circ + \cot 16^\circ$

Q.215. If $p \sin A - \cos A = 1$, then $p^2 - (1 + p^2) \cos A$ equals:

- (a) 1 (b) -1 (c) 2 (d) 0

Q.216. The value of

$$\left(\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta}\right)^3 \frac{1}{\tan \theta} - \sec^2 \theta$$
 is:

- (a) 2 (b) 1 (c) 0 (d) -1

Q.217. If $2 \operatorname{cosec}^2 \theta + 3 \cot^2 \theta = 17$, then the value of ' θ ' when $0^\circ \leq \theta \leq 90^\circ$ is:

- (a) 45° (b) 90° (c) 30° (d) 75°

Q.218. Solve the following:

$$\frac{1}{1 + \cos \theta} + \frac{1}{1 - \cos \theta} = ?$$

- (a) $2 \operatorname{cosec}^2 \theta$ (b) $2 \sin^2 \theta$
(c) $2 \sec^2 \theta$ (d) $2 \tan^2 \theta$

Q.219. If $7 \tan \theta = 3$, then

$$\frac{7 \sin \theta + 5 \cos \theta}{7 \sin \theta - 2 \cos \theta}$$
 is:

- (a) 7 (b) 5 (c) 8 (d) 6

Q.220. If $\cot^4 \phi - \cot^2 \phi = 1$, then the value

of $\cos^4 \phi + \cos^2 \phi$ is:

- (a) 2 (b) 1 (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{3}{2}$

Q.221. If $\tan 4\theta \cdot \tan 6\theta = 1$, where 6θ is an acute angle, then find the value of $\cot 5\theta$.

- (a) $\sqrt{3}$ (b) $-\sqrt{3}$ (c) -1 (d) 1

Q.222. If $7 \cos A = 6$, then the numerical value of $\frac{\operatorname{cosec} A + \cot A}{\operatorname{cosec} A - \cot A}$ is:

- (a) 1 (b) 0 (c) 13 (d) -13

Q.223. If $p(\sqrt{3} + \cot 30^\circ) = \tan^3 60^\circ - 2 \sin 60^\circ$, then the value of p is:

- (a) -1 (b) $2\sqrt{3}$ (c) 1 (d) $\sqrt{3}$

Q.224. If $\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} = 4 \sec \theta$, ($0 < \theta < 90^\circ$), then the value of $(\cot \theta + \operatorname{cosec} \theta)$ is:

- (a) $\sqrt{3}$ (b) $\sqrt{6}$ (c) $\sqrt{2}$ (d) $\sqrt{5}$

Q.225. If $\sin 31^\circ = \alpha$, then the value of $\cot 59^\circ$ is:

- (a) $\frac{\sqrt{1 - \alpha^2}}{2\alpha}$ (b) $\frac{\alpha}{\sqrt{1 - \alpha^2}}$
(c) $\frac{\sqrt{1 + \alpha^2}}{\alpha}$ (d) $\frac{\alpha}{\sqrt{1 + \alpha^2}}$

Q.226. If $\frac{\tan(x)}{\sec(x)} = \frac{1}{2}$, then the value of $(\sin(x) + \cos(x))^2$ is _____.

- (a) $\frac{2 + \sqrt{3}}{3}$ (b) $\frac{2 + \sqrt{3}}{2}$
(c) $\frac{2 - \sqrt{3}}{2}$ (d) $\frac{2 - \sqrt{3}}{3}$

Q.227. If $\tan A = \frac{5}{7}$, then find the value of $\frac{7 \sin A - 3 \cos A}{7 \sin A + 3 \cos A} + 4$.

- (a) $4\frac{1}{4}$ (b) $3\frac{5}{7}$ (c) $3\frac{1}{5}$ (d) $7\frac{1}{3}$

Q.228. If $\sin B = \frac{15}{17}$, what is the value of $\cos B(\sec B - \tan B)$? Given that $0 < B < \frac{\pi}{2}$

- (a) $\frac{1}{16}$ (b) $\frac{2}{15}$ (c) $\frac{8}{15}$ (d) $\frac{2}{17}$

Q.229. The value of $\left(\frac{\operatorname{cosec} A}{\operatorname{cosec} A - 1} + \frac{\operatorname{cosec} A}{\operatorname{cosec} A + 1}\right) \left(\frac{\sec A}{\sec A - 1} + \frac{\sec A}{\sec A + 1}\right)^{-1}$, when $A = 60^\circ$ is:

- (a) 4 (b) 3 (c) 1 (d) 2

Q.230. In a right angle $\triangle ABC$, $\angle B = 90^\circ$, if $\tan A = 1$, then $4 \sin A \cos A =$ _____.

- (a) 1 (b) 2 (c) $\frac{1}{2}$ (d) 4

Q.231. If $\sec \theta = \frac{29}{20}$ where $0 < \theta < 90^\circ$, then what is the value of $3 \operatorname{cosec} \theta + 3 \cot \theta$?

- (a) $\frac{7}{3}$ (b) 7 (c) 14 (d) 49

Q.232. If $\cos 24^\circ = \frac{m}{n}$, then the value of $(\operatorname{cosec} 24^\circ - \cos 66^\circ)$ is:

- (a) $\frac{m^2}{n\sqrt{m^2 - n^2}}$ (b) $\frac{n^2}{m\sqrt{m^2 - n^2}}$
(c) $\frac{n^2}{m\sqrt{n^2 - m^2}}$ (d) $\frac{m^2}{n\sqrt{n^2 - m^2}}$

Q.233. Find the value of

$$\left[\frac{4 \cos(90 - \theta) \sin^3(90 + \theta) - 4 \sin(90 + \theta) \cos^3(90 - \theta)}{\cos(-\frac{180 + 8\theta}{2})} \right]$$

- (a) -1 (b) 1 (c) 2 (d) 0

Q.234. If $\sin A + \cos A = \frac{1}{2\sqrt{2}}$, then find the value of $\sin^4 A + \cos^4 A$.

- (a) $\frac{79}{128}$ (b) $\frac{7}{13}$ (c) $\frac{79}{126}$ (d) $\frac{5}{13}$

Q.235. If $a = x \cos \theta + y \sin \theta$ and $b = x \sin \theta - y \cos \theta$, then $a^2 + b^2$ is equal to:

- (a) $x + y$ (b) $x^2 - y^2$ (c) $x^2 + y^2$ (d) $x - y$

Q.236. If $\operatorname{cosec} x = \frac{5}{4}$, then the value of

$$\frac{5 \sin^2 x + 8 \cot^2 x}{27 \tan^2 x - 125 \cos^2 x}$$
 is:

- (a) $\frac{33}{91}$ (b) $\frac{77}{30}$ (c) $\frac{55}{87}$ (d) $\frac{99}{47}$

Q.237. If $\sin \theta + \cos \theta = \sqrt{3} \cos \theta$, then the value of $\cot \theta$ is:

- (a) $\sqrt{3} - 1$ (b) $\frac{(\sqrt{3} + 1)}{2}$
(c) $\frac{\sqrt{3} - 1}{2}$ (d) $\sqrt{3} + 1$

Q.238. If $\sec(t) = \frac{x}{y}$, then $\cot(t) = ?$

- (a) $\frac{y}{\sqrt{x^2 - y^2}}$ (b) $\frac{x}{\sqrt{x^2 + y^2}}$
(c) $\frac{x}{\sqrt{x^2 - y^2}}$ (d) $\frac{y}{\sqrt{x^2 + y^2}}$

Q.239. At what angle θ , the value of $\sin(\theta)$ and $\cos(\theta)$ will have the same value?

- (a) 45° (b) 30° (c) 60° (d) 75°

Q.240. If $3 \sin A + 4 \cos A = 5$, then the value of $\tan A$ is:

- (a) $\frac{3}{4}$ (b) $\frac{3}{5}$ (c) $\frac{5}{4}$ (d) $\frac{4}{5}$

Q.241. Find the value of $\frac{\tan 55^\circ}{\cot 35^\circ}$

- (a) $\cos 35^\circ$ (b) 1 (c) $\sec 55^\circ$ (d) 0

Q.242. Simplify. $\frac{\sin A}{1 + \cos A} + \cot A$, where A is an acute angle.

- (a) $\sin A$ (b) $\cos A$
(c) $\sin A + \cos A$ (d) $\operatorname{cosec} A$

Q.243. If $\tan \theta = \frac{8}{15}$, and θ is an acute

angle, then the value of $\frac{\sqrt{1 - \sin \theta}}{\sqrt{1 + \sin \theta}}$ is:

- (a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) $\frac{1}{5}$ (d) $\frac{4}{5}$

Q.244. The given expression is equivalent to:

$$\sqrt{\frac{\sec \theta + 1}{\sec \theta - 1}} + \sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}}$$

- (a) $2 \sin \theta$ (b) $2 \operatorname{cosec} \theta$
(c) $2 \tan \theta$ (d) $2 \tan \theta \sec \theta$

Q.245. What is the value of

$$\sqrt{\frac{9 \sin^2 26^\circ + 9 \sin^2 64^\circ + 36}{32 - 7 \cos^2 32^\circ - 7 \cos^2 58^\circ}}?$$

- (a) $\frac{3\sqrt{5}}{5}$ (b) $\frac{5\sqrt{3}}{3}$ (c) $\frac{5\sqrt{5}}{3}$ (d) $\frac{3\sqrt{3}}{5}$

Q.246. The value of the expression $\sin x + \operatorname{cosec} x = 2$, then the value of $\sin^7 x + \operatorname{cosec}^7 x$ is:

- (a) 0 (b) 1 (c) 2 (d) 4

Q.247. If $\sin A + \cos A = \sqrt{3}$, then find the value of $\tan A + \cot A + 2 \sin A \cos A$.

- (a) 6 (b) 2 (c) 1 (d) 3

Q.248. If $3 \sin^2 \theta + 5 \cos^2 \theta = 4$ and θ is an acute angle, then the value of $\tan \theta$ is:

- (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\frac{1}{2}$ (d) 1

Q.249. The maximum value of $(2 \sin \theta + 3 \cos \theta)$ is:

- (a) $\sqrt{17}$ (b) $\sqrt{11}$ (c) 9 (d) $\sqrt{13}$

Q.250. Simplify $[1 - \sin^2 32^\circ + \frac{1}{1 + \tan^2 58^\circ}]$.

- (a) $\frac{1}{2}$ (b) $\sin 32^\circ$ (c) 1 (d) $\sqrt{3}$

Q.251. If $\tan \theta = \frac{5}{8}$, then find the value of

$$\frac{(1 + \cos \theta)(1 - \cos \theta)}{(1 + \sin \theta)(1 - \sin \theta)}$$

- (a) $\frac{5}{64}$ (b) $\frac{2}{25}$ (c) $\frac{25}{64}$ (d) $\frac{64}{25}$

Q.252. If $\tan \theta + \cot \theta = 2$, and θ is an acute angle, then the value of θ is:

- (a) 30° (b) 15° (c) 60° (d) 45°

Q.253. If A is an acute angle, then

$$\sqrt{\frac{1 - \cos A}{1 + \cos A}} + \sqrt{\frac{1 + \cos A}{1 - \cos A}}$$
 is equal to:

- (a) $2 \operatorname{cosec} A$ (b) $2 \cos A$
(c) $2 \sec A$ (d) $2 \sin A$

Q.254. The value of θ , when $\sqrt{3} \cos \theta + \sin \theta = 1$ ($1 \leq \theta \leq 90^\circ$), is:

- (a) 20° (b) 60° (c) 90° (d) 30°

Q.255. Let $x = r \cos(t)$, $y = r \sin(t) \cos(u)$, $z = r \sin(t) \sin(u)$. Then the value of $x^2 + y^2 + z^2$ is _____.

- (a) $r^2 \cos(u)$ (b) r^2 (c) $2r$ (d) $r^2 \sin(t)$

Q.256. If $4\tan\theta - 3 = 0$, then the value of $\frac{1 - \cos 2\theta}{1 + \cos 2\theta}$ is:

- (a) 1 (b) $\frac{7}{15}$ (c) $\frac{4}{3}$ (d) $\frac{9}{16}$

Q.257. If $\sin A - \cos A = \frac{\sqrt{3} - \sqrt{2}}{2}$, then the value of $\sin A \cdot \cos A$ is:

- (a) $\frac{2\sqrt{3} - 1}{4}$ (b) $\frac{2\sqrt{2} - 1}{8}$
(c) $\frac{2\sqrt{6} - 1}{8}$ (d) $\frac{3\sqrt{2} - 1}{4}$

Q.258. If $\cos A + \cos^2 A = 1$ then $\sin^2 A + \sin^4 A$ is equal to:

- (a) $\frac{\cos^2 A}{\cos A}$ (b) 0 (c) $\frac{\cos A}{\cos^2 A}$ (d) 1

Q.259. The value of $\frac{1}{\operatorname{cosec}\theta - \cot\theta} - \frac{1}{\sin\theta}$ is equal to:

- (a) $\tan\theta$ (b) $\operatorname{cosec}\theta$ (c) $\cot\theta$ (d) $\sec\theta$

Q.260. The value of $\frac{\sin^2\theta - 2\sin^4\theta}{2\cos^4\theta + \cos^2\theta}$ at $\theta = 45^\circ$

- (a) 1 (b) 0 (c) 3 (d) 2

Q.261. For what value of t is the value of $\sin^2(t)$, half of the value of $\tan(t)$?

- (a) 60° (b) 45° (c) 30° (d) 22.5°

Q.262. If $\tan A = \frac{1}{\sqrt{10}}$, A is an acute angle, then the value of $\sin A + \operatorname{cosec} A$ is:

- (a) $\frac{11}{\sqrt{10}}$ (b) $\frac{1}{\sqrt{10}}$ (c) $\frac{12}{\sqrt{11}}$ (d) $\frac{10}{\sqrt{11}}$

Q.263. If $\sec \theta = \sqrt{2}$, then $\operatorname{cosec} \theta$ is equal to:

- (a) 0 (b) $\sqrt{2}$ (c) -1 (d) 1

Q.264. If $2\sin^2\theta + 3\sin\theta - 2 = 0$, ($0 < \theta < 90^\circ$), then the value of θ is:

- (a) 15° (b) 30° (c) 60° (d) 45°

Q.265. If $\sin 3A = \cos(A - 26^\circ)$, then $A =$ _____, where $3A$ is an acute angle.

- (a) 29° (b) 52° (c) 26° (d) 78°

Q.266. The value of $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2$ is:

- (a) $3 + \tan^2 A + \cot^2 A$ (b) $7 + \tan^2 A + \cot^2 A$
(c) $5 + \tan^2 A + \cot^2 A$ (d) $1 + \tan^2 A + \cot^2 A$

Q.267. If $3\tan A = 4$ and A is an acute angle, then the value of $4\sin A + 3\cos A$ is:

- (a) 1 (b) 3 (c) 4 (d) 5

Q.268. If $10\sin^2\theta + 6\cos^2\theta = 7$, $0 < \theta < 90^\circ$ then find the value of $\tan\theta$.

- (a) $\sqrt{3}$ (b) 1 (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{2}$

Q.269. If $A = 15^\circ$, then What is the value

of $\frac{11\sqrt{2}\cos 3A + 10\sin 2A}{7\sqrt{3}\sin 4A}$?

- (a) $\frac{11}{7}$ (b) $\frac{21}{32}$ (c) $\frac{32}{21}$ (d) $\frac{7}{11}$

Q.270. Evaluate $2 \frac{\tan 54^\circ}{\cot 36^\circ} - \frac{\cot 41^\circ}{\tan 49^\circ}$.

- (a) -1 (b) 1 (c) 0 (d) 2

Q.271. Find the value of $(\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)(\tan \theta + \cot \theta)$.

- (a) 0 (b) $\frac{1}{2}$ (c) 1 (d) -1

Q.272. If $m + 5 = \sec^2 A \operatorname{cosec}^2 A (1 - \cos^2 A)$ ($1 - \sin^2 A$) + 5, then find the value of m :

- (a) 10 (b) 0 (c) 1 (d) 5

Q.273. If $\cot^2\theta - 2\cos^2\theta = 0$, ($0^\circ < \theta < 90^\circ$), then the value of θ is:

- (a) 45° (b) 60° (c) 90° (d) 30°

Q.274. If $x = p \sec A \cos B$, $y = q \sec A \sin B$ and $z = r \tan A$, what is the value of the following expression?

$$\frac{x^2}{p^2} + \frac{y^2}{q^2} - \frac{z^2}{r^2}$$

- (a) 1 (b) $p^2 + q^2 - r^2$ (c) $p^2 - q^2 + r^2$ (d) 0

Q.275. Find the value of $3\sin 15^\circ - 4\sin^3 15^\circ$

- (a) 2 (b) $\frac{1}{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$

Q.276. If $\sec\theta + \tan\theta = x$, then find $\sin\theta$.

- (a) $\frac{x^2 + 1}{1 - x^2}$ (b) $\frac{x^2 - 1}{1 + 2x^2}$
(c) $\frac{x^2 - 1}{1 + x^2}$ (d) $\frac{1 - x^2}{1 + x^2}$

Q.277. If $(48^\circ + k)$ is an acute angle and $\sin(48^\circ + k) = \cos 13^\circ$, what is the value of k (in $^\circ$)?

- (a) 17 (b) 23 (c) 29 (d) 37

Q.278. Evaluate the given expression.

$$\frac{5}{1 + \cot^2\theta} + \frac{3}{1 + \tan^2\theta} + 2\cos^2\theta$$

- (a) 0 (b) 3 (c) 5 (d) 2

Q.279. If A is an acute angle and $\tan A + \cot A = 2$, find the value of $7\tan^8 A - 6\cot^8 A + 8\sec^2 A$.

- (a) 6 (b) 17 (c) 16 (d) 7

Q.280. If $p = \frac{\sin A}{1 + \cos A}$, then $\frac{\sin A}{1 - \cos A} = ?$

- (a) $\frac{1}{p}$ (b) $\frac{1}{1 - p}$ (c) $\frac{1}{p + 1}$ (d) $p - 1$

Q.281. If $8\cot\theta = 7$, then the value of $\frac{1 + \sin\theta}{\cos\theta}$ is:

- (a) $\frac{8 + \sqrt{113}}{7}$ (b) $\frac{1 + \sqrt{113}}{8}$

- (c) $\frac{7 + \sqrt{113}}{8}$ (d) $\frac{1 + \sqrt{113}}{7}$

Q.282. In $\triangle ABC$, right angled at B if $\tan C = \sqrt{3}$, then find $\frac{\sin^2 C + \cos^2 C}{1 + \cot^2 C}$

- (a) $\frac{3}{4}$ (b) $\frac{16}{3}$ (c) $\frac{4}{15}$ (d) $\frac{3}{16}$

Q.283. The greatest value of $\sin^4\theta + \cos^4\theta$ is:

- (a) 2 (b) 1 (c) 3 (d) 4

Q.284. If $x = a \sec\theta \cos\phi$, $y = b \sec\theta \sin\phi$ and $z = c \tan\theta$, then the value of $(\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2})$ is equal to:

- (a) 2 (b) 1 (c) 3 (d) 0

Q.285. If $\cot \theta = \sqrt{11}$, then the value of $\frac{\operatorname{cosec}^2\theta - \sec^2\theta}{\operatorname{cosec}^2\theta + \sec^2\theta}$ is:

- (a) $\frac{3}{5}$ (b) $\frac{5}{6}$ (c) $\frac{4}{5}$ (d) $\frac{6}{7}$

Q.286. In right-angled triangle ABC , $\angle B = 90^\circ$ and angle A and angle C are acute angles.

If $\operatorname{cosec} A = 2\sqrt{2}$, then find the value of $\sin A \cdot \cos C + \cos A \cdot \sin C$.

- (a) $\sqrt{2}$ (b) 1 (c) 0 (d) $2\sqrt{2}$

Q.287. If $\sin\theta = \frac{2xy}{x^2 + y^2}$, then the value of $\tan\theta$ is:

- (a) $\frac{2xy}{x^2 - y^2}$ (b) $\frac{x^2 - y^2}{2xy}$
(c) $\frac{x^2 - y^2}{x^2 + y^2}$ (d) $\frac{x^2 + y^2}{x^2 - y^2}$

Q.288. $12\sin^2 A + 2\cos^2 A = 7$, where $0^\circ < A < 180^\circ$. Then $\cot A$ is:

- (a) $\pm \frac{1}{\sqrt{2}}$ (b) ± 1 (c) $\pm \frac{1}{\sqrt{3}}$ (d) $\pm \frac{1}{2}$

Q.289. What is the value of $\cos(x + y) - \sin(x - y) + \tan(2z)$?

- (a) $\cos x \cdot \cos y - \sin x \cdot \sin y - \sin x \cdot \cos y - \cos x \cdot \sin y + \frac{2 \tan z}{1 - \tan^2 z}$
(b) $\sin x \cdot \sin y + \frac{2 \tan z}{1 - \tan^2 z}$
(c) $\cos x \cdot \cos y - \sin x \cdot \sin y - \sin x \cdot \cos y + \cos x \cdot \sin y + \frac{2 \tan z}{1 - \tan^2 z}$
(d) $\cos x \cdot \cos y - \sin x \cdot \sin y - \cos x \cdot \sin y + \frac{\tan z}{1 - \tan^2 z}$

Q.290. Given that $\sin \theta = \frac{15}{17}$, Evaluate $\frac{15 \cot \theta + 17 \sin \theta}{8 \tan \theta + 16 \sec \theta}$, where $0^\circ < \theta < 90^\circ$.

- (a) $\frac{23}{49}$ (b) $\frac{49}{26}$ (c) $\frac{26}{23}$ (d) $\frac{49}{23}$

Q.291. If θ is an acute angle and $\tan\theta + \cot\theta = 2$, then find the value of $\tan^3\theta + \cot^3\theta + 6\tan^3\theta\cot^2\theta$.

- (a) 8 (b) 10 (c) 12 (d) 6

Q.292. If $\sqrt{13} \sin\theta = 2$, then the value of $\frac{3 \tan\theta + \sqrt{13} \sin\theta}{\sqrt{13} \cos\theta - 3 \tan\theta}$ is :

- (a) 2 (b) 3 (c) 0 (d) 4

Q.293. $(\cot\theta + \tan\theta)(\operatorname{cosec}\theta - \sin\theta)(\cos\theta - \sec\theta) =$ _____.

- (a) -1 (b) 2 (c) 0 (d) 1

Q.294. Find the value of the following expression.

$$\sin(30^\circ + \theta) - \cos(60^\circ - \theta)$$

- (a) $\frac{\sqrt{3}}{2}$ (b) $2\sin(30^\circ + \theta)$ (c) 1 (d) 0

Q.295. If $\alpha + \beta = 45^\circ$ and $(\tan\alpha + 1)(\tan\beta + 1) = 2x$, then x is:

- (a) -1 (b) 0 (c) 1 (d) 2

Q.296. What is the value of $\tan 570^\circ$?

- (a) $-\frac{1}{\sqrt{3}}$ (b) $-\sqrt{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$

Q.297. What is the value of

$$\frac{1}{\sin^4(90^\circ - 2\alpha)} + \frac{1}{\cos^2(90^\circ - 2\alpha) - 1} ?$$

- (a) $\sin^2 2\alpha + \tan^2 2\alpha$ (b) $\sec^2 2\alpha + \cot^2 2\alpha$
(c) $\cos^2 2\alpha \cot^2 2\alpha$ (d) $\sec^2 2\alpha \tan^2 2\alpha$

Q.298. If θ is an acute angle and $\cot\theta + \tan\theta = 2$, then find the value of $\tan^{12}\theta + \cot^{12}\theta + 2\tan^5\theta\cot^7\theta$.

- (a) 2 (b) 3 (c) 4 (d) 1

Q.299. If θ be acute angle and $\cos\theta = \frac{24}{25}$, then the value of $\cot(90^\circ - \theta)$ is equal to:

- (a) $\frac{6}{25}$ (b) $\frac{7}{24}$ (c) $\frac{7}{25}$ (d) $\frac{5}{24}$

Q.300. $\left[\frac{\operatorname{cosec} 31^\circ - \sec 59^\circ}{(\sin 18^\circ \div \cos 72^\circ)} \times \frac{\tan 26^\circ}{\cot 64^\circ} \right] + \left[\frac{\tan 30^\circ + \tan 15^\circ}{(1 - \tan 30^\circ \tan 15^\circ)} \right] =$ _____.

- (a) -1 (b) 0 (c) 1 (d) 2

Q.301. If $\cot A = \frac{5}{12}$, then find the value of $\frac{\sqrt{\sec A - \operatorname{cosec} A}}{\sqrt{\sec A + \operatorname{cosec} A}} - \sqrt{\frac{17}{7}}$.

- (a) $\frac{24}{\sqrt{119}}$ (b) $\frac{10}{\sqrt{119}}$ (c) 0 (d) $-\frac{10}{\sqrt{119}}$

Q.302. If $\tan(t) = \frac{3}{7}$, t is an acute angle, then what is the value of $\operatorname{cosec}(t)$?

- (a) $\frac{\sqrt{58}}{3}$ (b) $\frac{\sqrt{53}}{7}$ (c) $\frac{\sqrt{58}}{7}$ (d) $\frac{\sqrt{53}}{3}$

Q.303. If $\sin\theta = \frac{11}{12}$, then what will be

the value of $(\tan\theta + \sec\theta)$?

- (a) $\frac{12}{\sqrt{23}}$ (b) $\frac{11}{\sqrt{23}}$ (c) $\sqrt{23}$ (d) 23

Q.304. If $\cos^4\theta - \sin^4\theta = \frac{4}{5}$, then find the value of $\sin 4\theta$.

- (a) $\frac{24}{25}$ (b) $\frac{16}{25}$ (c) $\frac{21}{25}$ (d) $\frac{18}{25}$

Q.305. If $\cot\alpha = \frac{P}{Q}$, then find the value of

$$\frac{P \cos\alpha - Q \sin\alpha}{P \cos\alpha + Q \sin\alpha} - \frac{P^2 - Q^2}{P^2 + Q^2} + 3.$$

- (a) 0 (b) 3 (c) $\frac{P^2}{P^2 + Q^2}$ (d) $\frac{P}{Q}$

Q.306. If $\sin\theta = \frac{7}{25}$, then $\cos\theta$ and $\tan\theta$ are respectively _____. (where θ is in the first quadrant)

- (a) $\frac{25}{7}$ and $\frac{7}{24}$ (b) $\frac{24}{25}$ and $\frac{24}{7}$
(c) $\frac{24}{25}$ and $\frac{7}{24}$ (d) $\frac{25}{24}$ and $\frac{24}{7}$

Q.307. What is the maximum value of $7 \cos A + 24 \sin A + 32$?

- (a) 39 (b) 32 (c) 25 (d) 57

Q.308. If $\tan A = \sqrt{2} - 1$ then find the value of $\sin A \cos A + \tan A$.

- (a) $\frac{24\sqrt{2} + 15}{4}$ (b) $\frac{24\sqrt{2} - 15}{4}$
(c) $\frac{5\sqrt{2} + 4}{4}$ (d) $\frac{5\sqrt{2} - 4}{4}$

Q.309. If $4 \sin^2 \theta - 4\sqrt{3} \sin\theta + 3 = 0$, then find the value of θ . ($0^\circ < \theta < 90^\circ$)

- (a) 45° (b) 30° (c) 60° (d) 75°

Q.310. The value of $\left(\frac{\sin\theta}{1 + \cos\theta} + \frac{1 + \cos\theta}{\sin\theta} \right) \left(\frac{1}{\tan\theta + \cot\theta} \right)$ is equal to :

- (a) $\tan\theta$ (b) $2 \sin\theta$ (c) $2 \cos\theta$ (d) 1

Q.311. Simplify $\frac{\sin 60^\circ - \cos 60^\circ}{\sin 60^\circ + \cos 60^\circ}$

- (a) $2 + \sqrt{3}$ (b) $\sqrt{3} - 2$
(c) $\sqrt{3} - 1$ (d) $2 - \sqrt{3}$

Q.312. If $\cot\theta = \frac{28}{45}$, then find the value of $(\sin\theta - \cos\theta)$.

- (a) $\frac{45}{78}$ (b) $\frac{7}{53}$ (c) $\frac{28}{53}$ (d) $\frac{17}{53}$

Q.313. If $\sec 6A = \operatorname{cosec}(A - 29^\circ)$, where $2A$ is an acute angle, then the measure of $\angle A =$ _____.

- (a) 18 (b) 17 (c) 21 (d) 19

Q.314. Simplify: $(1 + \tan^2 A) + \left(1 + \frac{1}{\tan^2 A} \right)$

- (a) $\frac{1}{\sin^2 A - \sin^4 A}$ (b) $\frac{1}{\sin^2 A - \cos^4 A}$

- (c) $\frac{1}{\cos^2 A - \sin^4 A}$ (d) $\frac{1}{\cot^2 A - \tan^4 A}$

Q.315. If $\sin\alpha = \frac{5}{13}$, then the value of $\cos\alpha \cdot \operatorname{cosec}\alpha \cdot \cot\alpha$ is _____.

- (a) $\frac{12}{5}$ (b) $\frac{144}{25}$ (c) $\frac{5}{12}$ (d) $\frac{25}{144}$

Q.316. Find the value of $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 3(\sin^2 A + \cos^2 A)$.

- (a) -1 (b) 1 (c) 2 (d) -2

Q.317. Find the maximum value of $(19 \sin\theta + 6 \cot\theta \sin\theta)$.

- (a) $\sqrt{397}$ (b) $\sqrt{197}$ (c) $\sqrt{297}$ (d) $\sqrt{497}$

Q.318. If $\tan\theta = \frac{3}{10}$ then find the value of $\frac{(1 + \sin\theta)(1 - \sin\theta)}{(1 + \cos\theta)(1 - \cos\theta)}$

- (a) $\frac{100}{7}$ (b) $\frac{100}{3}$ (c) $\frac{100}{9}$ (d) $\frac{50}{3}$

Q.319. Which of the following is true when $x = \sin A + \cos A$; $y = \sec A + \operatorname{cosec} A$?

- (a) $y(1 + x^2) = 2x$ (b) $y - 2x = x^2 y$
(c) $y + 2x = x^2 y$ (d) $y(1 - 2x^2) = x$

Q.320. Solve the equation for A (in degrees):

$$2 \cos^2 A + 3 \cos A - 2 = 0, 0^\circ < A < 90^\circ$$

- (a) 30° (b) 80° (c) 60° (d) 45°

Q.321. Find the value of $\frac{1 - \cot^2 \theta}{\tan^2 \theta - 1}$

- (a) $\tan^2 \theta$ (b) $\cot \theta$ (c) $\tan \theta$ (d) $\cot^2 \theta$

Q.322. If $\frac{(\sec A + \tan A)}{\sec A - \tan A} = 2 \frac{51}{79}$

then the value of $\sin A$ is equal to:

- (a) $\frac{77}{144}$ (b) $\frac{61}{169}$ (c) $\frac{87}{169}$ (d) $\frac{65}{144}$

Q.323. If $4 \sin^2 \theta = 3(1 + \cos\theta)$, $0^\circ < \theta < 90^\circ$, then what is the value of $\sqrt{15} \tan\theta + \frac{4}{\sqrt{15}} \sin\theta + 2 \sec\theta$?

- (a) 24 (b) $4\sqrt{15}$ (c) $\frac{8}{\sqrt{15}}$ (d) 8

Q.324. If $\tan\theta + \cot\theta = 6$, then the value of $\tan^2\theta + \cot^2\theta$ is _____.

- (a) 24 (b) 34 (c) 20 (d) 22

Q.325. $(1 - \cos^2 A) \sec^2 A$ is equal to:

- (a) $\cos^2 A \cot^2 A$ (b) $\tan^2 A$
(c) $\cot^2 A$ (d) $\operatorname{cosec}^2 A$

Q.326. If $\sin x + \cos x = \sqrt{2} \sin x$, then the value of $\sin x - \cos x$ is:

- (a) $\sqrt{2} \sin x$ (b) $2 \sin x$
(c) $-\sqrt{2} \cos x$ (d) $\sqrt{2} \cos x$

Q.327. If A , B and C are the interior angles of $\triangle ABC$, then what is the value of

$$\left\{ \tan \frac{A}{2} + \operatorname{cosec} \frac{B+C}{2} \right\} \left\{ \tan \frac{A}{2} - \operatorname{cosec} \frac{B+C}{2} \right\} ?$$

- (a) 2 (b) -1 (c) -2 (d) 1

Q.328. Simplify $\frac{\sin^3 A + \cos^3 A}{\sin A + \cos A}$

- (a) $1 - \tan A \sec A$ (b) $1 - \cos A \cot A$
(c) $1 - \sin A \tan A$ (d) $1 - \sin A \cos A$

Q.329. If $\sin(x+y) = 1$ and $\cos(x-y)$

$= \frac{\sqrt{3}}{2}$, find the value of 'y'.

- (a) 90° (b) 60° (c) 30° (d) 20°

Q.330. The value of $\left(\frac{2 \cos^3 \theta - \cos \theta}{\sin \theta - 2 \sin^3 \theta} \right)^2$

+ 1, $\theta \neq 45^\circ$ is :

- (a) $\sin^2 \theta$ (b) $\sec^2 \theta$ (c) $\operatorname{cosec}^2 \theta$ (d) $\cot^2 \theta$

Q.331. If $\sin^4 \theta + \cos^4 \theta = 2 \sin^2 \theta \cos^2 \theta$, $0^\circ < \theta < 90^\circ$, then find the value of $\cot \theta$.

- (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{3}$ (d) 1

Q.332. $\tan^4 A + \tan^2 A$ is equal to :

- (a) $\sec^2 A - \sec^4 A$ (b) $-(\sec^2 A + \sec^4 A)$
(c) $\sec^4 A - \sec^2 A$ (d) $\sec^4 A + \sec^2 A$

Q.333. Solve the following :

$(\tan \theta + \sec \theta + 1)(\cot \theta - \operatorname{cosec} \theta + 1) = ?$

- (a) 2 (b) 1 (c) 0 (d) -1

Q.334. The value of $(\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)(\tan \theta + \cot \theta)$ is

- (a) -3 (b) 3 (c) $\sec \theta \operatorname{cosec} \theta$ (d) 1

Q.335. Find the value of the following expression.

$(1 + \tan \theta + \sec \theta) \times (1 + \cot \theta - \operatorname{cosec} \theta)$

- (a) 1 (b) $\cos \theta \sin \theta$ (c) 2 (d) $\tan \theta$

Q.336. Simplify $\frac{\tan 71^\circ + \tan 19^\circ}{1 - \tan 71^\circ \tan 19^\circ}$

- (a) Not defined (b) 1 (c) $\sqrt{3}$ (d) 0

Q.337. If $\sin(x+y) = \cos \{3(x+y)\}$, then, the value of $\cot \{2(x+y)\}$ is _____.

- (a) -1 (b) 1 (c) -2 (d) 2

Q.338. The value of $\frac{2 \cos 15^\circ \sin 15^\circ}{\cos^2 15^\circ - \sin^2 15^\circ}$ is:

- (a) $\frac{1}{\sqrt{3}}$ (b) ∞ (c) $\frac{\sqrt{5}-1}{\sqrt{5}+1}$ (d) $\sqrt{3}$

Q.339. The value of $\frac{17}{\sec^2 \theta} + 10 \sin^2 \theta +$

$\frac{7}{1 + \cot^2 \theta}$ is :

- (a) 34 (b) 20 (c) 17 (d) 19

Q.340. The value of $\sin^2 15^\circ + \sin^2 25^\circ + \sin^2 35^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 65^\circ + \sin^2 75^\circ$ is:

- (a) 4 (b) 7 (c) $\frac{7}{2}$ (d) $\frac{7}{3}$

Q.341. Evaluate $7 \left(\frac{\operatorname{cosec} 24^\circ}{\sec 66^\circ} \right)^3 + 8$

$\left(\frac{\cot 37^\circ}{\tan 53^\circ} \right)^4 - \left(2 \frac{\sec 14^\circ}{\operatorname{cosec} 76^\circ} \right)^2 + \left(-3 \frac{\tan 82^\circ}{\cot 8^\circ} \right)^3$

- (a) -16 (b) 40 (c) 10 (d) 2

Q.342. If $8 \tan A = 5$, what is the value of

$\frac{8 \sin A - 7 \cos A}{8 \sin A + 11 \cos A} ?$

- (a) $-\frac{1}{8}$ (b) $\frac{1}{19}$ (c) $\frac{1}{8}$ (d) $-\frac{1}{6}$

Q.343. Which of the following is NOT a correct formula ?

(a) $\cos^2 A \cot^2 A = \cot^2 A - \cos^2 A$

(b) $\frac{1 + \tan^2 A}{\tan^2 A} = \sin^2 A$

(c) $\sec A \sin A = \tan A$

(d) $\operatorname{Cosec}^2 A - \cot^2 A = 1$

Q.344. Evaluate the following.

$\frac{5 \cos^2 120^\circ + 4 \sec^2 30^\circ - \tan^2 135^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$

- (a) $\frac{37}{12}$ (b) $\frac{57}{5}$ (c) $\frac{67}{12}$ (d) $\frac{27}{5}$

Q.345. Evaluate

$\left(\frac{\sin 23^\circ \cos 67^\circ + \cos 23^\circ \sin 67^\circ}{\operatorname{cosec}^2 15^\circ - \tan^2 75^\circ} \right)^3$

- (a) 0 (b) 1 (c) 8 (d) -1

Q.346. If A is an acute angle, what is the

value of $\operatorname{cosec} A$, if $\cot A = \frac{2p}{p^2 - 1} ?$

(a) $\frac{p^2 + 1}{p^2 - 1}$ (b) $\frac{p^2 - 1}{p^2 + 1}$

(c) $\frac{p^2}{p^2 - 1}$ (d) $\frac{2p}{p^2 + 1}$

Q.347. The value of $\operatorname{cosec} \theta \left[\frac{1 + \cos \theta}{\sin \theta} + \right.$

$\left. \frac{\sin \theta}{1 + \cos \theta} \right] - 2 \cot^2 \theta$ is equal to:

- (a) 0 (b) 1 (c) 2 (d) 3

Q.348. If $\sin A + \cos A = \frac{4}{3}$, then find the

value of $\tan A + \cot A$.

- (a) $\frac{7}{18}$ (b) $\frac{3}{4}$ (c) $\frac{18}{7}$ (d) $\frac{4}{3}$

Q.349. The expression $\sin^2 \theta + \cos^2 \theta - 1 = 0$ is satisfied by how many values of θ ?

- (a) No value
(b) Only one value
(c) Infinitely many values
(d) Two values

Q.350. Find the value of $(\sin 75^\circ + \sin 15^\circ)$.

- (a) $\frac{\sqrt{6}}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\sqrt{6}$ (d) $\sqrt{3}$

Q.351. Find the value of $\frac{\cos 25^\circ - \sin 65^\circ}{\cos 25^\circ + \sin 65^\circ}$

- (a) -1 (b) 0 (c) 1 (d) $\frac{1}{2}$

Q.352. If $\sin(5x - 40^\circ) = \cos(5y + 40^\circ)$,

then the value of $x + y$ is:

- (a) 18° (b) 15° (c) 40° (d) 20°

Q.353. The sides of a right-angled triangle, right-angled at B, are 6, 8 and 10 units. C is the vertex opposite to the side with length 8 units. What is the value of $\tan^2 A + \cos^2 C$?

- (a) $\frac{369}{400}$ (b) $\frac{125}{400}$ (c) $\frac{321}{400}$ (d) $\frac{325}{400}$

Q.354. If $\sin \theta + \operatorname{cosec} \theta = \sqrt{5}$, then the value of $\sin^3 \theta + \operatorname{cosec}^3 \theta$ is:

- (a) $\frac{1}{\sqrt{5}}$ (b) $\sqrt{5}$ (c) $2\sqrt{5}$ (d) 0

Q.355. If $\cos 2\theta = 0$, where θ is an acute angle, then find the value of $\sin(75^\circ - \theta)$.

- (a) $\sqrt{2} + 1$ (b) $\frac{1}{2}$ (c) $\sqrt{2} - 1$ (d) $\frac{1}{\sqrt{2}}$

Q.356. If $\operatorname{Psin} 60^\circ = \operatorname{Qcosec} 45^\circ$, then the value of $\frac{P^2 + Q^2}{P^2 - Q^2}$ is equal to:

- (a) $\frac{5}{11}$ (b) $\frac{11}{7}$ (c) $\frac{11}{5}$ (d) $\frac{7}{11}$

Q.357. Find the value of $(\tan^2 \theta + \tan^4 \theta)$.

- (a) $\sec^2 \theta - \sec^4 \theta$ (b) $\cot^2 \theta - \tan^2 \theta$
(c) $\sec^4 \theta + \sec^2 \theta$ (d) $\sec^4 \theta - \sec^2 \theta$

Q.358. If $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$, then $(\cos \theta + \sin \theta)$ is:

- (a) $-\sqrt{2} \sin \theta$ (b) $\sqrt{2} \tan \theta$
(c) $\sqrt{2} \cos \theta$ (d) $-\sqrt{2} \cos \theta$

Q.359. The value of

$(1 + \sin^4 A - \cos^4 A) \operatorname{cosec}^2 A$ is:

- (a) -1 (b) 1 (c) -2 (d) 2

Q.360. If $\cos A = \frac{15}{17}$, $0 \leq A \leq 90^\circ$, then the value of $\cot(90^\circ - A)$ is :

- (a) $\frac{8}{15}$ (b) $\frac{7}{15}$ (c) $\frac{\sqrt{2}}{15}$ (d) $\frac{2\sqrt{2}}{15}$

Q.361. If $\sin 2A = \cos 15^\circ$, then the smallest positive value of A is:

- (a) 7.5° (b) 47.5° (c) 75° (d) 37.5°

Q.362. In $\triangle ABC$, $\angle B = 90^\circ$ and $AB : BC = 1 : 2$. The value of $\cos A + \tan C$ is:

- (a) $\frac{1 + \sqrt{5}}{2\sqrt{5}}$ (b) $\frac{2\sqrt{5}}{2 + \sqrt{5}}$
(c) $\frac{5 + \sqrt{5}}{2\sqrt{5}}$ (d) $\frac{2 + \sqrt{5}}{2\sqrt{5}}$

Q.363. If $\sec A - \tan A = p$, then find the value of $\sec A$.

- (a) $\frac{p^2 - 1}{p^2 + 1}$ (b) $\frac{p^2 + 1}{p^2 - 1}$
(c) $\frac{p^2 + 1}{2p}$ (d) $\frac{p^2 + 1}{p}$



Q.364. Find the value of $\sqrt{\frac{1-\tan A}{1+\tan A}}$.

- (a) $\sqrt{\frac{1+\sin A}{\cos A}}$ (b) $\sqrt{\frac{1+\sin 2A}{\cos 2A}}$
 (c) $\sqrt{\frac{1-\sin A}{\cos A}}$ (d) $\sqrt{\frac{1-\sin 2A}{\cos 2A}}$

Q.365. The value of $\sqrt{\frac{1+\sin A}{1-\sin A}}$ is :

- (a) $\operatorname{cosec} A + \cot A$ (b) $\operatorname{cosec} A - \cot A$
 (c) $\sec A - \tan A$ (d) $\sec A + \tan A$

Q.366. If $\sin A = \frac{2}{3}$, then find the value of $(7 - \tan A)(3 + \cos A)$.

- (a) $\frac{61}{3} + \frac{17}{\sqrt{5}}$ (b) $\frac{61}{3} - \frac{17}{3\sqrt{5}}$
 (c) $\frac{61}{3} + \frac{17}{3\sqrt{5}}$ (d) $\frac{61}{3\sqrt{5}} + \frac{17}{3}$

Q.367. If $\sin \beta = \frac{1}{3}$, $(\sec \beta - \tan \beta)^2$ is equal to :

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{3}{4}$

Q.368. Find the value of

$$\frac{\sin^2 39^\circ + \sin^2 (90^\circ - 39^\circ)}{\cos^2 35^\circ + \cos^2 (90^\circ - 35^\circ)} + 3 \tan 25^\circ \tan 65^\circ.$$

(a) 1 (b) 3 (c) 4 (d) 2

Q.369. If $\tan \theta + \cot \theta = 2$, θ is an acute angle, then find the value of $2 \tan^{25} \theta + 3 \cot^{20} \theta + 5 \tan^{30} \theta \cot^{15} \theta$

- (a) 12 (b) 10 (c) 8 (d) 6

Q.370. Find the value of $\sin (50^\circ + A) - \cos (40^\circ - A)$.

- (a) 1 (b) 0 (c) -1 (d) 2

Q.371. If $\sin \theta + \cos \theta = \sqrt{2} \cos \theta$, then find $\frac{\sin \theta - \cos \theta}{\sin \theta}$.

- (a) $-\sqrt{2}$ (b) -1 (c) 1 (d) $\sqrt{2}$

Q.372. $\frac{\cos \theta}{\sec \theta - 1} + \frac{\cos \theta}{\sec \theta + 1}$ is equal to :

- (a) $2 \sec^2 \theta$ (b) $2 \cot^2 \theta$
 (c) $2 \cos^2 \theta$ (d) $2 \sin^2 \theta$

Q.373. If $(\sin \theta - \cos \theta) = 0$, then the value of $\sin(\pi - \theta) + \sin(\frac{\pi}{2} - \theta)$ is:

- (a) 1 (b) 0 (c) $\sqrt{3}$ (d) $\sqrt{2}$

Q.374. If $\cos \theta + \sec \theta = \sqrt{3}$, then the value of $\cos^3 \theta + \sec^3 \theta$ is :

- (a) 0 (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{3}$ (d) $2\sqrt{3}$

Q.375. If $\cos A = \frac{63}{65}$, then find the value of $\tan A + \cot A$ (up to two places of decimal).

- (a) 4.19 (b) 2.76 (c) 3.19 (d) 5.23

Q.376. If $\frac{\cos \theta}{(1 + \sin \theta)} + \frac{\cos \theta}{(1 - \sin \theta)} = 4$

and θ is acute, then the value of θ is :

- (a) 60° (b) 15° (c) 45° (d) 30°

Q.377. If $\sin \alpha + \cos \alpha = \frac{2}{\sqrt{3}}$ then what

is $(\tan \alpha + \cot \alpha)$ equal to ?

- (a) 1 (b) 2 (c) 6 (d) 0

Q.378. The value of

$(\operatorname{cosec}^2 B - 1)(\sec^2 B - 1)$ is :

- (a) 4 (b) 3 (c) 2 (d) 1

Q.379. If $\sin \theta + \cos \theta = \frac{1}{29}$, find the value

of $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$.

- (a) $\frac{1}{41}$ (b) $\frac{43}{29}$ (c) $\frac{41}{29}$ (d) $\frac{1}{43}$

Q.380. The given expression is equal to :

$$\frac{(1 + \tan^2 A)}{\operatorname{cosec}^2 A \cdot \tan A}$$

- (a) $\sec^2 A$ (b) $\sec A$ (c) $\tan A$ (d) $\tan^2 A$

Q.381. $\cos^4 A - \sin^4 A$ is equal to :

- (a) $2 \cos^2 A + 1$ (b) $1 - 2 \sin^2 A$
 (c) $2 \sin^2 A - 1$ (d) $-(2 \sin^2 A + 1)$

Q.382. If $\operatorname{cosec} \theta = 1\frac{7}{22}$, find the value of $\cot^2 \theta$.

- (a) $\frac{49}{484}$ (b) $\frac{357}{484}$ (c) $\frac{225}{484}$ (d) $\frac{7}{22}$

Q.383. What is the value of the expression

$$\sin A \left(1 + \frac{\sin A}{\cos A}\right) + \cos A \left(1 + \frac{\cos A}{\sin A}\right) ?$$

- (a) $\sec A + \operatorname{cosec} A$ (b) $\sin A + \cos A$
 (c) $\sin A - \cos A$ (d) $\sec A - \operatorname{cosec} A$

Q.384. Simplify the expression:

$$\frac{3 - \sin^2 A + \cos^2 A}{2 + 2 \cos^2 A}$$

- (a) 1 (b) 0 (c) -1 (d) 2

Q.385. If $\sec \theta + \tan \theta = \sqrt{3}$, then the positive value of $\sin \theta + \cos \theta$ is:

- (a) $\frac{(1 + \sqrt{3})}{2}$ (b) $\frac{2}{\sqrt{3}}$
 (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{2}{(1 + \sqrt{3})}$

Q.386. If $2 \tan^2 A + 4 \cos^4 A = 3$, then the possible value of A is:

- (a) 45° (b) 60° (c) 0° (d) 30°

Q.387. If $a = \cot A + \cos A$ and $b = \cot A - \cos A$, then find the value of $a^2 - b^2 - 4\sqrt{ab}$.

- (a) 0 (b) -1 (c) 1 (d) -4

Q.388. The given expression is equal to :

$$1 - \frac{\tan^2 \phi}{\sec^2 \phi}$$

- (a) $\sin^2 \phi \cos^2 \phi$ (b) $\sin^2 \phi \cot^2 \phi$
 (c) $\cot^2 \phi \cos^2 \phi$ (d) $\tan^2 \phi \cos^2 \phi$

Q.389. Find the value of $\cot^2 B - \operatorname{cosec}^2 B$ for $0 < B < 90^\circ$.

- (a) 1 (b) 2 (c) -1 (d) -2

Q.390. In ΔOPQ , right-angled at P, $OP = 7$ cm and $OQ - PQ = 1$ cm. Determine the values of $\sin Q + \cos Q$.

- (a) $\frac{13}{25}$ (b) $\frac{21}{25}$ (c) $\frac{31}{25}$ (d) $\frac{33}{25}$

Q.391. The given expression is equal to:

$$\frac{2 \sec A}{\sec^2 A - 1}$$

- (a) $2 \operatorname{cosec} A \cot A$ (b) $2 \cos A \cot A$
 (c) $2 \operatorname{cosec} A \tan A$ (d) $2 \cos A \sin A$

Q.392. The given expression is equal to:

$$\frac{\cos^2 A}{(1 + \sin A)}$$

- (a) $1 - \sin A$ (b) $1 + \cos A$
 (c) $\sin A$ (d) $\cos A$

Q.393. The value of $\sin^2 30^\circ + \sin^2 40^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 35^\circ + \sin^2 45^\circ + \sin^2 50^\circ + \sin^2 60^\circ$ is:

- (a) 2 (b) 4 (c) 3 (d) 1

Q.394. Simplify the given expression.

$$\frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos 53^\circ \sin 37^\circ$$

- (a) 2 (b) -1 (c) -2 (d) 1

Q.395. The value of $\frac{1}{\sin \theta} - \frac{\cot^2 \theta}{1 + \operatorname{cosec} \theta}$ is

- (a) 2 (b) 1 (c) -1 (d) 0

Q.396. The value of $\frac{\sin 58^\circ}{\cos 32^\circ} +$

$$\frac{\sin 55^\circ \sec 35^\circ}{\tan 5^\circ \tan 45^\circ \tan 85^\circ}$$
 is equal to:

- (a) 2 (b) 1 (c) 0 (d) 3

Q.397. If $\sec \theta = 2\frac{4}{23}$ find the value of $\tan^2 \theta$.

- (a) $2\frac{16}{529}$ (b) $1\frac{200}{529}$ (c) $3\frac{384}{529}$ (d) $3\frac{177}{529}$

Q.398. Find the value of

$$\frac{\cos^2 25^\circ - \sin^2 65^\circ}{\cos^2 25^\circ + \sin^2 65^\circ} :$$

- (a) $\frac{1}{2}$ (b) 1 (c) -1 (d) 0

Q.399. Simplify the given equation.

$$\frac{\cot^3 A - 1}{\cot A - 1} = ?$$

- (a) $\operatorname{cosec}^2 A - \cot A$ (b) $\operatorname{cosec}^2 A + \cot A$
 (c) $\cot^2 A + \operatorname{cosec} A$ (d) $\cot^2 A - \operatorname{cosec} A$



Q.400. If $\tan \theta + \sec \theta = a$, then what is $\sec \theta$ equal to:

- (a) $\frac{a+1}{2a^2}$ (b) $\frac{(a-1)}{2a^2}$
(c) $\frac{a^2-1}{2a}$ (d) $\frac{a^2+1}{2a}$

Q.401. Find the value of

$$\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} + 2.$$

- (a) $1 + 2\sec A$ (b) $2 - \sec A$
(c) $2 + \sec A$ (d) $2(1 + \sec A)$

Q.402. If $\sec \theta + \tan \theta = \frac{1}{\sqrt{3}}$, then the positive value of $\cot \theta + \cos \theta$ is:

- (a) $\frac{3\sqrt{3}}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{2}{3\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}$

Q.403. If $\frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} = \frac{4}{5}$, then the value of $\frac{\csc^2 \theta}{2 - \csc^2 \theta}$ is:

- (a) $\frac{16}{25}$ (b) $\frac{40}{41}$ (c) $\frac{41}{40}$ (d) $\frac{31}{30}$

Q.404. ΔPQR is right angled at Q. If $PQ = 12$ cm $PR = 13$ cm, find $\tan P + \cot R$.

- (a) $\frac{9}{10}$ (b) 0 (c) $\frac{10}{12}$ (d) $\frac{12}{10}$

Q.405. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, then the value of $A + B$ is:

- (a) 60° (b) 15° (c) 30° (d) 45°

Q.406. If $\cot A = \frac{5}{12}$, find the value of the following expression.

$$\frac{5(1 - \cos^2 A)}{6(1 - \sin^2 A)}$$

- (a) $\frac{144}{5}$ (b) $\frac{24}{25}$ (c) $\frac{144}{25}$ (d) $\frac{24}{5}$

Q.407. If $\tan \theta \cdot \tan 2\theta = 1$ then the value of $\cot 5\theta$ is:

- (a) -1 (b) 1 (c) $-\sqrt{3}$ (d) $\sqrt{3}$

Q.408. If $4 \sin^2 A - 3 = 0$ and $0 \leq A \leq 90^\circ$, then $3 \sin A - 4 \sin^3 A$ is:

- (a) $\frac{1}{2}$ (b) $\sqrt{\frac{3}{2}}$ (c) 0 (d) 1

Q.409. ΔRMS is right-angled at M. The length of the base, $RM = 4$ cm and the length of perpendicular $MS = 3$ cm. Find the value of $\sec R$.

- (a) $\frac{3}{4}$ (b) $\frac{5}{4}$ (c) $\frac{4}{5}$ (d) $\frac{2}{5}$

Q.410. Find the value of the following expression.

$$2(\tan^2 A \cos^2 A + \cos^2 A)$$

- (a) 4 (b) 1 (c) 3 (d) 2

Q.411. If A, B, C are the acute angles,

such that $\tan(A + B - C) = \frac{1}{\sqrt{3}}$, $\cos(B +$

$C - A) = \frac{1}{2}$ and $\sin(C + A - B) = \frac{1}{\sqrt{2}}$. The

value of $A + B + C$ will be equal to:

- (a) 90° (b) 110° (c) 170° (d) 135°

Q.412. Find the value of the following expression.

$$\frac{(1 + \sec \theta)}{(\sec \theta)} (1 - \cos \theta)$$

- (a) $\cos^2 \theta$ (b) $\sec^2 \theta$ (c) $\cot^2 \theta$ (d) $\sin^2 \theta$

Q.413. Let ABC be a right-angled triangle where $\angle A = 90^\circ$ and $\angle C = 45^\circ$. Find the value of $\sec C + \sin C \sec C$.

- (a) 1 (b) $1 - \sqrt{2}$ (c) $1 + \sqrt{2}$ (d) $\sqrt{2} - 1$

Q.414. If $(4 \sin \theta + 5 \cos \theta) = 3$, then the value of $(4 \cos \theta - 5 \sin \theta)$ is:

- (a) $3\sqrt{2}$ (b) $4\sqrt{2}$ (c) $2\sqrt{3}$ (d) $2\sqrt{5}$

Q.415. Let ABC be a triangle right angled at B. If $\tan A = \frac{12}{5}$, then find the values of $\csc A$ and $\sec A$, respectively.

- (a) $\frac{13}{10}, \frac{5}{13}$ (b) $\frac{13}{12}, \frac{13}{5}$

- (c) $\frac{10}{13}, \frac{5}{13}$ (d) $\frac{12}{13}, \frac{5}{13}$

Q.416. If $\cos \theta - \sin \theta = 0$, then $(\sin^8 \theta + \cos^8 \theta)$ is:

- (a) $\frac{1}{8}$ (b) $\frac{1}{4}$ (c) $\frac{1}{6}$ (d) $\frac{1}{2}$

Q.417. Simplify the given expression

$$\frac{3(\sin^4 Z - \cos^4 Z + 1)}{\sin^2 Z}$$

- (a) 9 (b) 2 (c) 4 (d) 6

Q.418. Find the value of $\sin 47^\circ \cos 33^\circ - \cos 43^\circ \sin 57^\circ$.

- (a) -1 (b) 1 (c) 0 (d) $\frac{1}{2}$

Q.419. The value of $\frac{1 + \sin A}{\cos A} + \frac{\cos A}{1 + \sin A}$ is:

- (a) $2 \sec A$ (b) $2 \csc A$
(c) $\sec A$ (d) $\csc A$

Q.420. Simplify the given equation. $(1 + \tan^2 A)(1 + \cot^2 A) = ?$

- (a) $\frac{1}{\cos^2 A(1 + \sin^2 A)}$
(b) $\frac{1}{\sin^2 A(1 - \sin^2 A)}$
(c) $\frac{1}{\sin^2 A + 1 + \csc^2 A}$
(d) $\frac{1}{\sin^2 A(1 + \cos^2 A)}$

Q.421. The given expression is equal to:

$$(\cot B - \tan B) \cdot \sin B \cdot \cos B$$

- (a) $1 - 2\sec^2 B$ (b) $1 - 2\cos^2 B$

- (c) $2 \cos^2 B - 1$ (d) $2 \sec^2 B - 1$

Q.422. $\cot^2 \theta - \operatorname{cosec}^2 \theta$ equals to:

- (a) 0 (b) $\tan \theta$ (c) 1 (d) -1

Q.423. If $\sin \theta \cos \theta = \frac{1}{\sqrt{3}}$ then the value

of $(\sin^4 \theta + \cos^4 \theta)$ is:

- (a) 1 (b) $\frac{5}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{3}$

Q.424. Simplify the given expression.

$$\sqrt{\frac{1 + \cos P}{1 - \cos P}}$$

- (a) $\csc P - \cot P$ (b) $\sec P - \tan P$
(c) $\sec P + \tan P$ (d) $\csc P + \cot P$

Q.425. Find the value of $\tan(50^\circ + \theta) - \cot(40^\circ - \theta)$.

- (a) $\frac{1}{2}$ (b) -1 (c) 0 (d) 1

Q.426. What is the value of $(1 - \sin^2 A) \csc^2 A$?

Given A is an acute angle.

- (a) $\cot^2 A$ (b) $\cos^2 A$ (c) $\tan^2 A$ (d) $\sin^2 A$

Q.427. If $\tan 15^\circ = 2 - \sqrt{3}$, then the value of $\tan 15^\circ \cot 75^\circ + \tan 75^\circ \cot 15^\circ$ is:

- (a) 6 (b) 10 (c) 8 (d) 14

Q.428. If $\{(-\frac{3\sin \theta - \cos \theta}{\cos \theta + \sin \theta})\} = 1$, then the value of $\cot \theta$ is:

- (a) 3 (b) 0 (c) 1 (d) 2

Q.429. Evaluate the following: $\cos(36^\circ + A) \cdot \cos(36^\circ - A) + \cos(54^\circ + A) \cdot \cos(54^\circ - A)$

- (a) $\sin 2A$ (b) $\cos A$ (c) $\sin A$ (d) $\cos 2A$

Q.430. If $\sin 23^\circ = \frac{a}{b}$, then the value of $\sec 23^\circ - \sin 67^\circ$ is _____

- (a) $\frac{a^2}{\sqrt{b^2 - a^2}}$ (b) $\frac{b^2 - a^2}{ab}$
(c) $\frac{a^2}{b\sqrt{b^2 + a^2}}$ (d) $\frac{a^2}{b\sqrt{b^2 - a^2}}$

Q.431. If $\frac{\cos \beta}{\sec \alpha} = 15$ and $\frac{\sin \beta}{\sec \alpha} = 16$, then the value of $\sin^2 \beta$ is _____.

- (a) $\frac{256}{481}$ (b) $-\frac{256}{481}$ (c) $\frac{481}{256}$ (d) $-\frac{481}{256}$

Q.432. Simplify the following.

$$\frac{\sin^3 \alpha + \cos^3 \alpha}{\sin \alpha + \cos \alpha}$$

- (a) $1 + \sin \alpha \cos \alpha$ (b) $\tan \alpha$
(c) $1 - \sin \alpha \cos \alpha$ (d) $\sec \alpha$

Q.433. If $\tan \frac{\pi}{6} + \sec \frac{\pi}{6} = x$, then find x.

- (a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$ (c) $-\frac{1}{\sqrt{3}}$ (d) $\frac{2}{\sqrt{3}}$

Q.434. If $\cos x = -\frac{1}{2}$, x lies in third



quadrant, then $\tan x = ?$

- (a) $\sqrt{3}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{2}{\sqrt{3}}$ (d) $\frac{1}{\sqrt{3}}$

Q.435. Using $\cos(A + B) = \cos A \cos B - \sin A \sin B$, find the value of $\cos 75^\circ$.

- (a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}+1}{4}$
(c) $\frac{\sqrt{6}-\sqrt{2}}{4}$ (d) $\frac{\sqrt{6}+\sqrt{2}}{4}$

Q.436. $\frac{\sin^4 \theta + \cos^4 \theta}{1 - 2\sin^2 \theta \cos^2 \theta} = \underline{\hspace{2cm}}$

- (a) 1 (b) 2 (c) -1 (d) 0

Q.437. Using trigonometric formulas, find the value of $\left(\frac{\sin(x-y)}{\sin(x+y)}\right)\left(\frac{\tan x + \tan y}{\tan x - \tan y}\right)$

- (a) -2 (b) 2 (c) 0 (d) 1

Q.438. If $\tan \frac{A}{2} = x$, then find x .

- (a) $\frac{\sqrt{1+\cos A}}{\sqrt{1-\cos A}}$ (b) $\frac{\sqrt{1-\sin A}}{\sqrt{1+\cos A}}$
(c) $\frac{\sqrt{1-\cos A}}{\sqrt{1+\cos A}}$ (d) $\frac{\sqrt{\cos A - 1}}{\sqrt{1+\cos A}}$

Q.439. The value of

$$\left(\frac{1}{\sin \theta} + \frac{1}{\tan \theta}\right)\left(\frac{1}{\sin \theta} - \frac{1}{\tan \theta}\right) \text{ is :}$$

- (a) 0 (b) 2 (c) 3 (d) 1

Q.440. Simplify the following:

$$\sin 2x + 2 \sin 4x + \sin 6x$$

(a) $4 \cos^2 x \sin 4x$ (b) $4 \cos^2 x \sin x$
(c) $2 \cos^2 x \sin 4x$ (d) $4 \sin^2 x \sin 4x$

Q.441. If $\sin A = -\frac{3}{5}$, A lies in III quadrant, the value of $\sec A$ is:

- (a) $-\frac{4}{5}$ (b) $-\frac{5}{4}$ (c) $-\frac{3}{4}$ (d) $\frac{3}{4}$

Q.442. Find the value of the following expression. $12(\sin^4 \theta + \cos^4 \theta) + 18(\sin^6 \theta + \cos^6 \theta) + 78\sin^2 \theta \cos^2 \theta$

- (a) 30 (b) 40 (c) 10 (d) 20

Q.443. Using $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

find the value of $\tan 15^\circ$.

- (a) $\sqrt{3} + 1$ (b) $\sqrt{3} - 1$ (c) $2 - \sqrt{3}$ (d) $2 + \sqrt{3}$

Q.444. If $\frac{\sin x - \cos x}{\sin x + \cos x} = \frac{2}{5}$, then the

value of $\frac{1 + \cot^2 x}{1 - \cot^2 x}$ is:

- (a) 2.25 (b) 1.45 (c) 3.75 (d) 5.25

Q.445. Find the value of $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2$

- (a) 4 (b) 0 (c) 2 (d) 1

Q.446. If $\left(\frac{\cos A}{1 - \sin A}\right) + \left(\frac{\cos A}{1 + \sin A}\right) = 4$

then what will be the value of A? ($0^\circ < \theta < 90^\circ$)

- (a) 90° (b) 45° (c) 60° (d) 30°

Q.447. Using $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$, find the value of $\cot 15^\circ$.

- (a) $2 + \sqrt{3}$ (b) $2 - \sqrt{3}$
(c) $\sqrt{3} - 1$ (d) $\sqrt{3} + 1$

Q.448. If $P \cos \alpha = 3$ and $4 \tan \alpha = Q$, then what is the relation between P and Q, which is independent of α ?

- (a) $\frac{9}{P^2} + \frac{16}{Q^2} = 1$ (b) $\frac{9}{P^2} - \frac{16}{Q^2} = 1$
(c) $\frac{P^2}{9} - \frac{Q^2}{16} = 1$ (d) $\frac{P^2}{9} + \frac{Q^2}{16} = 1$

Q.449. For any acute angle θ , $\sin \theta + \sin^2 \theta = 1$, then the value of $\cos^2 \theta + \cos^4 \theta = \underline{\hspace{2cm}}$.

- (a) 0 (b) 1 (c) 2 (d) -1

Q.450. If $\cos x + \sin x = \sqrt{2} \cos x$, what is the value of $(\cos x - \sin x)^2 + (\cos x + \sin x)^2$?

- (a) 2 (b) 1 (c) 0 (d) $\frac{1}{\sqrt{2}}$

Q.451. Solve the given expression.

$$\frac{\sec^2(90^\circ - \theta) - \cot^2 \theta}{2(\sin^2 35^\circ + \sin^2 55^\circ)}$$

- (a) 0.5 (b) 2 (c) 1.5 (d) 1

Q.452. $\operatorname{cosec} 2910^\circ + \sec 4260^\circ + \tan 2565^\circ + \cot 1755^\circ = ?$

- (a) 3 (b) 1 (c) 4 (d) 0

Q.453. The maximum value of $(\sin^{12} \theta + \cos^{20} \theta)$ for all the real values of θ is:

- (a) 2 (b) 3 (c) 1 (d) 0

Q.454. What is the value of $\sin(-405^\circ)$?

- (a) $\frac{\sqrt{5}}{2}$ (b) $\frac{1}{2}$ (c) $-\frac{1}{2}$ (d) $-\frac{1}{\sqrt{2}}$

Q.455. Simplify the following expression

$$\frac{\sin \theta - 2345 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta}$$

- (a) $\tan \theta$ (b) $\sin \theta$ (c) $\sec \theta$ (d) $\cos \theta$

Q.456. Simplify $\frac{1 + \sin t}{4 - 4 \sin t} - \frac{1 - \sin t}{4 + 4 \sin t}$

- (a) $4 \tan t \cdot \sin t$ (b) $\tan t \cdot \sec t$
(c) $\tan t - \sin t$ (d) $\tan t + \sin t$

Q.457. PQR is a triangle right angled at Q and PQ : QR = 3 : 4. What is the value of $\sin P + \sin Q + \sin R$?

- (a) $\frac{4}{5}$ (b) $\frac{3}{5}$ (c) $\frac{12}{5}$ (d) $\frac{2}{5}$

Q.458. If $2 \cot \theta = 3$, then find the value

$$\text{of } \frac{\sqrt{13} \sin \theta - 3 \tan \theta}{3 \tan \theta + \sqrt{13} \cos \theta}$$

- (a) $\frac{1}{\sqrt{13}}$ (b) $\frac{2}{\sqrt{13}}$ (c) 0 (d) $\frac{2}{3}$

Q.459. Using the identity $\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$ find the value of $\tan 15^\circ$

correct to three decimal places. [Use $\sqrt{3} = 1.732$]

- (a) 0.268 (b) 0.27 (c) 0.267 (d) 0.269

Q.460. If $\operatorname{Cosec} \theta + \cot \theta = m$, find the

value of $\frac{m^2 - 1}{m^2 + 1}$

- (a) 1 (b) 0 (c) $\cos \theta$ (d) $-\sin \theta$

Q.461. If $\sin \theta + \operatorname{cosec} \theta = \sqrt{5}$, then the value of $\sin^3 \theta + \operatorname{cosec}^3 \theta = \underline{\hspace{2cm}}$.

- (a) 0 (b) $3\sqrt{5}$ (c) $\frac{1}{\sqrt{5}}$ (d) $2\sqrt{5}$

Q.462. If $a \cot \theta = b$, then what will be the

value of $\frac{b \cos \theta - a \sin \theta}{b \cos \theta + a \sin \theta}$?

- (a) $\frac{b^2 + a^2}{b^2 - a^2}$ (b) $b^2 + a^2$
(c) $\frac{b^2 - a^2}{b^2 + a^2}$ (d) 0

Q.463. Simplify the given expression.

$$\frac{1 + \sin^4 \theta + \cos^4 \theta}{\cos^2 \theta + \sin^4 \theta}$$

- (a) 3 (b) 1 (c) 2 (d) 4

Q.464. Using $\operatorname{cosec}(\alpha + \beta) = \frac{\sec \alpha \times \sec \beta \times \operatorname{cosec} \alpha \times \operatorname{cosec} \beta}{\sec \alpha \times \operatorname{cosec} \beta + \operatorname{cosec} \alpha \times \sec \beta}$, find the value of $\operatorname{cosec} 75^\circ$.

- (a) $\frac{\sqrt{6} + \sqrt{2}}{4}$ (b) $\frac{\sqrt{6} - \sqrt{2}}{4}$
(c) $\sqrt{6} - \sqrt{2}$ (d) $\sqrt{6} + \sqrt{2}$

Q.465. The value of $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$ is:

- (a) $\sec \theta + \tan \theta$ (b) $\operatorname{cosec} \theta - \cot \theta$
(c) $\operatorname{cosec} \theta + \cot \theta$ (d) $\sec \theta - \tan \theta$

Q.466. Find $\left(\frac{1}{\sin \theta}\right) - \sin \theta$.

- (a) $\cos \theta \cot \theta$ (b) $\cos \theta \sec \theta$
(c) $\cos \theta \operatorname{cosec} \theta$ (d) $\cos \theta \tan \theta$

Q.467. If $\frac{21 \cos A + 3 \sin A}{3 \cos A + 4 \sin A} = 2$, then find the value of $\cot A$

- (a) $\frac{9}{11}$ (b) $\frac{11}{9}$ (c) $\frac{1}{3}$ (d) $\frac{11}{10}$

Q.468. $\tan 4384^\circ + \cot 6814^\circ = ?$

- (a) -1 (b) 2 (c) 0 (d) 1

Q.469. If $A = 22.5^\circ$ what is the value of $(10\sqrt{2} \sin 2A - 7\sqrt{2} \cos 2A + 9 \tan 2A)$?

- (a) 12 (b) 15 (c) 10 (d) 6

Q.470. $\sin 600^\circ \cos 750^\circ + \sin 150^\circ \cos 240^\circ = ?$

- (a) $\frac{2}{3}$ (b) 1 (c) $\frac{1}{2}$ (d) -1

Q.471. What is the values of $\tan \frac{\theta}{2}$?

- (a) $\frac{\cos \theta}{1 - \sin \theta}$ (b) $\frac{\sin \theta}{1 - \cos \theta}$
(c) $\frac{\cos \theta}{1 - \cos \theta}$ (d) $\frac{\sin \theta}{1 + \cos \theta}$

Q.472. If $3x = \sec A$ and $\frac{3}{x} = \tan A$, then

$9(x^2 - \frac{1}{x^2})$ is ____.

- (a) 3 (b) 9 (c) 1 (d) $\frac{1}{9}$

Q.473. If $\tan^4 \theta + \tan^2 \theta = 1$, what is the value of $11(\cos^4 \theta + \cos^2 \theta)$?

- (a) -11 (b) 8 (c) 0 (d) 11

Q.474. The value of $\frac{32}{\sec^2 \theta} - \frac{20}{1 + \tan^2 \theta} +$

$12 \sin^2 \theta$ is :

- (a) 20 (b) 32 (c) 24 (d) 12

Q.475. If $\sin \theta + \cos \theta = \frac{\sqrt{11}}{3}$, then what

is $\tan \theta + \cot \theta$ equal to ?

- (a) 5 (b) 9 (c) 7 (d) 11

Q.476. If $1 + \sin^2 \theta = a \cos^2 \theta$, then find $\sec^4 \theta - \tan^4 \theta$.

- (a) a^{-2} (b) a^{-1} (c) a (d) a^2

Q.477. In a triangle PQR, if $QR = 3$ cm, $PR = 5$ cm and $\sin P = \frac{3}{5}$, then $\angle Q$ equals:

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

Q.478. The value of $\operatorname{cosec} 41^\circ - \sec 49^\circ$ is :

- (a) 1 (b) 0 (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

Q.479. If $\sin x + \sin^2 x = 1$, then what is the value of $\cos^8 x + 2 \cos^6 x + \cos^4 x$?

- (a) 2 (b) 0 (c) 4 (d) 1

Q.480. What is the value of

$$\left(\frac{1 + \sec^2 A}{1 + \cos^2 A} \right) \left(\frac{1 + \sin^2 A}{1 + \operatorname{cosec}^2 A} \right) ?$$

- (a) $\cos^2 A$ (b) $\operatorname{cosec}^2 A$
(c) $\sec^2 A$ (d) $\tan^2 A$

Q.481. What is $\frac{\sqrt{(1 - \sin A)}}{\sqrt{(1 + \sin A)}} ?$

- (a) $\sec^2 A$ (b) $\sec A + \tan A$
(c) $\tan^2 A$ (d) $\sec A - \tan A$

Q.482. What is the value of $(1 + \tan \theta - \sec \theta)(1 + \cot \theta + \operatorname{cosec} \theta)$?

- (a) -1 (b) 0 (c) 1 (d) 2

Q.483. What is the value of

$$\frac{\cos(50^\circ + A) - \sin(40^\circ - A)}{\cos 40^\circ \sec 40^\circ} ?$$

- (a) 1 (b) 2 (c) 0.5 (d) 0

Q.484. If $\tan A + \sec A = K$, then what is the value of $\sin A$?

- (a) $\frac{2K}{K^2 + 1}$ (b) $\frac{K^2 - 1}{2K}$
(c) $\frac{K^2 + 1}{K^2 - 1}$ (d) $\frac{K^2 - 1}{K^2 + 1}$

Q.485. What is $\frac{\operatorname{cosec} 90^\circ}{(1 - \cos^2 60^\circ)} ?$

- (a) 0 (b) $\frac{4}{3}$ (c) $\frac{1}{4}$ (d) $\frac{3}{4}$

Q.486. What is the value of

$$\sqrt{\frac{\operatorname{cosec} A + 1}{\operatorname{cosec} A - 1}} + \sqrt{\frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1}} ?$$

- (a) $2 \cos A$ (b) $\sec A$
(c) $\cos A$ (d) $2 \sec A$

Q.487. If $m = \sec \theta - \tan \theta$ and $n = \operatorname{cosec} \theta + \cot \theta$, then what is the value of $m + n$ ($m - 1$) ?

- (a) 0 (b) 2 (c) 1 (d) -1

Q.488. $\triangle ABC$ is a right angled triangle.

$\angle A = 90^\circ$ degree, $AB = 4$ cm, $BC = 5$ cm.

What is the value of $\cos B + \cot C$?

- (a) $\frac{17}{20}$ (b) $\frac{29}{20}$ (c) $\frac{23}{20}$ (d) $\frac{31}{20}$

Q.489. If $\triangle PQR$ is right angled at Q , then what is the value of $\cos(P + R)$?

- (a) 0.5 (b) 1 (c) 0.2 (d) 0

Q.490. If $\sqrt{3} \tan^2 \theta - 4 \tan \theta + \sqrt{3} = 0$, then what is the value of $\tan^2 \theta + \cot^2 \theta$?

- (a) $\frac{4}{3}$ (b) $\frac{10}{3}$ (c) 3 (d) $\frac{6}{5}$

Q.491. If $6 \cot \theta = 5$, then find the value of

$$\frac{(6 \cos \theta + \sin \theta)}{(6 \cos \theta - 4 \sin \theta)} ?$$

- (a) 5 (b) 1 (c) 6 (d) 0

Q.492. If $-\sin \theta + \operatorname{cosec} \theta = 6$, then what is the value of $\sin \theta + \operatorname{cosec} \theta$?

- (a) 6 (b) $\sqrt{40}$ (c) $\sqrt{34}$ (d) $\sqrt{38}$

Q.493. If $\sqrt{2} \sec^2 \theta - 4 \sec \theta + 2\sqrt{2} = 0$, then what is the value $\sin^2 \theta + \tan^2 \theta$?

- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{5}{2}$ (d) $\frac{3}{2}$

Q.494. What is the value of

$$\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} + \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} ?$$

- (a) $\frac{1}{(\sin^2 \theta - \cos^2 \theta)}$
(b) $2(\sin^2 \theta - \cos^2 \theta)$
(c) $\frac{2}{(\sin^2 \theta - \cos^2 \theta)}$
(d) $\sin^2 \theta - \cos^2 \theta$

Q.495. Simplify the following

$$\text{expression:- } \frac{1 - \sin A}{\cos A} + \frac{\cos A}{1 - \sin A} ?$$

- (a) $2 \sin A$ (b) $2 \cos A$ (c) $2 \sec A$ (d) $2 \tan A$

Q.496. If $\sec x + \tan x = 5$ and $\operatorname{cosec} y - \cot y = \frac{1}{3}$ then find the value of $(\sec x + \operatorname{cosec} y) - (\tan x - \cot y)$.

- (a) 2.2 (b) 3.2 (c) 4.2 (d) 3.1

Q.497. If $A = a \operatorname{cosec} \theta + b \cot \theta$ and $B = a \cot \theta + b \operatorname{cosec} \theta$, then what is the value of $A^2 - B^2$?

- (a) ab (b) $(a + b)$ (c) $(a^2 - b^2)$ (d) $(a - b)$

Q.498. If $\operatorname{cosec} \theta + \cot \theta = s$, then what is the value of $\cos \theta$?

- (a) $\frac{s^2 - 1}{s^2 + 1}$ (b) $\frac{s^2 + 1}{s^2 - 1}$
(c) $\frac{s^2 - 1}{2s}$ (d) $\frac{2s}{s^2 + 1}$

Q.499. Find the value of $\tan x$, if $\sec x + \tan x = \sqrt{3}$ and x lies between 0° and 90° .

- (a) $\sqrt{3}$ (b) 1 (c) $\frac{1}{\sqrt{3}}$ (d) 0

Q.500. Tan of complementary of an angle is same as which of the angles ?

- (a) $\sin$ of that angle
(b) $\cot$ of that angle
(c) $\cos$ of that angle
(d) cosec of that angle

Q.501. If $\sec \theta + \tan \theta = 2$, then what is the value of $\cot \theta$?

- (a) $\frac{5}{3}$ (b) $\frac{3}{5}$ (c) $\frac{4}{3}$ (d) $\frac{3}{4}$

Q.502. What is the value of $(\tan 10^\circ \tan 80^\circ + \tan 20^\circ \tan 70^\circ + \tan 30^\circ \tan 60^\circ + \tan 40^\circ \tan 50^\circ)$?

- (a) 4 (b) 3 (c) 5 (d) 2

Q.503. What is the value of $\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1} ?$

- (a) $2 \sec \theta$ (b) $2 \operatorname{cosec} \theta$
(c) $2 \cot \theta$ (d) $\cot \theta + \operatorname{cosec} \theta$

Q.504. If $6 \sec \theta = 10$, then find the value of $\frac{(5 \operatorname{cosec} \theta - 3 \cot \theta)}{(4 \cos \theta + 3 \sin \theta)} ?$

- (a) $\frac{2}{3}$ (b) $\frac{3}{2}$ (c) $\frac{5}{6}$ (d) $\frac{6}{5}$

Q.505. What is the value of $\operatorname{cosec} A(1 - \cos A)(\operatorname{cosec} A + \cot A)$?

- (a) 2 (b) 1 (c) 0 (d) 3

Q.506. If $\frac{(2 \cos^3 A - \cos A)}{(\sin A - 2 \sin^3 A)} = 1$, then what is the value of A ?

- (a) 90° (b) 30° (c) 45° (d) 60°



Q.507. $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = ?$

- (a) $5 + \tan^2\theta + \cot^2\theta$
 (b) $7 + \tan^2\theta - \cot^2\theta$
 (c) $7 + \tan^2\theta + \cot^2\theta$
 (d) $5 + \tan^2\theta - \cot^2\theta$

Q.508. The value of $\tan^2\theta + \cot^2\theta - \sec^2\theta \operatorname{cosec}^2\theta$ is:

- (a) 2 (b) -2 (c) 0 (d) $\frac{1}{2}$

Q.509. What is $\sin\alpha - \sin\beta$?

- (a) $2\cos\frac{\alpha+\beta}{2} \sin\frac{\alpha-\beta}{2}$
 (b) $2\sin\frac{\alpha+\beta}{2} \sin\frac{\alpha-\beta}{2}$
 (c) $2\cos\frac{\alpha-\beta}{2} \sin\frac{\alpha+\beta}{2}$
 (d) $2\cos\frac{\alpha+\beta}{2} \cos\frac{\alpha-\beta}{2}$

Q.510. What is the value of $\tan 240^\circ$?

- (a) $\sqrt{2}$ (b) $(-\sqrt{3})$ (c) $\sqrt{3}$ (d) 3

Q.511. If $\tan^2\theta = 1 - a^2$, then the value of $\sec\theta + \tan^3\theta \operatorname{cosec}\theta$ is:

- (a) $(2 - a)^{\frac{3}{2}}$ (b) $(a^2 - 1)^{\frac{3}{2}}$
 (c) $(2 - a^2)^{\frac{3}{2}}$ (d) $a^{\frac{3}{2}}$

Q.512. If $\sin(A+B) = \cos(A+B)$, what is the value of $\tan A$?

- (a) $\frac{1 - \tan B}{1 + \tan B}$ (b) $\frac{1 + \tan B}{1 - \tan B}$
 (c) $\frac{1 + \sec B}{1 - \sec B}$ (d) $\frac{1 - \operatorname{cosec} B}{1 + \operatorname{cosec} B}$

Q.513. In $\triangle ABC$, right-angled at B, AB = 7 cm and AC - BC = 1 cm. Find the value of $\sin C$.

- (a) $\frac{3}{7}$ (b) $\frac{12}{13}$ (c) $\frac{3}{25}$ (d) $\frac{7}{25}$

Q.514. If $\frac{(1 - \cos\theta)}{\sin\theta} = \frac{1}{5}$, then what will be the value of $\frac{(1 + \cos\theta)}{\sin\theta}$?

- (a) 5 (b) $\frac{2}{5}$ (c) $\frac{4}{5}$ (d) $\frac{1}{5}$

Q.515. Simplify: $\frac{\cos A}{1 + \tan A} - \frac{\sin A}{1 + \cot A}$

- (a) $\tan A$ (b) $\cos A - \sin A$
 (c) $\cos A \cdot \sin A$ (d) $\cos A + \sin A$

Q.516. $\frac{1}{1 + \cos(90^\circ - \theta)} + \frac{1}{1 - \cos(90^\circ - \theta)} = \underline{\hspace{2cm}}$

- (a) $2\sec^2\theta$ (b) 1 (c) 0 (d) $2\tan^2\theta$

Q.517. $\sin^4\theta + \cos^4\theta$ in terms of $\sin\theta$ can be written as:

- (a) $2\sin^4\theta + 2\sin^2\theta - 1$
 (b) $2\sin^4\theta - 2\sin^2\theta$

(c) $2\sin^4\theta - 2\sin^2\theta - 1$

(d) $2\sin^4\theta - 2\sin^2\theta + 1$

Q.518. Simplify the following :

$$\frac{\cos x - \sqrt{3}\sin x}{2}$$

(a) $\cos(\frac{\pi}{3} - x)$ (b) $\sin(\frac{\pi}{3} + x)$

(c) $\cos(\frac{\pi}{3} + x)$ (d) $\sin(\frac{\pi}{3} - x)$

Q.519. The value of $\sqrt{\frac{1 + \cos A}{1 - \cos A}}$ is:

- (a) $\sec A - \tan A$ (b) $\operatorname{cosec} A + \cot A$
 (c) $\sec A + \tan A$ (d) $\operatorname{cosec} A - \cot A$

Q.520. If $\sin\theta + \cos\theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$, then

what is the value of $\tan\theta + \cot\theta$?

- (a) $8(\sqrt{3}-2)$ (b) $12(\sqrt{3}-2)$
 (c) $12(\sqrt{3}+2)$ (d) $8(\sqrt{3}+2)$

Q.521. If $\sec 2\theta = \operatorname{cosec}(\theta - 36^\circ)$, where 2θ is an acute angle, find the value of θ .

- (a) 32° (b) 46° (c) 20° (d) 42°

Q.522. What will be the value of $\cos x \operatorname{cosec} x - \sin x \sec x$?

- (a) $\cot \frac{2x}{2}$ (b) $\tan 2x$
 (c) $\cot 2x$ (d) $2\cot 2x$

Q.523. $\cot^2 A - \cos^2 A$ is equal to :

- (a) $\cos^2 A \cdot \sin^2 A$ (b) $\cos^2 A \cdot \cot^2 A$
 (c) 1 (d) 0

Q.524. If $b \sin\theta = a$, then $\sec\theta + \tan\theta = ?$

- (a) $\sqrt{\frac{b+a}{b-a}}$ (b) $\sqrt{\frac{1}{b+a}}$
 (c) $\sqrt{\frac{1}{b-a}}$ (d) $\sqrt{\frac{b-a}{b+a}}$

Q.525. What will be the value of $\sin 10^\circ - \frac{4}{3} \sin^3 10^\circ$?

- (a) $\frac{1}{3\sqrt{3}}$ (b) $\frac{1}{6}$ (c) $\frac{1}{2\sqrt{3}}$ (d) $\frac{\sqrt{3}}{6}$

Q.526. What is the value of $\cos(-\frac{17\pi}{3})$?

- (a) 1 (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) 0

Q.527. Solve the following to find its value in terms of trigonometric ratios.

$$(\sin A + \cos A)(1 - \sin A \cos A)$$

- (a) $\sin^3 A + \cos^3 A$
 (b) $\sin^2 A - \cos^2 A$
 (c) $[\cos A - \sin A][\sin^2 A + \cos^2 A]$
 (d) $\sin^3 A - \cos^3 A$

Q.528. If $\sec\theta - 2\cos\theta = \frac{7}{2}$, where θ is a positive acute angle, then the value of $\sec\theta$ is:

- (a) 6 (b) 8 (c) 5 (d) 4

Q.529. Find the value of $\tan 3\theta$ if $\sec 3\theta = \operatorname{cosec}(4\theta - 15^\circ)$.

- (a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{3}$ (c) -1 (d) 1

Q.530. If $\tan A - \tan B - \tan C = \tan A \tan B \tan C$, what is the value of A in terms of B and C?

- (a) $A = B + C$ (b) $A = 2B - 2C$
 (c) $A = B - C$ (d) $A = \frac{B-C}{2}$

Q.531. If $\tan^2 \alpha = 3 + Q^2$, then $\sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha = ?$

- (a) $(3 + Q^2)^{\frac{3}{2}}$ (b) $(7 + Q^2)^{\frac{3}{2}}$
 (c) $(5 - Q^2)^{\frac{3}{2}}$ (d) $(4 + Q^2)^{\frac{3}{2}}$

Q.532. If $1 + \sin\theta = m \cos\theta$, then what is the value of $\sin\theta$?

- (a) $\frac{2m^2-1}{m^2+1}$ (b) $\frac{m^2-1}{m^2+1}$
 (c) $\frac{m^2+1}{2m^2-1}$ (d) $\frac{m^2+1}{m^2-1}$

Q.533. Find the value of the following expression.

$$5(\sin^4\theta + \cos^4\theta) + 3(\sin^6\theta + \cos^6\theta) + 19\sin^2\theta \cos^2\theta$$

- (a) 8 (b) 5 (c) 6 (d) 7

Q.534. If $\sin Y = x$, then what will be the value of $\cos 2Y$ (where $0 \leq Y \leq 90^\circ$)?

- (a) $(\sqrt{2}-1)x$ (b) $\sqrt{2}x$ (c) $1-2x$ (d) $1-2x^2$

Q.535. If $A + B = C$, then $\tan A \tan B \tan C = ?$

- (a) $\tan C + \tan A - \tan B$
 (b) $\tan C + \tan A + \tan B$
 (c) $\tan A - \tan B - \tan C$
 (d) $\tan C - \tan A - \tan B$

Q.536. If $x = 8(\sin\theta + \cos\theta)$ and $y = 9(\sin\theta - \cos\theta)$, then the value of $\frac{x^2}{8^2} + \frac{y^2}{9^2}$ is:

- (a) 4 (b) 6 (c) 8 (d) 2

Q.537. $\theta = 135^\circ$; $y = 15^\circ$. What is the value of $2\cos(\theta) \sin(y)$?

- (a) $\frac{\sqrt{3}-1}{2}$ (b) $\frac{1-\sqrt{3}}{2}$
 (c) $2 - \sqrt{3}$ (d) $\sqrt{3} - 2$

Q.538. If $\sec x - \cos x = 4$, then what will be the value of $\frac{(1 + \cos^2 x)}{\cos x}$?

- (a) $\frac{9}{4}$ (b) $\frac{1}{4}$ (c) $2\sqrt{5}$ (d) $\sqrt{5}$

Q.539. If $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$, then $x = ?$

- (a) B (b) A (c) $A+B$ (d) $A-B$



Q.540. $\cos^3 60^\circ - \cos^3 240^\circ - \cos^3 360^\circ =$

- (a) $\frac{-3}{4}$ (b) $\frac{-7}{5}$ (c) $\frac{-3}{5}$ (d) $\frac{-9}{7}$

Q.541. If $5 \sin^2 A + 3 \cos^2 A = 4$, $0 < A < 90^\circ$, then what is the value of $\tan A$?

- (a) 0 (b) 3 (c) 1 (d) 2

Q.542. If $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = \frac{3}{2}$, then the value of $\sin^4 \theta - \cos^4 \theta$ is

- (a) $\frac{5}{12}$ (b) $\frac{12}{13}$ (c) $\frac{11}{12}$ (d) $\frac{5}{13}$

Q.543. If $\sin^2 \theta \cos^2 \theta = \frac{2}{9}$, then what will be the value of $\operatorname{cosec}^2 \theta + \sec^2 \theta$?

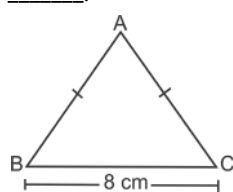
- (a) $\frac{7}{2}$ (b) $\frac{5}{2}$ (c) $\frac{9}{2}$ (d) $9\sqrt{2}$

Q.544. If $2 \frac{\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$,

then $a, b = ?$

- (a) $\frac{-3}{2}, \frac{-1}{2}$ (b) $\frac{3}{2}, \frac{1}{2}$ (c) -3, -1 (d) 3, 1

Q.545. In the given figure, ABC is an isosceles triangle with $BC = 8$ cm and $AB = AC = 5$ cm. The value of $\tan C - \cot B$ is



- (a) $-\frac{5}{12}$ (b) $-\frac{7}{12}$ (c) $\frac{7}{12}$ (d) $\frac{5}{12}$

Q.546. If $b \cos \theta = a$, then $\operatorname{cosec} \theta + \cot \theta =$

- (a) $\sqrt{\frac{1}{b+a}}$ (b) $\sqrt{\frac{b-a}{b+a}}$
(c) $\sqrt{\frac{b+a}{b-a}}$ (d) $\sqrt{\frac{1}{b-a}}$

Q.547. If $\operatorname{cosec} \theta + \cot \theta = p$, then the value of $\frac{p^2 - 1}{p^2 + 1}$ is:

- (a) $\cos \theta$ (b) $\sin \theta$ (c) $\cot \theta$ (d) $\operatorname{cosec} \theta$

Q.548. If $\sin^2 \theta - 3 \sin \theta + 2 = 0$, then find the value of θ ($0^\circ \leq \theta \leq 90^\circ$).

- (a) 45° (b) 0° (c) 60° (d) 90°

Q.549. If $\tan 3\theta \cdot \tan 7\theta = 1$, where 7θ is an acute angle, then find the value of $\cot 15\theta$.

- (a) 1 (b) -1 (c) $-\sqrt{3}$ (d) $\sqrt{3}$

Q.550. If $2 \sin \theta + 2 \sin^2 \theta = 2$, then the value of $2 \cos^4 \theta + 2 \cos^2 \theta$ is:

- (a) 4 (b) 2 (c) 1 (d) 0

Q.551. If $\cos \theta + \cos^2 \theta = 1$, find the value of $\sqrt{\sin^4 \theta + \cos^2 \theta}$

(a) $\sqrt{2} \cos \theta$ (b) $2 \cos \theta$

(c) $\sqrt{2} \sin \theta$ (d) $2 \sin \theta$

Q.552. If $\sec^2 A + \tan^2 A = 3$, then what is the value of $\cot A$?

- (a) $\frac{1}{\sqrt{3}}$ (b) 0 (c) 1 (d) $\sqrt{3}$

Q.553. What is the value of $\operatorname{cosec} 15^\circ \sec 15^\circ$?

- (a) 0.5 (b) 4 (c) 2 (d) 1

Q.554. $\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A} = ?$

- (a) $\tan A - \cot A$ (b) $\tan A + \cot A$
(c) $\sin A - \cos A$ (d) $\sin A + \cos A$

Q.555. If $\tan 40^\circ = a$, then find

$\frac{\tan 320^\circ - \tan 310^\circ}{1 + \tan 320^\circ \tan 310^\circ}$

- (a) $\frac{1-a^2}{a}$ (b) $\frac{1+a^2}{2a}$
(c) $\frac{1-a^2}{2a}$ (d) $\frac{1+a^2}{a}$

Q.556. If $\tan(\alpha + \beta) = a$, $\tan(\alpha - \beta) = b$ then the value of $\tan 2\alpha$ is:

- (a) $\frac{a+b}{1-ab}$ (b) $\frac{a+b}{1+ab}$
(c) $\frac{a-b}{1+ab}$ (d) $\frac{a-b}{1-ab}$

Q.557. If $x = \frac{2 \sin \theta}{(1 + \cos \theta + \sin \theta)}$ then the

value of $\frac{1 - \cos \theta + \sin \theta}{(1 + \sin \theta)}$ is:

- (a) $\frac{x}{(1+x)}$ (b) x (c) $\frac{1}{x}$ (d) $\frac{(1+x)}{x}$

Answer Key :-

1.(a)	2.(a)	3.(b)	4.(c)
5.(b)	6.(d)	7.(b)	8.(d)
9.(b)	10.(b)	11.(a)	12.(d)
13.(a)	14.(b)	15.(d)	16.(d)
17.(c)	18.(c)	19.(c)	20.(c)
21.(a)	22.(d)	23.(a)	24.(a)
25.(d)	26.(a)	27.(a)	28.(b)
29.(a)	30.(a)	31.(c)	32.(d)
33.(a)	34.(a)	35.(b)	36.(a)
37.(b)	38.(a)	39.(d)	40.(c)
41.(b)	42.(a)	43.(a)	44.(a)
45.(a)	46.(d)	47.(a)	48.(d)
49.(c)	50.(d)	51.(c)	52.(d)
53.(d)	54.(a)	55.(c)	56.(d)
57.(a)	58.(a)	59.(b)	60.(d)
61.(c)	62.(c)	63.(d)	64.(c)
65.(d)	66.(d)	67.(c)	68.(c)
69.(b)	70.(d)	71.(c)	72.(b)

73.(b)	74.(b)	75.(a)	76.(a)
77.(b)	78.(a)	79.(a)	80.(b)
81.(b)	82.(b)	83.(c)	84.(b)
85.(c)	86.(b)	87.(b)	88.(c)
89.(c)	90.(b)	91.(b)	92.(a)
93.(a)	94.(b)	95.(a)	96.(c)
97.(c)	98.(a)	99.(d)	100.(c)
101.(a)	102.(c)	103.(b)	104.(c)
105.(b)	106.(b)	107.(b)	108.(c)
109.(c)	110.(c)	111.(b)	112.(b)
113.(b)	114.(d)	115.(b)	116.(c)
117.(a)	118.(a)	119.(a)	120.(b)
121.(d)	122.(a)	123.(a)	124.(a)
125.(a)	126.(a)	127.(a)	128.(a)
129.(a)	130.(a)	131.(b)	132.(c)
133.(a)	134.(a)	135.(c)	136.(c)
137.(c)	138.(b)	139.(c)	140.(b)
141.(d)	142.(a)	143.(a)	144.(b)
145.(d)	146.(b)	147.(c)	148.(c)
149.(c)	150.(c)	151.(c)	152.(a)
153.(d)	154.(c)	155.(b)	156.(d)
157.(b)	158.(c)	159.(a)	160.(a)
161.(b)	162.(a)	163.(c)	164.(c)
165.(d)	166.(a)	167.(c)	168.(c)
169.(c)	170.(a)	171.(b)	172.(d)
173.(d)	174.(b)	175.(b)	176.(b)
177.(b)	178.(c)	179.(d)	180.(b)
181.(a)	182.(c)	183.(c)	184.(d)
185.(d)	186.(c)	187.(c)	188.(c)
189.(b)	190.(c)	191.(d)	192.(a)
193.(b)	194.(d)	195.(d)	196.(d)
197.(d)	198.(d)	199.(c)	200.(c)
201.(a)	202.(c)	203.(a)	204.(b)
205.(d)	206.(c)	207.(d)	208.(a)
209.(d)	210.(a)	211.(a)	212.(d)
213.(a)	214.(d)	215.(a)	216.(d)
217.(c)	218.(a)	219.(c)	220.(b)
221.(d)	222.(c)	223.(c)	224.(a)
225.(b)	226.(b)	227.(a)	228.(d)
229.(b)	230.(b)	231.(b)	232.(d)
233.(a)	234.(a)	235.(c)	236.(b)
237.(b)	238.(a)	239.(a)	240.(a)
241.(b)	242.(d)	243.(b)	244.(b)
245.(a)	246.(c)	247.(d)	248.(d)
249.(d)	250.(c)	251.(c)	252.(d)
253.(a)	254.(c)	255.(b)	256.(d)

257.(c)	258.(d)	259.(c)	260.(b)
261.(b)	262.(c)	263.(b)	264.(b)
265.(a)	266.(b)	267.(d)	268.(c)
269.(c)	270.(b)	271.(c)	272.(c)
273.(a)	274.(a)	275.(c)	276.(c)
277.(c)	278.(c)	279.(b)	280.(a)
281.(a)	282.(a)	283.(b)	284.(b)
285.(b)	286.(b)	287.(a)	288.(b)
289.(c)	290.(a)	291.(a)	292.(d)
293.(a)	294.(d)	295.(c)	296.(c)
297.(d)	298.(c)	299.(b)	300.(c)
301.(d)	302.(a)	303.(c)	304.(a)
305.(b)	306.(c)	307.(d)	308.(d)
309.(c)	310.(c)	311.(d)	312.(d)
313.(b)	314.(a)	315.(b)	316.(a)
317.(a)	318.(c)	319.(c)	320.(c)
321.(d)	322.(d)	323.(a)	324.(b)
325.(b)	326.(d)	327.(b)	328.(d)
329.(c)	330.(c)	331.(d)	332.(c)
333.(a)	334.(d)	335.(c)	336.(a)
337.(b)	338.(a)	339.(c)	340.(c)
341.(a)	342.(a)	343.(b)	344.(c)
345.(b)	346.(a)	347.(c)	348.(c)
349.(c)	350.(a)	351.(b)	352.(a)
353.(a)	354.(c)	355.(b)	356.(c)
357.(d)	358.(c)	359.(d)	360.(a)
361.(d)	362.(d)	363.(c)	364.(d)
365.(d)	366.(c)	367.(b)	368.(c)
369.(b)	370.(b)	371.(a)	372.(b)
373.(d)	374.(a)	375.(a)	376.(a)
377.(c)	378.(d)	379.(a)	380.(c)
381.(b)	382.(b)	383.(a)	384.(a)
385.(a)	386.(a)	387.(a)	388.(b)
389.(c)	390.(c)	391.(a)	392.(a)
393.(b)	394.(a)	395.(b)	396.(a)
397.(c)	398.(d)	399.(b)	400.(d)
401.(d)	402.(a)	403.(c)	404.(c)
405.(d)	406.(d)	407.(c)	408.(c)
409.(b)	410.(d)	411.(d)	412.(d)
413.(c)	414.(b)	415.(b)	416.(a)
417.(d)	418.(c)	419.(a)	420.(b)
421.(c)	422.(d)	423.(d)	424.(d)
425.(c)	426.(a)	427.(d)	428.(c)
429.(d)	430.(d)	431.(a)	432.(c)
433.(a)	434.(a)	435.(c)	436.(a)
437.(d)	438.(c)	439.(d)	440.(a)

441.(b)	442.(a)	443.(c)	444.(b)
445.(c)	446.(c)	447.(a)	448.(c)
449.(b)	450.(a)	451.(a)	452.(c)
453.(c)	454.(d)	455.(a)	456.(b)
457.(c)	458.(c)	459.(a)	460.(c)
461.(d)	462.(c)	463.(c)	464.(c)
465.(c)	466.(a)	467.(c)	468.(c)
469.(a)	470.(d)	471.(d)	472.(c)
473.(d)	474.(d)	475.(b)	476.(c)
477.(b)	478.(b)	479.(d)	480.(d)
481.(d)	482.(d)	483.(d)	484.(d)
485.(b)	486.(d)	487.(d)	488.(d)
489.(d)	490.(b)	491.(c)	492.(b)
493.(d)	494.(c)	495.(c)	496.(b)
497.(c)	498.(a)	499.(c)	500.(b)
501.(c)	502.(a)	503.(d)	504.(c)
505.(b)	506.(c)	507.(c)	508.(b)
509.(a)	510.(c)	511.(c)	512.(a)
513.(d)	514.(a)	515.(b)	516.(a)
517.(d)	518.(c)	519.(b)	520.(a)
521.(d)	522.(d)	523.(b)	524.(a)
525.(b)	526.(c)	527.(a)	528.(d)
529.(d)	530.(a)	531.(d)	532.(b)
533.(a)	534.(d)	535.(d)	536.(d)
537.(b)	538.(c)	539.(c)	540.(a)
541.(c)	542.(b)	543.(c)	544.(c)
545.(b)	546.(c)	547.(a)	548.(d)
549.(b)	550.(b)	551.(a)	552.(c)
553.(b)	554.(d)	555.(c)	556.(a)
557.(b)			

Solutions :-**Sol.1.(a)** According to question,

$$\sin(P) = \cos(Q) \Rightarrow P + Q = 90^\circ \dots (i)$$

$$\cos(P) = \sin(R) \Rightarrow P + R = 90^\circ \dots (ii)$$

Add eq. (i) and (ii),

$$\Rightarrow P + (P + Q + R) = 180^\circ$$

$$\Rightarrow P + 90^\circ = 180^\circ \quad (\text{given}) \Rightarrow P = 90^\circ$$

Sol.2.(a) We know that,

$$(\cot A + \operatorname{cosec} A)(\operatorname{cosec} A - \cot A) = 1$$

$$y(\operatorname{cosec} A - \cot A) = 1$$

$$(\operatorname{cosec} A - \cot A) = \frac{1}{y}$$

$$(\cot A + \operatorname{cosec} A) = y$$

$$2\operatorname{cosec} A = y + \frac{1}{y}$$

$$\operatorname{cosec} A = \frac{(y^2 + 1)}{2y}$$

$$\text{Sol.3.(b) Given :- } \cot(2y) = \frac{3}{4}$$

$$\begin{aligned} \cot 6y &= \cot 3 \times 2y = \frac{\cot^3 2y - 3\cot 2y}{3\cot^2 2y - 1} \\ &= \frac{\left(\frac{3}{4}\right)^3 - 3 \times \frac{3}{4}}{3 \times \left(\frac{3}{4}\right)^2 - 1} = \frac{\frac{27}{64} - \frac{9}{4}}{3 \times \frac{9}{16} - 1} \\ &= \frac{\frac{27 - 144}{64}}{\frac{27 - 16}{16}} = \frac{-117 \times 16}{11 \times 64} = -\frac{117}{44} \end{aligned}$$

Sol.4.(c) Given, $\cos x = -\frac{12}{13}$ (x lies in 3rd quadrant)

x lies in 3rd quadrant ($180^\circ + x$)

$$\frac{x}{2} \text{ lies in 2nd quadrant } (90^\circ + \frac{x}{2})$$

 $\therefore \sin$ is +ve

$$\text{Now we know, } \sin \frac{x}{2} = \sqrt{\frac{1 - \cos x}{2}}$$

$$\sin \frac{x}{2} = \sqrt{\frac{1 - \left(-\frac{12}{13}\right)}{2}}$$

$$\sin \frac{x}{2} = \sqrt{\frac{1 - \left(-\frac{12}{13}\right)}{2}} = \sqrt{\frac{13+12}{13 \times 2}}$$

$$\sin \frac{x}{2} = \frac{5}{\sqrt{26}}$$

Sol.5.(b)

$$\begin{aligned} &\tan 72^\circ - \tan 27^\circ - \tan 72^\circ \tan 27^\circ \\ &= \tan 72^\circ - \tan 72^\circ \tan 27^\circ - \tan 27^\circ \\ &= \tan 72^\circ (1 - \tan 27^\circ) - \tan 27^\circ \\ &= \tan(45^\circ + 27^\circ)(1 - \tan 27^\circ) - \tan 27^\circ \\ &= \left[\frac{\tan 45^\circ + \tan 27^\circ}{1 - \tan 45^\circ \tan 27^\circ} \right] (1 - \tan 27^\circ) - \tan 27^\circ \\ &= \frac{(1 + \tan 27^\circ)}{(1 - \tan 27^\circ)} (1 - \tan 27^\circ) - \tan 27^\circ \\ &= 1 + \tan 27^\circ - \tan 27^\circ = 1 \end{aligned}$$

$$\text{Sol.6.(d) } \frac{1 + \sin \theta - \cos \theta}{1 + \sin \theta + \cos \theta}$$

$$= \frac{1 + \sin \theta - (1 - 2\sin^2 \frac{\theta}{2})}{1 + \sin \theta + 2\cos^2 \frac{\theta}{2} - 1}$$

$$(\cos \theta = 1 - 2\sin^2 \frac{\theta}{2} = 2\cos^2 \frac{\theta}{2} - 1)$$

$$= \frac{1 + \sin \theta - 1 + 2\sin^2 \frac{\theta}{2}}{\sin \theta + 2\cos^2 \frac{\theta}{2}}$$

$$= \frac{\sin \theta + 2\sin^2 \frac{\theta}{2}}{\sin \theta + 2\cos^2 \frac{\theta}{2}}$$

$$= \frac{2\sin \frac{\theta}{2} \cos \frac{\theta}{2} + 2\sin^2 \frac{\theta}{2}}{2\sin \frac{\theta}{2} \cos \frac{\theta}{2} + 2\cos^2 \frac{\theta}{2}}$$

$$(\sin \theta = 2\sin \frac{\theta}{2} \cos \frac{\theta}{2})$$

$$= \frac{2\sin \frac{\theta}{2} (\cos \frac{\theta}{2} + \sin \frac{\theta}{2})}{2\cos \frac{\theta}{2} (\sin \frac{\theta}{2} + \cos \frac{\theta}{2})} = \tan \frac{\theta}{2}$$

Sol.7.(b) Identity used:

$\sin 2A = 2 \sin A \cos A$ and $\sin^2 A + \cos^2 A = 1$
Now,

$$= \frac{\cos^2 t + \sin^2 t - 2 \sin t \cos t}{\cos^2 t + \sin^2 t + 2 \sin t \cos t} \times \frac{\cos(t) + \sin(t)}{\cos(t) - \sin(t)}$$

$$= \frac{(\cos t - \sin t)^2}{(\cos t + \sin t)^2} \times \frac{\cos(t) + \sin(t)}{\cos(t) - \sin(t)}$$

$$= \frac{(\cos t - \sin t)}{(\cos t + \sin t)}$$

On dividing numerator and denominator by $\cos t$ we get,

$$= \frac{(1 - \tan t)}{(1 + \tan t)}$$

Sol.8.(d) By using identity:

$$\sin A \sin B = -\frac{1}{2} \{(\cos(A+B) - \cos(A-B))\}$$

hence,

$$\sin A \sin B = \frac{1}{2} \{(\cos(A-B) - \cos(A+B))\}$$

Sol.9.(b) Given: $a \sin^3 x + b \cos^3 x = \sin x$

$\cos x$ and $a \sin x = b \cos x$

$$a \sin^3 x + b \cos^3 x = \sin x \cos x$$

$$a \sin x \sin^2 x + b \cos^3 x = \sin x \cos x$$

$$b \cos x \sin^2 x + b \cos^3 x = \sin x \cos x$$

$$b \cos x (\sin^2 x + \cos^2 x) = \sin x \cos x$$

$$b \cos x = \sin x \cos x$$

$$b = \sin x$$

$$a \sin x = b \cos x$$

on putting value of b

$$a \sin x = \sin x \cos x \Rightarrow a = \cos x$$

Putting value of a and b

$$a^2 + b^2 = \sin^2 x + \cos^2 x = 1$$

Sol.10.(b) Given: $p \cos A = 2q \sin A$

$$\Rightarrow p = 2q \times \frac{\sin A}{\cos A} \dots\dots(i)$$

$$2p \operatorname{cosec} A - q \sec A = 3$$

$$\Rightarrow \frac{2p}{\sin A} - \frac{q}{\cos A} = 3$$

$$\Rightarrow 2p \times \cos A - q \times \sin A = 3 \times \sin A \cdot \cos A$$

$$\Rightarrow 2 \times 2q \times \frac{\sin A}{\cos A} \times \cos A - q \times \sin A = 3 \times \sin A \cdot \cos A$$

$$\sin A \cdot \cos A \quad [\text{from eq. (i)}]$$

$$\Rightarrow 4q \times \sin A - q \times \sin A = 3 \times \sin A \cdot \cos A$$

$$\Rightarrow 3q \times \sin A = 3 \times \sin A \cdot \cos A \Rightarrow q = \cos A$$

Put the value of q in eq. (i)

$$\Rightarrow p = 2 \times \cos A \times \frac{\sin A}{\cos A} \Rightarrow p = 2 \sin A$$

$$\text{Now, } p^2 + 4q^2$$

$$= (2 \sin A)^2 + 4(\cos A)^2$$

$$= 4(\sin^2 A + \cos^2 A) = 4$$

Sol.11.(a)

$$\text{Given:- } \tan A + \sin A = x$$

$$\tan A - \sin A = y$$

$$x^2 - y^2 = (\tan A + \sin A)^2 - (\tan A - \sin A)^2$$

$$= [(\tan A + \sin A) + (\tan A - \sin A)]$$

$$[(\tan A + \sin A) - (\tan A - \sin A)]$$

$$= (2 \tan A)(2 \sin A) = 4 \tan A \sin A$$

$$\text{Now, } xy = (\tan A + \sin A)(\tan A - \sin A) = \tan^2 A - \sin^2 A$$

$$xy = \frac{\sin^2 A}{\cos^2 A} - \sin^2 A = \sin^2 A \left(\frac{1}{\cos^2 A} - 1 \right) =$$

$$\sin^2 A \left(\frac{1 - \cos^2 A}{\cos^2 A} \right) = \sin^2 A \left(\frac{\sin^2 A}{\cos^2 A} \right)$$

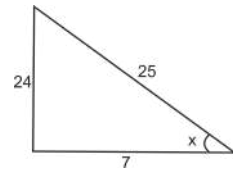
$$= \tan^2 A \sin^2 A$$

Hence, $\sqrt{xy} = \tan A \sin A$, and therefore

$$x^2 - y^2 = 4 \tan A \sin A = 4\sqrt{xy}$$

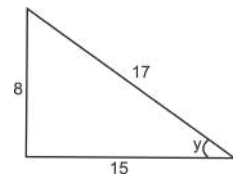
Sol.12.(d) Let $\cos^{-1}(\frac{7}{25}) = x$

$$\Rightarrow \cos x = \frac{7}{25}$$



$$\text{Then } \sin x = \frac{24}{25}$$

$$\text{and } \cos^{-1}(\frac{15}{17}) = y \Rightarrow \cos y = \frac{15}{17}$$



$$\text{Then, } \sin y = \frac{8}{17}$$

As we know, $\sin(x - y)$

$$= \sin x \cos y - \cos x \sin y$$

$$= \frac{24}{25} \times \frac{15}{17} - \frac{7}{25} \times \frac{8}{17}$$

$$= \frac{360 - 56}{425} = \frac{304}{425}$$

$$(x - y) = \sin^{-1}(\frac{304}{425})$$

$$\text{Then, } \cos^{-1}(\frac{7}{25}) - \cos^{-1}(\frac{15}{17})$$

$$= \sin^{-1}(\frac{304}{425})$$

Sol.13.(a) Take $x = 45^\circ$

$$6f(2) = 6 \times (\sin^{p+2} x + \cos^{p+2} x)$$

$$= 6 \times (\sin^4 x + \cos^4 x)$$

$$= 6 \times \left(\left(\frac{1}{\sqrt{2}} \right)^4 + \left(\frac{1}{\sqrt{2}} \right)^4 \right)$$

$$= 6 \times \left(\frac{1}{4} + \frac{1}{4} \right) = 6 \times \frac{1}{2} = 3$$

$$4f(4) = 4 \times (\sin^6 x + \cos^6 x)$$

$$= 4 \times ((\sin^2 x)^3 + (\cos^2 x)^3)$$

$$= 4 \times \left(\left(\frac{1}{\sqrt{2}} \right)^2 \right)^3 + \left(\left(\frac{1}{\sqrt{2}} \right)^2 \right)^3$$

$$= 4 \times \left(\frac{1}{8} + \frac{1}{8} \right) = 4 \times \frac{1}{4} = 1$$

$$10f(0) = 10 \times (\sin^2 x + \cos^2 x)$$

$$= 10 \times (1) = 10$$

$$\text{Hence, } 6f(2) - 4f(4) + 10f(0)$$

$$= 3 - 1 + 10 = 12$$

Sol.14.(b) Given :

$$\tan A = a \tan B \text{ and } \sin A = b \sin B$$

We put the value of $A = 60^\circ$ and $B = 30^\circ$

$$\tan 60^\circ = a \tan 30^\circ$$

$$\Rightarrow \sqrt{3} = a \times \frac{1}{\sqrt{3}} \Rightarrow a = 3$$

$$\sin 60^\circ = b \sin 30^\circ$$

$$\Rightarrow \frac{\sqrt{3}}{2} = b \times \frac{1}{2} \Rightarrow b = \sqrt{3}$$

$$\text{So, } \cos^2 A = (\cos 60^\circ)^2 = \left(\frac{1}{2} \right)^2 = \frac{1}{4}$$

Now, put the value a and b in all options one by one only option (b) satisfied.

$$\Rightarrow \frac{b^2 - 1}{a^2 - 1} = \frac{3 - 1}{9 - 1} = \frac{2}{8} = \frac{1}{4}$$

$$\text{Sol.15.(d)} \frac{\sin(A-B)}{\cos A \cos B} + \frac{\sin(B-C)}{\cos B \cos C}$$

$$\frac{\sin A \cos B - \cos A \sin B}{\cos A \cos B} + \frac{\sin B \cos C - \cos B \sin C}{\cos B \cos C}$$

$$\frac{\cos C (\sin A \cos B - \cos A \sin B) + \cos A (\sin B \cos C - \cos B \sin C)}{\cos A \cos B \cos C}$$

$$\frac{\sin A \cos B \cos C - \cos A \sin B \cos C + \cos A \sin B \cos C - \cos A \cos B \sin C}{\cos A \cos B \cos C}$$

$$= \frac{\sin A \cos B \cos C - \cos A \cos B \sin C}{\cos A \cos B \cos C}$$

$$= \frac{\cos B (\sin A \cos C - \cos A \sin C)}{\cos A \cos B \cos C}$$

$$= \frac{(\sin A \cos C - \cos A \sin C)}{\cos A \cos C} = \frac{\sin(A-C)}{\cos A \cos C}$$

$$\text{Sol.16.(d)} \frac{3 \cos^3 \theta - 2 \cos \theta}{\sin \theta - 3 \sin^3 \theta}$$

$$\frac{\cos \theta (3 \cos^2 \theta - 2)}{\sin \theta (1 - 3 \sin^2 \theta)}$$

$$\frac{\cos \theta (3 \cos^2 \theta - 2)}{\sin \theta (1 - 3 \{1 - \cos^2 \theta\})}$$

$$\frac{\cos \theta (3 \cos^2 \theta - 2)}{\sin \theta (3 \cos^2 \theta - 2)} = \frac{\cos \theta}{\sin \theta} = \cot \theta$$

$$\text{Sol.17.(c)} \cot A = \frac{7}{1} = \frac{\text{base}}{\text{perpendicular}}$$

$$\text{Hypotenuse} = \sqrt{(7)^2 + (1)^2}$$

$$\text{Hypotenuse} = \sqrt{50} = 5\sqrt{2}$$

Now,

$$\frac{5 \left(\frac{7}{5\sqrt{2}} \right) + 4 \left(\frac{1}{5\sqrt{2}} \right)}{\left(\frac{7}{5\sqrt{2}} \right)^3 + 7 \left(\frac{1}{5\sqrt{2}} \right)^3 + 6 \left(\frac{1}{5\sqrt{2}} \right)}$$

$$= \frac{\left(\frac{7}{\sqrt{2}} \right) + \left(\frac{4}{5\sqrt{2}} \right)}{\frac{343}{250\sqrt{2}} + \frac{7}{250\sqrt{2}} + \frac{6}{5\sqrt{2}}}$$

$$= \frac{\left(\frac{35 + 4}{5\sqrt{2}} \right)}{\frac{343 + 7 + 300}{250\sqrt{2}}} = \frac{39 \times 250\sqrt{2}}{5\sqrt{2} \times 650} = 3$$

$$\text{Alternate method:}$$

$$\frac{5 \cos A + 4 \sin A}{\cos^3 A + 7 \sin^3 A + 6 \sin A}$$

Divide numerator and denominator by $\sin A$

$$= \frac{5 \times \frac{\cos A}{\sin A} + 4 \times \frac{\sin A}{\sin A}}{\frac{\cos^3 A}{\sin A} + 7 \frac{\sin^3 A}{\sin A} + 6 \frac{\sin A}{\sin A}}$$

$$= \frac{5 \cot A + 4 \times 1}{\cos^2 A \cot A + 7 \sin^2 A + 6}$$

$$= \frac{5 \times 7 + 4 \times 1}{\cos^2 A \times 7 + 7 \sin^2 A + 6}$$

$$= \frac{39}{7 \cos^2 A + 7 \sin^2 A + 6}$$



$$= \frac{35 + 4}{7(\cos^2 A + \sin^2 A) + 6} = \frac{39}{7 + 6} = \frac{39}{13} = 3$$

Sol.18.(c) $\frac{[(\sin x + \sin y)(\sin y - \sin x)]}{[(\cos x + \cos y)(\cos y - \cos x)]}$

$$= \frac{-[(\sin x + \sin y)(\sin x - \sin y)]}{-[(\cos x + \cos y)(\cos x - \cos y)]}$$

$$= \frac{[(\sin^2 x - \sin^2 y)]}{[(\cos^2 x - \cos^2 y)]}$$

$$= \frac{[(1 - \cos^2 x) - (1 - \cos^2 y)]}{[(\cos^2 x - \cos^2 y)]}$$

$$= \frac{[1 - \cos^2 x - 1 + \cos^2 y]}{[(\cos^2 x - \cos^2 y)]}$$

$$= \frac{[-\cos^2 x + \cos^2 y]}{[(\cos^2 x - \cos^2 y)]} = \frac{-(\cos^2 x - \cos^2 y)}{[(\cos^2 x - \cos^2 y)]} = -1$$

Sol.19.(c) $\frac{\cos 13^\circ + \sin 13^\circ}{\cos 13^\circ - \sin 13^\circ}$

On dividing by $\cos 13^\circ$

$$\frac{\frac{\cos 13^\circ}{\cos 13^\circ} + \frac{\sin 13^\circ}{\cos 13^\circ}}{\frac{\cos 13^\circ}{\cos 13^\circ} - \frac{\sin 13^\circ}{\cos 13^\circ}} \Rightarrow \frac{1 + \tan 13^\circ}{1 - \tan 13^\circ}$$

$$\frac{\tan 45^\circ + \tan 13^\circ}{1 - \tan 45^\circ \cdot \tan 13^\circ} = \tan (45^\circ + 13^\circ)$$

$$= \tan 58^\circ$$

Sol.20.(c) $(\sin x + \sin y) = a$ ----- (i)

Squaring both side,

$$\sin^2 x + \sin^2 y + 2\sin x \sin y = a^2$$
 ----- (ii)

$$(\cos x + \cos y) = b$$
 ----- (iii)

Squaring both side,

$$\cos^2 x + \cos^2 y + 2\cos x \cos y = b^2$$
 ----- (iv)

Adding eqn (ii) & (iv) we get ;

$$(\sin^2 x + \cos^2 x) + (\sin^2 y + \cos^2 y) + 2\sin x \sin y + 2\cos x \cos y = a^2 + b^2$$

$$1 + 1 + 2(\sin x \sin y + \cos x \cos y) = a^2 + b^2$$

$$2(1 + \sin x \sin y + \cos x \cos y) = a^2 + b^2$$

$$(1 + \sin x \sin y + \cos x \cos y) = \frac{a^2 + b^2}{2}$$

$$\sin x \sin y + \cos x \cos y = \frac{a^2 + b^2}{2} - 1$$

$$= \frac{a^2 + b^2 - 2}{2}$$

Sol.21. (a) $\tan(t) + \cot(t) = 1$

$$\frac{\sin(t)}{\cos(t)} + \frac{\cos(t)}{\sin(t)} = 1$$

$$\frac{\sin^2(t) + \cos^2(t)}{\sin(t)\cos(t)} = 1$$

$$1 = \sin(t)\cos(t) \dots\dots(i)$$

$$[\sin(t) + \cos(t)]^2 = \sin^2(t) + \cos^2(t) + 2\sin(t)\cos(t)$$

$$= \sqrt{\sin^2(t) + \cos^2(t) + 2\sin(t)\cos(t)}$$

$$= \sqrt{1 + 2\sin(t)\cos(t)}$$

$$= \sqrt{1 + 2 \times 1} = \sqrt{3}$$

$$\text{Then, } \frac{1}{\sin(t) + \cos(t)} = \frac{1}{\sqrt{3}}$$

Sol.22.(d) $3 \sec^4 \theta + 8 = 10 \sec^2 \theta$

$$3 \sec^4 \theta - 10 \sec^2 \theta + 8 = 0$$

$$3 \sec^4 \theta - 6 \sec^2 \theta - 4 \sec^2 \theta + 8 = 0$$

$$3 \sec^2 \theta (\sec^2 \theta - 2) - 4(\sec^2 \theta - 2) = 0$$

$$(\sec^2 \theta - 2)(3 \sec^2 \theta - 4) = 0$$

$$\sec^2 \theta = 2 \text{ and } \frac{4}{3}$$

$$\Rightarrow \sec \theta = \sqrt{2} \text{ and } \frac{2}{\sqrt{3}}$$

$$\Rightarrow \theta = 45^\circ \text{ and } 30^\circ$$

$$\text{For } \theta = 45^\circ, \tan 45^\circ = 1$$

$$\text{For } \theta = 30^\circ, \tan 30^\circ = \frac{1}{\sqrt{3}}$$

Sol.23.(a) $\cot^4 A + \cot^2 A = 1$

$$\cot^2 A (\cot^2 A + 1) = 1$$

$$\cot A = \sin A \text{ or } \sin^2 A = \cos A$$

Now,

$$\cos^4 A - 3 \cos^2 A =$$

$$\left\{ (1 - \sin^2 A)^2 - 3 \sin^4 A \right\}$$

$$1 + \sin^4 A - 2 \sin A - 3 \sin^4 A$$

$$1 - 2(\sin^4 A + \sin^2 A) =$$

$$1 - 2(\cot^4 A + \cot^2 A) \Rightarrow 1 - 2 = -1$$

Sol.24.(a) According to the question,

$$\operatorname{cosec} \theta + \cot^3 \theta \sec \theta$$

$$\frac{1}{\sin \theta} + \frac{\cos^3 \theta}{\sin^3 \theta} \times \frac{1}{\cos \theta}$$

$$\frac{\sin^2 \theta}{\sin^3 \theta} + \frac{\cos^2 \theta}{\sin^3 \theta} = \frac{1}{\sin^3 \theta} = \operatorname{cosec}^3 \theta$$

now,

$$\cot^2 \theta = 1 - e^2 \quad (\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta)$$

$$\operatorname{cosec}^2 \theta = 2 - e^2$$

$$\operatorname{cosec} \theta = (2 - e^2)^{\frac{1}{2}}$$

(on cubing both sides)

$$\operatorname{cosec}^3 \theta = (2 - e^2)^{\frac{3}{2}}$$

Sol.25.(d) Minimum value of $a \sin^2 \theta + b$

$$\operatorname{cosec}^2 \theta = 2\sqrt{ab} \quad \{\text{when } a > b\}$$

$$\text{Hence, minimum value of } 16 \operatorname{cosec}^2 \theta + 25 \sin^2 \theta = 2\sqrt{16 \times 25} = 40$$

Sol.26.(a) $\operatorname{cosec} A + \cot A = a\sqrt{b}$

$$\left(\frac{1}{\sin A} + \frac{\cos A}{\sin A} \right)^2 = \left(\frac{1 + \cos A}{\sin A} \right)^2 = a^2 b$$

$$\frac{a^2 b - 1}{a^2 b + 1} = \frac{\left(\frac{1 + \cos A}{\sin A} \right)^2 - 1}{\left(\frac{1 + \cos A}{\sin A} \right)^2 + 1}$$

$$= \frac{1 + \cos^2 A + 2 \cos A - \sin^2 A}{1 + \cos^2 A + 2 \cos A + \sin^2 A}$$

$$= \frac{2 \cos^2 A + 2 \cos A}{2 + 2 \cos A} = \frac{2 \cos A (\cos A + 1)}{2 (\cos A + 1)}$$

$$= \cos A$$

Sol.27.(a) Put $\theta = 45^\circ$

Now, $\sin \theta + \tan \theta = p$ ----- (1)

$$\Rightarrow \frac{1}{\sqrt{2}} + 1 = p \Rightarrow \frac{1 + \sqrt{2}}{\sqrt{2}} = p$$

And, $\tan \theta - \sin \theta = q$ ----- (2)

$$\Rightarrow 1 - \frac{1}{\sqrt{2}} = q \Rightarrow \frac{\sqrt{2} - 1}{\sqrt{2}} = q$$

Now, checking through options, (a) satisfies the value of p and q.

$$p^2 - q^2 = 4\sqrt{pq} \Rightarrow \left(\frac{1 + \sqrt{2}}{\sqrt{2}} \right)^2 - \left(\frac{\sqrt{2} - 1}{\sqrt{2}} \right)^2$$

$$= 4 \sqrt{\frac{1 + \sqrt{2}}{\sqrt{2}} \times \frac{\sqrt{2} - 1}{\sqrt{2}}}$$

$$= \frac{(1 + \sqrt{2})^2 - (\sqrt{2} - 1)^2}{2} = 4 \times \sqrt{\frac{1}{2}}$$

$$\Rightarrow 2\sqrt{2} = 2\sqrt{2}$$

Sol.28.(b)

$$\sin \theta \cdot \cos \theta = 2 \cos^3 \theta - \frac{1}{4} \cos \theta$$

$$4 \sin \theta \cdot \cos \theta = 8 \cos^3 \theta - \cos \theta$$

$$4 \sin \theta = 8 \cos^2 \theta - 1$$

$$4 \sin \theta = 8 - 8 \sin^2 \theta - 1$$

$$8 \sin^2 \theta + 4 \sin \theta - 7 = 0$$

$$\sin \theta = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-4 \pm \sqrt{4^2 - 4 \times 8 \times (-7)}}{2 \times 8}$$

$$= \frac{-4 \pm \sqrt{240}}{2 \times 8} = \frac{-1 \pm \sqrt{15}}{4}$$

Here θ is an acute angle, so

$$\Rightarrow \sin \theta = \frac{\sqrt{15} - 1}{4}$$

Sol.29.(a)

If $A + B = 90^\circ$, then $\cot A \times \cot B = 1$

$$= 7(\cot 11^\circ \cot 13^\circ \cot 19^\circ \cot 29^\circ \cot 61^\circ \cot 71^\circ \cot 77^\circ \cot 79^\circ)$$

$$= 7(\cot 11^\circ \times \cot 79^\circ)(\cot 13^\circ \times \cot 77^\circ)$$

$$(\cot 19^\circ \times \cot 71^\circ)(\cot 29^\circ \times \cot 61^\circ)$$

$$= 7(1 \times 1 \times 1 \times 1) = 7$$

Sol.30.(a) $\sin^2 A + \cos^2 A = 1$

$$\sin^4 A + \cos^4 A + 2 \sin^2 A \cdot \cos^2 A = 1$$

$$\sin^4 A + \cos^4 A = 1 - 2 \sin^2 A \cdot \cos^2 A$$

$$\frac{\sin^4 A + \cos^4 A}{1 - 2 \sin^2 A \cos^2 A} = 1$$

Sol.31.(c) $pq = \cot^2 A - \cos^2 A$

$$= \cot^2 A \times \cos^2 A \dots (\text{identity})$$

$$\sqrt{pq} = \cot A \cdot \cos A$$

$$p^2 = \cot^2 A + \cos^2 A + 2 \times \cot A \cdot \cos A$$

$$q^2 = \cot^2 A + \cos^2 A - 2 \times \cot A \cdot \cos A$$

$$p^2 - q^2 = 4 \times \cot A \cdot \cos A$$

Now,

$$p^2 - q^2 = 4 \times \cot A \cdot \cos A = \sqrt{16pq}$$

Sol.32.(d)

$$4(\sin^6 A + \cos^6 A) - 6(\sin^4 A + \cos^4 A) + 8$$

$$[\because a^3 + b^3 = (a + b)^3 - 3ab(a + b) \text{ and } a^2 + b^2 = (a + b)^2 - 2ab]$$

$$= 4(1 - 3 \sin^2 A \cdot \cos^2 A) - 6(1 - 2 \sin^2 A$$



$$\begin{aligned}
 & \cos^2 A + 8 \\
 & = 4 - 12\sin^2 A \cos^2 A - 6 + 12\sin^2 A \\
 & \cos^2 A + 8 = 6
 \end{aligned}$$

Sol.33.(a) Concept used :

$$\begin{aligned}
 (1 + \cos 2\theta) &= 2\cos^2 \theta \\
 \sqrt{2 + \sqrt{2 + \sqrt{2 + 2\cos 8\theta}}} \\
 &= \sqrt{2 + \sqrt{2 + \sqrt{2(1 + \cos 8\theta)}}} \\
 &= \sqrt{2 + \sqrt{2 + \sqrt{2 \times 2\cos^2 4\theta}}} \\
 &= \sqrt{2 + \sqrt{2 + 2\cos 4\theta}} \\
 &= \sqrt{2 + \sqrt{2(1 + \cos 4\theta)}} \\
 &= \sqrt{2 + \sqrt{2 \times 2\cos^2 2\theta}} = \sqrt{2 + 2\cos 2\theta} \\
 &= \sqrt{2(1 + \cos 2\theta)} = \sqrt{2 \times 2\cos^2 \theta} = 2\cos \theta
 \end{aligned}$$

Short trick -

$$2\cos\left[\frac{8\theta}{2^3}\right] = 2\cos \theta$$

$$\text{Sol.34.(a)} \sqrt{\frac{1 - \sin 3\theta}{1 + \sin 3\theta}}$$

After rationalizing,

$$\begin{aligned}
 &= \sqrt{\frac{(1 - \sin 3\theta) \times (1 - \sin 3\theta)}{(1 + \sin 3\theta) \times (1 - \sin 3\theta)}} \\
 &= \sqrt{\frac{(1 - \sin 3\theta) \times (1 - \sin 3\theta)}{(1 - \sin^2 3\theta)}} \\
 &= \sqrt{\left(\frac{1 - \sin 3\theta}{\cos 3\theta}\right)^2} \\
 &= \left(\frac{1 - \sin 3\theta}{\cos 3\theta}\right) = \sec 3\theta - \tan 3\theta
 \end{aligned}$$

$$\text{Sol.35.(b)} \cos^2 \theta + \sin \theta + 1 = 0$$

$$1 - \sin^2 \theta + \sin \theta + 1 = 0$$

$$\sin^2 \theta - \sin \theta - 2 = 0$$

$$\sin^2 \theta - 2\sin \theta + \sin \theta - 2 = 0$$

$$\sin \theta (\sin \theta - 2) + 1(\sin \theta - 2) = 0$$

$$(\sin \theta - 2)(\sin \theta + 1) = 0$$

$$\sin \theta = -1 \text{ or } +2$$

Since $\sin \theta$ has a range of -1 to +1.

$$\text{So, } \sin \theta = -1 \text{ or } \sin \theta = \sin \frac{3\pi}{2}$$

$$\text{Comparing both sides } \Rightarrow \theta = \frac{3\pi}{2}$$

Therefore solution of $\cos^2 \theta + \sin \theta + 1$

$$= 0 \text{ lies between } \left(\frac{5\pi}{4}, \frac{7\pi}{4}\right).$$

$$\text{Sol.36.(a)} (1 + \cot A + \tan A)$$

$$(\sin A - \cos A) \frac{\sin A \cos A}{(\sin^3 A - \cos^3 A)}$$

$$= (1 + \cot A + \tan A) \frac{\sin A \cos A}{\sin A \cos A}$$

$$\frac{(\sin A - \cos A)(\sin^2 A + \cos^2 A + \sin A \cos A)}{\sin A \cos A}$$

$$= (1 + \cot A + \tan A) \frac{\sin A \cos A}{(1 + \sin A \cos A)}$$

Now, put $A = 45^\circ$

$$(3) \times \frac{\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}}{\left(1 + \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right)} = 3 \times \frac{1}{2} \times \frac{2}{3} = 1$$

$$\text{Sol.37.(b)} \sin m + \sin n = p$$

$$\Rightarrow (\sin m + \sin n)^2 = p^2$$

$$\Rightarrow \sin^2 m + \sin^2 n + 2 \sin m \sin n = p^2$$

$$\text{-----(eq. 1)}$$

$$\text{And } \cos m + \cos n = q$$

$$\Rightarrow (\cos m + \cos n)^2 = q^2$$

$$\Rightarrow \cos^2 m + \cos^2 n + 2 \cos m \cos n = q^2$$

$$\text{-----(eq. 2)}$$

After adding (eq. 1 and 2)

Then,

$$(\sin^2 m + \cos^2 m) + (\sin^2 n + \cos^2 n) +$$

$$2(\sin m \sin n + \cos m \cos n) = p^2 + q^2$$

$$\Rightarrow 2 + 2(\sin m \sin n + \cos m \cos n) = p^2 + q^2$$

$$\Rightarrow (\sin m \sin n + \cos m \cos n)$$

$$= \frac{(p^2 + q^2 - 2)}{2}$$

$$\text{Sol.38.(a)}$$

$$b \operatorname{cosec} \theta + a \cot \theta = p$$

$$b \cot \theta + a \operatorname{cosec} \theta = q$$

$$\text{-----}$$

$$b^2 - a^2 = p^2 - q^2$$

$$\text{Sol.39.(d)}$$

$$\cos(\pi - A) \cdot \cot\left(\frac{\pi}{2} + A\right) \cos(-A)$$

$$\tan(\pi + A) \tan\left(\frac{3\pi}{2} + A\right) \sin(2\pi - A)$$

$$= \frac{-\cos A \cdot (-\tan A) \cos A}{\tan A \cdot (-\cot A) \cdot (-\sin A)} = \frac{\cos^2 A}{\sin A \cot A}$$

$$= \cos A$$

$$\text{Sol.40.(c)} \cos 2A \cos 2B + \sin^2(A - B) -$$

$$\sin^2(A + B) = \cos 2A \cos 2B +$$

$$\{\sin(A - B) + \sin(A + B)\} \times$$

$$\{\sin(A - B) - \sin(A + B)\}$$

$$= \cos 2A \cos 2B + \{2 \sin A \cos B \times (-2 \cos A$$

$$\cdot \sin B)\} = \cos 2A \cos 2B - \sin 2A \sin 2B$$

$$= \cos(2A + 2B)$$

$$\text{Sol.41.(b)} \frac{k - k \cot^2 30^\circ}{1 + \cot^2 30^\circ}$$

$$= \sin^2 60^\circ + 4 \tan^2 45^\circ - \operatorname{cosec}^2 60^\circ$$

$$\frac{k(1 - 3)}{(1 + 3)} = \frac{3}{4} + 4 \times 1 - \frac{4}{3}$$

$$k = -2 \left\{ \frac{3}{4} + 4 - \frac{4}{3} \right\} = -2 \times \left(\frac{41}{12} \right) = -6.83$$

$$\text{Sol.42.(a)} \cos \theta + \sin \theta = \sqrt{2} \cos \theta$$

$$\sin \theta = \sqrt{2} \cos \theta - \cos \theta$$

$$\sin \theta = \cos \theta (\sqrt{2} - 1)$$

$$\sin \theta \times \frac{1}{\sqrt{2} - 1} = \cos \theta$$

$$(\sqrt{2} + 1) \sin \theta = \cos \theta$$

Subtracting $\sin \theta$ from it

$$\sin \theta (\sqrt{2} + 1 - 1) = \cos \theta - \sin \theta$$

$$\sqrt{2} \sin \theta = \cos \theta - \sin \theta$$

$$\text{Sol.43.(a)} \sin 73^\circ + \cos 137^\circ$$

$$\cos 137^\circ = \cos(90^\circ + 47^\circ) = -\sin 47^\circ$$

$$\sin 73^\circ - \sin 47^\circ = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$= 2 \cos\left(\frac{73+47}{2}\right) \cdot \sin\left(\frac{73-47}{2}\right)$$

$$= 2 \cos\left(\frac{120^\circ}{2}\right) \sin\left(\frac{26^\circ}{2}\right)$$

$$= 2 \cos 60^\circ \cdot \sin 13^\circ = 2 \times \frac{1}{2} \times \sin 13^\circ = \sin 13^\circ$$

Sol.44.(a) As we know that,

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\cos^2 18^\circ = 1 - \left(\frac{\sqrt{5}-1}{4}\right)^2$$

$$= \frac{16 - (5 + 1 - 2\sqrt{5})}{16} = \frac{10 + 2\sqrt{5}}{16}$$

$$\cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}$$

$$\text{Sol.45.(a)} \frac{\sin 4\theta + \sin 2\theta}{\cos 4\theta + \cos 2\theta}$$

$$= \frac{2 \sin\left(\frac{4\theta+2\theta}{2}\right) \cos\left(\frac{4\theta-2\theta}{2}\right)}{2 \cos\left(\frac{4\theta+2\theta}{2}\right) \cos\left(\frac{4\theta-2\theta}{2}\right)}$$

$$= \frac{\sin 3\theta \cos \theta}{\cos 3\theta \cos \theta} = \frac{\sin 3\theta}{\cos 3\theta} = \tan 3\theta$$

$$\text{Sol.46.(d)}$$

$$\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ$$

$$\text{No. of terms} = \frac{\text{Last term} - \text{1st term}}{\text{common difference}} + 1$$

$$= \frac{89 - 1}{2} + 1 = 45$$

$$\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ$$

$$= \frac{45}{2} = 22.5$$

$$\cot^2 46^\circ - \sec^2 44^\circ + (\sin^2 1^\circ + \sin^2 3^\circ + \sin^2 5^\circ + \dots + \sin^2 89^\circ)$$

$$= \tan^2 44^\circ - \sec^2 44^\circ + (22.5) = -1 + 22.5$$

$$= 21.5$$

Sol.47.(a) $\sin A = 0.5$ is true for all thevalues of $(2\pi n + 30^\circ)$ and $(2n + 1)\pi - 30^\circ$ For the $n = 0$

$$2\pi n + 30^\circ = 30^\circ \text{ and } (2n + 1)\pi - 30^\circ$$

$$= 150^\circ$$

And for $n = 1$

$$2\pi n + 30^\circ = 390^\circ \text{ and } (2n + 1)\pi - 30^\circ$$

$$= 510^\circ$$

 $\sin A = 0.5$ is true for $150^\circ, 390^\circ$ and 510°

$$\text{So, } \frac{A}{3} = 50^\circ, 130^\circ \text{ and } 170^\circ$$

Only 170° is given in the answer so it is the correct answer.

$$\text{Sol.48.(d)} \cos 27^\circ = x$$

$$\sin 63^\circ = \cos 27^\circ = x$$

$$\cos 63^\circ = \sqrt{1 - x^2}$$

$$\Rightarrow \tan 63^\circ = \frac{x}{\sqrt{1 - x^2}}$$

$$\text{Sol.49.(c)} x \sin \theta = y \cos \theta$$

$$= x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$$

$$= x \sin^3 \theta + \frac{x \sin \theta}{\cos \theta} \times \cos^3 \theta = \sin \theta \cos \theta$$

$$= x \sin^3 \theta + x \sin \theta \times \cos^2 \theta = \sin \theta \cos \theta$$



$$\begin{aligned}
 &= x \sin \theta (\sin^2 \theta + \cos^2 \theta) = \sin \theta \cos \theta \\
 &= x \sin \theta = \sin \theta \cos \theta \Rightarrow x = \cos \theta \\
 &= x \sin \theta = y \cos \theta \\
 &= \cos \theta \cdot \sin \theta = y \cos \theta \Rightarrow y = \sin \theta \\
 &\Rightarrow x^2 + y^2 = \sin^2 \theta + \cos^2 \theta = 1
 \end{aligned}$$

Short trick:- In such type of questions, value of $x^2 + y^2$ is the square of constant term before $\sin \theta \cos \theta$

Sol.50.(d) $\cos^2 \theta + \cos^4 \theta = 1$
 $\cos^4 \theta = 1 - \cos^2 \theta \Rightarrow \cos^4 \theta = \sin^2 \theta$ Or
 $\cos^2 \theta = \sin \theta$
 $\sin \theta + \sin^2 \theta = \cos^2 \theta + \sin^2 \theta = 1$

Sol.51.(c) $\sin \theta - \cos \theta = \frac{7}{17}$

$\frac{p}{h} - \frac{b}{h} = \frac{7}{17}$ (here, H = 17 or its multiple) & We know, 15, 8 and 17 make right angle triangles. Put, p = 15 and b = 8

$$\sin \theta + \cos \theta = \frac{p}{h} + \frac{b}{h} = \frac{15+8}{17} = \frac{23}{17}$$

Short trick :- $\sin \theta - \cos \theta = \frac{7}{17}$

$$\begin{aligned}
 \sin \theta + \cos \theta &= \sqrt{a^2 + b^2 - m^2} \\
 &= \sqrt{1^2 + 1^2 - \left(\frac{7}{17}\right)^2} = \sqrt{\frac{529}{289}} = \frac{23}{17}
 \end{aligned}$$

Sol.52.(d) sum of roots

$$= \sec \theta + \sin \theta = \frac{-b}{a} = \frac{k}{\sqrt{6}}$$

product of roots = $\sec \theta \times \sin \theta$

$$= \tan \theta = \frac{c}{a} = \frac{\sqrt{6}}{\sqrt{6}} = 1 \Rightarrow \tan \theta = 1$$

it means, $\theta = 45^\circ$

$$\sec \theta + \sin \theta = \frac{k}{\sqrt{6}} \Rightarrow \sqrt{2} + \frac{1}{\sqrt{2}} = \frac{k}{\sqrt{6}}$$

$$\frac{3}{\sqrt{2}} = \frac{k}{\sqrt{6}} \Rightarrow k = 3\sqrt{3}$$

Sol.53.(d) If $\cos A$, $\sin A$ and $\cot A$ are in geometric progression, then

$$\sin^2 A = \cos A \times \cot A$$

$$\sin^2 A = \cos A \times \frac{\cos A}{\sin A} \Rightarrow \sin^3 A = \cos^2 A$$

After dividing by $\cos^3 A$

$$\frac{\sin^3 A}{\cos^3 A} = \frac{1}{\cos A} \Rightarrow \tan^3 A = \sec A$$

$$\tan^6 A - \tan^2 A = \sec^2 A - \tan^2 A = 1$$

Sol.54.(a) $\cos 225^\circ = \cos (180 + 45)^\circ$

$$= -\cos 45^\circ = -\frac{1}{\sqrt{2}} = -\frac{1}{1.414} = -0.7071$$

Sol.55.(c) $\tan 2A = \tan[(A+B) + (A-B)]$

$$= \frac{\tan(A+B) + \tan(A-B)}{1 - \tan(A+B) \tan(A-B)}$$

$$\cos(A+B) = \frac{24}{25} \Rightarrow \tan(A+B) = \frac{7}{24}$$

$$\sin(A-B) = \frac{15}{17} \Rightarrow \tan(A-B) = \frac{15}{8}$$

$$\tan[(A+B) + (A-B)]$$

$$= \frac{\frac{7}{24} + \frac{15}{8}}{1 - \frac{7}{24} \times \frac{15}{8}} = \frac{416}{87}$$

Sol.56.(d) $\cos^2 \theta - \sin^2 \theta = \tan^2 \theta$

$$\Rightarrow \frac{\cos^2 \theta - \sin^2 \theta}{1} = \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$\Rightarrow \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} = \frac{\sin^2 \theta}{\cos^2 \theta}$$

By using componendo and Dividendo rule,

$$\Rightarrow \frac{\cos^2 \theta}{-\sin^2 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$\Rightarrow \frac{-\sin^2 \theta}{\cos^2 \theta} = \frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \theta + \cos^2 \theta}$$

$$\Rightarrow \tan^2 \theta = \cos^2 \theta - \sin^2 \theta$$

Sol.57.(a) There are a total 20 terms and the value of $\sin \theta$ can't be greater than 1. Clearly for the sum of 20 terms being 20 values of each term must be 1. So each angle must be 90°

$$(\alpha_1 + \alpha_2 + \alpha_3 \dots \alpha_{20}) = 90 \times 20 = 1800^\circ$$

Sol.58.(a) Given, $\sin(\alpha + \beta) = 1$

$$\sin(\alpha + \beta) = \sin 90^\circ \Rightarrow \alpha + \beta = 90^\circ \dots (1)$$

$$\cos(\alpha - \beta) = \frac{1}{2}$$

Similarly, $\alpha - \beta = 60^\circ \dots (2)$

Adding (1) and (2)

$$2\alpha = 150^\circ \Rightarrow \alpha = 75^\circ$$

$$\text{From eq (1)} \Rightarrow \beta = 90^\circ - 75^\circ = 15^\circ$$

Sol.59.(b) Given $\theta = 9^\circ$

$$\cot \theta \cdot \cot 2\theta \cdot \cot 3\theta \cdot \cot 4\theta \cdot \cot 5\theta \cdot \cot 6\theta \cdot \cot 7\theta \cdot \cot 8\theta \cdot \cot 9\theta$$

$$= \cot 9 \cdot \cot 18 \cdot \cot 27 \cdot \cot 36 \cdot \cot 45 \cdot \cot 54 \cdot \cot 63 \cdot \cot 72 \cdot \cot 81 = 1 \times 1 \times 1 \times 1 \times 1 \times 1$$

$$\dots (\cot A \cot B = 1, \text{ if } A + B = 90^\circ) = 1$$

Sol.60.(d)

$$\tan 30^\circ \cdot \tan 60^\circ \Rightarrow \frac{1}{\sqrt{3}} \times \sqrt{3} = 1$$

$$\sin 30^\circ \Rightarrow \frac{1}{2}; \tan 45^\circ \Rightarrow 1 \Rightarrow \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Clearly $\cos 30^\circ$ is an irrational quantity.

Sol.61.(c) Values of $\cos \theta$, $\cot \theta$ and $\operatorname{cosec} \theta$ continuously decreases. Only value of $\tan \theta$ is continuously increases.

Sol.62.(c) Given,

$$\sec \theta = 4x \text{ and } \tan \theta = \frac{4}{x}$$

Squaring both sides

$$\sec^2 \theta = 16x^2 \dots (1) \text{ and } \tan^2 \theta = \frac{16}{x^2} \dots (2)$$

Subtracting (2) from (1)

$$\sec^2 \theta - \tan^2 \theta = 16x^2 - \frac{16}{x^2}$$

$$\Rightarrow 1 = 2(8x^2 - \frac{8}{x^2}) \Rightarrow 8(x^2 - \frac{1}{x^2}) = \frac{1}{2}$$

Sol.63.(d)

$$\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \sqrt{\frac{\operatorname{cosec} \theta - \cot \theta}{\operatorname{cosec} \theta + \cot \theta}} = \frac{1 - r}{1 + r}$$

$$\Rightarrow \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \sqrt{\frac{\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta}}{\frac{1}{\sin \theta} + \frac{\cos \theta}{\sin \theta}}}$$

$$= \frac{1 - r}{1 + r} \Rightarrow \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} \times \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

$$= \frac{1 - r}{1 + r} \Rightarrow \frac{(1 - \cos \theta)}{(1 + \cos \theta)} = \frac{1 - r}{1 + r}$$

Comparing both the terms $\Rightarrow r = \cos \theta$

Sol.64.(c)

$$\frac{(\sin \theta - \operatorname{cosec} \theta)(\cos \theta - \sec \theta)}{\tan^2 \theta - \sin^2 \theta} = r^3$$

$$\text{Put } \theta = 30^\circ \Rightarrow \frac{\left(\frac{1}{2} - 2\right)\left(\frac{\sqrt{3}}{2} - \frac{2}{\sqrt{3}}\right)}{\left(\frac{1}{\sqrt{3}}\right)^2 - \left(\frac{1}{2}\right)^2} = r^3$$

$$\Rightarrow \frac{\left(\frac{-3}{2}\right)\left(\frac{3-4}{2\sqrt{3}}\right)}{\frac{1}{3} - \frac{1}{4}} = r^3 \Rightarrow \frac{\left(\frac{3}{4\sqrt{3}}\right)}{\frac{1}{12}} = r^3$$

$$\Rightarrow r^3 = 3\sqrt{3} \Rightarrow r = \sqrt{3}$$

through options $\Rightarrow \cot \theta \Rightarrow \cot 30^\circ = \sqrt{3}$

Clearly $r = \cot \theta$

Sol.65.(d) $3 - 2\sin^2 \theta - 3\cos \theta = 0$

$$= 3 - 2(1 - \cos^2 \theta) - 3\cos \theta = 0$$

$$= 2\cos^2 \theta - 3\cos \theta + 1 = 0$$

$$= 2\cos^2 \theta - 2\cos \theta - \cos \theta + 1 = 0$$

$$= 2\cos \theta (\cos \theta - 1) - 1(\cos \theta - 1) = 0$$

$$= \cos \theta = 1 \text{ or } \frac{1}{2} \Rightarrow \text{Clearly } \theta =$$

$$60^\circ (2\operatorname{Cosec} \theta + \tan \theta) \Rightarrow 2\left(\frac{2}{\sqrt{3}}\right) + \sqrt{3}$$

$$\Rightarrow \frac{4}{\sqrt{3}} + \sqrt{3} = \frac{7\sqrt{3}}{3}$$

Short trick:- put $\theta = 60^\circ$

$$3 - 2\sin^2 \theta - 3\cos \theta = 3 - 2 \times \frac{3}{4} - 3 \times \frac{1}{2}$$

$$= 3 - \frac{3}{2} - \frac{3}{2} = 0$$

Condition satisfied. Clearly $\theta = 60^\circ$

$$\text{So, } (2\operatorname{Cosec} \theta + \tan \theta) = 2\left(\frac{2}{\sqrt{3}}\right) + \sqrt{3}$$

$$= \frac{4}{\sqrt{3}} + \sqrt{3} = \frac{7\sqrt{3}}{3}$$

Sol.66.(d)

$$\left(\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta}\right) \left(\frac{1}{\tan \theta + \cot \theta}\right)$$

$$= \left(\frac{\sin^2 \theta + (1 + \cos \theta)^2}{\sin \theta (1 + \cos \theta)}\right) \left(\frac{1}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}}\right)$$

$$= \left(\frac{\sin^2 \theta + 1 + \cos^2 \theta + 2\cos \theta}{\sin \theta (1 + \cos \theta)}\right) \left(\frac{1}{\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}}\right)$$

$$= \frac{2 + 2\cos \theta}{\sin \theta (1 + \cos \theta)} \left(\frac{\sin \theta \cos \theta}{1}\right)$$

$$= \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)} \times \sin \theta \cos \theta = 2\cos \theta$$

Sol.67.(c)

$$\frac{(\cos 9^\circ + \sin 81^\circ)(\sec 9^\circ + \operatorname{cosec} 81^\circ)}{\sin 56^\circ \sec 34^\circ + \cos 25^\circ \operatorname{cosec} 65^\circ}$$

$$\frac{[\cos(90^\circ - 81^\circ) + \sin 81^\circ][(\sec(90^\circ - 81^\circ) + \operatorname{cosec} 81^\circ)]}{\sin 56^\circ \sec(90^\circ - 56^\circ) + \cos(90^\circ - 65^\circ) \operatorname{cosec} 65^\circ} =$$

$$\frac{(2\sin 81^\circ)(2\operatorname{cosec} 81^\circ)}{\sin 56^\circ \operatorname{cosec} 56^\circ + \sin 65^\circ \operatorname{cosec} 65^\circ}$$

$$= \frac{4 \times 1}{1 + 1} = 2$$

Sol.68.(c)

$$\frac{(\sec \theta + \tan \theta)(1 - \sin \theta)}{\operatorname{cosec} \theta(1 + \cos \theta)(\operatorname{cosec} \theta - \cot \theta)}$$

$$= \frac{(\frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta})(1 - \sin \theta)}{(\operatorname{cosec} \theta + \operatorname{cosec} \theta \cos \theta)(\operatorname{cosec} \theta - \cot \theta)}$$

$$= \frac{(\frac{1}{\cos \theta})(1 + \sin \theta)(1 - \sin \theta)}{(\operatorname{cosec} \theta + \cot \theta)(\operatorname{cosec} \theta - \cot \theta)}$$

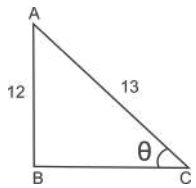
$$= \frac{\frac{\cos^2 \theta}{\cos \theta}}{\operatorname{cosec}^2 \theta - \cot^2 \theta} = \cos \theta$$

Sol.69.(b) $3\cos^2 A + 7\sin^2 A = 4$

$$= 3\cos^2 A + 7\sin^2 A = 4(\cos^2 A + \sin^2 A)$$

$$= 3\sin^2 A = \cos^2 A$$

$$= \cot^2 A = 3 \Rightarrow \cot A = \sqrt{3}$$

Sol.70.(d)

$$\operatorname{Cosec} \theta = \frac{AC}{AB} = \frac{13}{12}$$

$$AB^2 + BC^2 = AC^2 \Rightarrow BC = \sqrt{13^2 - 12^2} = 5$$

$$\sin \theta = \frac{12}{13}, \cos \theta = \frac{5}{13} \text{ and } \tan \theta = \frac{12}{5}$$

$$\sin \theta + \cos \theta - \tan \theta$$

$$= \frac{12}{13} + \frac{5}{13} - \frac{12}{5} = -\frac{71}{65}$$

Sol.71.(c) Given, $\sec 4\theta = \operatorname{Cosec}(\theta + 20^\circ)$

$$= \operatorname{Cosec}(90^\circ - 4\theta) = \operatorname{Cosec}(\theta + 20^\circ)$$

$$\text{Clearly, } (4\theta + \theta + 20^\circ) = 90^\circ$$

$$5\theta = 70^\circ \Rightarrow \theta = 14^\circ$$

Sol.72.(b) Put $\theta = 30^\circ$

$$3\sin \theta = 2\cos^2 \theta = 3 \times \frac{1}{2} = 2 \times \left(\frac{\sqrt{3}}{2}\right)^2$$

$$= \frac{3}{2} = \frac{3}{2} \text{ (Condition satisfied)}$$

$$(\tan^2 \theta + \sec^2 \theta - \operatorname{cosec}^2 \theta)$$

$$= \left(\frac{1}{\sqrt{3}}\right)^2 + \left(\frac{2}{\sqrt{3}}\right)^2 - (2)^2$$

$$= \frac{1}{3} + \frac{4}{3} - 4 = -\frac{7}{3}$$

Sol.73.(b)

Applying componendo and dividendo

$$\frac{\tan \theta + \sin \theta}{\tan \theta - \sin \theta} = \frac{k+1}{k-1} \Rightarrow \frac{\tan \theta}{\sin \theta} = \frac{k}{1}$$

$$\Rightarrow \frac{\sin \theta}{\cos \theta \sin \theta} = k \Rightarrow \sec \theta = k$$

Sol.74.(b)

$$\frac{1}{\operatorname{cosec} \theta - 1} + \frac{1}{\operatorname{cosec} \theta + 1} = 2 \sec \theta$$

$$= \frac{(\operatorname{cosec} \theta + 1) + (\operatorname{cosec} \theta - 1)}{(\operatorname{cosec} \theta + 1)(\operatorname{cosec} \theta - 1)} = 2 \sec \theta$$

$$= \frac{2\operatorname{cosec} \theta}{(\operatorname{cosec}^2 \theta - 1)} = 2 \sec \theta$$

$$= \frac{2\operatorname{cosec} \theta}{\cot^2 \theta} = 2 \sec \theta$$

$$= \frac{2\sin^2 \theta}{\sin \theta \cos^2 \theta} = 2 \times \frac{1}{\cos \theta}$$

$$= \tan \theta = 1 \Rightarrow \theta = 45^\circ$$

$$(\cot \theta + \cos \theta) \Rightarrow 1 + \frac{1}{\sqrt{2}} = \frac{\sqrt{2} + 1}{\sqrt{2}}$$

$$= \frac{2 + \sqrt{2}}{2}$$

Short trick :- Put $\theta = 45^\circ$

$$\text{LHS} \Rightarrow \frac{1}{\operatorname{cosec} \theta - 1} + \frac{1}{\operatorname{cosec} \theta + 1}$$

$$= \frac{1}{\sqrt{2} - 1} + \frac{1}{\sqrt{2} + 1}$$

$$= \sqrt{2} + 1 + \sqrt{2} - 1 = 2\sqrt{2}$$

$$\text{RHS} = 2\sec \theta = 2\sqrt{2}$$

$$\cot \theta + \cos \theta \Rightarrow 1 + \frac{1}{\sqrt{2}} = \frac{\sqrt{2} + 1}{\sqrt{2}}$$

$$= \frac{2 + \sqrt{2}}{2}$$

$$\text{Sol.75.(a)} \sin \theta = \frac{a}{\sqrt{a^2 + b^2}}$$

$$\cos \theta = \sqrt{1 - \left(\frac{a}{\sqrt{a^2 + b^2}}\right)^2} = \frac{b}{\sqrt{a^2 + b^2}}$$

$$\sec \theta + \tan \theta$$

$$= \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} \Rightarrow \frac{1 + \sin \theta}{\cos \theta}$$

$$= \frac{1 + \frac{a}{\sqrt{a^2 + b^2}}}{\frac{b}{\sqrt{a^2 + b^2}}} = \frac{\sqrt{a^2 + b^2} + a}{b}$$

Sol.76.(a) $\cos^2 \theta - 3\cos \theta + 2 = \sin^2 \theta$

$$\Rightarrow \cos^2 \theta - 3\cos \theta + 2 = 1 - \cos^2 \theta$$

$$= 2\cos^2 \theta - 3\cos \theta + 1 = 0$$

$$= 2\cos^2 \theta - 2\cos \theta - \cos \theta + 1 = 0$$

$$= 2\cos \theta(\cos \theta - 1) - 1(\cos \theta - 1) = 0$$

$$\Rightarrow \cos \theta = 1 \text{ or } \frac{1}{2}$$

$$\text{Clearly } \theta = 60^\circ, \text{ (as } 0^\circ < \theta < 90^\circ)$$

$$2\operatorname{cosec} \theta + 4\cot \theta$$

$$= 2\left(\frac{2}{\sqrt{3}}\right) + 4\left(\frac{1}{\sqrt{3}}\right) = \frac{8\sqrt{3}}{3}$$

$$\text{Sol.77.(b)} \frac{1}{\sec x - \tan x} - \frac{1}{\cos x}$$

$$= \sec x + \tan x - \sec x \dots \{ \because \sec x + \tan x \}$$

$$= \frac{1}{\sec x - \tan x} = \tan x$$

Sol.78.(a) $\tan^2 \theta - 3\sec \theta + 3 = 0$

$$\Rightarrow \sec^2 \theta - 1 - 3\sec \theta + 3 = 0$$

$$\Rightarrow \sec^2 \theta - 3\sec \theta + 2 = 0$$

$$\Rightarrow \sec^2 \theta - 2\sec \theta - \sec \theta + 2 = 0$$

$$\Rightarrow \sec \theta(\sec \theta - 2) - 1(\sec \theta - 2) = 0$$

$$\Rightarrow (\sec \theta - 1)(\sec \theta - 2) = 0$$

$$\Rightarrow \sec \theta = 1 \text{ or } 2, \text{ Clearly } \theta = 60^\circ \text{ or } 0^\circ$$

$$\Rightarrow \text{But } \theta \neq 0, \text{ so } \theta = 60^\circ$$

$$\sin \theta + \cot \theta = \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{3}} = \frac{3+2}{2\sqrt{3}} = \frac{5\sqrt{3}}{6}$$

Sol.79.(a)

$$(1 + \tan^2 \theta) + (1 + (\tan^2 \theta)^{-1}) = k$$

$$\Rightarrow (1 + \tan^2 \theta) + (1 + \cot^2 \theta) = k$$

$$\Rightarrow \sec^2 \theta + \operatorname{cosec}^2 \theta = k$$

$$\Rightarrow \frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta} = k$$

$$\Rightarrow \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta} = k \Rightarrow \sec^2 \theta \operatorname{cosec}^2 \theta = k$$

$$\Rightarrow \sec \theta \operatorname{cosec} \theta = \sqrt{k}$$

Short-Trick: Put $\theta = 45^\circ$

$$(1 + \tan^2 \theta) + (1 + (\tan^2 \theta)^{-1}) = k$$

$$\Rightarrow (1 + 1) + (1 + 1) = k$$

$$\Rightarrow 4 = k \Rightarrow 2 = \sqrt{k}$$

Going through options, Option A :

$$\operatorname{cosec} \theta \sec \theta = \sqrt{2} \cdot \sqrt{2} = 2$$

Clearly option (a) is the right answer.

$$\text{Sol.80.(b)} \frac{1}{\sin \theta} - \frac{\cot^2 \theta}{1 + \operatorname{cosec} \theta}$$

$$\Rightarrow \operatorname{cosec} \theta - \frac{(\operatorname{cosec}^2 \theta - 1)}{1 + \operatorname{cosec} \theta}$$

$$\Rightarrow \operatorname{cosec} \theta - \frac{(\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1)}{1 + \operatorname{cosec} \theta}$$

$$\Rightarrow \operatorname{cosec} \theta - \operatorname{cosec} \theta + 1 = 1$$

$$\text{Sol.81.(b)} \left(\frac{1}{1 + \sin^2 \theta} + \frac{1}{1 + \operatorname{cosec}^2 \theta} \right)$$

$$= \left(\frac{1}{1 + \sin^2 \theta} + \frac{1}{1 + \frac{1}{\sin^2 \theta}} \right)$$

$$\Rightarrow \left(\frac{1}{1 + \sin^2 \theta} + \frac{\sin^2 \theta}{1 + \sin^2 \theta} \right) = 1$$

$$\text{Sol.82.(b)} \frac{\cos \theta}{1 - \sin \theta} + \frac{\cos \theta}{1 + \sin \theta} = 4$$

$$= \cos \theta \left(\frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} \right) = 4$$

$$= \cos \theta \left(\frac{1 + \sin \theta + 1 - \sin \theta}{1 - \sin^2 \theta} \right) = 4$$

$$\Rightarrow \cos \theta = \frac{1}{2}, \text{ Clearly } \theta = 60^\circ$$

$$\text{So, } \tan \theta + \operatorname{cosec} \theta \Rightarrow \sqrt{3} + \frac{2}{\sqrt{3}}$$

$$= \frac{3+2}{\sqrt{3}} = \frac{5\sqrt{3}}{3}$$

Sol.83.(c) Put $\theta = 45^\circ$,

$$\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} + 1}{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1}$$

$$= \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1$$

Going through the options



$$\sec\theta + \tan\theta = \sqrt{2} + 1$$

Sol.84.(b) Put $\theta = 45^\circ$

$$\sqrt{\sec^2\theta + \csc^2\theta} \times \sqrt{\tan^2\theta - \sin^2\theta}$$

$$\Rightarrow \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2} \times \sqrt{1 - \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{2}$$

On checking all the options one by one, option (b) satisfies.

$$\sin\theta \sec^2\theta \Rightarrow \frac{1}{\sqrt{2}} \times (\sqrt{2})^2 = \sqrt{2}$$

Sol.85.(c) $\sec\theta - \tan\theta = P$

$$\sec\theta + \tan\theta = \frac{1}{P}$$

$$2\sec\theta = P + \frac{1}{P} \Rightarrow \cos\theta = \frac{2P}{P^2 + 1}$$

$$\cos^2\theta = \left[\frac{2P}{P^2 + 1}\right]^2$$

$$1 - \sin^2\theta = \frac{4P^2}{2P^2 + 1 + P^4}$$

$$\sin^2\theta = 1 - \frac{4P^2}{2P^2 + 1 + P^4}$$

$$= \frac{2P^2 + 1 + P^4 - 4P^2}{2P^2 + 1 + P^4}$$

$$\sin\theta = \frac{1 - P^2}{P^2 + 1}, \csc\theta = \frac{P^2 + 1}{1 - P^2}$$

Alternate :-

$$\sec\theta - \tan\theta = P \Rightarrow \sec\theta + \tan\theta = \frac{1}{P}$$

$$2\sec\theta = P + \frac{1}{P}$$

$$\cos\theta = \frac{2P}{P^2 + 1} = \frac{\text{base}}{\text{hypotenuse}}$$

$$\text{Perpendicular} = 1 - P^2$$

$$\csc\theta = \frac{\text{hypotenuse}}{\text{Perpendicular}} = \frac{P^2 + 1}{1 - P^2}$$

Sol.86.(b) $\cos^2\theta = 3\cot^2\theta - 3\cos^2\theta$

$$\Rightarrow 4\cos^2\theta = 3\cot^2\theta \Rightarrow \sin^2\theta = \frac{3}{4}$$

$$\Rightarrow \sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 60^\circ$$

$$\left(\frac{1}{2}\sec\theta + \sin\theta\right)^{-1} = \frac{1}{\left(\frac{1}{2} \times 2 + \frac{\sqrt{3}}{2}\right)}$$

$$= \frac{2}{\sqrt{3} + 2} = 2(2 - \sqrt{3})$$

Sol.87.(b) $\left[\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta}\right]^2 + 1$

$$= \left[\frac{\sin\theta(1 - 2\sin^2\theta)}{\cos\theta(2\cos^2\theta - 1)}\right]^2 + 1$$

$$= \left[\frac{\sin\theta \cdot \cos 2\theta}{\cos\theta \cdot \cos 2\theta}\right]^2 + 1$$

$$= \tan^2\theta + 1 = \sec^2\theta$$

Sol.88.(c) Let $A = 45^\circ$

$$(\csc A - \sin A)^2 + (\sec A - \cos A)^2 -$$

$$(\cot A - \tan A)^2$$

$$= (\csc 45^\circ - \sin 45^\circ)^2 + (\sec 45^\circ - \cos 45^\circ)^2 - (\cot 45^\circ - \tan 45^\circ)^2$$

$$= \left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)^2 + \left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)^2 - (1 - 1)^2$$

$$= \frac{1}{2} + \frac{1}{2} - 0 = 1$$

Sol.89.(c) $1 + \frac{\tan^2 A}{1 + \sec A} = 1 + \frac{\sec^2 A - 1}{1 + \sec A}$

$$= 1 + \sec A - 1 = \sec A$$

Sol.90.(b) $\tan(90^\circ - \theta) = \sqrt{3}$,

$$\tan(90^\circ - A) = \tan 60^\circ$$

$$(90^\circ - \theta) = 60^\circ$$

$$\theta = 90^\circ - 60^\circ = 30^\circ$$

$$\text{So, } \cos\theta = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Sol.91.(b) Given :- $\sec A = \frac{13}{5}$

We know that (5, 12, 13) is a Pythagorean triplet

$$\text{Hence, } \tan A = \frac{12}{5}$$

Sol.92.(a)

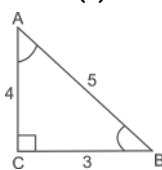
If we use the cofunction form, then

$$\cos A = \sin B \Rightarrow A + B = 90^\circ$$

$$\text{Given:- } A = 90^\circ - x \text{ and } B = 2x$$

$$(90^\circ - x) + (2x) = 90^\circ \Rightarrow x = 90^\circ - 90^\circ = 0^\circ$$

Sol.93.(a)



$$\text{Now, } \tan B = \frac{4}{3}$$

Sol.94.(b) $\sin^2 A - \cos^2 A = \frac{1}{4}$ ----(i)

We know that,

$$\sin^2 A + \cos^2 A = 1 \text{ ----(ii)}$$

Form, equation (i) - equation (ii)

$$-2\cos^2 A = \frac{1}{4} - 1 = \frac{-3}{4} \Rightarrow \cos^2 A = \frac{3}{8}$$

$$\text{Since } \frac{1}{\sec^2 A} = \cos^2 A = \frac{3}{8}$$

Sol.95.(a) $\cos(90^\circ - A) = \cos(2A - 30^\circ)$

$$(90^\circ - A) = (2A - 30^\circ)$$

$$3A = 120^\circ \Rightarrow A = 40^\circ$$

Sol.96.(c) $\sin A + \cos A = \sqrt{2} \cos A$

$$\frac{\sin A}{\cos A} + 1 = \sqrt{2}$$

$$\tan A = \sqrt{2} - 1 \Rightarrow \cot A = \frac{1}{\sqrt{2} - 1}$$

$$= \frac{\sqrt{2} + 1}{1} = \sqrt{2} + 1$$

Sol.97.(c) $\sin\theta = \frac{15}{17}$

(8, 15, 17) is a Pythagorean triplet.

$$\therefore \tan\theta = \frac{15}{8}$$

Sol.98.(a) Given :- $\tan X = 3$,

$$\cot X = \frac{1}{\tan X} = \frac{1}{3}$$

$$\tan X + \cot X = 3 + \frac{1}{3} = \frac{10}{3}$$

Sol.99.(d) $\sin B + \cos B = \frac{13}{12}$

Squaring on both sides

$$(\sin^2 B + \cos^2 B + 2\sin B \cos B) = \frac{169}{144}$$

$$2\sin B \cos B = \frac{169}{144} - 1 \Rightarrow \sin 2B = \frac{25}{144}$$

Sol.100.(c) Given,

$$\sin A = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{5}{13}$$

By using pythagorean triplet (5, 12, 13), base = 12

$$\Rightarrow \cos A = -\frac{12}{13} \text{ (Since A is in the second quadrant)}$$

Now, $(\sin A + \cos A)^2 = \sin^2 A + \cos^2 A$

$$+ 2\sin A \times \cos A$$

$$= 1 + 2 \times \frac{5}{13} \times -\frac{12}{13} = \frac{49}{169}$$

Sol.101.(a) Given, $\sin A = \frac{p}{q}$

By using pythagorean triplet (p, q,

$$\sqrt{q^2 - p^2}), \text{ base} = \sqrt{q^2 - p^2}$$

$$1 + \tan^2 A = \sec^2 A$$

$$\Rightarrow \left(\frac{1}{\cos A}\right)^2 = \left(\frac{q}{\sqrt{q^2 - p^2}}\right)^2 = \frac{q^2}{q^2 - p^2}$$

Sol.102.(c) We know: $\sec^2 A = 1 + \tan^2 A$

$$\Rightarrow 1 + x^2 + \frac{1}{x^2} + 2 = x^2 + \frac{1}{x^2} + 3$$

Sol.103.(b) $\sin^4 A - \cos^4 A = \frac{1}{2}$

$$(\sin^2 A - \cos^2 A)(\sin^2 A + \cos^2 A) = \frac{1}{2}$$

$$(\sin^2 A - \cos^2 A) \times 1 = \frac{1}{2}$$

$$\sin^2 A - \cos^2 A = \frac{1}{2}$$

$$1 - \cos^2 A - \cos^2 A = \frac{1}{2}$$

$$-2\cos^2 A = -\frac{1}{2} \Rightarrow \cos^2 A = \frac{1}{4}$$

Sol.104.(c) Given :- $\cot A = x + \frac{1}{x}$

$$\sec^2 A = 1 + \tan^2 A = 1 + \frac{1}{\cot^2 A}$$

$$= 1 + \left(x + \frac{1}{x}\right)^2$$

Sol.105.(b) $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$$= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}} = \frac{\frac{5}{6}}{\frac{5}{6}} = 1$$

$$\tan(A + B) = 1$$

Sol.106.(b) $\sec A = 2$

$$\sec A = \sec 60^\circ \Rightarrow A = 60^\circ$$

$$\tan A - \operatorname{cosec} A$$

$$= \tan 60^\circ - \operatorname{cosec} 60^\circ$$

$$= \sqrt{3} - \frac{2}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\text{Sol.107.(b)} \tan B + \cot B = 4$$

$$\Rightarrow \tan B + \frac{1}{\tan B} = 4$$

$$\Rightarrow \tan^2 B - 4 \tan B + 1 = 0$$

By using quadratic formula,

$$\Rightarrow \tan B = \frac{4 \pm \sqrt{16 - 4 \times 1 \times 1}}{2 \times 1}$$

$$\Rightarrow \tan B = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

Now,

$$\Rightarrow \tan^2 B + \sec^2 B = 2 \tan^2 B + 1$$

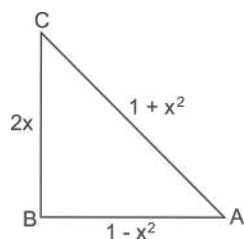
$$= 2(2 \pm \sqrt{3})^2 + 1$$

$$= 2(4 \pm 3 \pm 4\sqrt{3}) + 1 = 15 \pm 8\sqrt{3}$$

So, the according to options,

$$\Rightarrow \tan^2 B + \sec^2 B = 15 + 8\sqrt{3}$$

$$\text{Sol.108.(c)} \sin A = \frac{2x}{1+x^2}$$



$$\tan A = \frac{BC}{AB} = \frac{2x}{1-x^2}$$

$$\text{Sol.109.(c)} \sin A + \cos A = \frac{3}{2}$$

Squaring on both sides

$$\sin^2 A + \cos^2 A + 2 \sin A \cos A = \frac{9}{4}$$

$$1 + 2 \sin A \cos A = \frac{9}{4}$$

$$2 \sin A \cos A = \frac{9}{4} - 1 \Rightarrow \sin A \cos A = \frac{5}{8}$$

$$\text{Sol.110.(c)}$$

$$\sin 40^\circ \cos 20^\circ + \cos 40^\circ \sin 20^\circ$$

We know that :- $\sin(A+B) = \sin A \cos B + \sin B \cos A$

$$\therefore \sin(40^\circ + 20^\circ) \Rightarrow \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\text{Sol.111.(b)} \text{ Given :- } \sin x = \frac{7}{25}$$

As $7^2 + 24^2 = 25^2$, the set (7, 24, 25) is a Pythagorean triplet

$$\therefore \tan x = \frac{7}{24}$$

$$\text{Now, } \frac{1 + \tan x}{1 - \tan x} = \frac{1 + \frac{7}{24}}{1 - \frac{7}{24}} = \frac{31}{17}$$

$$\text{Sol.112.(b)} \text{ Given, } \sin A + \cos A = Y$$

$$1 + 2 \sin A \times \cos A = Y^2$$

$$\Rightarrow 2 \sin A \times \cos A = Y^2 - 1$$

$$\text{Now, } \sin^2 A + \cos^2 A + 2 \sin A \cos A$$

$$\Rightarrow 1 + Y^2 - 1 = Y^2$$

$$\text{Sol.113.(b)} \text{ Given, } \sin x = \sqrt{3} \cos x$$

$$\Rightarrow \tan x = \sqrt{3} \Rightarrow x = 60^\circ$$

$$\text{Now, } \sin^4 x + \cos^4 x = \left(\frac{\sqrt{3}}{2}\right)^4 + \left(\frac{1}{2}\right)^4$$

$$= \frac{9}{16} + \frac{1}{16} \Rightarrow \frac{10}{16} = \frac{5}{8}$$

$$\text{Sol.114.(d)} \cot A = \sqrt{3}$$

$$\cot A = \cot 30^\circ \Rightarrow A = 30^\circ$$

$$(1 + \sec A)(1 + \operatorname{cosec} A)$$

$$= (1 + \sec 30^\circ)(1 + \operatorname{cosec} 30^\circ)$$

$$= \left(1 + \frac{2}{\sqrt{3}}\right)(1 + 2) = \frac{3\sqrt{3} + 6}{\sqrt{3}}$$

$$\text{Sol.115.(b)} \text{ Given: } \sec x + \tan x = \sqrt{2}$$

We know that

$$\sec^2 x - \tan^2 x = 1$$

$$(\sec x + \tan x)(\sec x - \tan x) = 1$$

$$(\sec x - \tan x) = \frac{1}{\sqrt{2}}$$

$$\text{Sol.116.(c)} \text{ Given: } \operatorname{cosec} A = y$$

We know

$$\operatorname{cosec}^2 A - \cot^2 A = 1$$

$$\operatorname{cosec}^2 A - 1 = \cot^2 A$$

$$y^2 - 1 = \cot^2 A$$

$$\text{Sol.117.(a)} \text{ Given :- } \cot A = \frac{11}{60}$$

$$\operatorname{cosec}^2 A - \cot^2 A = 1$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A = \sqrt{1 + \left(\frac{11}{60}\right)^2}$$

$$\sqrt{\frac{3600 + 121}{3600}} = \sqrt{\frac{3721}{3600}} = \frac{61}{60}$$

$$\text{Sol.118.(a)} \text{ Given : } \sin x = 0.8$$

$$\Rightarrow \sin x = \frac{4}{5}, \cos x = \frac{3}{5}$$

$$\Rightarrow \sin 2x = 2 \sin x \cos x$$

$$\Rightarrow \cos 2x = 1 - 2 \sin^2 x$$

Now,

$$\sin 2x + \cos 2x$$

$$\Rightarrow 2 \sin x \cos x + 1 - 2 \sin^2 x$$

$$= 2 \times \frac{4}{5} \times \frac{3}{5} + 1 - 2 \times \left(\frac{4}{5}\right)^2$$

$$= \frac{24}{25} + 1 - \frac{32}{25} \Rightarrow 1 - \frac{8}{25} = \frac{17}{25}$$

$$= 0.68$$

Sol.119.(a) The value of $\tan \theta$ and $\cot \theta$ becomes same when $\theta = 45^\circ$.

$$\text{Sol.120.(b)} \cos B = 1 - 2 \sin^2 B$$

$$\text{We know: } \sin^2 B + \cos^2 B = 1$$

$$\Rightarrow \cos B = \sin^2 B + \cos^2 B - 2 \sin^2 B$$

$$\Rightarrow \cos B = \cos^2 B - \sin^2 B = \cos 2B$$

$$\text{So, } \cos 2B = 1 - 2 \sin^2 B$$

$$\text{Sol.121.(d)} A/Q,$$

$$2x + 30^\circ + 5x - 3^\circ = 90^\circ$$

$$27^\circ + 7x = 90^\circ \Rightarrow 7x = 63^\circ \Rightarrow x = 9^\circ$$

$$\text{Sol.122.(a)} \text{ Given :- } \sin A = \frac{5}{13}$$

5, 12, 13 is a Pythagorean triplet.

$$\therefore \cos A = \frac{12}{13}$$

Sol.123.(a) 3, 4, 5 is a Pythagorean triplet.

$$\therefore \sin A = \frac{3}{5}, \cos A = \frac{4}{5}$$

$$\sin^4 A + \cos^4 A = \left(\frac{3}{5}\right)^4 + \left(\frac{4}{5}\right)^4$$

$$= \frac{81}{625} + \frac{256}{625} = \frac{337}{625}$$

$$\text{Sol.124.(a)} \text{ Given : } \sin A + \cos A = \sqrt{2}$$

Squaring both sides

$$(\sin A + \cos A)^2 = 2$$

$$\sin^2 A + \cos^2 A + 2 \sin A \cos A = 2$$

$$1 + 2 \sin A \cos A = 2$$

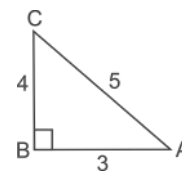
$$2 \sin A \cos A = 1 \Rightarrow \sin 2A = 1$$

$$\text{Sol.125.(a)} \text{ According to question}$$

$$\sin A = \frac{3}{5} \text{ and } \cos A = \frac{4}{5}$$

$$\text{Then, } \tan A = \frac{\sin A}{\cos A} = \frac{\frac{3}{5}}{\frac{4}{5}} = \frac{3}{4}$$

$$\text{Sol.126.(a)} \cos A = \frac{3}{5}$$



$$\tan(90^\circ - A) = \cot A$$

$$\cot A = \frac{3}{4}$$

$$\text{Sol.127.(a)} \text{ Given : } \tan A = x + \frac{1}{x}$$

We know that,

$$\Rightarrow \sec^2 A = 1 + \tan^2 A$$

$$\Rightarrow \sec^2 A = 1 + \left(x + \frac{1}{x}\right)^2$$

$$\Rightarrow \sec^2 A = 1 + \left(x^2 + \frac{1}{x^2} + 2\right)$$

$$\Rightarrow \sec^2 A = x^2 + \frac{1}{x^2} + 3$$

$$\text{Sol.128.(a)} \text{ Given: } \sec A + \tan A = \frac{3}{2}$$

$$\text{We know: } \sec^2 A - \tan^2 A = 1$$

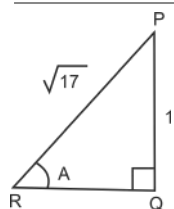
$$(\sec A + \tan A)(\sec A - \tan A) = 1$$

$$\frac{3}{2} \times (\sec A - \tan A) = 1$$

$$\sec A - \tan A = \frac{2}{3}$$

$$\text{Sol.129.(a)} \sin A = \frac{1}{\sqrt{17}}$$





$$\Rightarrow RQ = \sqrt{RP^2 - PQ^2} = \sqrt{17 - 1}$$

$$= \sqrt{16} = 4$$

Now, $\tan A + \sec A$

$$= \frac{1}{4} + \frac{\sqrt{17}}{4} = \frac{(1 + \sqrt{17})}{4}$$

Sol. 130.(a) $\cos^2 90^\circ + \sin^2 60^\circ$

$$= 0^2 + \left(\frac{\sqrt{3}}{2}\right)^2 = 0 + \frac{3}{4} = \frac{3}{4}$$

Sol. 131.(b) $\sin = \frac{9}{41}$

We know that, (9, 40, 41) is triplet,
Base = 40

$\sec A = -\frac{41}{40}$ (Since, A lies in second quadrant)

Now, $\sin A \times \sec A = \frac{9}{41} \times -\frac{41}{40} = -\frac{9}{40}$

Sol. 132.(c) $\tan A = \frac{x}{y} = \frac{P}{B}$

So, $H = \sqrt{x^2 + y^2}$

Then, $\sin A = \frac{P}{H} = \frac{x}{\sqrt{x^2 + y^2}}$

Sol. 133.(a)

$$(15\sin A + 20\cos A)^2 + (5\sin A + 10\cos A)^2$$

$$\left(15 \times \frac{3}{5} + 20 \times \frac{4}{5}\right)^2 + \left(5 \times \frac{3}{5} + 10 \times \frac{4}{5}\right)^2$$

$$(9 + 16)^2 + (3 + 8)^2$$

$$625 + 121 = 746$$

Sol. 134.(a)

Applying the value putting method :-

$$\sin \beta = \cos \beta$$

Putting, $\beta = 45^\circ$

$$\sin 45 = \cos 45^\circ$$

$$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \text{ (LHS = RHS)}$$

So, $\beta = 45^\circ$

$$\sin^3 \beta + \cos^3 \beta = \left(\frac{1}{\sqrt{2}}\right)^3 + \left(\frac{1}{\sqrt{2}}\right)^3$$

$$= \frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{2}$$

Sol. 135.(c) $\tan 2\theta = \cot(3\theta + 10^\circ)$

$$\Rightarrow \cot(90^\circ - 2\theta) = \cot(3\theta + 10^\circ)$$

$$\Rightarrow -2\theta - 3\theta = 10^\circ - 90^\circ$$

$$\Rightarrow 5\theta = 80^\circ \Rightarrow \theta = 16^\circ$$

Sol. 136.(c) Given that: $\sin \theta + \cos \theta = \frac{\sqrt{3}}{2}$

We know that,

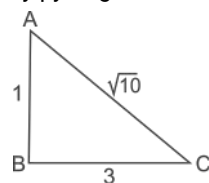
$$\Rightarrow \sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta$$

$$= (\sin \theta + \cos \theta)^2 = \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}$$

Sol. 137.(c) given

$$\sin C = \frac{1}{\sqrt{10}}$$

by pythagoras theorem



$$\Rightarrow \cot C = \frac{3}{1} = 3$$

Sol. 138.(b) Given, $\sin A = 0.8$

We know that

$$\sin^2 A + \cos^2 A = 1$$

$$\cos^2 A = 1 - (0.8)^2 = 1 - 0.64 = 0.36$$

$$\cos A = \sqrt{0.36} = 0.6$$

Then,

$$\tan A = \frac{\sin A}{\cos A} = \frac{0.8}{0.6} = 1.33$$

Sol. 139.(c) Given, $\operatorname{cosec} A + \cot A = r$

Now multiply both sides by $\operatorname{cosec} A - \cot A$

$$(\operatorname{cosec} A + \cot A)(\operatorname{cosec} A - \cot A)$$

$$= r(\operatorname{cosec} A - \cot A)$$

$$1 = r(\operatorname{cosec} A - \cot A)$$

$$(\operatorname{cosec} A - \cot A) = \frac{1}{r}$$

$$(\operatorname{cosec} A + \cot A) + (\operatorname{cosec} A - \cot A)$$

$$= r + \frac{1}{r}$$

$$2 \operatorname{cosec} A = r + \frac{1}{r}$$

$$\operatorname{cosec} A = \frac{r^2 + 1}{2r}$$

Sol. 140.(b) $\cos(90^\circ - A) = \sin A$

Sol. 141.(d) given, $\cos \theta = \frac{5}{13}$

Where θ lies in 2nd quadrant

Then,

$$\text{Perpendicular (p)} = 12, \text{Base (b)} = 5,$$

$$H = 13$$

Now

$$(1 + \sin \theta)(1 - \sin \theta)$$

$$= \left(1 + \frac{12}{13}\right)\left(1 - \frac{12}{13}\right) = \frac{25}{13} \times \frac{1}{13} = \frac{25}{169}$$

Sol. 142.(a) Given, $\cos x = \sin(3x + 10^\circ)$

We know that,

$$\sin \theta = \cos(90^\circ - \theta)$$

Then comparing,

$$\cos x = \cos(90^\circ - (3x - 10^\circ))$$

$$x = 90^\circ - (3x - 10^\circ)$$

$$x = 90^\circ - 3x + 10^\circ$$

$$x = 100^\circ - 3x$$

$$4x = 100^\circ \Rightarrow x = 25^\circ$$

Sol. 143.(a) Given, $\cos A = x \Rightarrow \cos A$

$$= \frac{x \rightarrow b}{1 \rightarrow h} \text{ then, } p = \sqrt{1 - x^2}$$

$$\text{So, } \cot A = \frac{x}{\sqrt{1 - x^2}}$$

Sol. 144.(b)

According to pythagoras theorem

$$\cos A = \frac{3 \rightarrow b}{5 \rightarrow h} \text{ Then } p = 4$$

$$\cot A = \frac{b \rightarrow}{p \rightarrow} = \frac{3}{4}$$

Sol. 145.(d) We know that:-

$$\tan(A + B) = 90^\circ$$

$$x + 2x - 10^\circ = 90^\circ$$

$$3x = 100 \Rightarrow x = 33.33$$

Sol. 146.(b) $\sin A = \frac{35}{37} = \frac{\text{perpendicular}}{\text{hypotenous}}$

By pythagoras theorem

$$\text{base}^2 = 37^2 - 35^2$$

$$\text{base}^2 = 1369 - 1225 \Rightarrow 144$$

$$\text{base} = 12$$

$$\cot A = \frac{12}{35}$$

Sol. 147.(c) Given: $A + B = 90^\circ$

$$B = 90^\circ - A$$

$$\sin B = \sin(90^\circ - A) = \cos A$$

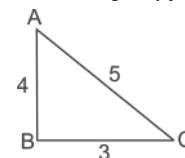
$$\sin A = \frac{3}{5}, \cos A = \frac{4}{5} \dots \dots \text{(using triplet 3, 4, 5)}$$

$$\sin A \times \sin B = \sin A \times \cos A$$

$$= \frac{3}{5} \times \frac{4}{5} = \frac{12}{25}$$

Sol. 148.(c) $\cot A = \frac{4}{3}$

According to pythagoras



then,

$$\operatorname{cosec} A = \frac{1}{\sin A} = \frac{5}{3}$$

Sol. 149.(c) Given: $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

$$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

According to the question,

$$= 3\operatorname{cosec}^2 30^\circ - 3\cot^2 30^\circ$$

$$= 3(\operatorname{cosec}^2 30^\circ - \cot^2 30^\circ) = 3 \times 1 = 3$$

Sol. 150.(c) According to the question,

$$\sec(90^\circ - \theta) = y$$

$$\Rightarrow \operatorname{cosec} \theta = y \Rightarrow \frac{1}{\sin \theta} = y \Rightarrow \sin \theta = \frac{1}{y}$$

Sol. 151.(c) $\sec^2 A - \tan^2 A = 1$

$$(\sec A - \tan A)(\sec A + \tan A) = 1$$

$$\frac{1}{4} \times (\sec A + \tan A) = 1$$

$$\sec A + \tan A = \frac{1}{\frac{1}{4}} = 4$$

Sol. 152.(a) We know that $\sin A = \cos B$

when, $A + B = 90^\circ$

Now,

$$2x + (3x - 60^\circ) = 90^\circ$$

$$\Rightarrow 5x - 60^\circ = 90^\circ \Rightarrow x = 30^\circ$$



Sol.153.(d) $\tan A + \cot A = \frac{4}{5}$

Squaring on both side

$$\tan^2 A + \cot^2 A + 2 \tan A \cot A = \left(\frac{4}{5}\right)^2$$

$$\tan^2 A + \cot^2 A + 2 \tan A \cot A = \frac{16}{25}$$

Sol.154.(c) $\cos 2A = 3$

We know that: $\cos 2A = 2\cos^2 A - 1$

$$2\cos^2 A - 1 = 3 \Rightarrow 2\cos^2 A = 4$$

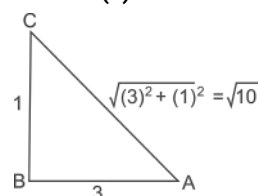
$$\cos^2 A = 2 \Rightarrow \cos A = \sqrt{2}$$

Sol.155.(b) $\operatorname{cosec} x = \sec(4x - 40^\circ)$

$$\operatorname{cosec} x = \operatorname{cosec}(90^\circ - 4x + 40^\circ)$$

$$x = 130^\circ - 4x \Rightarrow x = 26^\circ$$

Sol.156.(d)



$$\cos A = \frac{3}{\sqrt{10}}$$

Sol.157.(b)

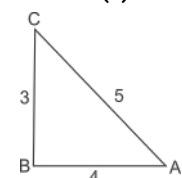
We know that $\cot Q = \tan(90^\circ - Q)$

So, $\tan P = \tan(90^\circ - Q)$

$\Rightarrow P = 90^\circ - Q$ (since both are acute angles)

$$P + Q = 90^\circ$$

Sol.158.(c)



$$\cos A = \frac{4}{5}, \sin A = \frac{3}{5}$$

$$\sin^2 A + \cos^2 A - 2\sin A \cos A$$

$$= 1 - 2\sin A \cos A \quad [\sin^2 A + \cos^2 A = 1]$$

$$= 1 - 2 \times \frac{3}{5} \times \frac{4}{5} = 1 - \frac{24}{25} = \frac{1}{25}$$

Sol.159.(a) Given, $\sec A - \cos A = 2x$

Let, $\cos A = x$

$$\text{Now, } \Rightarrow \frac{1}{\cos A} - \cos A = 2$$

$$\Rightarrow \frac{1}{x} - x = 2 \Rightarrow 1 - x^2 = 2x$$

$$\Rightarrow x^2 + 2x - 1 \Rightarrow x = \frac{-2 \pm \sqrt{4 + 4}}{2}$$

$$x = -1 + \sqrt{2}$$

Hence, value of $\cos A = -1 + \sqrt{2}$

Sol.160.(a) $\cos \theta = \frac{8}{17} \Rightarrow \sin \theta = \frac{15}{17}$

if θ lies in the 3rd quadrant only $\tan \theta$ and $\cot \theta$ are positive other trigonometric function is negative

$$\text{Hence, } \sin \theta = -\frac{15}{17}$$

Sol.161.(b) Given, $\cos \theta = \frac{3}{5}$

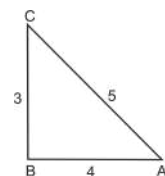
$$\cot \theta = \frac{3}{4}$$

Now, $\cot \theta \times \tan(90^\circ - \theta)$

$$= \cot^2 \theta \Rightarrow \cot^2 \theta = \frac{9}{16}$$

Sol.162.(a) Given:

$$4 \tan A = 3 \text{ or } \tan A = \frac{3}{4} = \frac{P}{B}$$



$$\begin{aligned} \text{Now, } 3 \sin A - 4 \cos A &= 3 \times \frac{3}{5} - 4 \times \frac{4}{5} \\ &= \frac{9}{5} - \frac{16}{5} = \frac{-7}{5} \end{aligned}$$

Sol.163.(c) We know that,

$$\sin A = \cos B \Rightarrow A + B = 90^\circ$$

$$\text{So, } 2x + (3x - 20^\circ) = 90^\circ$$

$$5x = 110^\circ \Rightarrow x = 22^\circ$$

Sol.164.(c) Given:-

$$\tan A + \sec A = 5 \text{ ---(i)}$$

We know the identity: $\sec^2 A - \tan^2 A = 1$

$$(\sec A - \tan A)(\sec A + \tan A) = 1$$

$$\sec A - \tan A = \frac{1}{5} \text{ ---(ii)}$$

[(i) - (ii)],

$$2 \tan A = 5 - \frac{1}{5} \Rightarrow \tan A = \frac{24}{5 \times 2} = \frac{12}{5}$$

Sol.165.(d) $\sin A = \frac{2x}{(1+x^2)}$

(Hence, $2x, (1-x^2), (1+x^2)$ is triplet)

$$\cot A = \frac{B}{P} = \frac{(1-x^2)}{2x}$$

Sol.166.(a) $160^\circ = \frac{\pi}{180^\circ} \times 160^\circ = \frac{8\pi}{9}$

Sol.167.(c) Put $x = y = 45^\circ$

$$\tan 45^\circ = \cot 45^\circ = 1$$

$$\therefore x + y = 45^\circ + 45^\circ = 90^\circ$$

Sol.168.(c) In a right-angled triangle $\triangle ABC$ (right-angled at C):

We know that:

$$\sin A = \cos B = \frac{3}{5}$$

$$\text{So, } \cos B = \frac{3}{5}$$

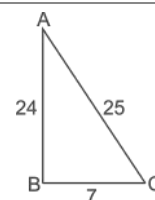
Sol.169.(c) Given: $A + B = 90^\circ$

so, $\sin A = \cos B$

$$\sin A = \frac{35}{37}$$

$$\text{Hence, } \cos B = \frac{35}{37}$$

Sol.170.(a) Given:- $\tan A = \frac{7}{24}$



$$\sin A = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{7}{25}$$

Sol.171.(b) Given: $\sec x - \tan x = \frac{1}{3}$

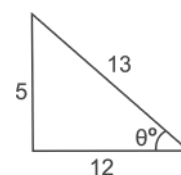
We know that: $\sec^2 x - \tan^2 x = 1$

$$(\sec x - \tan x)(\sec x + \tan x) = 1$$

$$\frac{1}{3} \times (\sec x + \tan x) = 1$$

$$\sec x + \tan x = \frac{1}{\frac{1}{3}} = 3$$

Sol.172.(d) $\tan \theta = \frac{5}{12}$



$$\sec \theta = \frac{h}{b} = \frac{13}{12}$$

Sol.173.(d) We know that,

$$\sin A = \cos B \Rightarrow A + B = 90^\circ$$

So,

$$(2x + 15^\circ) + (x - 5^\circ) = 90^\circ$$

$$3x + 10^\circ = 90^\circ \Rightarrow x = 26.67^\circ$$

Sol.174.(b) $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

$$= \frac{\frac{1}{2} - \frac{1}{3}}{1 + \frac{1}{2} \times \frac{1}{3}} \Rightarrow \frac{\frac{1}{6}}{\frac{7}{6}} = \frac{1}{7}$$

$$\tan(A - B) = \frac{1}{7}$$

Sol.175.(b)

$$\cos^4 \theta = (\cos^2 \theta)^2 = \left(\frac{1 + \cos 2\theta}{2}\right)^2$$

$$= \frac{(1 + \cos 2\theta)^2}{2^2} = \frac{(1 + \cos 2\theta)^2}{4}$$

Sol.176.(b) $\cos A = \frac{4}{5}$

3, 4, and 5 form a Pythagorean triplet.

$$\sin A + \cot A = \frac{3}{5} + \frac{4}{3} = \frac{29}{15}$$

Sol.177.(b) $\sin A = \frac{5}{13}$

5, 12, and 13 form a Pythagorean triplet.

$$\therefore \cot A = \frac{12}{5}$$

Sol.178.(c) Given:

$$\tan P = \cot Q = \tan(90^\circ - Q)$$

Since both are acute angles,

$$P = 90^\circ - Q$$

Therefore, $P + Q = 90^\circ$

Sol.179.(d) $(\sec^2 \theta - \tan^2 \theta)^2$
 $= \sec^4 \theta + \tan^4 \theta - 2\sec^2 \theta \tan^2 \theta$
 $\Rightarrow 1 = \sec^4 \theta + \tan^4 \theta - 2\sec^2 \theta \tan^2 \theta$
 $\Rightarrow 1 + 2\sec^2 \theta \tan^2 \theta = \sec^4 \theta + \tan^4 \theta$

Sol.180.(b) Given:

$\sin(x-y) = 1 \Rightarrow x-y = 90^\circ$ ----(i)
 $\cos(x+y) = 0 \Rightarrow x+y = 90^\circ$ ----(ii)

Add both equations:

$x-y = 90^\circ, x+y = 90^\circ$

$\Rightarrow x = 90^\circ, y = 0^\circ$

Now evaluate:

$(\cos x + \cos y)^2 + (\sin x + \sin y)^2$
 $= (\cos 90^\circ + \cos 0^\circ)^2 + (\sin 90^\circ + \sin 0^\circ)^2$
 $= (0+1)^2 + (1+0)^2 = 1+1 = 2$

Sol.181.(a) $\tan \theta - \cot \theta = k$

Squaring both sides:

$(\tan \theta - \cot \theta)^2 = k^2$
 $\Rightarrow \tan^2 \theta + \cot^2 \theta - 2\tan \theta \cdot \cot \theta = k^2$
 $\Rightarrow \tan^2 \theta + \cot^2 \theta - 2\tan \theta \cdot \frac{1}{\tan \theta} = k^2$
 $\Rightarrow \tan^2 \theta + \cot^2 \theta = k^2 + 2$

Sol.182.(c)

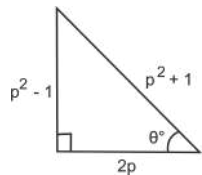
$\sec \theta + \tan \theta = p$ ----(i)

$\sec \theta - \tan \theta = \frac{1}{p}$ ----(ii)

$[\therefore \sec^2 \theta - \tan^2 \theta = 1]$

(i) + (ii)

$2\sec \theta = p + \frac{1}{p} \Rightarrow \sec \theta = \frac{p^2 + 1}{2p}$



$\sin \theta = \frac{(p^2 - 1)}{(p^2 + 1)}$

Sol.183.(c) $5\cos^2 \theta + 4\sin^2 \theta = \frac{17}{4}$

$\Rightarrow 5\cos^2 \theta + 4(1 - \cos^2 \theta) = \frac{17}{4}$

$\Rightarrow 5\cos^2 \theta + 4 - 4\cos^2 \theta = \frac{17}{4}$

$\Rightarrow \cos^2 \theta = \frac{17}{4} - 4 = \frac{1}{4}$

Sol.184.(d) Given:- $\sin \theta + \cos \theta = \frac{7}{5}$

$\Rightarrow \sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta = \frac{49}{25}$

$\Rightarrow 2\sin \theta \cos \theta = \frac{49}{25} - 1 = \frac{24}{25}$

$\Rightarrow \sin \theta \cos \theta = \frac{12}{25}$

$\sin^4 \theta + \cos^4 \theta = (\sin^2 \theta + \cos^2 \theta)^2 - 2\sin^2 \theta \cos^2 \theta$

$\Rightarrow \sin^4 \theta + \cos^4 \theta = 1 - 2 \times \frac{144}{625} = \frac{337}{625}$

Sol.185.(d)

$\sin(90^\circ - \theta) = \cos \theta$ (Using the Identity)

$\Rightarrow \cos \theta = \frac{3}{4}$

Sol.186.(c)

Given:- $\sec(90^\circ - x) = \operatorname{cosec}(2x - 30^\circ)$

$\sec(90^\circ - x) = \operatorname{cosec} x$ (Using the complementary angle identity)

$\operatorname{cosec} x = \operatorname{cosec}(2x - 30^\circ)$

$x = 2x - 30^\circ \Rightarrow x = 30^\circ$

Sol.187.(c) $\sin 45^\circ \times \cos 45^\circ$

$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{2}$

Sol.188.(c) $\sin \theta = \frac{8}{17} \Rightarrow \cos \theta = \frac{15}{17}$

So, $\cos \theta + \sin \theta = \frac{15}{17} + \frac{8}{17} = \frac{23}{17}$

Sol.189.(b) Given, $\cos \theta = \sin 2\theta$

$\Rightarrow \cos \theta = 2\sin \theta \cos \theta$

$\Rightarrow 1 = 2\sin \theta$

Then, $\sin \theta = \frac{1}{2}$

Sol.190.(c) $(\sin \theta - \cos \theta) = \frac{1}{5}$

Squaring both sides

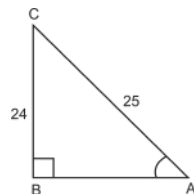
$\sin^2 \theta + \cos^2 \theta - 2\sin \theta \cos \theta = \frac{1}{25}$

$1 - \frac{1}{25} = 2\sin \theta \cos \theta$

$\frac{24}{25} = 2\sin \theta \cos \theta$ ($\because 2\sin \theta = 2\sin \theta \cos \theta$)

$\sin 2\theta = \frac{24}{25}$

Sol.191.(d)



In $\triangle ABC$,

From pythagorean triplet = (7, 24, 25)

So, BA = 7 units

$\Rightarrow \cot A = \frac{7}{24}$

Sol.192.(a) $\sec A + \tan A = 8$

We know,

$(\sec A + \tan A)(\sec A - \tan A) = 1$

$\Rightarrow 8 \times (\sec A - \tan A) = 1$

$\Rightarrow (\sec A - \tan A) = \frac{1}{8} = 0.125$

Sol.193.(b) $\cot \theta = \frac{40}{41}$

$(\sec^2 \theta - 1) = \tan^2 \theta = \frac{1681}{1600}$

Sol.194.(d) We know that,
 $\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$

$\frac{9}{16} = 1 + \cot^2 \theta$

Sol.195.(d) $\operatorname{cosec} A + \cot A = 5$ (i)

We know: $\operatorname{cosec}^2 A - \cot^2 A = 1$

$(\operatorname{cosec} A + \cot A)(\operatorname{cosec} A - \cot A) = 1$

$\operatorname{cosec} A - \cot A = \frac{1}{5}$ (ii)

After solving eq.(i) and (ii) we get

$\operatorname{cosec} A = \frac{26}{5 \times 2} = 2.6$

Sol.196.(d) Given,

$(\sin^6 A + \cos^6 A + 3\sin^2 A \cdot \cos^2 A) - (\sin^4 A - \cos^4 A + 1) \operatorname{cosec}^2 A$

Put $\angle A = 90^\circ$

$\Rightarrow (1 + 0 + 0) - (1 - 0 + 1)1 = 1 - 2 = -1$

Sol.197.(d) Given, $3(\sec^2 A - 1)(\operatorname{cosec} A - 1)(\operatorname{cosec} A + 1) + 4(\operatorname{cosec}^2 A - \cot^2 A)$

Using identities : $\sec^2 A - 1 = \tan^2 A$,
 $\operatorname{cosec}^2 A - \cot^2 A = 1$,

$a^2 - b^2 = (a - b)(a + b)$

$= 3 \tan^2 A (\operatorname{Cosec}^2 A - 1^2) + 4(1)$

$= 3 \tan^2 A \cot^2 A + 4$ (as $\operatorname{Cosec}^2 A - 1 = \cot^2 A$)

$= 3 \tan^2 A \times \frac{1}{\tan^2 A} + 4$ (as $\cot^2 A = \frac{1}{\tan^2 A}$)

$= 3 + 4 = 7$

Sol.198.(d) $\sin^2 30^\circ - \sin^2 40^\circ + \sin^2 45^\circ - \sin^2 55^\circ - \sin^2 35^\circ + \sin^2 45^\circ - \sin^2 50^\circ + \sin^2 60^\circ$

$= (\sin^2 30^\circ + \sin^2 60^\circ) - (\sin^2 40^\circ + \sin^2 50^\circ)$

$+ (\sin^2 45^\circ + \sin^2 45^\circ) - (\sin^2 55^\circ + \sin^2 35^\circ)$

$= 1 - 1 + 1 - 1$ ($\because \sin^2 \theta + \sin^2(90^\circ - \theta) = 1$)

$= 0$

Sol.199.(c) $\cos^2 29^\circ + \cos^2 61^\circ$

$= \cos^2 29^\circ + \cos^2(90^\circ - 29^\circ)$

$= \cos^2 29^\circ + \sin^2 29^\circ = 1$

Sol.200.(c) Given : $\operatorname{cosec} \theta = \frac{5}{3}$

$\Rightarrow (\sec^2 \theta - 1) \times \cot^2 \theta \times (1 + \cot^2 \theta)$

We know,

$\sec^2 \theta = 1 + \tan^2 \theta$, $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$

and $\tan^2 \theta \times \cot^2 \theta = 1$

Then, $(1 + \tan^2 \theta - 1) \times \cot^2 \theta \times (\operatorname{cosec}^2 \theta)$

$= \tan^2 \theta \times \cot^2 \theta \times \operatorname{cosec}^2 \theta$

$= \operatorname{cosec}^2 \theta = \left(\frac{5}{3}\right)^2 = \frac{25}{9}$

Sol.201.(a) Identity used:

$a^3 - b^3 = (a - b)(a^2 + b^2 + ab)$

$\frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} =$

$\frac{(\sin A - \cos A)[\sin^2 A + \cos^2 A + \sin A \cdot \cos A]}{\sin A - \cos A}$

$= \frac{(\sin A - \cos A)[1 + \sin A \cdot \cos A]}{\sin A - \cos A}$

$= 1 + \sin A \cdot \cos A$

Sol.202.(c) $\frac{\sec \alpha + \tan \alpha}{\sec \alpha - \tan \alpha} = \frac{7}{4}$



By componendo and dividendo we get;

$$\frac{\sec \alpha}{\tan \alpha} = \frac{11}{3} \Rightarrow \sec \alpha \times \cot \alpha = \frac{11}{3}$$

$$\frac{1}{\cos \alpha} \times \frac{\cos \alpha}{\sin \alpha} = \frac{11}{3}$$

$$\frac{1}{\sin \alpha} = \frac{11}{3} \Rightarrow \operatorname{cosec} \alpha = \frac{11}{3}$$

$$\begin{aligned} \text{Sol.203.(a)} \quad & \frac{\sin A}{1 + \cos A} \\ &= \frac{\sin A}{1 + \cos A} \times \frac{(1 - \cos A)}{(1 - \cos A)} \\ &= \frac{\sin A(1 - \cos A)}{1 - \cos^2 A} \\ &= \frac{\sin A(1 - \cos A)}{\sin^2 A} = \frac{1 - \cos A}{\sin A} \end{aligned}$$

$$\text{Sol.204.(b)} \quad \frac{7\sin\theta + 4\cos\theta}{9\sin\theta - 2\cos\theta} = \frac{5}{4}$$

$$28\sin\theta + 16\cos\theta = 45\sin\theta - 10\cos\theta$$

$$17\sin\theta = 26\cos\theta$$

$$\frac{\sin\theta}{\cos\theta} = \frac{26}{17} \Rightarrow \tan\theta = \frac{26}{17}$$

$$\begin{aligned} \text{Now, } \frac{\tan^2\theta + 5}{\tan^2\theta - 5} &= \frac{\left(\frac{26}{17}\right)^2 + 5}{\left(\frac{26}{17}\right)^2 - 5} \\ &= \frac{\frac{676}{289} + 5}{\frac{676}{289} - 5} = \frac{\frac{676 + 1445}{289}}{\frac{676 - 1445}{289}} = \frac{2121}{-769} \\ &= \frac{2121}{289} \times \frac{289}{-769} = -\frac{2121}{769} \end{aligned}$$

Sol.205.(d) Formula used :

$$\sec^2\theta = 1 + \tan^2\theta \text{ and } \operatorname{cosec}^2\theta = 1 + \cot^2\theta$$

now,

$$\begin{aligned} & \frac{2 + \tan^2 2A + \cot^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A} \\ &= \frac{1 + \tan^2 2A + 1 + \cot^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A} \\ &= \frac{\sec^2 2A + \operatorname{cosec}^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A} \\ &= \frac{\sec^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A} + \frac{\operatorname{cosec}^2 2A}{\sec 2A \cdot \operatorname{cosec} 2A} \\ &= \frac{\sec 2A}{\operatorname{cosec} 2A} + \frac{\operatorname{cosec} 2A}{\sec 2A} \\ &= \frac{1}{\frac{\cos 2A}{\sin 2A}} + \frac{1}{\frac{\sin 2A}{\cos 2A}} = \frac{\sin 2A}{\cos 2A} + \frac{\cos 2A}{\sin 2A} \\ &= \frac{\sin^2 2A + \cos^2 2A}{\cos 2A \cdot \sin 2A} = \frac{1}{\cos 2A \cdot \sin 2A} \\ &= \sec 2A \cdot \operatorname{cosec} 2A \end{aligned}$$

Sol.206.(c)

Using the given options, we have ;

$$(a) \text{ when } t = 45^\circ, \sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}},$$

$\sin(t) \neq \cos(t)$ [false]

$$(b) \text{ when } t = 30^\circ, \sin 30^\circ = \frac{1}{2}, \cos 30^\circ$$

$$= \frac{\sqrt{3}}{2}, \sin 30^\circ > \cos 30^\circ \text{ [false]}$$

$$(c) \text{ when } t = 30^\circ, \sin 30^\circ = \frac{1}{2}, \cos 30^\circ$$

$$= \frac{\sqrt{3}}{2}, \sin 30^\circ < \cos 30^\circ \text{ [true]}$$

$$(d) \text{ when } t = 60^\circ, \sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 60^\circ$$

$$= \frac{1}{2}; \sin 60^\circ < \cos 60^\circ \text{ [false]}$$

Clearly, we can see that only option(c) is the correct one.

Sol.207.(d) Given: $2 \tan \theta = 3$

$$\text{then } \frac{\sin \theta}{\cos \theta} = \frac{3}{2} \Rightarrow \sin \theta = \frac{3}{2} \cos \theta$$

Now,

$$\begin{aligned} & \frac{3 \sin \theta - 2 \cos \theta}{3 \sin \theta + 2 \cos \theta} \\ &= \frac{3 \times \frac{3}{2} \cos \theta - 2 \cos \theta}{3 \times \frac{3}{2} \cos \theta + 2 \cos \theta} \\ &= \frac{\frac{9 \cos \theta - 4 \cos \theta}{2}}{\frac{9 \cos \theta + 4 \cos \theta}{2}} = \frac{5 \cos \theta}{13 \cos \theta} = \frac{5}{13} \end{aligned}$$

$$\text{Sol.208.(a) Given : } \cot \theta = \frac{4}{3}$$

$$\begin{aligned} &= \frac{[\sec \theta (1 + \cot^2 \theta) (\operatorname{cosec}^2 \theta - \cot^2 \theta)]}{\operatorname{cosec}^3 \theta} \\ &= \frac{\left[\frac{1}{\cos \theta} \times \left(1 + \frac{\cos^2 \theta}{\sin^2 \theta}\right) \left(\frac{1}{\sin^2 \theta} - \frac{\cos^2 \theta}{\sin^2 \theta}\right)\right]}{\left(\frac{1}{\sin \theta}\right)^3} \\ &= \frac{\left[\frac{1}{\cos \theta} \times \left(\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta}\right) \left(\frac{1 - \cos^2 \theta}{\sin^2 \theta}\right)\right]}{\frac{1}{\sin^3 \theta}} \end{aligned}$$

$$\begin{aligned} \therefore \sin^2 \theta + \cos^2 \theta &= 1, 1 - \cos^2 \theta = \sin^2 \theta \\ &= \frac{\sin^2 \theta \times \sin^3 \theta}{\cos \theta \times \sin^4 \theta} = \frac{\sin \theta}{\cos \theta} \\ &= \tan \theta = \frac{1}{\cot \theta} = \frac{3}{4} \end{aligned}$$

$$\begin{aligned} \text{Sol.209.(d)} \quad & \frac{3 \sin 58^\circ}{5 \cos 32^\circ} + \frac{2 \cos 62^\circ}{5 \sin 28^\circ} \\ &= \frac{3 \sin (90^\circ - 32^\circ)}{5 \cos 32^\circ} + \frac{2 \cos (90^\circ - 28^\circ)}{5 \sin 28^\circ} \\ &= \frac{3 \cos 32^\circ}{5 \cos 32^\circ} + \frac{2 \sin 28^\circ}{5 \sin 28^\circ} = \frac{3}{5} + \frac{2}{5} = 1 \end{aligned}$$

$$\text{Sol.210.(a)} \quad 7 \cos^2 \theta + 5 \sin^2 \theta - 6 = 0$$

Let $\theta = 45^\circ$

$$7 \times \frac{1}{2} + 5 \times \frac{1}{2} - 6 = \frac{12}{2} - 6 = 6 - 6 = 0$$

Now,

$$\begin{aligned} & 1 + \sqrt{\frac{\sec \theta + \tan \theta}{\sec \theta - \tan \theta}} \\ &= 1 + \sqrt{\frac{\sec 45^\circ + \tan 45^\circ}{\sec 45^\circ - \tan 45^\circ}} \\ &= 1 + \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1}} \\ \text{Now, } & 1 + \sqrt{\frac{\sqrt{2} + 1}{\sqrt{2} - 1}} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1} \\ &= 1 + \sqrt{2} + 1 = \sqrt{2} + 2 \end{aligned}$$

$$\text{Sol.211.(a)} \quad \frac{\sec 35^\circ}{\operatorname{cosec} 55^\circ} - \frac{\cos 25^\circ}{\sin 65^\circ}$$

$$\frac{\sec(90^\circ - 55^\circ)}{\operatorname{cosec} 55^\circ} - \frac{\cos(90^\circ - 65^\circ)}{\sin 65^\circ}$$

$$\frac{\operatorname{cosec} 55^\circ}{\operatorname{cosec} 55^\circ} - \frac{\sin 65^\circ}{\sin 65^\circ} = 1 - 1 = 0$$

Sol.212.(d) Concept used :

$$\operatorname{cosec}(90^\circ - \theta) = \sec \theta$$

$$\cos(90^\circ - \theta) = \sin \theta$$

Now,

$$\begin{aligned} & \operatorname{cosec} 56^\circ \cos 34^\circ - \cos 59^\circ \operatorname{cosec} 31^\circ \\ & \operatorname{cosec}(90^\circ - 34^\circ) \cdot \cos 34^\circ - \cos 59^\circ \operatorname{cosec}(90^\circ - 59^\circ) \\ & \sec 34^\circ \cdot \cos 34^\circ - \cos 59^\circ \cdot \sec 59^\circ \\ & 1 - 1 = 0 \end{aligned}$$

$$\text{Sol.213.(a)} \quad \left[\frac{4}{3} \tan^2 60^\circ + 3 \cos^2 30^\circ - 2\right]$$

$$\sec^2 30^\circ - \frac{3}{4} \cot^2 60^\circ \div [\sin 60^\circ \cos 30^\circ - \cos 60^\circ \sin 30^\circ]$$

$$\left[\frac{4}{3} \times (\sqrt{3})^2 + 3 \times \left(\frac{\sqrt{3}}{2}\right)^2 - 2 \times \left(\frac{2}{\sqrt{3}}\right)^2 - \frac{3}{4} \times \left(\frac{1}{\sqrt{3}}\right)^2\right] \div \sin(60^\circ - 30^\circ)$$

$$\left[4 + \frac{9}{4} - \frac{8}{3} - \frac{1}{4}\right] \div \sin 30^\circ$$

$$\frac{10}{3} \div \frac{1}{2} = \frac{20}{3} = 6\frac{2}{3}$$

$$\text{Sol.214.(d)} \quad \sin(90^\circ - \theta) = \cos \theta$$

$$\tan(90^\circ - \theta) = \cot \theta$$

Now, $\sin 74^\circ + \tan 74^\circ$

$$\sin(90^\circ - 16^\circ) + \tan(90^\circ - 16^\circ)$$

$$\cos 16^\circ + \cot 16^\circ$$

$$\text{Sol.215.(a)} \quad p \sin A - \cos A = 1$$

Put $A = 90^\circ$, we have ;

$$p - 0 = 1 \Rightarrow p = 1$$

$$\text{Now, } p^2 - (1 + p^2) \cos A = 1 - (1 + 1) \cos 90^\circ = 1 - 0 = 1$$

Sol.216.(d)

$$\begin{aligned} & \left(\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta}\right)^3 \frac{1}{\tan \theta} - \sec^2 \theta \\ &= \left(\frac{\sin \theta (1 - 2 \sin^2 \theta)}{\cos \theta (2 \cos^2 \theta - 1)}\right)^3 \frac{1}{\tan \theta} - \sec^2 \theta \\ &[\because \cos 2\theta = (2 \cos^2 \theta - 1) = (1 - 2 \sin^2 \theta)] \\ &= \left(\frac{\sin \theta \cdot \cos 2\theta}{\cos \theta \cdot \cos 2\theta}\right)^3 \times \frac{1}{\tan \theta} - \sec^2 \theta \\ &= \left(\frac{\sin \theta}{\cos \theta}\right)^3 \times \frac{1}{\tan \theta} - \sec^2 \theta \\ &= \tan^3 \theta \times \frac{1}{\tan \theta} - \sec^2 \theta \\ &= \tan^2 \theta - \sec^2 \theta = -1 \end{aligned}$$

$$\text{Sol.217.(c)} \quad 2 \operatorname{cosec}^2 \theta + 3 \cot^2 \theta = 17$$

By putting the options one by one we get, $\theta = 30^\circ$ satisfies the equation:

$$\begin{aligned} & 2 \operatorname{cosec}^2 \theta + 3 \cot^2 \theta = 2 \operatorname{cosec}^2 30^\circ + 3 \\ & \cot^2 30^\circ = 2 \times 4 + 3 \times 3 = 8 + 9 = 17 \end{aligned}$$

$$\begin{aligned} \text{Sol.218.(a)} \quad & \frac{1}{1 + \cos \theta} + \frac{1}{1 - \cos \theta} \\ \Rightarrow & \frac{1 - \cos \theta + 1 + \cos \theta}{1 - \cos^2 \theta} \end{aligned}$$

$$\Rightarrow \frac{2}{\sin^2 \theta} = 2 \operatorname{cosec}^2 \theta$$

Sol.219.(c) $\frac{7 \sin \theta + 5 \cos \theta}{7 \sin \theta - 2 \cos \theta}$

$$\Rightarrow \frac{7 \tan \theta + 5}{7 \tan \theta - 2} = \frac{7 \times \frac{3}{7} + 5}{7 \times \frac{3}{7} - 2}$$

$$= \frac{3+5}{3-2} = 8$$

Sol.220.(b) $\cot^4 \phi \cdot \cot^2 \phi = 1$

$$\Rightarrow \cot^4 \phi = 1 + \cot^2 \phi$$

$$\Rightarrow \cot^4 \phi = \operatorname{cosec}^2 \phi \dots (\operatorname{cosec}^2 \phi = 1 + \cot^2 \phi)$$

$$\Rightarrow 111 = \frac{1}{\sin^2 \phi}$$

$$\Rightarrow \cos^4 \phi = \sin^2 \phi \dots (i)$$

Now,

$$\cos^4 \phi + \cos^2 \phi = \sin^2 \phi + \cos^2 \phi = 1$$

Sol.221.(d) Given:- $\tan 4\theta \cdot \tan 6\theta = 1$

$$\Rightarrow \tan 4\theta = \frac{1}{\tan 6\theta}$$

$$\Rightarrow \tan 4\theta = \cot 6\theta$$

$$\Rightarrow \tan 4\theta = \tan(90^\circ - 6\theta)$$

$$\Rightarrow 4\theta = 90^\circ - 6\theta \Rightarrow 10\theta = 90^\circ \Rightarrow \theta = 9^\circ$$

Put the value of θ in $\cot 5\theta$

$$\cot 45^\circ = 1$$

Sol.222.(c) Given, $7 \cos A = 6, \cos A = \frac{6}{7}$

Here $B = 6, H = 7, P = \sqrt{49 - 36} = \sqrt{13}$

Now, $\frac{\operatorname{cosec} A + \cot A}{\operatorname{cosec} A - \cot A} = \frac{\frac{7}{\sqrt{13}} + \frac{6}{\sqrt{13}}}{\frac{7}{\sqrt{13}} - \frac{6}{\sqrt{13}}}$

$$= \frac{13}{1} = 13$$

Sol.223.(c)

$$p(\sqrt{3} + \cot 30^\circ) = \tan^3 60^\circ - 2 \sin 60^\circ$$

Put the value of $\cot 30^\circ = \sqrt{3}, \tan 60^\circ$

$$= \sqrt{3} \text{ and } \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\Rightarrow p(\sqrt{3} + \sqrt{3}) = (\sqrt{3})^3 - 2 \times \frac{\sqrt{3}}{2}$$

$$\Rightarrow 2\sqrt{3}p = 3\sqrt{3} - \sqrt{3} \Rightarrow 2\sqrt{3}p = 2\sqrt{3} \Rightarrow p = 1$$

Sol.224.(a)

$$\Rightarrow \frac{1}{1 - \sin \theta} + \frac{1}{1 + \sin \theta} = 4 \sec \theta$$

$$\Rightarrow \frac{1 + \sin \theta + 1 - \sin \theta}{1 - \sin^2 \theta} = \frac{4}{\cos \theta}$$

$$\Rightarrow \frac{2}{\cos^2 \theta} = \frac{4}{\cos \theta}$$

$$\Rightarrow \cos \theta = \frac{1}{2} = \cos 60^\circ \Rightarrow \theta = 60^\circ$$

Now,

$$(\cot \theta + \operatorname{cosec} \theta)$$

$$= \cot 60^\circ + \operatorname{cosec} 60^\circ$$

$$= \frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} = \frac{3}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \sqrt{3}$$

Sol.225.(b) $\sin 31^\circ = \alpha$

$$\sin(90^\circ - 59^\circ) = \alpha$$

$$\cos 59^\circ = \alpha$$

$$\sin 59^\circ = \sqrt{1 - \cos^2 59^\circ}$$

$$= \sqrt{1 - \alpha^2}$$

Now,

$$\cot 59^\circ = \frac{\cos 59^\circ}{\sin 59^\circ} = \frac{\alpha}{\sqrt{1 - \alpha^2}}$$

Sol.226.(b) $\frac{\tan(x)}{\sec(x)} = \frac{1}{2}$

$$\frac{\frac{\sin x}{\cos x}}{\frac{1}{\cos x}} = \frac{1}{2} \Rightarrow \sin x = \frac{1}{2}$$

$$\sin x = \sin 30^\circ \Rightarrow x = 30^\circ$$

Now,

$$(\sin x + \cos x)^2$$

$$\sin^2 x + \cos^2 x + 2 \sin x \cdot \cos x$$

$$1 + 2 \sin x \cdot \cos x$$

$$1 + 2 \times \sin 30^\circ \times \cos 30^\circ$$

$$1 + 2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2} \Rightarrow \frac{2 + \sqrt{3}}{2}$$

Sol.227.(a) $\tan A = \frac{5}{7}$

Now, $\frac{7 \sin A - 3 \cos A}{7 \sin A + 3 \cos A} + 4$

On dividing by $\cos A$ in fraction we get,

$$\frac{7 \tan A - 3}{7 \tan A + 3} + 4 = \frac{7 \times \frac{5}{7} - 3}{7 \times \frac{5}{7} + 3} + 4$$

$$= \frac{5 - 3}{5 + 3} + 4 = \frac{2}{8} + 4 = 4 \frac{1}{4}$$

Sol.228.(d) $\sin B = \frac{15}{17}$

now, $\cos B(\sec B - \tan B)$

$$\cos B \times \frac{1}{\cos B} - \cos B \times \frac{\sin B}{\cos B}$$

$$= 1 - \sin B = 1 - \frac{15}{17} = \frac{2}{17}$$

Sol.229.(b) $\left(\frac{\operatorname{cosec} A}{\operatorname{cosec} A - 1} + \frac{\operatorname{cosec} A}{\operatorname{cosec} A + 1} \right)$

$$\left(\frac{\sec A}{\sec A - 1} + \frac{\sec A}{\sec A + 1} \right)^{-1}$$

$$\frac{\operatorname{cosec}^2 A + \operatorname{cosec} A + \operatorname{cosec}^2 A - \operatorname{cosec} A}{\operatorname{cosec}^2 A - 1^2}$$

$$= \frac{\sec^2 A + \sec A + \sec^2 A - \sec A}{\sec^2 A - 1^2}$$

$$= \frac{2 \operatorname{cosec}^2 A}{\operatorname{cosec}^2 A - 1} \times \frac{\sec^2 A - 1}{2 \sec^2 A}$$

$$= \frac{\operatorname{cosec}^2 A}{\cot^2 A} \times \frac{\tan^2 A}{\sec^2 A}$$

Now, $A = 60^\circ$

$$= \frac{4}{\frac{1}{3}} \times \frac{3}{4} = 4 \times \frac{3}{4} = 3$$

Sol.230.(b) In right $\triangle ABC, \angle B = 90^\circ$

$$\Rightarrow \tan A = 1 \Rightarrow \tan A = \tan 45^\circ \Rightarrow A = 45^\circ$$

Now,

$$4 \sin A \cdot \cos A$$

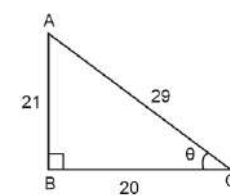
$$= 4 \times \sin 45^\circ \times \cos 45^\circ$$

$$= 4 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{4}{2} = 2$$

Sol.231.(b) Given :

$$\sec \theta = \frac{29}{20} = \frac{\text{hypotenuse}}{\text{base}}$$

By pythagorean triplet (20, 21, 29)



Then,

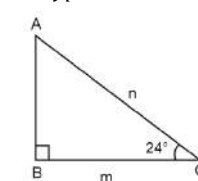
$$3 \operatorname{cosec} \theta + 3 \cot \theta$$

$$= 3 \times \frac{29}{21} + 3 \times \frac{20}{21}$$

$$= \frac{87 + 60}{21} = \frac{147}{21} = 7$$

Sol.232.(d) Given: $\cos 24^\circ = \frac{m}{n}$

$$= \frac{\text{base}}{\text{hypotenuse}}$$



By using pythagoras theorem,

$$AB^2 = AC^2 - BC^2 \Rightarrow AB^2 = n^2 - m^2$$

$$AB = \sqrt{n^2 - m^2}$$

Now,

$$(\operatorname{cosec} 24^\circ - \cos 66^\circ)$$

$$= \operatorname{cosec} 24^\circ - \cos(90^\circ - 24^\circ)$$

$$= \operatorname{cosec} 24^\circ - \sin 24^\circ$$

$$= \frac{\text{hypotenuse}}{\text{perpendicular}} - \frac{\text{perpendicular}}{\text{hypotenuse}}$$

$$= \frac{n}{\sqrt{n^2 - m^2}} - \frac{\sqrt{n^2 - m^2}}{n}$$

$$= \frac{n^2 - n^2 + m^2}{n\sqrt{n^2 - m^2}} = \frac{m^2}{n\sqrt{n^2 - m^2}}$$

Sol.233.(a)

$$\left[\frac{4 \cos(90 - \theta) \sin^3(90 + \theta) - 4 \sin(90 + \theta) \cos^3(90 - \theta)}{\cos\left(\frac{180 + 8\theta}{2}\right)} \right]$$

$$= \frac{4 \sin \theta \times \cos^3 \theta - 4 \cos \theta \times \sin^3 \theta}{-\sin 4\theta}$$

$$= \frac{4 \sin \theta \cos \theta (\cos^2 \theta - \sin^2 \theta)}{-\sin 2 \times (2\theta)}$$

$$= \frac{4 \sin \theta \cos \theta (\cos^2 \theta - \sin^2 \theta)}{-2 \sin 2\theta \cos 2\theta}$$

$$(\because \cos 2\theta = \cos^2 \theta - \sin^2 \theta)$$

$$= \frac{2 \sin \theta \cos \theta}{-\sin 2\theta} = \frac{2 \sin \theta \cos \theta}{-2 \sin \theta \cos \theta}$$

$$= -1 (\because \sin 2\theta = 2 \sin \theta \cos \theta)$$

Sol.234.(a) $\sin A + \cos A = \frac{1}{2\sqrt{2}}$

Squaring both side, we get

$$\Rightarrow \sin^2 A + \cos^2 A + 2 \sin A \cos A = \frac{1}{8}$$

$$\Rightarrow 2 \sin A \cos A = \frac{1}{8} - 1 = \frac{-7}{8}$$

$$\Rightarrow \sin A \cos A = \frac{-7}{16}$$

Now, we know that

$$\sin^2 A + \cos^2 A = 1$$

Again Squaring both side, we get

$$\Rightarrow \sin^4 A + \cos^4 A + 2 \sin^2 A \cos^2 A = 1$$

$$\Rightarrow \sin^4 A + \cos^4 A + 2 \times \left(\frac{49}{256}\right) = 1$$

$$\Rightarrow \sin^4 A + \cos^4 A = 1 - \frac{49}{128} = \frac{79}{128}$$

Sol.235.(c) if

$$a = x \cos \theta + y \sin \theta \text{ and } b = x \sin \theta - y \cos \theta$$

Then $a^2 + b^2 = x^2 + y^2$ (by using direct result)

Sol.236.(b) Given, cosec x

$$= \frac{\text{hypotenuse}}{\text{perpendicular}} = \frac{5}{4}$$

By using pythagorean triplet (4, 3, 5),

base = 3

Now,

$$\begin{aligned} & \frac{5 \sin^2 x + 8 \cot^2 x}{27 \tan^2 x - 125 \cos^2 x} \\ &= \frac{5 \times \left(\frac{4}{5}\right)^2 + 8 \times \left(\frac{3}{4}\right)^2}{27 \times \left(\frac{4}{3}\right)^2 - 125 \times \left(\frac{3}{5}\right)^2} \\ &= \frac{5 \times \frac{16}{25} + 8 \times \frac{9}{16}}{27 \times \frac{16}{9} - 125 \times \frac{9}{25}} \\ &= \frac{\frac{16}{5} + \frac{9}{2}}{48 - 45} = \frac{32 + 45}{3 \times 10} = \frac{77}{30} \end{aligned}$$

Sol.237.(b) $\sin \theta + \cos \theta = \sqrt{3} \cos \theta$

$$\Rightarrow \sin \theta = \sqrt{3} \cos \theta - \cos \theta$$

$$\Rightarrow \sin \theta = \cos \theta (\sqrt{3} - 1)$$

$$\Rightarrow \cot \theta = \frac{1}{\sqrt{3} - 1}$$

$$\Rightarrow \cot \theta = \frac{1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = \frac{(\sqrt{3} + 1)}{2}$$

Sol.238.(a) $\sec(t) = \frac{x}{y} = \frac{\text{Hypotenuse}}{\text{Base}}$

$$\text{Perpendicular} = \sqrt{x^2 - y^2}$$

$$\cot(t) = \frac{\text{Base}}{\text{Perpendicular}} = \frac{y}{\sqrt{x^2 - y^2}}$$

Sol.239.(a) As we know that, $\sin 45^\circ$ and

$\cos 45^\circ$ have the same value i.e. $\frac{1}{\sqrt{2}}$.

Sol.240.(a) Given: $3 \sin A + 4 \cos A = 5$

Squaring both side, we have

$$\Rightarrow (3 \sin A + 4 \cos A)^2 = 5^2$$

$$\Rightarrow 9 \sin^2 A + 16 \cos^2 A + 24 \sin A \cos A = 25$$

$$\Rightarrow 9 (1 - \cos^2 A) + 16 (1 - \sin^2 A) + 24 \sin A \cos A = 25$$

$$\Rightarrow 16 \sin^2 A + 9 \cos^2 A - 24 \sin A \cos A = 0$$

$$\Rightarrow (4 \sin A - 3 \cos A)^2 = 0$$

$$\Rightarrow 4 \sin A = 3 \cos A \Rightarrow \tan A = \frac{3}{4}$$

Short trick: Given: $3 \sin A + 4 \cos A = 5$

By using pythagorean triplet = (3, 4, 5)

$$\tan A = \frac{\text{Perpendicular}}{\text{Base}} = \frac{3}{4}$$

Sol.241.(b) $\frac{\tan 55^\circ}{\cot 35^\circ}$

$$= \frac{\tan 55^\circ}{\cot (90^\circ - 55^\circ)} = \frac{\tan 55^\circ}{\tan 55^\circ} = 1$$

Sol.242.(d) $\frac{\sin A}{1 + \cos A} + \cot A$

$$= \frac{\sin A}{1 + \cos A} + \frac{\cos A}{\sin A}$$

$$= \frac{\sin^2 A + \cos A + \cos^2 A}{(1 + \cos A) \cdot \sin A}$$

$$= \frac{(1 + \cos A)}{(1 + \cos A) \cdot \sin A} = \frac{1}{\sin A} = \text{cosec } A$$

Sol.243.(b) $\tan \theta = \frac{8}{15} = \frac{\text{perpendicular}}{\text{base}}$

By pythagorean triplet = (8, 15, 17)

Now,

$$\begin{aligned} \frac{\sqrt{1 - \sin \theta}}{\sqrt{1 + \sin \theta}} &= \frac{\sqrt{1 - \frac{8}{17}}}{\sqrt{1 + \frac{8}{17}}} = \frac{\sqrt{\frac{17 - 8}{17}}}{\sqrt{\frac{17 + 8}{17}}} \\ &= \sqrt{\frac{9}{25}} = \frac{3}{5} \end{aligned}$$

Sol.244.(b) $\sqrt{\frac{\sec \theta + 1}{\sec \theta - 1}} + \sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}}$

$$= \sqrt{\frac{\sec \theta + 1}{\sec \theta - 1}} \times \frac{\sec \theta + 1}{\sec \theta + 1} +$$

$$\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} \times \frac{\sec \theta - 1}{\sec \theta - 1}$$

$$= \frac{\sec \theta + 1}{\tan \theta} + \frac{\sec \theta - 1}{\tan \theta}$$

$$= \frac{2 \sec \theta}{\tan \theta} = 2 \text{ cosec } \theta$$

Sol.245.(a) $\sqrt{\frac{9 \sin^2 26^\circ + 9 \sin^2 64^\circ + 36}{32 - 7 \cos^2 32^\circ - 7 \cos^2 58^\circ}}$

$$= \sqrt{\frac{9 \cos^2 64^\circ + 9 \sin^2 64^\circ + 36}{32 - (7 \sin^2 58^\circ + 7 \cos^2 58^\circ)}}$$

$$= \sqrt{\frac{9 + 36}{32 - 7}} = \sqrt{\frac{45}{25}} = \frac{3\sqrt{5}}{5}$$

Sol.246.(c) $\sin x + \text{cosec } x = 2$

$$\text{Put } x = 90^\circ \Rightarrow 1 + 1 = 2$$

$$\Rightarrow 2 = 2 \text{ (LHS = RHS)}$$

$$\text{Now, } \sin^7 x + \text{cosec}^7 x = 1^7 + 1^7 = 2$$

Sol.247.(d) $\sin A + \cos A = \sqrt{3}$

Squaring on both side, we get

$$\Rightarrow 1 + 2 \sin A \cos A = 3$$

$$\Rightarrow \sin A \cos A = 1$$

$$\tan A + \cot A + 2 \sin A \cos A$$

$$= \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} + 2 \sin A \cos A$$

$$= \frac{\sin^2 A + \cos^2 A}{\cos A \sin A} + 2 \sin A \cos A$$

$$= 1 + 2 = 3$$

Sol.248.(d) $3 \sin^2 \theta + 5 \cos^2 \theta = 4$

$$\Rightarrow 3(\sin^2 \theta + \cos^2 \theta) + 2 \cos^2 \theta = 4$$

$$\Rightarrow 2 \cos^2 \theta = 1 \Rightarrow \cos \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

$$\therefore \tan \theta = \tan 45^\circ = 1$$

Sol.249.(d) Maximum value of $a \sin x + b$

$$\cos x = \sqrt{a^2 + b^2}$$

$$\text{maximum value of } (2 \sin \theta + 3 \cos \theta) = \sqrt{4 + 9} = \sqrt{13}$$

Sol.250.(c) $[1 - \sin^2 32^\circ + \frac{1}{1 + \tan^2 58^\circ}]$

$$= \cos^2 32^\circ + \frac{1}{\sec^2 58^\circ}$$

$$[\because 1 - \sin^2 A = \cos^2 A, 1 + \tan^2 A = \sec^2 A]$$

$$= \cos^2 32^\circ + \cos^2 58^\circ$$

$$= \cos^2 32^\circ + \sin^2 32^\circ = 1$$

Sol.251.(c) Given, $\tan \theta = \frac{5}{8}$

$$\frac{(1 + \cos \theta)(1 - \cos \theta)}{(1 + \sin \theta)(1 - \sin \theta)}$$

$$\frac{1 - \cos^2 \theta}{1 - \sin^2 \theta} = \frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta = \left(\frac{5}{8}\right)^2 = \frac{25}{64}$$

Sol.252.(d) $\tan \theta + \cot \theta = 2$

Taking $\theta = 45^\circ$ then satisfies the equation,

$$\text{LHS} = \tan 45^\circ + \cot 45^\circ = 1 + 1 = 2$$

= RHS

So, the value of θ will be 45°

Sol.253.(a) $\sqrt{\frac{1 - \cos A}{1 + \cos A}} + \sqrt{\frac{1 + \cos A}{1 - \cos A}}$

$$= \sqrt{\frac{1 - \cos A}{1 + \cos A}} \times \frac{1 - \cos A}{1 - \cos A} +$$

$$\sqrt{\frac{1 + \cos A}{1 - \cos A}} \times \frac{1 + \cos A}{1 + \cos A}$$

$$= \sqrt{\frac{(1 - \cos A)^2}{1 - \cos^2 A}} + \sqrt{\frac{(1 + \cos A)^2}{1 - \cos^2 A}}$$

$$= \sqrt{\frac{(1 - \cos A)^2}{\sin^2 A}} + \sqrt{\frac{(1 + \cos A)^2}{\sin^2 A}}$$

$$= \frac{1 - \cos A}{\sin A} + \frac{1 + \cos A}{\sin A}$$

$$= \frac{1 - \cos A + 1 + \cos A}{\sin A} = 2 \text{ cosec } A$$

Sol.254.(c) $\sqrt{3} \cos \theta + \sin \theta = 1$

Putting value of $\theta = 90^\circ$ satisfies the given equation,

$$\text{LHS} = \sqrt{3} \cos \theta + \sin \theta$$

$$= \sqrt{3} \times \cos 90^\circ + \sin 90^\circ$$

$$= \sqrt{3} \times 0 + 1 = 1 = \text{RHS}$$

Sol.255.(b) According to question,

$$x^2 + y^2 + z^2$$

$$= (r \cos t)^2 + (r \sin t \cos u)^2$$

$$+ (r \sin t \sin u)^2$$

$$= r^2 \cos^2 t + r^2 \cos^2 u \sin^2 t + r^2 \sin^2 t \sin^2 u$$

$$= r^2 \cos^2 t + r^2 \sin^2 t (\cos^2 u + \sin^2 u)$$

$$= r^2 (\cos^2 t + \sin^2 t) = r^2$$

Sol.256.(d) $4\tan\theta - 3 = 0$

$$\Rightarrow 4\tan\theta = 3 \Rightarrow \tan\theta = \frac{3}{4}$$

Now,

$$\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \frac{1 - 1 + 2\sin^2\theta}{1 + 2\cos^2\theta - 1} = \frac{2\sin^2\theta}{2\cos^2\theta}$$

$$= \tan^2\theta = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$$

Sol.257.(c) $\sin A - \cos A = \frac{\sqrt{3} - \sqrt{2}}{2}$

Taking square both side,

$$(\sin A - \cos A)^2 = \left(\frac{\sqrt{3} - \sqrt{2}}{2}\right)^2$$

$$\sin^2 A + \cos^2 A - 2\sin A \cos A = \frac{3 + 2 - 2\sqrt{6}}{4}$$

$$1 - 2\sin A \cos A = \frac{5 - 2\sqrt{6}}{4}$$

$$2\sin A \cos A = 1 - \frac{5 - 2\sqrt{6}}{4}$$

$$2\sin A \cos A = \frac{4 - 5 + 2\sqrt{6}}{4}$$

$$\sin A \cos A = \frac{-1 + 2\sqrt{6}}{8} = \frac{2\sqrt{6} - 1}{8}$$

Sol.258.(d) $\cos A + \cos^2 A = 1$

$$\Rightarrow \cos A = 1 - \cos^2 A \Rightarrow \cos A = \sin^2 A$$

Then,

$$\sin^2 A + \sin^4 A = \sin^2 A + \cos^2 A = 1$$

Sol.259.(c) $\frac{1}{\csc\theta - \cot\theta} - \frac{1}{\sin\theta}$

$$= \frac{1}{\frac{1}{\sin\theta} - \frac{\cos\theta}{\sin\theta}} - \frac{1}{\sin\theta}$$

$$= \frac{\sin\theta}{1 - \cos\theta} - \frac{1}{\sin\theta} = \frac{\sin^2\theta - 1 + \cos\theta}{(1 - \cos\theta)\sin\theta}$$

$$= \frac{-(1 - \sin^2\theta) + \cos\theta}{(1 - \cos\theta)\sin\theta} = \frac{-\cos^2\theta + \cos\theta}{(1 - \cos\theta)\sin\theta}$$

$$= \frac{\cos\theta}{\sin\theta} = \cot\theta$$

Sol.260.(b) $\frac{\sin^2\theta - 2\sin^4\theta}{2\cos^4\theta + \cos^2\theta}$

$$= \frac{(\sin 45^\circ)^2 - 2(\sin 45^\circ)^4}{2(\cos 45^\circ)^4 + (\cos 45^\circ)^2}$$

$$= \frac{\left(\frac{1}{\sqrt{2}}\right)^2 - 2\left(\frac{1}{\sqrt{2}}\right)^4}{2\left(\frac{1}{\sqrt{2}}\right)^4 + \left(\frac{1}{\sqrt{2}}\right)^2} = \frac{\frac{1}{2} - 2 \times \frac{1}{4}}{2 \times \frac{1}{4} + \frac{1}{2}}$$

$$= \frac{0}{1} = 0$$

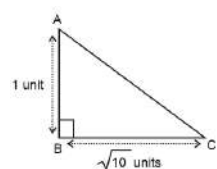
Sol.261.(b) According to the question,

$$\sin^2 t = \frac{1}{2} \times \tan^2 t$$

$$\sin^2 t = \frac{1}{2} \times \frac{\sin^2 t}{\cos^2 t} \Rightarrow \cos^2 t = \frac{1}{2}$$

$$\cos t = \frac{1}{\sqrt{2}} = \cos 45^\circ \Rightarrow t = 45^\circ$$

Sol.262.(c) $\tan A = \frac{1}{\sqrt{10}}$



By using pythagoras theorem,

$$AB^2 + BC^2 = AC^2$$

$$AC^2 = 1^2 + (\sqrt{10})^2 \Rightarrow AC^2 = 1 + 10$$

$$AC = \sqrt{11} \text{ units}$$

Now,

$$\sin A + \csc A$$

$$\frac{1}{\sqrt{11}} + \frac{\sqrt{11}}{1} \Rightarrow \frac{1 + 11}{\sqrt{11}} = \frac{12}{\sqrt{11}}$$

Sol.263.(b) $\sec\theta = \sqrt{2}$

$$\Rightarrow \sec\theta = \sec 45^\circ \Rightarrow \theta = 45^\circ$$

$$\therefore \csc 45^\circ = \sqrt{2}$$

Sol.264.(b) $2\sin^2\theta + 3\sin\theta - 2 = 0$

$$\Rightarrow 2\sin^2\theta + 3\sin\theta = 2$$

$$\text{if } \theta = 30^\circ$$

$$\Rightarrow 2 \times \frac{1}{4} + 3 \times \frac{1}{2} = 2 \Rightarrow 2 = 2$$

$$\therefore \theta = 30^\circ$$

Sol.265.(a) We know that

$$\sin\theta = \cos(90^\circ - \theta)$$

$$\sin 3A = \cos(90^\circ - 3A)$$

$$\Rightarrow \cos(A - 26^\circ) = \cos(90^\circ - 3A)$$

$$\Rightarrow A - 26^\circ = 90^\circ - 3A \Rightarrow 4A = 116^\circ \Rightarrow A = 29^\circ$$

Sol.266.(b)

$$(\sin A + \csc A)^2 + (\cos A + \sec A)^2$$

By value putting method, Let $A = 45^\circ$

$$= \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)^2 + \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)^2$$

$$= 2 \left(\frac{1}{\sqrt{2}} + \sqrt{2}\right)^2 = 2 \times \frac{9}{2} = 9$$

By checking options one by one, Option

(b) is satisfy

$$\text{i.e. } 7 + \tan^2 A + \cot^2 A = 7 + 1 + 1 = 9$$

Sol.267.(d) Given : $3 \tan A = 4$

$$\tan A = \frac{4}{3} = \frac{P}{B}$$

$$\therefore H = 5 \text{ (3, 4, 5 is a triplet)}$$

$$\text{Now, } 4\sin A + 3\cos A = 4 \times \frac{4}{5} + 3 \times \frac{3}{5}$$

$$= \frac{16}{5} + \frac{9}{5} = 5$$

Sol.268.(c) $10\sin^2\theta + 6\cos^2\theta = 7$

$$4\sin^2\theta + 6\sin^2\theta + 6\cos^2\theta = 7$$

$$4\sin^2\theta + 6(\sin^2\theta + \cos^2\theta) = 7$$

$$4\sin^2\theta = 1$$

$$\sin^2\theta = \frac{1}{4} \Rightarrow \sin\theta = \frac{1}{2}$$

$$\sin\theta = \sin 30^\circ \dots (\sin 30^\circ = \frac{1}{2}) \Rightarrow \theta = 30^\circ$$

$$\text{Now, } \tan 30^\circ = \frac{1}{\sqrt{3}}$$

Sol.269.(c) $\frac{11\sqrt{2} \cos 3A + 10 \sin 2A}{7\sqrt{3} \sin 4A}$

$$\text{Put } A = 15^\circ$$

$$\frac{11\sqrt{2} \cos 3 \times 15^\circ + 10 \sin 2 \times 15^\circ}{7\sqrt{3} \sin 4 \times 15^\circ}$$

$$\frac{11\sqrt{2} \cos 45^\circ + 10 \sin 30^\circ}{7\sqrt{3} \sin 60^\circ}$$

$$\frac{11\sqrt{2} \times \frac{1}{\sqrt{2}} + 10 \times \frac{1}{2}}{7\sqrt{3} \times \frac{\sqrt{3}}{2}} \Rightarrow \frac{11 + 5}{\frac{21}{2}}$$

$$= \frac{32}{21}$$

Sol.270.(b) $2 \frac{\tan 54^\circ}{\cot 36^\circ} - \frac{\cot 41^\circ}{\tan 49^\circ}$

$$2 \frac{\tan 54^\circ}{\cot 36^\circ} - \frac{\cot 41^\circ}{\tan 49^\circ} = 2 - 1 = 1$$

Sol.271.(c) $(\csc\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta)$

$$\left(\frac{1}{\sin\theta} - \sin\theta\right)\left(\frac{1}{\cos\theta} - \cos\theta\right)\left(\frac{1}{\cot\theta} + \cot\theta\right)$$

$$\left(\frac{1 - \sin^2\theta}{\sin\theta}\right)\left(\frac{1 - \cos^2\theta}{\cos\theta}\right)\left(\frac{1 + \cot^2\theta}{\cot\theta}\right)$$

$$\left(\frac{\cos^2\theta}{\sin\theta}\right)\left(\frac{\sin^2\theta}{\cos\theta}\right)\left(\frac{\csc^2\theta}{\cot\theta}\right)$$

$$\frac{1}{(\cos\theta)(\sin\theta)} \left(\frac{\sin^2\theta}{\cos\theta} \cdot \frac{\csc^2\theta}{\cot\theta}\right)$$

$$(\cos\theta)(\sin\theta) \left(\frac{1}{\sin\theta \cos\theta}\right) = 1$$

Sol.272.(c) $\Rightarrow m + 5$

$$= \sec^2 A \cdot \csc^2 A (1 - \cos^2 A)(1 - \sin^2 A) + 5$$

$$\Rightarrow m + 5 = \sec^2 A \cdot \csc^2 A \cdot \sin^2 A \cdot \cos^2 A + 5$$

$$\Rightarrow m + 5 = \frac{1}{\cos^2 A} \cdot \frac{1}{\sin^2 A} \cdot \sin^2 A \cdot \cos^2 A + 5$$

$$5 \Rightarrow m = 1$$

Sol.273.(a) $\cot^2\theta - 2\cos^2\theta = 0$

$$\cot^2\theta = 2\cos^2\theta \Rightarrow \frac{\cos^2\theta}{\sin^2\theta} = 2\cos^2\theta$$

$$\sin^2\theta = \frac{1}{2} \Rightarrow \sin\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

Sol.274.(a) $x = p \sec A \cos B$

$$\frac{x}{p} = \sec A \cos B \dots (i)$$

$$y = q \sec A \sin B$$

$$\frac{y}{q} = \sec A \sin B \dots (ii)$$

$$z = r \tan A$$

$$\frac{z}{r} = \frac{\sin A}{\cos A} \dots (iii)$$

On squaring eq. (i), (ii) and (iii) and then adding (i), (ii) and subtracting (iii) we get,

$$\frac{x^2}{p^2} + \frac{y^2}{q^2} - \frac{z^2}{r^2}$$

$$= \sec^2 A \cos^2 B + \sec^2 A \sin^2 B - \frac{\sin^2 A}{\cos^2 A}$$

$$= \sec^2 A (\cos^2 B + \sin^2 B) - \frac{\sin^2 A}{\cos^2 A}$$

$$= \frac{1}{\cos^2 A} - \frac{\sin^2 A}{\cos^2 A} = \frac{1 - \sin^2 A}{\cos^2 A} = \frac{\cos^2 A}{\cos^2 A} = 1$$

Short trick: Put $A = B = 0^\circ$

then



$$x = p \sec A \cos B$$

$$\frac{x}{p} = 1 \dots (i)$$

$$y = q \sec A \sin B$$

$$\frac{y}{q} = 0 \dots (ii)$$

$$z = r \tan A$$

$$\frac{z}{r} = 0 \dots (iii)$$

hence,

$$\frac{x}{p} + \frac{y}{q} - \frac{z}{r} = 1 + 0 - 0 = 1$$

$$\text{Sol.275.(c)} \quad 3 \sin 15^\circ - 4 \sin^3 15^\circ$$

$$= \sin 3A = \sin 3 \times 15^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\text{Sol.276.(c)} \quad \sec \theta + \tan \theta = x \dots (i)$$

$$\text{and, } \sec \theta - \tan \theta = \frac{1}{x} \dots (ii)$$

On adding (i) and (ii) we get,

$$2 \sec \theta = x + \frac{1}{x}$$

$$\sec \theta = \frac{x^2 + 1}{2x} = \frac{\text{hypotenuse}}{\text{base}}$$

$$\text{then, perpendicular} = x^2 - 1$$

$$\sin \theta = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{x^2 - 1}{x^2 + 1}$$

$$\text{Sol.277.(c)} \quad \sin(48^\circ + k) = \cos 13^\circ$$

$$\sin(48^\circ + k) = \cos(90^\circ - 77^\circ)$$

$$\sin(48^\circ + k) = \sin 77^\circ$$

$$(48^\circ + k) = 77^\circ$$

$$k = 77^\circ - 48^\circ = 29^\circ$$

$$\text{Sol.278.(c)}$$

$$\frac{5}{1 + \cot^2 \theta} + \frac{3}{1 + \tan^2 \theta} + 2 \cos^2 \theta$$

$$\frac{5}{\operatorname{cosec}^2 \theta} + \frac{3}{\sec^2 \theta} + 2 \cos^2 \theta$$

$$5 \sin^2 \theta + 3 \cos^2 \theta + 2 \cos^2 \theta$$

$$5 \sin^2 \theta + 5 \cos^2 \theta \Rightarrow 5(\sin^2 \theta + \cos^2 \theta) = 5$$

$$\text{Sol.279.(b)} \quad \tan A + \cot A = 2$$

$$\text{Put } A = 45^\circ$$

$$\text{LHS} = \tan 45^\circ + \cot 45^\circ = 1 + 1 = \text{RHS}$$

Now,

$$7 \tan^8 45^\circ - 6 \cot^8 45^\circ + 8 \sec^2 45^\circ$$

$$7 \times (1)^8 - 6 \times (1)^8 + 8 \times (\sqrt{2})^2$$

$$\Rightarrow 7 - 6 + 16 = 17$$

$$\text{Sol.280.(a)} \Rightarrow p = \frac{\sin A}{1 + \cos A}$$

$$\Rightarrow p = \frac{\sin A}{1 + \cos A} \times \frac{1 - \cos A}{1 - \cos A}$$

$$\Rightarrow p = \frac{\sin A(1 - \cos A)}{1 - \cos^2 A}$$

$$\Rightarrow p = \frac{\sin A(1 - \cos A)}{\sin^2 A}$$

$$\Rightarrow p = \frac{(1 - \cos A)}{\sin A} \Rightarrow \frac{\sin A}{(1 - \cos A)} = \frac{1}{p}$$

$$\text{Sol.281.(a)} \quad 8 \cot \theta = 7$$

$$\cot \theta = \frac{7}{8} = \frac{\text{Base}}{\text{Perpendicular}}$$

$$\text{Hence, Hypotenuse} = \sqrt{113}$$

$$\frac{1 + \sin \theta}{\cos \theta} = \frac{1 + \frac{\text{perpendicular}}{\text{Hypotenuse}}}{\frac{\text{Base}}{\text{Hypotenuse}}}$$

$$= \frac{1 + \frac{8}{\sqrt{113}}}{\frac{7}{\sqrt{113}}} = \frac{\frac{\sqrt{113} + 8}{\sqrt{113}}}{\frac{7}{\sqrt{113}}} = \frac{\sqrt{113} + 8}{7}$$

$$\text{Sol.282.(a)} \quad \frac{\sin^2 C + \cos^2 C}{1 + \cot^2 C}$$

$$= \frac{1}{1 + \frac{1}{\tan^2 C}} = \frac{1}{1 + \frac{1}{3}} = \frac{1}{\frac{4}{3}} = \frac{3}{4}$$

$$\text{Sol.283.(b)} \quad \sin^4 \theta + \cos^4 \theta$$

$$(\sin^2 \theta + \cos^2 \theta)^2 - 2 \sin^2 \theta \cos^2 \theta$$

$$1 - 2 \sin^2 \theta \cos^2 \theta$$

$$\text{As } 2 \sin^2 \theta \cos^2 \theta \geq 0$$

$$\text{hence, } \sin^4 \theta + \cos^4 \theta \leq 1$$

$$\text{Sol.284.(b)} \quad x = a \sec \theta \cos \theta$$

On squaring both side

$$\frac{x^2}{a^2} = \sec^2 \theta \cos^2 \theta \dots (i)$$

$$y = b \sec \theta \sin \theta$$

On squaring both side

$$\frac{y^2}{b^2} = \sec^2 \theta \sin^2 \theta \dots (ii)$$

$$z = c \tan \theta$$

On squaring both side

$$\frac{z^2}{c^2} = \tan^2 \theta \dots (iii)$$

On adding (i), (ii) and then subtracting (iii)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2}$$

$$= \sec^2 \theta \cos^2 \theta + \sec^2 \theta \sin^2 \theta - \tan^2 \theta$$

$$= \sec^2 \theta (\cos^2 \theta + \sin^2 \theta) - \tan^2 \theta$$

$$= \sec^2 \theta - \tan^2 \theta = 1$$

$$\text{Sol.285.(b)} \quad \cot \theta = \frac{\sqrt{11}}{1} = \frac{\text{base}}{\text{perpendicular}}$$

$$\text{hypotenuse}(h) = \sqrt{(\sqrt{11})^2 + 1^2} = \sqrt{12}$$

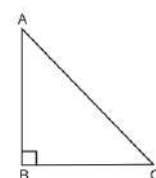
Now,

$$\frac{\operatorname{cosec}^2 \theta - \sec^2 \theta}{\operatorname{cosec}^2 \theta + \sec^2 \theta}$$

$$\frac{\left(\frac{h}{p}\right)^2 - \left(\frac{h}{b}\right)^2}{\left(\frac{h}{p}\right)^2 + \left(\frac{h}{b}\right)^2} \Rightarrow \frac{\left(\frac{\sqrt{12}}{1}\right)^2 - \left(\frac{\sqrt{12}}{\sqrt{11}}\right)^2}{\left(\frac{\sqrt{12}}{1}\right)^2 + \left(\frac{\sqrt{12}}{\sqrt{11}}\right)^2}$$

$$\frac{12 - \frac{12}{11}}{12 + \frac{12}{11}} = \frac{132 - 12}{132 + 12} = \frac{120}{144} = \frac{5}{6}$$

$$\text{Sol.286.(b)}$$



$$\operatorname{cosec} A = \frac{AC}{BC} = \frac{2\sqrt{2}}{1}$$

$$AB = \sqrt{(2\sqrt{2})^2 - (1)^2} = \sqrt{8 - 1} = \sqrt{7}$$

$$\sin A \cdot \cos C + \cos A \cdot \sin C$$

$$= \frac{BC}{AC} \times \frac{BC}{AC} + \frac{AB}{AC} \times \frac{AB}{AC}$$

$$= \frac{1}{2\sqrt{2}} \times \frac{1}{2\sqrt{2}} + \frac{\sqrt{7}}{2\sqrt{2}} \times \frac{\sqrt{7}}{2\sqrt{2}}$$

$$= \frac{1}{8} + \frac{7}{8} = 1$$

Sol.287.(a) $(2xy : x^2 + y^2 : x^2 - y^2)$ are pythagorean triplet

$$\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{2xy}{x^2 + y^2}$$

$$\text{So, } \tan \theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{2xy}{x^2 - y^2}$$

$$\text{Sol.288.(b)}$$

$$12 \sin^2 A + 2 \cos^2 A = 7 \dots (i)$$

$$12(1 - \cos^2 A) + 2 \cos^2 A = 7$$

$$12 - 12 \cos^2 A + 2 \cos^2 A = 7$$

$$-10 \cos^2 A = -5$$

$$\cos^2 A = \frac{1}{2} \dots (ii)$$

$$\cos A = \pm \frac{1}{\sqrt{2}}$$

From equation (i) and (ii)

$$12 \sin^2 A + 2 \times \frac{1}{2} = 7 \Rightarrow 12 \sin^2 A = 6$$

$$\sin^2 A = \frac{1}{2} \Rightarrow \sin A = \pm \frac{1}{\sqrt{2}}$$

$$\text{Now, } \cot A = \frac{\cos A}{\sin A} = \frac{\pm \frac{1}{\sqrt{2}}}{\pm \frac{1}{\sqrt{2}}} = \pm 1$$

Sol.289.(c) As we know,

$$\{\cos(x + y) = \cos x \cdot \cos y - \sin x \cdot \sin y\}$$

$$\{\sin(x - y) = \sin x \cdot \cos y - \cos x \cdot \sin y\}$$

$$\{\tan(2z) = \frac{2 \tan z}{1 - \tan^2 z}\}$$

$$\text{So, } \cos(x + y) - \sin(x - y) + \tan(2z)$$

$$= \cos x \cdot \cos y - \sin x \cdot \sin y - (\sin x \cdot \cos y -$$

$$\cos x \cdot \sin y) + \frac{2 \tan z}{1 - \tan^2 z}$$

$$= \cos x \cdot \cos y - \sin x \cdot \sin y - \sin x \cdot \cos y +$$

$$\cos x \cdot \sin y + \frac{2 \tan z}{1 - \tan^2 z}$$

$$\text{Sol.290.(a)} \quad \sin \theta = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{15}{17}$$

$$\text{Base} = \sqrt{(17)^2 - (15)^2}$$

$$= \sqrt{289 - 225} = \sqrt{64} = 8 \text{ unit}$$

$$\text{now, } \cot \theta = \frac{\text{base}}{\text{perpendicular}} = \frac{8}{15}$$

$$\sec \theta = \frac{\text{base}}{\text{hypotenuse}} = \frac{17}{8}$$

$$\frac{15 \cot \theta + 17 \sin \theta}{8 \tan \theta + 16 \sec \theta}$$

$$= \frac{15 \cot \theta + 17 \sin \theta}{8 \times \frac{1}{\cot \theta} + 16 \sec \theta}$$

$$= \frac{15 \times \frac{8}{15} + 17 \times \frac{15}{17}}{8 \times \frac{15}{8} + 16 \times \frac{17}{8}} = \frac{8 + 15}{15 + 34} = \frac{23}{49}$$

Sol.291.(a) According to the question,

$$\tan \theta + \cot \theta = 2$$

Let $\theta = 45^\circ$ then,

$$1 + 1 = 2 \text{ (satisfied)}$$

Now, put the value of $\theta = 45^\circ$ we get,

$$\Rightarrow \tan^3 \theta + \cot^3 \theta + 6 \tan \theta \cot^2 \theta$$

$$\Rightarrow 1 + 1 + 6 = 8$$

Sol.292.(d) $\sqrt{13} \sin \theta = 2$

$$\sin \theta = \frac{2}{\sqrt{13}} = \frac{p}{h}$$

$$b = \sqrt{(\sqrt{13})^2 - (2)^2} = 3$$

According to the question,

$$\Rightarrow \frac{3 \times \frac{2}{3} + \sqrt{13} \times \frac{2}{\sqrt{13}}}{\sqrt{13} \times \frac{3}{\sqrt{13}} - 3 \times \frac{2}{3}} = \frac{2+2}{3-2} = 4$$

Sol.293.(a)

On putting $\theta = 45^\circ$ then we get,

$$\Rightarrow (\cot \theta + \tan \theta)(\operatorname{cosec} \theta - \sin \theta)(\cos \theta - \sec \theta)$$

$$\Rightarrow (1+1)(\sqrt{2} - \frac{1}{\sqrt{2}})(\frac{1}{\sqrt{2}} - \sqrt{2})$$

$$\Rightarrow (2)(\frac{1}{\sqrt{2}})(-\frac{1}{\sqrt{2}}) \Rightarrow -1$$

Sol.294.(d) Put $\theta = 0^\circ$ we get,

$$\sin(30+0) - \cos(60-0) \Rightarrow \frac{1}{2} - \frac{1}{2} = 0$$

Sol.295.(c) $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta}$

Given that $\alpha + \beta = 45^\circ$

$$\text{Then, } \tan 45^\circ = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \cdot \tan \beta} = 1$$

$$\tan \alpha + \tan \beta = 1 - \tan \alpha \cdot \tan \beta$$

$$\tan \alpha + \tan \beta + \tan \alpha \cdot \tan \beta = 1 \text{ -----(i)}$$

$$\text{Now, } (\tan \alpha + 1)(\tan \beta + 1) = 2x$$

$$\Rightarrow \tan \alpha + \tan \beta + \tan \alpha \cdot \tan \beta + 1 = 2x$$

Putting the value from equation (i) we get;

$$1 + 1 = 2x \Rightarrow x = 1$$

Sol.296.(c) According to the question,

$$\tan 570^\circ = \tan(180^\circ \times 3 + 30^\circ)$$

[$\because$ the value of $\tan \theta$ in 3rd quadrant is positive]

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

Sol.297.(d)

$$\frac{1}{\sin^4(90^\circ - 2\alpha)} + \frac{1}{\cos^2(90^\circ - 2\alpha) - 1}$$

$$\frac{1}{\cos^4 2\alpha} - \frac{1}{1 - \sin^2 2\alpha} (\because \sin(90^\circ - \theta) = \cos \theta)$$

$$\frac{1}{\cos^4 2\alpha} - \frac{1}{\cos^2 2\alpha}$$

$$\frac{1}{\cos^2 2\alpha} \left(\frac{1}{\cos^2 2\alpha} - 1 \right)$$

$$\frac{1}{\cos^2 2\alpha} \left(\frac{1 - \cos^2 2\alpha}{\cos^2 2\alpha} \right)$$

$$\frac{1}{\cos^2 2\alpha} \left(\frac{\sin^2 2\alpha}{\cos^2 2\alpha} \right) = \sec^2 2\alpha \tan^2 2\alpha$$

Sol.298.(c) $\cot \theta + \tan \theta = 2$

Let $\theta = 45^\circ$

$$\cot 45^\circ + \tan 45^\circ = 2$$

$$1 + 1 = 2 \Rightarrow 2 = 2 \text{ (LHS = RHS)}$$

now, $\tan^{12} \theta + \cot^{12} \theta + 2 \tan^5 \theta \cot^7 \theta$

$$\Rightarrow (\tan 45^\circ)^{12} + (\cot 45^\circ)^{12} + 2 (\tan 45^\circ)^5$$

$$(\cot 45^\circ)^7 \Rightarrow 1 + 1 + 2(1)(1) = 4$$

Sol.299.(b) $\cos \theta = \frac{\text{base}}{\text{hypotenuse}} = \frac{24}{25}$

$$\text{Perpendicular} = \sqrt{(25)^2 - (24)^2}$$

$$= \sqrt{625 - 576} = \sqrt{49} = 7$$

Now, $\cot(90^\circ - \theta) = \tan \theta$

$$= \frac{\text{perpendicular}}{\text{base}} = \frac{7}{24}$$

Sol.300.(c) $\left[\frac{\operatorname{cosec} 31^\circ - \sec 59^\circ}{(\sin 18^\circ \div \cos 72^\circ)} \times \frac{\tan 26^\circ}{\cot 64^\circ} \right]$

$$+ \left[\frac{\tan 30^\circ + \tan 15^\circ}{(1 - \tan 30^\circ \tan 15^\circ)} \right]$$

$$= \left[\frac{\operatorname{cosec} 31^\circ - \sec(90^\circ - 31^\circ)}{(\sin 18^\circ \div \cos 72^\circ)} \times \frac{\tan(90^\circ - 64^\circ)}{\cot 64^\circ} \right]$$

$$+ \left[\frac{\tan 30^\circ + \tan 15^\circ}{(1 - \tan 30^\circ \tan 15^\circ)} \right]$$

$$= \left[\frac{\operatorname{cosec} 31^\circ - \operatorname{cosec} 31^\circ}{(\sin 18^\circ \div \cos 72^\circ)} \times \frac{\cot 64^\circ}{\cot 64^\circ} \right] + \tan$$

$$(30^\circ + 15^\circ)$$

$$= \left[\frac{0}{(\sin 18^\circ \div \cos 72^\circ)} \times \frac{\cot 64^\circ}{\cot 64^\circ} \right] + \tan 45^\circ$$

$$= 0 + 1 = 1$$

Sol.301.(d) $\cot A = \frac{5}{12} = \frac{b}{p}$

$$h = \sqrt{12^2 + 5^2} \Rightarrow h = 13$$

According to the question,

$$\Rightarrow \frac{\sqrt{\sec A - \operatorname{cosec} A}}{\sqrt{\sec A + \operatorname{cosec} A}} - \sqrt{\frac{17}{7}}$$

$$\Rightarrow \frac{\sqrt{\frac{13}{5} - \frac{13}{12}}}{\sqrt{\frac{13}{5} + \frac{13}{12}}} - \sqrt{\frac{17}{7}}$$

$$\Rightarrow \frac{\sqrt{\frac{156 - 65}{60}}}{\sqrt{\frac{156 + 65}{60}}} - \sqrt{\frac{17}{7}}$$

$$\Rightarrow \sqrt{\frac{91}{221}} - \sqrt{\frac{17}{7}} \Rightarrow \frac{\sqrt{13 \times 7}}{\sqrt{13 \times 17}} - \frac{\sqrt{17}}{\sqrt{7}}$$

$$\Rightarrow \frac{\sqrt{7}}{\sqrt{17}} - \frac{\sqrt{17}}{\sqrt{7}} \Rightarrow -\frac{10}{\sqrt{119}}$$

Sol.302.(a) $\tan(t) = \frac{\text{perpendicular}}{\text{base}} = \frac{3}{7}$

$$\text{Hypotenuse} = \sqrt{(3)^2 + (7)^2} = \sqrt{58}$$

$$\operatorname{cosec}(t) = \frac{\text{hypotenuse}}{\text{perpendicular}} = \frac{\sqrt{58}}{3}$$

Sol.303.(c) $\sin \theta = \frac{11}{12} = \frac{p}{h}$

$$b = \sqrt{(12)^2 - (11)^2}$$

$$b = \sqrt{144 - 121} = \sqrt{23}$$

According to the question,

$$\Rightarrow (\tan \theta + \sec \theta) = \frac{p}{b} + \frac{h}{b} = \frac{p+h}{b}$$

$$\Rightarrow \frac{11+12}{\sqrt{23}} = \frac{23}{\sqrt{23}} = \sqrt{23}$$

Sol.304.(a) $\cos^4 \theta - \sin^4 \theta = \frac{4}{5}$

$$(\cos^2 \theta)^2 - (\sin^2 \theta)^2 = \frac{4}{5}$$

$$(\cos^2 \theta + \sin^2 \theta)(\cos^2 \theta - \sin^2 \theta) = \frac{4}{5}$$

$$(\cos^2 \theta - \sin^2 \theta) = \frac{4}{5}$$

$$\cos 2\theta = \frac{4}{5} \quad [\cos 2\theta = \cos^2 \theta - \sin^2 \theta]$$

$$\cos^2 2\theta = \frac{16}{25}$$

$$\text{Now, } \sin^2 2\theta = 1 - \cos^2 2\theta = 1 - \frac{16}{25}$$

$$\sin^2 2\theta = \frac{9}{25} \Rightarrow \sin 2\theta = \frac{3}{5}$$

Now, $\sin 4\theta = 2 \sin 2\theta \cos 2\theta$

Putting the value of $\sin 2\theta$ and $\cos 2\theta$

$$\sin 4\theta = 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$$

Sol.305.(b) $\frac{P \cos \alpha - Q \sin \alpha}{P \cos \alpha + Q \sin \alpha}$

Divide numerator and denominator by $\sin \alpha$

$$\frac{P \times \frac{\cos \alpha}{\sin \alpha} - Q \times \frac{\sin \alpha}{\sin \alpha}}{P \times \frac{\cos \alpha}{\sin \alpha} + Q \times \frac{\sin \alpha}{\sin \alpha}}$$

$$\frac{P \times \frac{P}{Q} - Q}{P \times \frac{P}{Q} + Q} = \frac{P^2 - Q^2}{P^2 + Q^2}$$

$$= \frac{P \times \frac{P}{Q} - Q}{P \times \frac{P}{Q} + Q} = \frac{P^2 - Q^2}{P^2 + Q^2}$$

$$\text{So, } \frac{P \cos \alpha - Q \sin \alpha}{P \cos \alpha + Q \sin \alpha} - \frac{P^2 - Q^2}{P^2 + Q^2} + 3$$

$$= \frac{P^2 - Q^2}{P^2 + Q^2} - \frac{P^2 - Q^2}{P^2 + Q^2} + 3 = 3$$

Sol.306.(c) $\sin \theta = \frac{7}{25} = \frac{\text{perpendicular}}{\text{hypotenuse}}$

$$\text{base} = \sqrt{(25)^2 - (7)^2}$$

$$\text{base} = 24$$

$$\cos \theta = \frac{24}{25}, \tan \theta = \frac{7}{24}$$

Sol.307.(d) Concept :- maximum value

$$\text{of } A \cos x + B \sin x = \sqrt{A^2 + B^2}$$

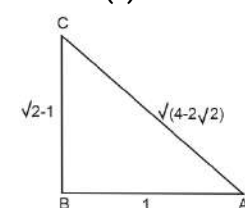
Maximum value $(7 \cos A + 24 \sin A)$

$$= \sqrt{(7)^2 + (24)^2} = \sqrt{625} = 25$$

So, maximum value of

$$7 \cos A + 24 \sin A + 32 = 25 + 32 = 57$$

Sol.308.(d)



$$\begin{aligned}
 AC &= \sqrt{(\sqrt{2}-1)^2 + 1^2} = \sqrt{3-2\sqrt{2}+1} \\
 &= \sqrt{4-2\sqrt{2}} \\
 \sin A \cos A + \tan A \\
 &= \frac{\sqrt{2}-1}{\sqrt{4-2\sqrt{2}}} \times \frac{1}{\sqrt{4-2\sqrt{2}}} + \sqrt{2}-1 \\
 &= \frac{\sqrt{2}-1}{4-2\sqrt{2}} + \sqrt{2}-1 \\
 &= \frac{\sqrt{2}-1}{2\sqrt{2}(\sqrt{2}-1)} + \sqrt{2}-1 \\
 &= \frac{1}{2\sqrt{2}} + \sqrt{2}-1 = \frac{\sqrt{2}}{4} + \sqrt{2}-1 \\
 &= \frac{\sqrt{2}+4\sqrt{2}-4}{4} = \frac{5\sqrt{2}-4}{4}
 \end{aligned}$$

Sol.309.(c) $4 \sin^2 \theta - 4\sqrt{3} \sin \theta + 3 = 0$
 $4 \sin^2 \theta - 2\sqrt{3} \sin \theta - 2\sqrt{3} \sin \theta + 3 = 0$
 $2 \sin \theta (2 \sin \theta - \sqrt{3}) - \sqrt{3} (2 \sin \theta - \sqrt{3}) = 0$
 $(2 \sin \theta - \sqrt{3})(2 \sin \theta - \sqrt{3}) = 0$
 $(2 \sin \theta - \sqrt{3})^2 = 0$
 $2 \sin \theta = \sqrt{3} \Rightarrow \sin \theta = \frac{\sqrt{3}}{2}$
 $\sin \theta = \sin 60^\circ$ So, $\theta = 60^\circ$

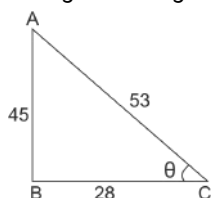
Sol.310.(c)

$$\begin{aligned}
 &\left(\frac{\sin \theta}{1+\cos \theta} + \frac{1+\cos \theta}{\sin \theta} \right) \left(\frac{1}{\tan \theta + \cot \theta} \right) \\
 &= \frac{\sin^2 \theta + (1+\cos \theta)^2}{(1+\cos \theta) \sin \theta} \times \left(\frac{1}{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}} \right) \\
 &= \frac{\sin^2 \theta + 1 + \cos^2 \theta + 2 \cos \theta}{(1+\cos \theta) \sin \theta} \times \frac{1}{\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}} \\
 &= \frac{2(1+\cos \theta)}{(1+\cos \theta) \sin \theta} \times \cos \theta \sin \theta = 2 \cos \theta
 \end{aligned}$$

Sol.311.(d) $\frac{\sin 60^\circ - \cos 60^\circ}{\sin 60^\circ + \cos 60^\circ}$

$$\begin{aligned}
 \frac{\frac{\sqrt{3}}{2} - \frac{1}{2}}{\frac{\sqrt{3}}{2} + \frac{1}{2}} &= \frac{\sqrt{3}-1}{\sqrt{3}+1} \\
 &= \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} \\
 &= \frac{(\sqrt{3}-1)^2}{(\sqrt{3})^2 - (1)^2} = \frac{(\sqrt{3})^2 + (1)^2 - 2\sqrt{3}}{3-1} \\
 &= \frac{3+1-2\sqrt{3}}{2} = \frac{4-2\sqrt{3}}{2} = 2-\sqrt{3}
 \end{aligned}$$

Sol.312.(d) According to question, we have given triangle :



$$\begin{aligned}
 AC &= \sqrt{45^2 + 28^2} = \sqrt{2809} = 53 \\
 \text{Now, } \sin \theta - \cos \theta &= \frac{45}{53} - \frac{28}{53} = \frac{17}{53}
 \end{aligned}$$

Sol.313.(b) $\sec 6A = \operatorname{cosec} (A - 29^\circ)$
 $\operatorname{cosec}(90^\circ - 6A) = \operatorname{cosec} (A - 29^\circ)$

$$\begin{aligned}
 \Rightarrow 90^\circ - 6A &= A - 29^\circ \\
 7A &= 119 \Rightarrow A = \frac{119}{7} = 17^\circ
 \end{aligned}$$

Sol.314.(a)

Formula used : $\sin^2 A + \cos^2 A = 1$

$$\sec^2 A - \tan^2 A = 1$$

$$\operatorname{cosec}^2 A - \cot^2 A = 1$$

$$\text{Now, } (1 + \tan^2 A) + (1 + \cot^2 A)$$

$$= \sec^2 A + \operatorname{cosec}^2 A$$

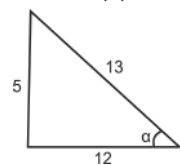
$$\begin{aligned}
 \frac{1}{\cos^2 A} + \frac{1}{\sin^2 A} &= \frac{\sin^2 A + \cos^2 A}{\sin^2 A \cos^2 A} \\
 &= \frac{1}{\sin^2 A \cos^2 A}
 \end{aligned}$$

Now, solving option (a) we get ;

$$\begin{aligned}
 \frac{1}{\sin^2 A - \sin^4 A} &= \frac{1}{\sin^2 A (1 - \sin^2 A)} \\
 &= \frac{1}{\sin^2 A \cos^2 A}
 \end{aligned}$$

So, the correct option is (a).

Sol.315.(b) According to question,



$$\begin{aligned}
 \cos \alpha \cdot \operatorname{cosec} \alpha \cdot \cot \alpha &= \cot^2 \alpha = \left(\frac{12}{5} \right)^2 \\
 &= \frac{144}{25}
 \end{aligned}$$

Sol.316.(a) $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) - 3(\sin^2 A + \cos^2 A)$

$$\begin{aligned}
 &\left(1 + \frac{\cos A}{\sin A} - \frac{1}{\sin A} \right) \left(1 + \frac{\sin A}{\cos A} + \frac{1}{\cos A} \right) - 3 \times 1 \\
 &= \left(\frac{\sin A + \cos A - 1}{\sin A} \right) \left(\frac{\cos A + \sin A + 1}{\cos A} \right) - 3 \\
 &= \frac{(\sin A + \cos A)^2 - 1}{\sin A \cos A} - 3 \\
 &= \frac{\sin^2 A + \cos^2 A + 2 \sin A \cos A - 1}{\sin A \cos A} - 3 \\
 &= \frac{1 + 2 \sin A \cos A - 1}{\sin A \cos A} - 3 = 2 - 3 = -1
 \end{aligned}$$

Sol.317.(a) Given:

$$(19 \sin \theta + 6 \cot \theta \sin \theta)$$

$$= (19 \sin \theta + 6 \cos \theta)$$

We know that,

$$y = (A \sin \theta + B \cos \theta) \text{ where, } A > B$$

$$y_{\max} = \sqrt{A^2 + B^2}$$

So,

$$\text{Maximum value of } (19 \sin \theta + 6 \cos \theta)$$

$$= \sqrt{19^2 + 6^2} = \sqrt{361 + 36} \Rightarrow \sqrt{397}$$

Sol.318.(c) Given: $\tan \theta = \frac{3}{10}$

$$\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)} = \frac{(1)^2 - (\sin \theta)^2}{(1)^2 - (\cos \theta)^2}$$

$$\frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\tan^2 \theta} = \left(\frac{10}{3} \right)^2 = \frac{100}{9}$$

Sol.319.(c) $x = \sin A + \cos A \dots$ (i)

On squaring both side

$$x^2 = \sin^2 A + \cos^2 A + 2 \sin A \cdot \cos A$$

$$x^2 = 1 + 2 \sin A \cdot \cos A$$

$$\frac{x^2 - 1}{2} = \sin A \cdot \cos A \dots$$
 (ii)

Now,

$$y = \sec A + \operatorname{cosec} A$$

$$y = \frac{\sin A + \cos A}{\sin A \cdot \cos A}$$

$$y = \frac{x}{\frac{x^2 - 1}{2}} \text{ from (i) and (ii)}$$

$$y = \frac{2x}{x^2 - 1} \Rightarrow (x^2 - 1)y = 2x$$

$$(x^2 y - y) = 2x \Rightarrow x^2 y = 2x + y$$

Short trick : Put $A = 45^\circ$

Sol.320.(c) $2 \cos^2 A + 3 \cos A - 2 = 0$

$$\Rightarrow 2 \cos^2 A + 4 \cos A - \cos A - 2 = 0$$

$$\Rightarrow 2 \cos A (\cos A + 2) - 1 (\cos A + 2) = 0$$

$$\Rightarrow (\cos A + 2)(2 \cos A - 1) = 0$$

$$\Rightarrow (\cos A + 2) = 0 \text{ and } (2 \cos A - 1) = 0$$

$$\Rightarrow \cos A = -2 \text{ and } \cos A = \frac{1}{2}$$

$$\cos A = \frac{1}{2} \Rightarrow \cos A = \cos 60^\circ \Rightarrow A = 60^\circ$$

Sol.321.(d) $\frac{1 - \cot^2 \theta}{\tan^2 \theta - 1}$

$$\begin{aligned}
 \frac{1 - \frac{\cos^2 \theta}{\sin^2 \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta} - 1} &\Rightarrow \frac{\frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\cos^2 \theta}} \\
 &= \frac{\cos^2 \theta}{\sin^2 \theta} = \cot^2 \theta
 \end{aligned}$$

Sol.322.(d) $\frac{(\sec A + \tan A)}{\sec A - \tan A} = 2 \frac{51}{79}$

$$\frac{(\sec A + \tan A)}{\sec A - \tan A} = \frac{209}{79}$$

By applying componendo and dividendo rule

$$\frac{\sec A}{\tan A} = \frac{209 + 79}{209 - 79}$$

$$\frac{\sec A}{\tan A} = \frac{288}{130} \Rightarrow \frac{\frac{1}{\cos A}}{\frac{\sin A}{\cos A}} = \frac{288}{130}$$

$$\frac{1}{\sin A} = \frac{144}{65} \Rightarrow \sin A = \frac{65}{144}$$

Sol.323.(a) $4 \sin^2 \theta = 3(1 + \cos \theta)$

$$4(1 - \cos^2 \theta) = 3(1 + \cos \theta)$$

$$4(1 - \cos \theta)(1 + \cos \theta) = 3(1 + \cos \theta)$$

$$(1 - \cos \theta) = \frac{3}{4}$$

$$\cos \theta = 1 - \frac{3}{4} = \frac{1}{4} = \frac{\text{base}}{\text{hypotenuse}}$$

$$\text{Perpendicular} = \sqrt{15}$$

Now,

$$\sqrt{15} \tan \theta + \frac{4}{\sqrt{15}} \sin \theta + 2 \sec \theta$$

$$\sqrt{15} \times \frac{\text{Perpendicular}}{\text{base}} + \frac{4}{\sqrt{15}} \times$$

$$\frac{\text{Perpendicular}}{\text{hypotenuse}} + 2 \times \frac{\text{hypotenuse}}{\text{base}}$$



$$\sqrt{15} \times \frac{\sqrt{15}}{1} + \frac{4}{\sqrt{15}} \times \frac{\sqrt{15}}{4} + 2 \times \frac{4}{1}$$

$$15 + 1 + 8 = 24$$

Sol.324.(b) $\tan \theta + \cot \theta = 6$

On squaring both side,

$$(\tan \theta + \cot \theta)^2 = 36$$

$$\tan^2 \theta + \cot^2 \theta + 2 \tan \theta \cot \theta = 36$$

We know that, $\tan \theta \cot \theta = 1$

Hence,

$$\tan^2 \theta + \cot^2 \theta = 36 - 2 = 34$$

Sol.325.(b) $(1 - \cos^2 A) \sec^2 A$

Using identity $\therefore \sin^2 A = 1 - \cos^2 A$ and

$$\sec^2 A = \frac{1}{\cos^2 A}$$

$$\text{then, } \sin^2 A \times \frac{1}{\cos^2 A} = \tan^2 A$$

Sol.326.(d) $\sin x + \cos x = \sqrt{2} \sin x$

On squaring both side,

$$(\sin x + \cos x)^2 = (\sqrt{2} \sin x)^2$$

$$\sin^2 x + \cos^2 x + 2 \sin x \cos x = 2 \sin^2 x$$

$$= 1 + 2 \sin x \cos x = 2 \sin^2 x$$

$$= 2 \sin x \cos x = 2 \sin^2 x - 1 \dots (i)$$

Now, $(\sin x - \cos x)^2$

$$= \sin^2 x + \cos^2 x - 2 \sin x \cos x$$

$$= 1 - 2 \sin x \cos x \dots \text{from (i)}$$

$$= 1 - (2 \sin^2 x - 1) = 1 - 2 \sin^2 x + 1$$

$$= 2 - 2 \sin^2 x = 2(1 - \sin^2 x)$$

$$(\sin x - \cos x)^2 = 2 \cos^2 x$$

$$\sin x - \cos x = \sqrt{2} \cos x$$

Sol.327.(b) In $\triangle ABC$

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\frac{\angle A}{2} + \frac{\angle B + \angle C}{2} = 90^\circ$$

$$\text{Now, } \left\{ \tan \frac{A}{2} + \operatorname{cosec} \frac{B+C}{2} \right\} \left\{ \tan \frac{A}{2} - \operatorname{cosec} \frac{B+C}{2} \right\}$$

$$\left\{ \tan \frac{A}{2} + \operatorname{cosec} (90^\circ - \frac{A}{2}) \right\}$$

$$\left\{ \tan \frac{A}{2} - \operatorname{cosec} (90^\circ - \frac{A}{2}) \right\}$$

$$\left(\tan \frac{A}{2} + \sec \frac{A}{2} \right) \left(\tan \frac{A}{2} - \sec \frac{A}{2} \right)$$

$$\tan^2 \frac{A}{2} - \sec^2 \frac{A}{2} = -1$$

$$\text{Sol.328.(d)} = \frac{\sin^3 A + \cos^3 A}{\sin A + \cos A} =$$

$$\frac{(\sin A + \cos A)(\sin^2 A + \cos^2 A - \sin A \cos A)}{\sin A + \cos A}$$

$$= (1 - \sin A \cos A)$$

Sol.329.(c) $\sin(x + y) = 1$

$$x + y = 90^\circ \dots (\text{as } \sin 90^\circ = 1) \dots (i)$$

$$\cos(x - y) = \frac{\sqrt{3}}{2}$$

$$x - y = 30^\circ \dots (\text{as } \cos 30^\circ = \frac{\sqrt{3}}{2}) \dots (ii)$$

From eq. (i) and (ii)

$$x = \frac{120^\circ}{2} = 60^\circ \text{ and } y = 30^\circ$$

$$\text{Sol.330.(c)} \left(\frac{2 \cos^3 \theta - \cos \theta}{\sin \theta - 2 \sin^3 \theta} \right)^2 + 1$$

$$= \left[\frac{\cos \theta (2 \cos^2 \theta - 1)}{\sin \theta (1 - 2 \sin^2 \theta)} \right]^2 + 1$$

$$(\because \cos 2A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A)$$

$$= \cot^2 \theta + 1 = \operatorname{cosec}^2 \theta$$

$$\text{Sol.331.(d)} \sin^4 \theta + \cos^4 \theta = 2 \sin^2 \theta \cos^2 \theta$$

$$\sin^4 \theta + \cos^4 \theta - 2 \sin^2 \theta \cos^2 \theta = 0$$

$$(\sin^2 \theta - \cos^2 \theta)^2 = 0$$

Taking square root

$$\sin^2 \theta - \cos^2 \theta = 0 \Rightarrow \sin^2 \theta = \cos^2 \theta$$

$$1 = \frac{\cos^2 \theta}{\sin^2 \theta} \Rightarrow 1 = \cot^2 \theta \Rightarrow \cot \theta = 1$$

$$\text{Sol.332.(c)} \tan^4 A + \tan^2 A$$

$$\tan^2 A (\tan^2 A + 1)$$

$$(\sec^2 A - 1)(\sec^2 A) \Rightarrow \sec^4 A - \sec^2 A$$

Sol.333.(a)

$$[\tan \theta + \sec \theta + 1][\cot \theta - \operatorname{cosec} \theta + 1]$$

$$\left[\frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta} + 1 \right] \left[\frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta} + 1 \right]$$

$$\left[\frac{(\sin \theta + \cos \theta) + 1}{\cos \theta} \right] \left[\frac{(\sin \theta + \cos \theta) - 1}{\sin \theta} \right]$$

$$\left[\frac{(\cos \theta + \sin \theta)^2 - 1^2}{\cos \theta \sin \theta} \right]$$

$$\left[\frac{\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta - 1}{\cos \theta \sin \theta} \right]$$

$$\left[\frac{1 - 1 + 2 \sin \theta \cos \theta}{\cos \theta \sin \theta} \right] = \left[\frac{2 \sin \theta \cos \theta}{\cos \theta \sin \theta} \right] = 2$$

Sol.334.(d)

$$(\operatorname{cosec} \theta - \sin \theta)(\sec \theta - \cos \theta)$$

$$(\tan \theta + \cot \theta)$$

$$\left(\frac{1}{\sin \theta} - \sin \theta \right) \left(\frac{1}{\cos \theta} - \cos \theta \right) \left(\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right)$$

$$\left(\frac{1 - \sin^2 \theta}{\sin \theta} \right) \left(\frac{1 - \cos^2 \theta}{\cos \theta} \right) \left(\frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta} \right)$$

$$\left(\frac{\cos^2 \theta}{\sin \theta} \right) \left(\frac{\sin^2 \theta}{\cos \theta} \right) \left(\frac{1}{\cos \theta \sin \theta} \right)$$

$$\left(\frac{\cos^2 \theta}{\sin \theta} \right) \left(\frac{\sin^2 \theta}{\cos \theta} \right) \left(\frac{1}{\cos \theta \sin \theta} \right) = 1$$

Short trick :-

Put $\theta = 45^\circ$

Sol.335.(c)

$$(1 + \tan \theta + \sec \theta) \times (1 + \cot \theta - \operatorname{cosec} \theta)$$

$$= \left(1 + \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta} \right) \times \left(1 + \frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta} \right)$$

$$= \left(\frac{\sin \theta + \cos \theta + 1}{\cos \theta} \right) \times \left(\frac{\sin \theta + \cos \theta - 1}{\sin \theta} \right)$$

$$= \left(\frac{(\sin \theta + \cos \theta)^2 - (1)^2}{\cos \theta \sin \theta} \right)$$

$$= \left(\frac{\sin^2 \theta + \cos^2 \theta + 2 \cos \theta \sin \theta - 1}{\cos \theta \sin \theta} \right)$$

$$= \left(\frac{1 + 2 \cos \theta \sin \theta - 1}{\cos \theta \sin \theta} \right) = \left(\frac{2 \cos \theta \sin \theta}{\cos \theta \sin \theta} \right)$$

$$= 2$$

Short trick -

Put $\theta = 45^\circ$

Sol.336.(a) Formula used $\therefore \tan(A + B)$

$$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{Then, } \frac{\tan 71^\circ + \tan 19^\circ}{1 - \tan 71^\circ \tan 19^\circ}$$

$$= \tan(71^\circ + 19^\circ) = \tan(90^\circ) = \text{not defined}$$

Sol.337.(b) $\sin(x + y) = \cos\{3(x + y)\}$

When, $\sin a = \cos b$ then $a + b = 90^\circ$

Hence,

$$(x + y) + 3(x + y) = 90^\circ$$

$$4(x + y) = 90^\circ \Rightarrow 2(x + y) = 45^\circ$$

Now,

$$\cot 2(x + y) = \cot(45^\circ) = \cot 45^\circ = 1$$

Sol.338.(a)

Identity used :- $\sin 2A = 2 \sin A \cos A$ and

$$\cos 2A = 1 - 2 \sin^2 A$$

$$\text{Now, } \frac{2 \cos 15^\circ \sin 15^\circ}{\cos^2 15^\circ - \sin^2 15^\circ}$$

$$= \frac{\sin 2(15^\circ)}{1 - \sin^2 15^\circ - \sin^2 15^\circ} = \frac{\sin 30^\circ}{1 - 2 \sin^2 15^\circ}$$

$$= \frac{\sin 30^\circ}{\cos 2(15^\circ)} = \frac{\sin 30^\circ}{\cos 30^\circ} = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$$

$$\text{Sol.339.(c)} \frac{17}{\sec^2 \theta} + 10 \sin^2 \theta + \frac{7}{1 + \cot^2 \theta}$$

$$= 17 \cos^2 \theta + 10 \sin^2 \theta + \frac{7}{1 + \frac{\cos^2 \theta}{\sin^2 \theta}}$$

$$= 17 \cos^2 \theta + 10 \sin^2 \theta + \frac{7 \sin^2 \theta}{\sin^2 \theta + \cos^2 \theta}$$

$$= 17 \cos^2 \theta + 10 \sin^2 \theta + 7 \sin^2 \theta$$

$$= 17(\cos^2 \theta + \sin^2 \theta) \Rightarrow 17$$

Sol.340.(c) $\sin^2 15^\circ + \sin^2 25^\circ + \sin^2 35^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 65^\circ + \sin^2 75^\circ$

$$= \sin^2(90^\circ - 75^\circ) + \sin^2(90^\circ - 65^\circ) + \sin^2(90^\circ - 55^\circ) + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 65^\circ + \sin^2 75^\circ$$

$$= \cos^2 75^\circ + \cos^2 65^\circ + \cos^2 55^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 65^\circ + \sin^2 75^\circ$$

$$= \cos^2 75^\circ + \sin^2 75^\circ + \cos^2 65^\circ + \sin^2 65^\circ + \cos^2 55^\circ + \sin^2 55^\circ + \sin^2 45^\circ$$

$$= 1 + 1 + 1 + \frac{1}{2} \Rightarrow 3 + \frac{1}{2} = \frac{7}{2}$$

$$\text{Sol.341.(a)} 7 \left(\frac{\operatorname{cosec} 24^\circ}{\sec 66^\circ} \right)^3 + 8 \left(\frac{\cot 37^\circ}{\tan 53^\circ} \right)^4 -$$

$$\left(2 \frac{\sec 14^\circ}{\operatorname{cosec} 76^\circ} \right)^2 + \left(-3 \frac{\tan 82^\circ}{\cot 8^\circ} \right)^3$$

$$\Rightarrow 7 \left(\frac{\operatorname{cosec} 24^\circ}{\sec(90^\circ - 24^\circ)} \right)^3 + 8 \left(\frac{\cot 37^\circ}{\tan(90^\circ - 37^\circ)} \right)^4$$

$$- \left(2 \frac{\sec 14^\circ}{\operatorname{cosec}(90^\circ - 14^\circ)} \right)^2 + \left(-3 \frac{\tan 82^\circ}{\cot(90^\circ - 82^\circ)} \right)^3$$

$$\Rightarrow 7 \left(\frac{\operatorname{cosec} 24^\circ}{\operatorname{cosec} 24^\circ} \right)^3 + 8 \left(\frac{\cot 37^\circ}{\cot 37^\circ} \right)^4 -$$



$$\begin{aligned} & \left(2 \frac{\sec 14^\circ}{\sec 14^\circ}\right)^2 + \left(-3 \frac{\tan 82^\circ}{\tan 82^\circ}\right)^3 \\ & \Rightarrow 7(1)^3 + 8(1)^4 - (2)^2 + (-3)^3 \\ & \Rightarrow 7 + 8 - 4 - 27 = -16 \end{aligned}$$

Sol.342.(a) Given, $8 \tan A = 5$

$$\Rightarrow \tan A = \frac{5}{8}$$

$$\text{Now, } \frac{8 \sin A - 7 \cos A}{8 \sin A + 11 \cos A}$$

On dividing numerator and denominator by $\cos A$, we get

$$\begin{aligned} & \Rightarrow \frac{8 \frac{\sin A}{\cos A} - 7}{8 \frac{\sin A}{\cos A} + 11} = \frac{8 \tan A - 7}{8 \tan A + 11} \\ & \Rightarrow \frac{8 \times \frac{5}{8} - 7}{8 \times \frac{5}{8} + 11} = \frac{5 - 7}{5 + 11} = \frac{-2}{16} = \frac{-1}{8} \end{aligned}$$

Sol.343.(b) Let $A = 45^\circ$

(a) LHS = $\cos^2 A \cot^2 A$

$$= \cos^2 45^\circ \cot^2 45^\circ = \frac{1}{2}$$

RHS = $\cot^2 A - \cos^2 A$

$$= \cot^2 45^\circ - \cos^2 45^\circ = 1 - \frac{1}{2} = \frac{1}{2}$$

$$(b) \frac{1 + \tan^2 A}{\tan^2 A} = \sin^2 A$$

$$\text{LHS} = \frac{1 + \tan^2 A}{\tan^2 A} = \frac{1 + 1}{1} = 2$$

$$\text{RHS} = \sin^2 A = \sin^2 45^\circ = \frac{1}{2}$$

Hence LHS $\neq$ RHS

$$\Rightarrow \frac{1 + \tan^2 A}{\tan^2 A} = \sin^2 A \text{ (not the correct formula).}$$

Sol.344.(c) Given expression

$$\begin{aligned} & \frac{5 \cos^2 120^\circ + 4 \sec^2 30^\circ - \tan^2 135^\circ}{\sin^2 30^\circ + \cos^2 30^\circ} \\ & \Rightarrow \frac{5 \left(\frac{1}{2}\right)^2 + 4 \left(\frac{2}{\sqrt{3}}\right)^2 - (-1)(-1)}{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ & \Rightarrow \frac{\frac{5}{4} + \frac{16}{3} - 1}{\frac{1}{4} + \frac{3}{4}} = \frac{\frac{67}{12}}{\frac{4}{4}} = \frac{67}{12} \end{aligned}$$

Sol.345.(b)

$$\begin{aligned} & \left(\frac{\sin 23^\circ \cos 67^\circ + \cos 23^\circ \sin 67^\circ}{\csc^2 15^\circ - \tan^2 75^\circ} \right)^3 \\ & \Rightarrow \left(\frac{\sin 23^\circ \cos (90-23)^\circ + \cos 23^\circ \sin (90-23)^\circ}{\csc^2 15^\circ - \tan^2 (90-15)^\circ} \right)^3 \\ & \Rightarrow \left(\frac{\sin 23^\circ \sin 23^\circ + \cos 23^\circ \cos 23^\circ}{\csc^2 15^\circ - \cot^2 15^\circ} \right)^3 \\ & \Rightarrow \left(\frac{\sin^2 23^\circ + \cos^2 23^\circ}{1} \right)^3 \text{ (since } \csc^2 \theta - \cot^2 \theta = 1 \text{ and } \sin^2 \theta + \cos^2 \theta = 1) \\ & = 1^3 = 1 \end{aligned}$$

Sol.346.(a) We know

$$\operatorname{Cosec}^2 A - \cot^2 A = 1$$

$$\Rightarrow \operatorname{Cosec}^2 A - \left(\frac{2p}{p^2 - 1}\right)^2 = 1$$

$$\Rightarrow \operatorname{Cosec}^2 A = 1 + \left(\frac{2p}{p^2 - 1}\right)^2$$

$$\Rightarrow \operatorname{Cosec}^2 A = \frac{p^4 + 1 - 2p^2 + 4p^2}{(p^2 - 1)^2}$$

$$\Rightarrow \operatorname{Cosec}^2 A = \frac{p^4 + 1 + 2p^2}{(p^2 - 1)^2}$$

$$\Rightarrow \operatorname{Cosec}^2 A = \frac{(p^2 + 1)^2}{(p^2 - 1)^2}$$

$$\Rightarrow \operatorname{Cosec} A = \frac{p^2 + 1}{p^2 - 1}$$

Sol.347.(c)

$$\begin{aligned} & \operatorname{cosec} \theta \left[\frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta} \right] - 2 \cot^2 \theta \\ & = \operatorname{cosec} \theta \left[\frac{(1 + \cos \theta)^2 + \sin^2 \theta}{(\sin \theta)(1 + \cos \theta)} \right] - 2 \cot^2 \theta \\ & = \operatorname{cosec} \theta \left[\frac{1 + \cos^2 \theta + 2 \cos \theta + \sin^2 \theta}{(\sin \theta)(1 + \cos \theta)} \right] - 2 \cot^2 \theta \\ & = \operatorname{cosec} \theta \left[\frac{1 + 1 + 2 \cos \theta}{(\sin \theta)(1 + \cos \theta)} \right] - 2 \cot^2 \theta \\ & (\because \sin^2 \theta + \cos^2 \theta = 1) \\ & = \operatorname{cosec} \theta \times \frac{2}{\sin \theta} - 2 \cot^2 \theta \\ & = 2 \operatorname{cosec}^2 \theta - 2 \cot^2 \theta = 2 \\ & (\because \operatorname{cosec}^2 \theta - \cot^2 \theta = 1) \end{aligned}$$

Sol.348.(c) $(\sin A + \cos A) = \frac{4}{3}$

Squaring both side, we have;

$$\sin^2 A + \cos^2 A + 2 \sin A \cos A = \frac{16}{9}$$

$$1 + 2 \sin A \cos A = \frac{16}{9} \text{ (As, } \sin^2 A + \cos^2 A = 1)$$

$$2 \sin A \cos A = \frac{16}{9} - 1 = \frac{7}{9}$$

$$\Rightarrow \sin A \cos A = \frac{7}{9 \times 2} = \frac{7}{18}$$

$$\tan A + \cot A = \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}$$

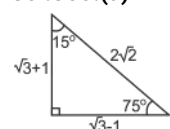
$$= \frac{\sin^2 A + \cos^2 A}{\cos A \sin A} = \frac{1}{\sin A \cos A} = \frac{1}{\frac{7}{18}} = \frac{18}{7}$$

Sol.349.(c) $\sin^2 \theta + \cos^2 \theta - 1 = 0$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Clearly we can see that there are infinitely many values which satisfy the given expression.

Sol.350.(a)



$$\sin 75^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$\sin 15^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$\sin 75^\circ + \sin 15^\circ = \frac{\sqrt{3}+1}{2\sqrt{2}} + \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}+1+\sqrt{3}-1}{2\sqrt{2}} = \frac{2\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{3} \times \sqrt{2}}{\sqrt{2} \times \sqrt{2}} = \frac{\sqrt{6}}{2}$$

Sol.351.(b) As we know,

$$\sin(90^\circ - \theta) = \cos \theta$$

$$\cos(90^\circ - \theta) = \sin \theta$$

$$\begin{aligned} & \frac{\cos 25^\circ - \sin 65^\circ}{\cos 25^\circ + \sin 65^\circ} = \frac{\sin 65^\circ - \sin 65^\circ}{\sin 65^\circ + \sin 65^\circ} \\ & = \frac{0}{2 \sin 65^\circ} = 0 \end{aligned}$$

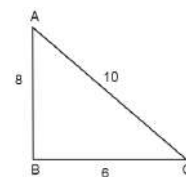
Sol.352.(a) $\sin(5x - 40^\circ) = \cos(5y + 40^\circ)$

$$\cos[90^\circ - (5x - 40^\circ)] = \cos(5y + 40^\circ)$$

$$130^\circ - 5x = 5y + 40^\circ \Rightarrow 5y + 5x = 90^\circ$$

$$x + y = 18^\circ$$

Sol.353.(a) According to the question, we have given triangle;



$$\tan^2 A = \left(\frac{6}{8}\right)^2 = \frac{36}{64} = \frac{9}{16}$$

$$\cos^2 C = \left(\frac{6}{10}\right)^2 = \frac{36}{100} = \frac{9}{25}$$

$$\tan^2 A + \cos^2 C = \frac{9}{16} + \frac{9}{25} = 9 \left(\frac{25+16}{400} \right)$$

$$= 9 \times \frac{41}{400} = \frac{369}{400}$$

Sol.354.(c) $\sin \theta + \frac{1}{\sin \theta} = \sqrt{5}$

$$\text{Then, } \sin^3 \theta + \frac{1}{\sin^3 \theta} = \left(\sin \theta + \frac{1}{\sin \theta} \right)^3 -$$

$$3 \left(\sin \theta + \frac{1}{\sin \theta} \right)$$

$$= (\sqrt{5})^3 - 3\sqrt{5} = 5\sqrt{5} - 3\sqrt{5} = 2\sqrt{5}$$

Sol.355.(b) $\cos 2\theta = 0$

$$\cos 2\theta = \cos 90^\circ \Rightarrow 2\theta = 90^\circ$$

$$\theta = \left(\frac{90^\circ}{2} \right) = 45^\circ$$

$$\sin(75^\circ - \theta) = \sin(75^\circ - 45^\circ) = \sin 30^\circ = \frac{1}{2}$$

Sol.356.(c) $\operatorname{Psin} 60^\circ = \operatorname{Qcosec} 45^\circ$

$$P \times \frac{\sqrt{3}}{2} = Q \times \sqrt{2} \Rightarrow \frac{P}{Q} = \frac{2\sqrt{2}}{\sqrt{3}}$$

$$\frac{P^2 + Q^2}{P^2 - Q^2} = \frac{(2\sqrt{2})^2 + (\sqrt{3})^2}{(2\sqrt{2})^2 - (\sqrt{3})^2} = \frac{8+3}{8-3} = \frac{11}{5}$$

Sol.357.(d)

$$\tan^2 \theta + \tan^4 \theta = \tan^2 \theta (1 + \tan^2 \theta)$$

$$= \tan^2 \theta \cdot \sec^2 \theta = (\sec^2 \theta - 1) \cdot \sec^2 \theta$$

$$= \sec^4 \theta - \sec^2 \theta$$

Sol.358.(c) $\cos \theta - \sin \theta = \sqrt{2} \sin \theta$

Squaring both sides we get,

$$\Rightarrow (\cos \theta - \sin \theta)^2 = 2 \sin^2 \theta$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta - 2 \cos \theta \sin \theta = 2 \sin^2 \theta$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta - 2 \sin^2 \theta = 2 \cos \theta \sin \theta$$

$$\Rightarrow \cos^2 \theta - \sin^2 \theta = 2 \cos \theta \sin \theta$$

$$\Rightarrow (\cos \theta - \sin \theta)(\cos \theta + \sin \theta)$$

$$= 2 \cos \theta \sin \theta$$

$$\Rightarrow \sqrt{2} \sin \theta (\cos \theta + \sin \theta) = 2 \cos \theta \sin \theta$$

$$\Rightarrow (\cos \theta + \sin \theta) = \sqrt{2} \cos \theta$$

Sol.359.(d) According to the question,

$$(1 + \sin^4 A - \cos^4 A) \operatorname{cosec}^2 A$$

On Putting $A = 90^\circ$

$$(1 + 1^4 - 0^4) 1^2 = 2$$

Sol.360.(a) According to the question,

$$\cos A = \frac{\text{base}}{\text{hypotenuse}} = \frac{15}{17}$$

So the perpendicular = 8 (8, 15, 17 pythagorean triples)

$$\cot (90^\circ - A) = \tan A$$

$$\tan A = \frac{\text{perpendicular}}{\text{base}} = \frac{8}{15}$$

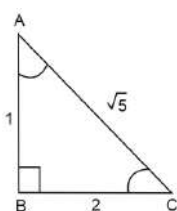
Sol.361.(d)

If $\sin A = \cos B$ then, $A + B = 90^\circ$

$$\sin 2A = \cos 15^\circ$$

$$\Rightarrow 2A + 15^\circ = 90^\circ \Rightarrow A = 37.5^\circ$$

Sol.362.(d)



$$\cos A + \tan C = \frac{1}{\sqrt{5}} + \frac{1}{2} = \frac{2 + \sqrt{5}}{2\sqrt{5}}$$

Sol.363.(c) According to the question, $\sec A - \tan A = p$

$$\sec A + \tan A = \frac{1}{p} \Rightarrow \sec A = \frac{p^2 + 1}{2p}$$

$$\text{Sol.364.(d)} \sqrt{\frac{1 - \tan A}{1 + \tan A}}$$

$$= \sqrt{\frac{1 - \frac{\sin A}{\cos A}}{1 + \frac{\sin A}{\cos A}}} = \sqrt{\frac{\frac{\cos A - \sin A}{\cos A}}{\frac{\cos A + \sin A}{\cos A}}}$$

$$= \sqrt{\frac{\cos A - \sin A}{\cos A + \sin A}} \times \frac{\cos A - \sin A}{\cos A - \sin A}$$

$$= \sqrt{\frac{(\cos A - \sin A)^2}{\cos^2 A - \sin^2 A}}$$

$$= \sqrt{\frac{1 - 2 \sin A \cos A}{\cos^2 A - \sin^2 A}} = \sqrt{\frac{1 - \sin 2A}{\cos 2A}}$$

Sol.365.(d)

$$\sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sqrt{\frac{1 + \sin A}{1 - \sin A}} \times \frac{1 + \sin A}{1 + \sin A}$$

$$= \sqrt{\frac{(1 + \sin A)^2}{(1 - \sin A)(1 + \sin A)}}$$

$$= \frac{(1 + \sin A)}{\sqrt{1 - \sin^2 A}} = \frac{(1 + \sin A)}{\cos A} = \sec A + \tan A$$

$$\text{Sol.366.(c)} \sin A = \frac{2}{3} = \frac{\text{perpendicular}}{\text{hypotenuse}}$$

$$\text{Then, base} = \sqrt{9 - 4} = \sqrt{5}$$

Now, $(7 - \tan A)(3 + \cos A)$

$$= (7 - \frac{2}{\sqrt{5}})(3 + \frac{\sqrt{5}}{3})$$

$$= (21 - \frac{2}{3}) + (\frac{7\sqrt{5}}{3} - \frac{6}{\sqrt{5}})$$

$$= \frac{61}{3} + \frac{17}{3\sqrt{5}}$$

$$\text{Sol.367.(b)} \sin \beta = \frac{1}{3}$$

Then, $(\sec \beta - \tan \beta)^2$

$$= (\frac{1}{\cos \beta} - \frac{\sin \beta}{\cos \beta})^2 = (\frac{1 - \sin \beta}{\cos \beta})^2$$

$$= \frac{(1 - \sin \beta)^2}{\cos^2 \beta} = \frac{(1 - \sin \beta)^2}{1 - \sin^2 \beta} = \frac{(1 - \frac{1}{3})^2}{1 - \frac{1}{9}}$$

$$= \frac{1}{2}$$

$$\text{Sol.368.(c)} \frac{\sin^2 39^\circ + \sin^2 (90^\circ - 39^\circ)}{\cos^2 35^\circ + \cos^2 (90^\circ - 35^\circ)} +$$

$$3 \tan 25^\circ \tan 65^\circ$$

$$= \frac{\sin^2 39^\circ + \cos^2 39^\circ}{\cos^2 35^\circ + \sin^2 35^\circ} + 3 \tan 25^\circ \times \frac{1}{\tan 25^\circ}$$

$$= 1 + 3 = 4$$

Sol.369.(b) According to the question,

$$\tan \theta + \cot \theta = 2$$

On Putting $\theta = 45^\circ$

$$1 + 1 = 2 \text{ (satisfied)}$$

Now,

$$2 \tan^{25} \theta + 3 \cot^{20} \theta + 5 \tan^{30} \theta \cot^{15} \theta$$

$$2 \times 1 + 3 \times 1 + 5 \times 1 \times 1 = 10$$

Sol.370.(b) According to the question,

$$\sin(50^\circ + A) - \cos(40^\circ - A)$$

Let $A = 10^\circ$

$$\text{Then } \sin 60^\circ - \cos 30^\circ = \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} = 0$$

Sol.371.(a) According to the question,

$$\sin \theta + \cos \theta = \sqrt{2} \cos \theta$$

$$\sin \theta = \cos \theta (\sqrt{2} - 1)$$

$$\text{now, } \frac{\sin \theta - \cos \theta}{\sin \theta} = \frac{\sin \theta}{\sin \theta} - \frac{\cos \theta}{\sin \theta}$$

$$1 - \frac{\cos \theta}{\cos \theta (\sqrt{2} - 1)}$$

$$= 1 - \frac{1}{\sqrt{2} - 1} \times \frac{(\sqrt{2} + 1)}{(\sqrt{2} + 1)} = -\sqrt{2}$$

$$\text{Sol.372.(b)} \frac{\cos \theta}{\sec \theta - 1} + \frac{\cos \theta}{\sec \theta + 1}$$

$$= \cos \theta \left(\frac{\sec \theta + 1 + \sec \theta - 1}{\tan^2 \theta} \right)$$

$$= \cos \theta \left(\frac{2 \sec \theta}{\tan^2 \theta} \right) = 2 \cot^2 \theta$$

Sol.373.(d) $\sin \theta - \cos \theta = 0$

$$\sin \theta = \cos \theta \Rightarrow \theta = 45^\circ$$

$$\text{Now, } \sin(\pi - \theta) + \sin\left(\frac{\pi}{2} - \theta\right)$$

$$= \sin(\theta) + \cos(\theta) = \sin(45^\circ) + \cos(45^\circ)$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

$$\text{Sol.374.(a)} \cos \theta + \sec \theta = \sqrt{3}$$

$$\cos^3 \theta + \sec^3 \theta = (\sqrt{3})^3 - 3(\sqrt{3}) = 0$$

Sol.375.(a)

Pythagoras triplets :- (16, 63, 65)

$$\cos A = \frac{63}{65} \Rightarrow \sin A = \frac{16}{65}$$

$$\text{Now, } \tan A + \cot A = \frac{1}{\sin A \cos A}$$

$$= \frac{65 \times 65}{16 \times 63} = 4.19$$

$$\text{Sol.376.(a)} \frac{\cos \theta}{(1 + \sin \theta)} + \frac{\cos \theta}{(1 - \sin \theta)} = 4$$

$$\frac{2}{(\cos^2 \theta)} = \frac{4}{\cos \theta} \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

$$\text{Sol.377.(c)} \sin \alpha + \cos \alpha = \frac{2}{\sqrt{3}}$$

$$1 + 2 \sin \alpha \cos \alpha = \frac{4}{3}$$

$$\sin 2\alpha = \frac{4}{3} - 1 = \frac{1}{3}$$

$$\text{Now, } \tan \alpha + \cot \alpha = \frac{2}{\sin 2\alpha} = 6$$

Sol.378.(d) $(\operatorname{cosec}^2 B - 1)(\sec^2 B - 1)$

$$(\cot^2 B)(\tan^2 B) = 1$$

$$\text{Sol.379.(a)} \sin \theta + \cos \theta = \frac{1}{29} \Rightarrow 2 \sin \theta$$

$$\cos \theta = \left(\frac{1}{29}\right)^2 - 1 = \left(\frac{-840}{841}\right)$$

$$\sin \theta - \cos \theta = \sqrt{1 - \left(\frac{-840}{841}\right)^2}$$

$$= \frac{41}{29}$$

$$\text{Now, } \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = \frac{1}{29} \times \frac{29}{41} = \frac{1}{41}$$

Sol.380.(c)

$$\frac{(1 + \tan^2 A)}{\operatorname{cosec}^2 A \times \tan A} = \frac{(\sec^2 A)}{\operatorname{cosec}^2 A \times \tan A}$$

$$= \frac{(\sin^2 A)}{\cos^2 A \times \tan A} = \tan A$$

Sol.381.(b) $\cos^4 A - \sin^4 A$

$$= (\cos^2 A - \sin^2 A)(\cos^2 A + \sin^2 A)$$

$$= 1 - 2 \sin^2 A$$

$$\text{Sol.382.(b)} \operatorname{cosec} \theta = \frac{29}{22}$$

$$\cot^2 \theta = \operatorname{cosec}^2 \theta - 1 = \frac{841}{484} - 1 = \frac{357}{484}$$

Sol.383.(a)

$$\sin A \left(1 + \frac{\sin A}{\cos A}\right) + \cos A \left(1 + \frac{\cos A}{\sin A}\right)$$

$$= \sin A + \frac{\sin^2 A}{\cos A} + \cos A + \frac{\cos^2 A}{\sin A}$$

$$= \frac{\sin^2 A + \cos^2 A}{\cos A} + \frac{\sin^2 A + \cos^2 A}{\sin A}$$

$$= \frac{1}{\cos A} + \frac{1}{\sin A} = \sec A + \operatorname{cosec} A$$

$$\begin{aligned}\text{Sol.384.(a)} \quad & \frac{3 - \sin^2 A + \cos^2 A}{2 + 2\cos^2 A} \\ &= \frac{\sin^2 A + \cos^2 A + 2 - \sin^2 A + \cos^2 A}{2 + 2\cos^2 A} \\ &= \frac{\cos^2 A + 2 + \cos^2 A}{2 + 2\cos^2 A} = 1\end{aligned}$$

$$\begin{aligned}\text{Sol.385.(a)} \quad & \sec\theta + \tan\theta = \sqrt{3} \\ \sec\theta - \tan\theta &= \frac{1}{\sqrt{3}} \\ \sec\theta &= \frac{\sqrt{3} + \frac{1}{\sqrt{3}}}{2} = \frac{2}{\sqrt{3}} \Rightarrow \theta = 30^\circ \\ \text{Now, } \sin\theta + \cos\theta &= \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1 + \sqrt{3}}{2}\end{aligned}$$

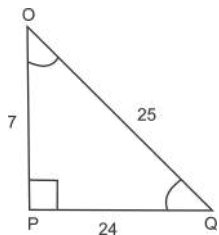
$$\begin{aligned}\text{Sol.386.(a)} \quad & 2\tan^2 A + 4\cos^4 A = 3 \\ \text{Put } A &= 45^\circ \\ 2\tan^2 45^\circ + 4\cos^4 45^\circ \\ &= 2 \times 1 + 4 \times \left(\frac{1}{\sqrt{2}}\right)^4 = 2 + 1 = 3\end{aligned}$$

$$\begin{aligned}\text{Sol.387.(a)} \quad & \cot^2 A - \cos^2 A = \\ & \cot^2 A \cdot \cos^2 A \dots (\text{identity}) \\ a^2 &= \cot^2 A + \cos^2 A + 2 \cdot \cot A \cdot \cos A \\ b^2 &= \cot^2 A + \cos^2 A - 2 \cdot \cot A \cdot \cos A \\ ab &= \cot^2 A - \cos^2 A \\ \text{Now,} \\ a^2 - b^2 - 4\sqrt{ab} &= 4 \cdot \cot A \cdot \cos A - 4 \\ \sqrt{\cot^2 A - \cos^2 A} \\ &= 4 \cdot \cot A \cdot \cos A - 4 \cdot \cot A \cdot \cos A = 0\end{aligned}$$

$$\begin{aligned}\text{Sol.388.(b)} \quad & 1 - \frac{\tan^2 \phi}{\sec^2 \phi} = \frac{\sec^2 \phi - \tan^2 \phi}{\sec^2 \phi} \\ &= \frac{1}{\sec^2 \phi} = \cos^2 \phi = \frac{\sin^2 \phi \cdot \cos^2 \phi}{\sin^2 \phi} \\ &= \sin^2 \phi \cdot \cot^2 \phi\end{aligned}$$

$$\begin{aligned}\text{Sol.389.(c)} \quad & 1 + \cot^2 B = \operatorname{cosec}^2 B \\ \cot^2 B - \operatorname{cosec}^2 B &= -1\end{aligned}$$

Sol.390.(c)
Pythagoras triplets:- (7, 24, 25)



$$\sin Q + \cos Q = \frac{7}{25} + \frac{24}{25} = \frac{31}{25}$$

$$\begin{aligned}\text{Sol.391.(a)} \quad & \frac{2 \sec A}{\sec^2 A - 1} = \frac{2 \cot^2 A}{\cos A} \\ &= \frac{2 \cos A}{\sin^2 A} = \frac{2 \cot A}{\sin A} = 2 \operatorname{cosec} A \cdot \cot A\end{aligned}$$

$$\text{Sol.392.(a)} \quad \frac{\cos^2 A}{(1 + \sin A)} = \frac{1 - \sin^2 A}{(1 + \sin A)}$$

$$= \frac{(1 + \sin A)(1 - \sin A)}{(1 + \sin A)} = (1 - \sin A)$$

$$\begin{aligned}\text{Sol.393.(b)} \quad & \text{If } A + B = 90^\circ \text{ then } \sin^2 A + \sin^2 B = 1 \\ \sin^2 30^\circ + \sin^2 40^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \\ & \sin^2 35^\circ + \sin^2 45^\circ + \sin^2 50^\circ + \sin^2 60^\circ \\ &= \sin^2 30^\circ + \sin^2 60^\circ + \sin^2 40^\circ + \sin^2 50^\circ \\ &+ \sin^2 45^\circ + \sin^2 45^\circ + \sin^2 55^\circ + \sin^2 35^\circ \\ &= 1 + 1 + 1 + 1 = 4\end{aligned}$$

$$\begin{aligned}\text{Sol.394.(a)} \quad & \text{If } A + B = 90^\circ, \text{ Then } \sin^2 A + \sin^2 B = 1 \\ \text{and } \cos^2 A + \cos^2 B &= 1 \\ \frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos^2 53^\circ \\ \sin 37^\circ \\ &= \frac{\sin^2 32^\circ + \sin^2 58^\circ}{\cos^2 32^\circ + \cos^2 58^\circ} + \sin^2 53^\circ + \cos^2 53^\circ \\ &= \frac{1}{1} + 1 = 2\end{aligned}$$

$$\begin{aligned}\text{Sol.395.(b)} \quad & \frac{1}{\sin \theta} - \frac{\cot^2 \theta}{1 + \operatorname{cosec} \theta} \\ &= \operatorname{cosec} \theta - \frac{(\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1)}{1 + \operatorname{cosec} \theta} \\ &= \operatorname{cosec} \theta - \operatorname{cosec} \theta + 1 = 1\end{aligned}$$

$$\begin{aligned}\text{Sol.396.(a)} \quad & \frac{\sin 58^\circ}{\cos 32^\circ} + \frac{\sin 55^\circ \cdot \sec 35^\circ}{\tan 5^\circ \cdot \tan 45^\circ \cdot \tan 85^\circ} \\ &= \frac{\cos 32^\circ}{\cos 32^\circ} + \frac{\sin 55^\circ \cdot \operatorname{cosec} 55^\circ}{\tan 5^\circ \cdot \tan 85^\circ} \\ &= 1 + \frac{1}{\tan 5^\circ \cdot \cot 5^\circ} = 1 + 1 = 2\end{aligned}$$

$$\begin{aligned}\text{Sol.397.(c)} \quad & \sec \theta = \frac{50}{23} \Rightarrow \sec^2 \theta = \frac{2500}{529} \\ \tan^2 \theta &= \sec^2 \theta - 1 = \frac{2500 - 529}{529} \\ &= \frac{1971}{529} = 3 \frac{384}{529}\end{aligned}$$

$$\begin{aligned}\text{Sol.398.(d)} \quad & \frac{\cos^2 25^\circ - \sin^2 65^\circ}{\cos^2 25^\circ + \sin^2 65^\circ} = \\ & \frac{\cos^2 25^\circ - \cos^2 25^\circ}{\cos^2 25^\circ + \sin^2 65^\circ} = 0\end{aligned}$$

$$\begin{aligned}\text{Sol.399.(b)} \quad & \frac{\cot^3 A - 1}{\cot A - 1} \\ &= \frac{(\cot A - 1)(\cot^2 A + \cot A + 1)}{\cot A - 1} \\ &= (\cot^2 A + \cot A + 1)\end{aligned}$$

$$\begin{aligned}\text{Sol.400.(d)} \quad & \text{If } \sec \theta + \tan \theta = a \\ \text{Then, } \sec \theta - \tan \theta &= \frac{1}{a} \\ \text{on solving above equation, we get} \\ \sec \theta &= \frac{a + \frac{1}{a}}{2} = \frac{a^2 + 1}{2a}\end{aligned}$$

$$\text{Sol.401.(d)} \quad \frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} + 2$$

$$\begin{aligned}&= \frac{1 - \sin A}{\cos A} + \frac{1 + \sin A}{\cos A} + 2 \\ &= \frac{1 - \sin A + 1 + \sin A}{\cos A} + 2 \\ &= 2 \sec A + 2 = 2(1 + \sec A)\end{aligned}$$

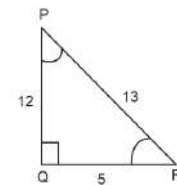
$$\begin{aligned}\text{Sol.402.(a)} \quad & \sec \theta + \tan \theta = \frac{1}{\sqrt{3}} \\ \sec \theta - \tan \theta &= \sqrt{3} \\ \sec \theta &= \frac{\frac{1}{\sqrt{3}} + \sqrt{3}}{2} = \frac{2}{\sqrt{3}} \Rightarrow \cos \theta = \frac{\sqrt{3}}{2} \\ \text{Therefore, } \theta &= 30^\circ\end{aligned}$$

$$\text{Now, } \cot \theta + \cos \theta = \sqrt{3} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

$$\begin{aligned}\text{Sol.403.(c)} \quad & \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} = \frac{4}{5} \\ \frac{\sin \theta - \cos \theta + \sin \theta + \cos \theta}{\sin \theta - \cos \theta - \sin \theta - \cos \theta} &= \frac{4 + 5}{4 - 5} \\ \frac{2 \sin \theta}{-2 \cos \theta} &= \frac{9}{-1} \Rightarrow \cot \theta = \frac{1}{9} \\ \operatorname{cosec}^2 \theta &= 1 + \left(\frac{1}{9}\right)^2 = \left(\frac{82}{81}\right)\end{aligned}$$

$$\begin{aligned}\text{Now, } \frac{\operatorname{cosec}^2 \theta}{2 - \operatorname{cosec}^2 \theta} &= \frac{\frac{82}{81}}{2 - \frac{82}{81}} \\ &= \frac{82}{80} = \frac{41}{40}\end{aligned}$$

Sol.404.(c)



$$\tan P + \cot R = \frac{5}{12} + \frac{5}{12} = \frac{10}{12}$$

$$\begin{aligned}\text{Sol.405.(d)} \quad & \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \\ &= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}} = \frac{\frac{5}{6}}{1 - \frac{1}{6}} = 1 \\ \Rightarrow A + B &= 45^\circ\end{aligned}$$

$$\begin{aligned}\text{Sol.406.(d)} \quad & \tan A = \frac{12}{5} \\ \frac{5(1 - \cos^2 A)}{6(1 - \sin^2 A)} &= \frac{5(\sin^2 A)}{6(\cos^2 A)} = \frac{5}{6} \times \tan^2 A \\ &= \frac{5}{6} \times \frac{144}{25} = \frac{24}{5}\end{aligned}$$

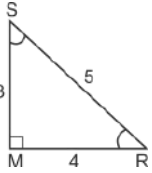
$$\begin{aligned}\text{Sol.407.(c)} \quad & \tan \theta \cdot \tan 2\theta = 1 \\ \text{Put } \theta &= 30^\circ, \text{ Then} \\ \tan 30^\circ \cdot \tan 60^\circ &= \frac{1}{\sqrt{3}} \times \sqrt{3} = 1\end{aligned}$$

$$\begin{aligned}& \dots (\text{satisfied}) \\ \cot 5\theta &= \cot(150^\circ) = \cot(90^\circ + 60^\circ) \\ &= -\tan 60^\circ = -\sqrt{3}\end{aligned}$$

$$\begin{aligned}\text{Sol.408.(c)} \quad & 4\sin^2 A - 3 = 0 \\ \text{Multiply by } \sin A, \text{ both sides} \\ 4\sin^3 A - 3\sin A &= 0 \text{ or} \\ 3\sin A - 4\sin^3 A &= 0\end{aligned}$$



Sol.409.(b) Pythagoras triplets :- (3, 4, 5)



$$\sec R = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{5}{4}$$

$$\begin{aligned} \text{Sol.410.(d)} \quad & 2(\tan^2 A \cdot \cos^2 A + \cos^2 A) \\ &= 2\cos^2 A(\tan^2 A + 1) \\ &= 2\cos^2 A(\sec^2 A) = 2 \end{aligned}$$

Sol.411.(d)

$$\tan(A + B - C) = \frac{1}{\sqrt{3}} = \tan 30^\circ$$

$$\Rightarrow (A + B - C) = 30^\circ \quad \dots \text{e.q. (1)}$$

$$\cos(B + C - A) = \frac{1}{2} = \cos 60^\circ$$

$$\Rightarrow (B + C - A) = 60^\circ \quad \dots \text{e.q. (2)}$$

$$\sin(C + A - B) = \frac{1}{\sqrt{2}} = \sin 45^\circ$$

$$\Rightarrow (C + A - B) = 45^\circ \quad \dots \text{e.q. (3)}$$

Adding e.q. (1), e.q. (2) and e.q. (3), we get

$$A + B + C = 30^\circ + 60^\circ + 45^\circ = 135^\circ$$

$$\text{Sol.412.(d)} \quad \frac{(1 + \sec \phi)}{\sec \phi} \times (1 - \cos \phi)$$

$$= (1 + \cos \phi)(1 - \cos \phi)$$

$$= (1 - \cos^2 \phi) = \sin^2 \phi$$

$$\text{Sol.413.(c)} \quad \sec C + \sin C \cdot \sec C$$

$$= \sec C + \tan C = \sec 45^\circ + \tan 45^\circ$$

$$= \sqrt{2} + 1$$

$$\text{Sol.414.(b)} \quad 4\sin \theta + 5\cos \theta = 3$$

$$4\cos \theta - 5\sin \theta = x$$

$$\text{-----}$$

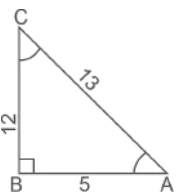
$$(4)^2 + (5)^2 = (3)^2 + (x)^2$$

$$x = \sqrt{16 + 25 - 9} = \sqrt{32} = 4\sqrt{2}$$

Sol.415.(b)

Pythagorean triplets:- (5, 12, 13)

$$\tan A = \frac{\text{Perpendicular}}{\text{Base}} = \frac{12}{5}$$



$$\csc A = \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{13}{12}$$

$$\sec A = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{13}{5}$$

$$\text{Sol.416.(a)} \quad \cos \theta = \sin \theta \Rightarrow \theta = 45^\circ$$

$$\text{Now, } \sin^8 \theta + \cos^8 \theta = \left(\frac{1}{\sqrt{2}}\right)^8 + \left(\frac{1}{\sqrt{2}}\right)^8$$

$$= \frac{1}{16} + \frac{1}{16} = \frac{1}{8}$$

$$\text{Sol.417.(d)} \quad \frac{3(\sin^4 Z - \cos^4 Z + 1)}{\sin^2 Z}$$

$$= \frac{3[(\sin^2 Z + \cos^2 Z)(\sin^2 Z - \cos^2 Z) + 1]}{\sin^2 Z} \because$$

$$(\sin^2 Z + \cos^2 Z = 1)$$

$$= \frac{3(\sin^2 Z - \cos^2 Z + 1)}{\sin^2 Z}$$

$$= \frac{3(\sin^2 Z + \sin^2 Z)}{\sin^2 Z} \because (1 - \cos^2 Z = \sin^2 Z)$$

$$= \frac{3 \times 2\sin^2 Z}{\sin^2 Z} = 6$$

Short trick : Put $Z = 90^\circ$

$$\frac{3(\sin^4 Z - \cos^4 Z + 1)}{\sin^2 Z} = \frac{3(1 - 0 + 1)}{1} = 6$$

Sol.418.(c)

$$\sin 47^\circ \cos 33^\circ - \cos 43^\circ \sin 57^\circ$$

$$= \sin(90^\circ - 43^\circ) \cos 33^\circ - \cos 43^\circ$$

$$\sin(90^\circ - 33^\circ)$$

$$= \cos 43^\circ \cos 33^\circ - \cos 43^\circ \cos 33^\circ = 0$$

$$\text{Sol.419.(a)} \quad \frac{1 + \sin A}{\cos A} + \frac{\cos A}{1 + \sin A}$$

$$= \frac{(1 + \sin A)^2 + (\cos A)^2}{(1 + \sin A)\cos A}$$

$$= \frac{1 + \sin^2 A + 2\sin A + \cos^2 A}{(1 + \sin A)\cos A}$$

$$= \frac{2 + 2\sin A}{(1 + \sin A)\cos A}$$

$$= \frac{2(1 + \sin A)}{(1 + \sin A)\cos A} = 2 \sec A$$

$$\text{Sol.420.(b)} \quad (1 + \tan^2 A)(1 + \cot^2 A)$$

$$(\sec^2 A)(\operatorname{cosec}^2 A) = \frac{1}{\cos^2 A} \times \frac{1}{\sin^2 A}$$

$$= \frac{1}{\sin^2 A(1 - \sin^2 A)}$$

Sol.421.(c) According to the question,

$$\frac{\cos B}{\sin B} - \frac{\sin B}{\cos B} \times (\sin B \cdot \cos B)$$

$$= \frac{(\cos B)^2 - (\sin B)^2}{\sin B \cdot \cos B} \times (\sin B \cdot \cos B)$$

$$= 2\cos^2 B - 1$$

$$\text{Sol.422.(d)} \quad \text{Given:- } \cot^2 \theta - \operatorname{cosec}^2 \theta = ?$$

$$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$-(\cot^2 \theta - \operatorname{cosec}^2 \theta) = 1$$

$$\cot^2 \theta - \operatorname{cosec}^2 \theta = -1$$

Sol.423.(d) According to the question,

$$(\sin^4 \theta + \cos^4 \theta) = (\sin^2 \theta + \cos^2 \theta)^2 - 2\sin^2 \theta \cdot \cos^2 \theta$$

$$\text{We know that } (\sin^2 \theta + \cos^2 \theta) = 1$$

$$1^2 - 2 \times \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$

Sol.424.(d) According to the question,

$$\sqrt{\frac{1 + \cos P \times (1 + \cos P)}{1 - \cos P \times (1 + \cos P)}} = \frac{1 + \cos P}{\sin P}$$

$$= \operatorname{cosec} P + \cot P$$

$$\text{Sol.425.(c)} \quad \tan(50 + \theta) - \cot(40 - \theta)$$

$$= \tan(50 + \theta) - \cot[90 - (50 + \theta)] \{ \cot(90 - \theta) = \tan \theta \}$$

$$= \tan(50 + \theta) - \tan(50 + \theta) = 0$$

$$\text{Sol.426.(a)} \quad (1 - \sin^2 A) \operatorname{cosec}^2 A$$

$$= (\cos^2 A) \operatorname{cosec}^2 A = \frac{\cos^2 A}{\sin^2 A} = \cot^2 A$$

Sol.427.(d)

$$\tan 15^\circ = 2 - \sqrt{3} \Rightarrow \cot 15^\circ = 2 + \sqrt{3}$$

$$\tan 15^\circ \cot 75^\circ + \tan 75^\circ \cot 15^\circ$$

$$= \tan 15^\circ \cot(90^\circ - 15^\circ) + \tan(90^\circ - 15^\circ)$$

$$\cot 15^\circ$$

$$= \tan 15^\circ \tan 15^\circ + \cot 15^\circ \cot 15^\circ$$

$$= (\tan 15^\circ)^2 + (\cot 15^\circ)^2$$

$$= (2 - \sqrt{3})^2 + (2 + \sqrt{3})^2$$

$$= 4 + 3 - 4\sqrt{3} + 4 + 3 + 4\sqrt{3} = 14$$

$$\text{Sol.428.(c)} \quad \frac{3\sin \theta - \cos \theta}{\cos \theta + \sin \theta} = 1$$

$$2\sin \theta = 2\cos \theta \Rightarrow \frac{\cos \theta}{\sin \theta} = \frac{2}{2} \Rightarrow \cot \theta = 1$$

Sol.429.(d) Formula used : $\cos(A - B)$

$$= \cos A \cdot \cos B + \sin A \cdot \sin B$$

$$\cos(36^\circ + A) \cdot \cos(36^\circ - A) + \cos(54^\circ + A) \cdot$$

$$\cos(54^\circ - A)$$

$$\Rightarrow \cos(36^\circ + A) \cdot \cos(36^\circ - A) + \cos[90 - (36^\circ - A)] \cdot \cos[90 - (36^\circ + A)]$$

$$\Rightarrow \cos(36^\circ + A) \cdot \cos(36^\circ - A) + \sin(36^\circ + A) \sin(36^\circ - A)$$

$$\Rightarrow \cos[36^\circ + A - (36^\circ - A)] \Rightarrow \cos(2A)$$

Sol.430.(d)

$$\sin 23^\circ = \frac{a}{b} = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

$$\text{Base} = \sqrt{b^2 - a^2}$$

$$\sec 23^\circ - \sin 67^\circ = \sec 23^\circ - \cos 23^\circ$$

$$= \frac{b}{\sqrt{b^2 - a^2}} - \frac{\sqrt{b^2 - a^2}}{b}$$

$$= \frac{b^2 - b^2 + a^2}{b\sqrt{b^2 - a^2}} = \frac{a^2}{b\sqrt{b^2 - a^2}}$$

$$\text{Sol.431.(a)} \quad \frac{\cos \beta}{\sec \alpha} = 15 \text{ and } \frac{\sin \beta}{\sec \alpha} = 16$$

$$\frac{\cos \beta}{\sec \alpha} \times \frac{\sec \alpha}{\sin \beta} = \frac{15}{16}$$

$$\Rightarrow \cot \beta = \frac{15}{16} = \frac{\text{Base}}{\text{Perpendicular}}$$

$$\text{Hypotenuse} = \sqrt{225 + 256} = \sqrt{481}$$

$$\sin^2 \beta = \left(\frac{\text{Perpendicular}}{\text{hypotenuse}}\right)^2$$

$$= \left(\frac{16}{\sqrt{481}}\right)^2 = \frac{256}{481}$$

$$\text{Sol.432.(c)} \quad \frac{\sin^3 \alpha + \cos^3 \alpha}{\sin \alpha + \cos \alpha}$$

$$= \frac{(\sin \alpha + \cos \alpha)(\sin^2 \alpha + \cos^2 \alpha - \sin \alpha \cos \alpha)}{\sin \alpha + \cos \alpha}$$

$$= 1 - \sin \alpha \cos \alpha$$

Sol.433.(a) According to question,

$$\tan \frac{\pi}{6} + \sec \frac{\pi}{6} = x \Rightarrow \tan 30^\circ + \sec 30^\circ = x$$



$$\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} \Rightarrow x = \sqrt{3}$$

Sol.434.(a) x lies in third quadrant In third quadrant only $\tan\theta$ and $\cot\theta$ are positive

$$\text{Now, } x = \cos^{-1}\left(-\frac{1}{2}\right) = 60^\circ$$

$$\text{Then, } \tan x = \tan(60^\circ) = \sqrt{3}$$

Sol.435.(c) According to the question,

$$\cos 75^\circ = \cos(45^\circ + 30^\circ)$$

$$= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4}$$

Sol.436.(a) According to the question,

$$\begin{aligned} & \frac{\sin^4\theta + \cos^4\theta}{1-2\sin^2\theta\cos^2\theta} \\ &= \frac{(\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta}{1-2\sin^2\theta\cos^2\theta} \\ &= \frac{1-2\sin^2\theta\cos^2\theta}{1-2\sin^2\theta\cos^2\theta} = 1 \end{aligned}$$

$$\begin{aligned} & \text{Sol.437.(d)} \left(\frac{\sin(x-y)}{\sin(x+y)}\right)\left(\frac{\tan x + \tan y}{\tan x - \tan y}\right) \\ &= \left(\frac{\sin x \cos y - \cos x \sin y}{\sin x \cos y + \cos x \sin y}\right)\left(\frac{\tan x + \tan y}{\tan x - \tan y}\right) \\ &= \left(\frac{\sin x \cos y - \cos x \sin y}{\cos x \cos y}\right)\left(\frac{\tan x + \tan y}{\tan x - \tan y}\right) \\ &= \left(\frac{\tan x - \tan y}{\tan x + \tan y}\right)\left(\frac{\tan x + \tan y}{\tan x - \tan y}\right) = 1 \end{aligned}$$

$$\text{Sol.438.(c)} \sin\left(\frac{A}{2}\right) = \sqrt{\frac{(1-\cos A)}{2}}$$

$$\text{And, } \cos\left(\frac{A}{2}\right) = \sqrt{\frac{(1+\cos A)}{2}}$$

$$\text{Given, } \tan \frac{A}{2} = x$$

$$= \sqrt{\frac{(1-\cos A)}{2}} \div \sqrt{\frac{(1+\cos A)}{2}}$$

$$= \sqrt{\frac{(1-\cos A)}{2}} \times \sqrt{\frac{2}{(1+\cos A)}}$$

$$= \sqrt{\frac{(1-\cos A)}{(1+\cos A)}}$$

$$\text{Formula :- } \tan \frac{A}{2} = \sqrt{\frac{(1-\cos A)}{(1+\cos A)}}$$

Sol.439.(d)

$$\left(\frac{1}{\sin\theta} + \frac{1}{\tan\theta}\right)\left(\frac{1}{\sin\theta} - \frac{1}{\tan\theta}\right)$$

$$= \left(\frac{1}{\sin^2\theta} - \frac{1}{\tan^2\theta}\right) \Rightarrow (\operatorname{cosec}^2\theta - \cot^2\theta) = 1$$

Sol.440.(a) $\sin 2x + 2 \sin 4x + \sin 6x$

$$2 \sin\left(\frac{6x+2x}{2}\right) \cos\left(\frac{6x-2x}{2}\right) + 2 \sin 4x$$

$$= 2 \sin 4x \cos 2x + 2 \sin 4x$$

$$= 2 \sin 4x (\cos 2x + 1)$$

$$= 2 \sin 4x \{2 \cos^2 x - 1 + 1\}$$

$$= 2 \sin 4x (2 \cos^2 x) = 4 \cos^2 x \sin 4x$$

Sol.441.(b) In third quadrant, only value of $\tan\theta$ and $\cot\theta$ is positive.

$$\text{Given that, } \sin A = -\frac{3}{5}$$

$$= \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

Using pythagoras theorem,

$$H^2 = P^2 + B^2 \Rightarrow B^2 = 25 - 9 = 16$$

$$B = \sqrt{16} = \pm 4$$

$$\text{Now, } \sec A = \frac{\text{Hypotenuse}}{\text{Base}} = -\frac{5}{4}$$

$$\begin{aligned} & \text{Sol.442.(a)} 12(\sin^4\theta + \cos^4\theta) + \\ & 18(\sin^6\theta + \cos^6\theta) + 78\sin^2\theta\cos^2\theta \\ & \text{Put } \theta = 0^\circ \\ & = 12(0+1) + 18(0+1) + 78 \times 0 \times 1 = 30 \end{aligned}$$

Sol.443.(c) Let $A = 45^\circ$ and $B = 30^\circ$

According to question,

$$\tan(45^\circ - 30^\circ) = \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ}$$

$$\tan 15^\circ = \frac{1 - \frac{1}{\sqrt{3}}}{1 + 1 \times \frac{1}{\sqrt{3}}} = \frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$$

$$= \frac{3+1-2\sqrt{3}}{3-1} = 2 - \sqrt{3}$$

$$\text{Sol.444.(b)} \frac{\sin x - \cos x}{\sin x + \cos x} = \frac{2}{5}$$

$$5 \sin x - 5 \cos x = 2 \sin x + 2 \cos x$$

$$3 \sin x = 7 \cos x$$

$$\frac{\cos x}{\sin x} = \frac{3}{7} \Rightarrow \cot x = \frac{3}{7} \Rightarrow \cot^2 x = \frac{9}{49}$$

$$\text{Now, } \frac{1 + \cot^2 x}{1 - \cot^2 x} = \frac{1 + \frac{9}{49}}{1 - \frac{9}{49}} = \frac{\frac{49+9}{49}}{\frac{49-9}{49}}$$

$$= \frac{58}{49} \times \frac{49}{40} = \frac{58}{40} = 1.45$$

Sol.445.(c) $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$

$$= \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta + \sin^2\theta +$$

$$\cos^2\theta - 2\sin\theta\cos\theta$$

$$= 2(\sin^2\theta + \cos^2\theta) = 2 \times 1 = 2$$

$$\text{Sol.446.(c)} \left(\frac{\cos A}{1 - \sin A}\right) + \left(\frac{\cos A}{1 + \sin A}\right) = 4$$

$$\cos A \left(\frac{1 + \sin A + 1 - \sin A}{(1 - \sin A)(1 + \sin A)}\right) = 4$$

$$\cos A \left(\frac{2}{(1 - \sin^2 A)}\right) = 4$$

$$\cos A \left(\frac{2}{\cos^2 A}\right) = 4$$

$$\cos A = \frac{1}{2} \Rightarrow A = 60^\circ$$

$$\text{Sol.447.(a)} \cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$$

$$\cot 15^\circ = \cot(45^\circ - 30^\circ)$$

$$= \frac{\cot 45^\circ \cot 30^\circ + 1}{\cot 30^\circ - \cot 45^\circ}$$

$$= \frac{1 \times \sqrt{3} + 1}{\sqrt{3} - 1} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{4 + 2\sqrt{3}}{3 - 1} = 2 + \sqrt{3}$$

$$\text{Sol.448.(c)} P \cos \alpha = 3 \Rightarrow \sec \alpha = \frac{P}{3}$$

taking square both sides,

$$\sec^2 \alpha = \frac{P^2}{9} \text{ -----(1)}$$

$$\text{Now, } 4 \tan \alpha = Q \Rightarrow \tan \alpha = \frac{Q}{4}$$

taking square both sides

$$\tan^2 \alpha = \frac{Q^2}{16} \text{ -----(2)}$$

from equation (1) and (2), we get

$$\sec^2 \alpha - \tan^2 \alpha = \frac{P^2}{9} - \frac{Q^2}{16}$$

$$\therefore \frac{P^2}{9} - \frac{Q^2}{16} = 1$$

Sol.449.(b) $\sin\theta + \sin^2\theta = 1$

$$\sin\theta = 1 - \sin^2\theta$$

$$\sin\theta = \cos^2\theta \Rightarrow \sin^2\theta = \cos^4\theta$$

Now,

$$\cos^2\theta + \cos^4\theta = \cos^2\theta + \sin^2\theta = 1$$

Sol.450.(a) $(a-b)^2 + (a+b)^2 = 2(a^2 + b^2)$

$$(\cos x - \sin x)^2 + (\cos x + \sin x)^2$$

$$= 2(\cos^2 x + \sin^2 x) = 2$$

$$\text{Sol.451.(a)} \frac{\sec^2(90^\circ - \theta) - \cot^2\theta}{2(\sin^2 35^\circ + \sin^2 55^\circ)}$$

$$\frac{\operatorname{cosec}^2(\theta) - \cot^2\theta}{2(\sin^2 35^\circ + \cos^2 35^\circ)} = \frac{1}{2} = 0.5$$

Sol.452.(c) $\operatorname{cosec} 2910^\circ = \operatorname{cosec} (360^\circ \times$

$$8 + 30^\circ) = \operatorname{cosec} 30^\circ = 2$$

$$\sec 4260^\circ = \sec (360^\circ \times 11 + 300^\circ)$$

$$= \sec 300^\circ = \sec(360^\circ - 60^\circ) = \sec 60^\circ = 2$$

$$\tan 2565^\circ = \tan (360^\circ \times 7 + 45^\circ)$$

$$= \tan (45^\circ) = 1$$

$$\cot 1755^\circ = \cot (360^\circ \times 4 + 315^\circ)$$

$$= \cot (315^\circ) = \cot (270^\circ + 45^\circ)$$

$$= -\tan 45^\circ = -1$$

Now,

$$\operatorname{cosec} 2910^\circ + \sec 4260^\circ + \tan 2565^\circ +$$

$$\cot 1755^\circ \Rightarrow 2 + 2 + 1 - 1 = 4$$

Sol.453.(c) For the maximum value of

$$\sin^{12}\theta + \cos^{20}\theta, \text{ Put } \theta = 90^\circ,$$

$$\sin^{12}(90^\circ) + \cos^{20}(90^\circ) = 1 + 0 = 1$$

Sol.454.(d) $\sin(-405^\circ) = -\sin(360^\circ + 45^\circ)$

$$= -\sin 45^\circ = \frac{-1}{\sqrt{2}}$$

Sol.455.(a) $\cos 2\theta = (2\cos^2\theta - 1)$

$$= (1 - 2\sin^2\theta)$$

$$\text{Now, } \frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \frac{\sin\theta(1 - 2\sin^2\theta)}{\cos\theta(2\cos^2\theta - 1)}$$

$$= \tan\theta$$

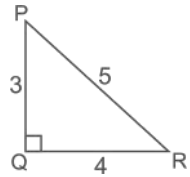
Sol.456.(b) $\frac{1 + \sin t}{4 - 4 \sin t} - \frac{1 - \sin t}{4 + 4 \sin t}$

$$= \frac{1}{4} \left[\frac{1 + \sin t}{(1 - \sin t)} - \frac{1 - \sin t}{(1 + \sin t)} \right]$$

$$= \frac{1}{4} \left[\frac{(1 + \sin t)^2 - (1 - \sin t)^2}{(1 - \sin t)(1 + \sin t)} \right]$$

$$= \frac{1}{4} \left[\frac{4 \sin t}{\cos^2 t} \right] = \frac{\sin t}{\cos t} \times \frac{1}{\cos t} = \tan t \sec t$$

Sol.457.(c)



$\sin P = \frac{4}{5}$, $\sin Q = 90^\circ = 1$ And, $\sin R = \frac{3}{5}$

Hence, $\sin P + \sin Q + \sin R$

$$= \frac{4}{5} + 1 + \frac{3}{5} = \frac{12}{5}$$

Sol.458.(c) Given, $\cot \theta = \frac{3}{2}$

$$= \frac{\text{Base}}{\text{Perpendicular}}$$

By using pythagoras theorem, we get

Hypotenuse = $\sqrt{13}$

Now, $\frac{\sqrt{13} \sin \theta - 3 \tan \theta}{3 \tan \theta + \sqrt{13} \cos \theta}$

$$= \frac{\sqrt{13} \times \frac{2}{\sqrt{13}} - 3 \times \frac{2}{3}}{3 \times \frac{2}{3} + \sqrt{13} \times \frac{3}{\sqrt{13}}} = 0$$

Sol.459.(a)

$\tan(45^\circ - 30^\circ) = \frac{\tan 45^\circ - \tan 30^\circ}{(1 + \tan 45^\circ \tan 30^\circ)}$

$$= \frac{1 - \frac{1}{\sqrt{3}}}{(1 + 1 \times \frac{1}{\sqrt{3}})} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} = 0.268$$

Short trick :-

$\tan 15^\circ = (2 - \sqrt{3}) = 2 - 1.732 = 0.268$

Sol.460.(c) If $\operatorname{Cosec} \theta + \cot \theta = m$,

Then $\operatorname{Cosec} \theta - \cot \theta = \frac{1}{m}$

Now, $\frac{m^2 - 1}{m^2 + 1} = \frac{m - \frac{1}{m}}{m + \frac{1}{m}}$

$$= \frac{\operatorname{Cosec} \theta + \cot \theta - (\operatorname{Cosec} \theta - \cot \theta)}{\operatorname{Cosec} \theta + \cot \theta + (\operatorname{Cosec} \theta - \cot \theta)}$$

$$= \frac{2 \cot \theta}{2 \operatorname{Cosec} \theta} = \cos \theta$$

Sol.461.(d) $\sin \theta + \operatorname{cosec} \theta = \sqrt{5}$

$\sin \theta + \frac{1}{\sin \theta} = \sqrt{5}$ Then, $\sin^3 \theta + \frac{1}{\sin^3 \theta}$

$$= (\sqrt{5})^3 - 3 \times \sqrt{5} = 2\sqrt{5}$$

Sol.462.(c)

Given, $\cot \theta = \frac{b}{a} = \frac{\text{Base}}{\text{Perpendicular}}$

By using pythagoras theorem,

Hypotenuse = $\sqrt{b^2 + a^2}$

Now, $\frac{b \cos \theta - a \sin \theta}{b \cos \theta + a \sin \theta}$

$$= \frac{\frac{b \times b}{\sqrt{b^2 + a^2}} - \frac{a \times a}{\sqrt{b^2 + a^2}}}{\frac{b \times b}{\sqrt{b^2 + a^2}} + \frac{a \times a}{\sqrt{b^2 + a^2}}} = \frac{b^2 - a^2}{b^2 + a^2}$$

Sol.463.(c) Let $\theta = 45^\circ$

According to question,

$$\frac{1 + \sin^4 \theta + \cos^4 \theta}{\cos^2 \theta + \sin^4 \theta} = \frac{1 + \sin^4 45^\circ + \cos^4 45^\circ}{\cos^2 45^\circ + \sin^4 45^\circ}$$

$$= \frac{1 + \left(\frac{1}{\sqrt{2}}\right)^4 + \left(\frac{1}{\sqrt{2}}\right)^4}{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^4}$$

$$= \frac{1 + \frac{1}{4} + \frac{1}{4}}{\frac{1}{2} + \frac{1}{4}} = \frac{\frac{6}{4}}{\frac{3}{4}} = 2$$

Sol.464.(c) $\operatorname{cosec}(\alpha + \beta)$

$$= \frac{\sec \alpha \times \sec \beta \times \operatorname{cosec} \alpha \times \operatorname{cosec} \beta}{\sec \alpha \times \operatorname{cosec} \beta + \operatorname{cosec} \alpha \times \sec \beta}$$

$\operatorname{cosec}(45^\circ + 30^\circ)$

$$= \frac{\sec 45^\circ \times \sec 30^\circ \times \operatorname{cosec} 45^\circ \times \operatorname{cosec} 30^\circ}{\sec 45^\circ \times \operatorname{cosec} 30^\circ + \operatorname{cosec} 45^\circ \times \sec 30^\circ}$$

$$= \frac{\sqrt{2} \times \frac{2}{\sqrt{3}} \times \sqrt{2} \times 2}{\sqrt{2} \times 2 + \sqrt{2} \times \frac{2}{\sqrt{3}}}$$

$$= \frac{\sqrt{2} \times \frac{2}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{2\sqrt{2}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3} + 1}$$

$$= \frac{2\sqrt{2}}{\sqrt{3} + 1} \times \frac{\sqrt{3} - 1}{\sqrt{3} - 1}$$

$$= \sqrt{2}(\sqrt{3} - 1) = \sqrt{6} - \sqrt{2}$$

Sol.465.(c) $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}}$

$$= \sqrt{\frac{(1 + \cos \theta) \times (1 + \cos \theta)}{(1 - \cos \theta) \times (1 + \cos \theta)}}$$

$$= \sqrt{\frac{(1 + \cos \theta)^2}{1 - \cos^2 \theta}} = \sqrt{\frac{(1 + \cos \theta)^2}{\sin^2 \theta}}$$

$$= \frac{1 + \cos \theta}{\sin \theta} = \operatorname{cosec} \theta + \cot \theta$$

Sol.466.(a) $\frac{1}{\sin \theta} - \sin \theta = \frac{1 - \sin^2 \theta}{\sin \theta}$

$$= \frac{\cos^2 \theta}{\sin \theta} = \cos \theta \cdot \cot \theta$$

Sol.467.(c) $\frac{21 \cos A + 3 \sin A}{3 \cos A + 4 \sin A} = 2$

$$= 21 \cos A + 3 \sin A = 6 \cos A + 8 \sin A$$

$$= 15 \cos A = 5 \sin A$$

$$\Rightarrow \cot A = \frac{5}{15} = \frac{1}{3}$$

Sol.468.(c) $\tan 4384^\circ + \cot 6814^\circ$

$$= \tan(360^\circ \times 12 + 64^\circ) + \cot(360^\circ \times 18 + 334^\circ) = \tan 64^\circ + \cot(270^\circ + 64^\circ)$$

$$= \tan 64^\circ - \tan 64^\circ = 0$$

Sol.469.(a)

$$10\sqrt{2} \sin 45^\circ - 7\sqrt{2} \cos 45^\circ + 9 \tan 45^\circ$$

$$10 - 7 + 9 = 12$$

Sol.470.(d)

$$\sin 600^\circ \cos 750^\circ + \sin 150^\circ \cos 240^\circ$$

$$= \sin(180^\circ \times 3 + 60^\circ) \cos(180^\circ \times 4 + 30^\circ) + \sin(180^\circ - 30^\circ) \cos(180^\circ + 60^\circ)$$

$$= -\sin 60^\circ \cos 30^\circ - \sin 30^\circ \cos 60^\circ$$

$$= -\sin(60^\circ + 30^\circ) = -1$$

Sol.471.(d) $\sin 2\theta = 2 \sin \theta \cos \theta$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\frac{\sin \theta}{1 + \cos \theta} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = \tan\left(\frac{\theta}{2}\right)$$

$$\Rightarrow \tan\left(\frac{\theta}{2}\right) = \frac{\sin \theta}{1 + \cos \theta}$$

Short trick : put $\theta = 30^\circ$

Sol.472.(c)

$$3x = \sec A \Rightarrow 9x^2 = \sec^2 A \dots\dots\dots (1)$$

$$\frac{3}{x} = \tan A \Rightarrow \frac{9}{x^2} = \tan^2 A \dots\dots\dots (2)$$

Subtracting eqn.(2) from (1), we get

$$9x^2 - \frac{9}{x^2} = 1 \Rightarrow 9\left(x^2 - \frac{1}{x^2}\right) = 1$$

Sol.473.(d) $\tan^4 \theta + \tan^2 \theta = 1$

$$\tan^2 \theta (\tan^2 \theta + 1) = 1 \Rightarrow \tan^2 \theta \times \sec^2 \theta = 1$$

$$\frac{\sin^2 \theta}{\cos^2 \theta} \times \frac{1}{\cos^2 \theta} = 1 \Rightarrow \sin^2 \theta = \cos^4 \theta$$

Now, $11(\cos^4 \theta + \cos^2 \theta)$

$$= 11(\sin^2 \theta + \cos^2 \theta) = 11$$

Sol.474.(d) $\frac{32}{\sec^2 \theta} - \frac{20}{1 + \tan^2 \theta} + 12 \sin^2 \theta$

Put, $\theta = 0^\circ$ Then

$$\frac{32}{\sec^2 0^\circ} - \frac{20}{1 + \tan^2 0^\circ} + 12 \sin^2 0^\circ$$

$$= 32 - 20 + 0 = 12$$

Sol.475.(b) $\sin \theta + \cos \theta = \frac{\sqrt{11}}{3}$

$$\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = \frac{11}{9}$$

$$2 \sin \theta \cos \theta = \frac{11}{9} - 1 = \frac{2}{9}$$

$$\sin \theta \cos \theta = \frac{1}{9}$$

Now, $\tan \theta + \cot \theta = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = 1 \times \frac{9}{1} = 9$$

Sol.476.(c) $1 + \sin^2 \theta = a \cos^2 \theta$

$$\frac{1}{\cos^2 \theta} + \frac{\sin^2 \theta}{\cos^2 \theta} = a$$

$$\sec^2 \theta + \tan^2 \theta = a$$

and $\sec^2 \theta - \tan^2 \theta = 1$ (identity)

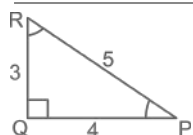
Now, $\sec^4 \theta - \tan^4 \theta$

$$= (\sec^2 \theta + \tan^2 \theta)(\sec^2 \theta - \tan^2 \theta)$$

$$\sec^4 \theta - \tan^4 \theta = a \times 1 = a$$

Sol.477.(b)





Pythagoras triplets : - (3, 4, 5)

$$\sin P = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{3}{5}$$

$$\text{base} = 4 \text{ So, } \angle Q = 90^\circ = \frac{\pi}{2}$$

Sol. 478.(b) $\operatorname{cosec} 41^\circ - \sec 49^\circ$

$$= \operatorname{cosec} 41^\circ - \sec (90^\circ - 41^\circ)$$

$$= \operatorname{cosec} 41^\circ - \operatorname{cosec} 41^\circ = 0$$

Sol. 479.(d) $\sin x + \sin^2 x = 1$

$$\sin x = \cos^2 x \Rightarrow \cos^4 x = \sin^2 x. \text{ Now,}$$

$$\cos^8 x + 2\cos^6 x + \cos^4 x = (\cos^4 x + \cos^2 x)^2$$

$$= (\sin^2 x + \cos^2 x)^2 = 1$$

Sol. 480.(d) $\left(\frac{1 + \sec^2 A}{1 + \cos^2 A}\right) \left(\frac{1 + \sin^2 A}{1 + \operatorname{cosec}^2 A}\right)$

$$= \left(\frac{1 + \frac{1}{\cos^2 A}}{1 + \cos^2 A}\right) \left(\frac{1 + \frac{1}{\operatorname{cosec}^2 A}}{1 + \operatorname{cosec}^2 A}\right)$$

$$\left(\frac{1}{\cos^2 A}\right) \left(\frac{1}{\operatorname{cosec}^2 A}\right) = \tan^2 A$$

Sol. 481.(d) $\frac{\sqrt{1 - \sin A}}{\sqrt{1 + \sin A}}$

$$= \frac{\sqrt{1 - \sin A}}{\sqrt{1 + \sin A}} \times \frac{\sqrt{1 - \sin A}}{\sqrt{1 - \sin A}}$$

$$= \frac{(1 - \sin A)}{\sqrt{(1 + \sin A)} \sqrt{(1 - \sin A)}}$$

$$= \frac{(1 - \sin A)}{\sqrt{1 - \sin^2 A}} = \frac{(1 - \sin A)}{\cos A}$$

$$= \sec A - \tan A$$

Sol. 482.(d)

$$(1 + \tan \theta - \sec \theta)(1 + \cot \theta + \operatorname{cosec} \theta)$$

$$= \left(1 + \frac{\sin \theta}{\cos \theta} - \frac{1}{\cos \theta}\right) \left(1 + \frac{\cos \theta}{\sin \theta} + \frac{1}{\sin \theta}\right)$$

$$= \left(1 + \frac{\sin \theta - 1}{\cos \theta}\right) \left(1 + \frac{\cos \theta + 1}{\sin \theta}\right)$$

$$= \left(\frac{\cos \theta + \sin \theta - 1}{\cos \theta}\right) \left(\frac{\cos \theta + \sin \theta + 1}{\sin \theta}\right)$$

$$= \frac{(\cos \theta + \sin \theta)^2 - 1^2}{\sin \theta \cos \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta - 1}{\sin \theta \cos \theta}$$

$$= \frac{2\sin \theta \cos \theta}{\sin \theta \cos \theta} = 2$$

Sol. 483.(d) $\frac{\cos(50^\circ + A) - \sin(40^\circ - A)}{\cos 40^\circ \cdot \sec 40^\circ}$

$$= \frac{\cos(50^\circ + A) - \sin(40^\circ - A)}{\cos 40^\circ \times \frac{1}{\cos 40^\circ}}$$

Let $A = 40^\circ$ then,

$$\frac{\cos(50^\circ + 40^\circ) - \sin(40^\circ - 40^\circ)}{1}$$

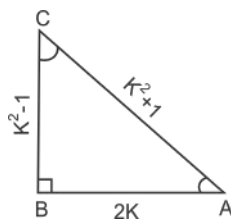
$$= \frac{\cos(90^\circ)}{1} = 0$$

Sol. 484.(d) $\sec A + \tan A = K \dots \text{E.q. (1)}$

$$\sec A - \tan A = \frac{1}{K} \dots \text{E.q. (2)}$$

On adding e.q. (1) and e.q. (2), we get

$$2 \sec A = \frac{K^2 + 1}{K} \Rightarrow \cos A = \frac{2K}{K^2 + 1}$$



$$\text{Now, } \sin A = \frac{K^2 - 1}{K^2 + 1}$$

Sol. 485.(b) $\frac{\operatorname{cosec} 90^\circ}{(1 - \cos^2 60^\circ)}$

$$= \frac{1}{(1 - \frac{1}{4})} = \frac{1}{(\frac{3}{4})} = \frac{4}{3}$$

Sol. 486.(d)

$$\sqrt{\frac{\operatorname{cosec} A + 1}{\operatorname{cosec} A - 1}} + \sqrt{\frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1}}$$

$$= \frac{\sqrt{\operatorname{cosec} A + 1}}{\sqrt{\operatorname{cosec} A - 1}} + \frac{\sqrt{\operatorname{cosec} A - 1}}{\sqrt{\operatorname{cosec} A + 1}}$$

$$= \frac{(\operatorname{cosec} A + 1) + (\operatorname{cosec} A - 1)}{\sqrt{\operatorname{cosec}^2 A - 1}}$$

$$= \frac{(2 \operatorname{cosec} A)}{\sqrt{\cot^2 A}} = \frac{(2 \operatorname{cosec} A)}{\cot A}$$

$$= \frac{2(\frac{1}{\sin A})}{\frac{\cos A}{\sin A}} = 2 \sec A$$

Sol. 487.(d) Putting $\theta = 45^\circ$

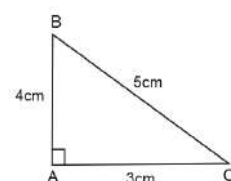
$$m = \sec \theta - \tan \theta = \sqrt{2} - 1$$

$$n = \operatorname{cosec} \theta + \cot \theta = \sqrt{2} + 1$$

Putting the value of m and n in $m + n(m-1)$

$$\sqrt{2} - 1 + (\sqrt{2} + 1)(\sqrt{2} - 1 - 1) = -1$$

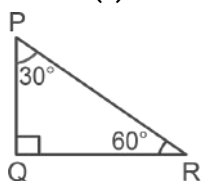
Sol. 488.(d)



$$\cos B = \frac{4}{5} \text{ and } \cot C = \frac{3}{4}$$

$$\cos B + \cot C \Rightarrow \frac{4}{5} + \frac{3}{4} = \frac{31}{20}$$

Sol. 489.(d)



$$\cos(30^\circ + 60^\circ) = \cos 90^\circ = 0$$

Sol. 490.(b) $\sqrt{3} \tan^2 \theta - 4 \tan \theta + \sqrt{3} = 0$

$$= \sqrt{3} \tan^2 \theta - \tan \theta - 3 \tan \theta + \sqrt{3} = 0$$

$$= \tan \theta (\sqrt{3} \tan \theta - 1) - \sqrt{3} (\sqrt{3} \tan \theta - 1) = 0$$

$$= (\sqrt{3} \tan \theta - 1)(\tan \theta - \sqrt{3})$$

$$\Rightarrow \tan \theta = \left(\frac{1}{\sqrt{3}} \text{ and } \sqrt{3}\right) \text{ [here both value exists]} \Rightarrow \theta = 30^\circ \text{ and } 60^\circ$$

Then, by putting $\theta = 30^\circ$

$$\tan^2 \theta + \cot^2 \theta \Rightarrow \frac{1}{3} + 3 = \frac{10}{3}$$

Sol. 491.(c) $6 \cot \theta = 5 \Rightarrow \cot \theta = \frac{5}{6}$

$$\text{Now, } \frac{(6 \cos \theta + \sin \theta)}{(6 \cos \theta - 4 \sin \theta)} = \frac{(6 \cot \theta + 1)}{(6 \cot \theta - 4)}$$

$$= \frac{(6 \times \frac{5}{6} + 1)}{(6 \times \frac{5}{6} - 4)} = 6$$

Sol. 492.(b) $\operatorname{cosec} \theta - \sin \theta = 6$ Then, formula:- $(x + y)^2 = (x - y)^2 + 4xy$

$$\operatorname{cosec} \theta + \sin \theta = \sqrt{(6^2) + 4} = \sqrt{40}$$

Sol. 493.(d) $\sqrt{2} \sec^2 \theta - 4 \sec \theta + 2\sqrt{2} = 0$

$$\sqrt{2} \sec^2 \theta - 2 \sec \theta - 2 \sec \theta + 2\sqrt{2} = 0$$

$$\sqrt{2} \sec \theta (\sec \theta - \sqrt{2}) - 2(\sec \theta - \sqrt{2}) = 0$$

$$\sec \theta = \sqrt{2}, \sec \theta = \frac{2}{\sqrt{2}} = \frac{\sqrt{2}}{1} = \frac{H}{B}$$

So, $P = 1$

$$\sin^2 \theta + \tan^2 \theta = \frac{1}{2} + 1 = \frac{3}{2}$$

Short trick :-

$$\sqrt{2} \sec^2 \theta - 4 \sec \theta + 2\sqrt{2} = 0$$

Put $\theta = 45^\circ$

$$\text{So value of } \sin^2 \theta + \tan^2 \theta = \frac{1}{2} + 1 = \frac{3}{2}$$

Sol. 494.(c) $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} + \frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta}$

$$= \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} \times \frac{\sin \theta + \cos \theta}{\sin \theta + \cos \theta} +$$

$$\frac{\sin \theta - \cos \theta}{\sin \theta + \cos \theta} \times \frac{\sin \theta - \cos \theta}{\sin \theta - \cos \theta}$$

$$= \frac{(\sin \theta + \cos \theta)^2}{\sin^2 \theta - \cos^2 \theta} + \frac{(\sin \theta - \cos \theta)^2}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta + \sin^2 \theta + \cos^2 \theta - 2\sin \theta \cos \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{2}{(\sin^2 \theta - \cos^2 \theta)}$$

Sol. 495.(c) $\frac{1 - \sin A}{\cos A} + \frac{\cos A}{1 - \sin A}$

$$= \frac{(1 - \sin A)^2 + \cos^2 A}{(\cos A)(1 - \sin A)}$$

$$= \frac{(1)^2 + \sin^2 A - 2\sin A + \cos^2 A}{(\cos A)(1 - \sin A)}$$

$$= \frac{2(1 - \sin A)}{(\cos A)(1 - \sin A)} = 2 \sec A$$

Sol. 496.(b) $\sec x + \tan x = 5$

$$\text{Then, } \sec x - \tan x = \frac{1}{5}$$

$$\operatorname{cosec} y - \cot y = \frac{1}{3}$$

then, $\operatorname{cosec} y + \cot y = 3$



Now,

$$\begin{aligned} & (\sec x + \operatorname{cosec} y) - (\tan x - \cot y) \\ &= (\sec x - \tan x + \operatorname{cosec} y + \cot y) \\ &= \frac{1}{5} + 3 = \frac{16}{5} = 3.2 \end{aligned}$$

Sol.497.(c) Put the value of $\theta = 45^\circ$,

$$A = a\sqrt{2} + b \text{ and } B = a + b\sqrt{2}$$

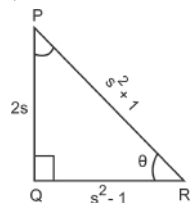
$$\text{Then, } A^2 = 2a^2 + b^2 + 2\sqrt{2}ab$$

$$B^2 = a^2 + 2b^2 + 2\sqrt{2}ab$$

$$A^2 - B^2 = 2a^2 + b^2 + 2\sqrt{2}ab - a^2 - 2b^2 - 2\sqrt{2}ab \Rightarrow A^2 - B^2 = (a^2 - b^2)$$

Sol.498.(a) Pythagoras triplets :-

$$(2s, s^2 + 1, s^2 - 1)$$



$$\operatorname{cosec} \theta + \cot \theta = s$$

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{s}$$

$$\operatorname{Cosec} \theta = \frac{s^2 + 1}{2s} \Rightarrow \sin \theta = \frac{2s}{s^2 + 1}$$

$$\text{So that, } \cos \theta = \frac{s^2 - 1}{s^2 + 1}$$

Sol.499.(c) $x = 30^\circ$ (put)

$$\sec x + \tan x = \sqrt{3}$$

$$\Rightarrow \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} = \sqrt{3} \text{ (satisfied)}$$

According to the question,

$$\tan x = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

Sol.500.(b)

Complementary angle of $\tan = \cot$

Sol.501.(c) $\sec \theta + \tan \theta = 2$ ----- (eq. 1)

$$\text{As we know } \sec^2 \theta - \tan^2 \theta = 1$$

$$\text{Then, } (\sec \theta + \tan \theta) (\sec \theta - \tan \theta) = 1$$

$$\Rightarrow (\sec \theta - \tan \theta) = \frac{1}{2} \text{ ----- (eq. 2)}$$

By (eq. 1 and 2)

$$2 \sec \theta = 2 + \frac{1}{2}$$

$$\Rightarrow \sec \theta = \frac{\text{hypotenuse}}{\text{base}} = \frac{5}{4}$$

$$\text{then, perpendicular} = \sqrt{(5)^2 - (4)^2} = 3$$

$$\text{Therefore, } \cot \theta = \frac{\text{base}}{\text{perpendicular}} = \frac{4}{3}$$

Sol.502.(a) $\tan 10^\circ \tan 80^\circ + \tan 20^\circ \tan 70^\circ + \tan 30^\circ \tan 60^\circ + \tan 40^\circ \tan 50^\circ$
 $= \tan 10^\circ \tan (90^\circ - 10^\circ) + \tan 20^\circ \tan (90^\circ - 20^\circ) + \tan 30^\circ \tan (90^\circ - 30^\circ) + \tan 40^\circ \tan (90^\circ - 40^\circ)$
 $= \tan 10^\circ \cot 10^\circ + \tan 20^\circ \cot 20^\circ + \tan 30^\circ \cot 30^\circ + \tan 40^\circ \cot 40^\circ$
 $= 1 + 1 + 1 + 1 = 4$

Sol.503.(d) Let $\theta = 45^\circ$

$$\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1} = \frac{1 + \sqrt{2} - 1}{1 - \sqrt{2} + 1} = \frac{\sqrt{2}}{2 - \sqrt{2}}$$

$$\frac{\sqrt{2}}{\sqrt{2}(\sqrt{2} - 1)} \text{ (multiply by } (\sqrt{2} + 1) \text{ in numerator and denominator)}$$

$$\frac{(\sqrt{2} + 1)}{1} = (\sqrt{2} + 1)$$

According to the question,

In option d, put $\theta = 45^\circ$

$$(\cot \theta + \operatorname{cosec} \theta) \Rightarrow (1 + \sqrt{2}) \text{ (satisfied)}$$

Sol.504.(c) $6 \sec \theta = 10$

$$\sec \theta = \frac{10}{6} = \frac{5}{3} = \frac{\text{Hypotenuse}}{\text{Base}}$$

Using Pythagoras theorem;

$$\text{Perpendicular} = \sqrt{5^2 - 3^2} = 4,$$

$$\operatorname{cosec} \theta = \frac{5}{4} \text{ and } \cot \theta = \frac{3}{4}$$

$$\frac{(5 \operatorname{cosec} \theta - 3 \cot \theta)}{(4 \cos \theta + 3 \sin \theta)}$$

$$= \frac{(5 \times \frac{5}{4} - 3 \times \frac{3}{4})}{(4 \times \frac{3}{5} + 3 \times \frac{4}{5})} = \frac{(\frac{25}{4} - \frac{9}{4})}{(\frac{12}{5} + \frac{12}{5})}$$

$$= \frac{(\frac{25-9}{4})}{(\frac{12+12}{5})} = \frac{20}{24} = \frac{5}{6}$$

Sol.505.(b)

$$\operatorname{cosec} A (1 - \cos A) (\operatorname{cosec} A + \cot A)$$

$$\Rightarrow \frac{1}{\sin A} (1 - \cos A) \left(\frac{1 + \cos A}{\sin A} \right)$$

$$\Rightarrow \frac{(1 - \cos^2 A)}{\sin^2 A} \Rightarrow \frac{(1 - \cos^2 A)}{(1 - \cos^2 A)} = 1$$

$$\text{Sol.506.(c)} \frac{(2 \cos^3 A - \cos A)}{(\sin A - 2 \sin^3 A)} = 1$$

$$\Rightarrow \frac{\cos A (2 \cos^2 A - 1)}{\sin A (1 - 2 \sin^2 A)} = 1 \Rightarrow \frac{\cos A \cdot \cos 2A}{\sin A \cdot \cos 2A}$$

$$= 1 \Rightarrow \cot A = 1 \Rightarrow A = 45^\circ$$

Sol.507.(c)

$$(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2$$

Putting the value of $\theta = 45^\circ$

$$\left(\frac{1}{\sqrt{2}} + \sqrt{2} \right)^2 + \left(\frac{1}{\sqrt{2}} + \sqrt{2} \right)^2 = 9$$

By putting value of $\theta = 45^\circ$ in the option (c)

$$\text{we get, } 7 + \tan^2 \theta + \cot^2 \theta = 7 + 1 + 1 = 9.$$

Sol.508.(b)

$$\tan^2 \theta + \cot^2 \theta - \sec^2 \theta \cdot \operatorname{cosec}^2 \theta$$

Put $\theta = 45^\circ$

$$\tan^2 45^\circ + \cot^2 45^\circ - \sec^2 45^\circ \cdot \operatorname{cosec}^2 45^\circ \Rightarrow 1 + 1 - 2 \times 2 \Rightarrow -2$$

Sol.509.(a) Formula :-

$$\sin a - \sin b = 2 \sin \left(\frac{a-b}{2} \right) \cos \left(\frac{a+b}{2} \right)$$

$$\sin \alpha - \sin \beta = 2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha - \beta}{2}$$

Sol.510.(c) In third quadrant value of $\tan$

θ and $\cot \theta$ is +ve.

$$\begin{aligned} \tan 240^\circ &= \tan (180^\circ + 60^\circ) \\ &= \tan 60^\circ = \sqrt{3} \end{aligned}$$

Sol.511.(c) $\tan^2 \theta = 1 - a^2$

Adding 1 both side,

$$1 + \tan^2 \theta = 1 - a^2 + 1$$

$$\sec^2 \theta = 2 - a^2$$

$$\sec \theta = \sqrt{(2 - a^2)} \text{ (eq. 1)}$$

Now, $\sec \theta + \tan^3 \theta \times \operatorname{Cosec} \theta$

$$= \frac{1}{\cos \theta} + \frac{\sin^3 \theta}{\cos^3 \theta} \times \frac{1}{\sin \theta}$$

$$= \frac{\cos^2 \theta + \sin^2 \theta}{\cos^3 \theta} = \sec^3 \theta \text{ (eq. 2)}$$

Putting the value of (eq. 1) in (eq. 2),

$$\text{we get, } \sec^3 \theta = (2 - a^2)^{\frac{3}{2}}$$

Sol.512.(a) Formula :-

$$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

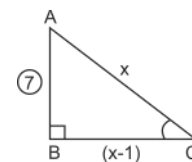
$$\sin(A + B) = \cos(A - B) \text{ (given)}$$

Clearly value of $\sin \theta$ and $\cos \theta$ are equal at 45° , $(A + B) = 45^\circ \Rightarrow A = 45^\circ - B$

Now, $\tan A = \tan(45^\circ - B)$

$$= \frac{\tan 45^\circ - \tan B}{1 + \tan 45^\circ \tan B} = \frac{1 - \tan B}{1 + \tan B}$$

Sol.513.(d) Let, the length of side AC = x cm, then BC = (x - 1) cm



Now, using pythagoras theorem :-

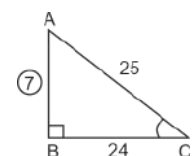
$$AC^2 = AB^2 + BC^2 \Rightarrow x^2 = 7^2 + (x - 1)^2$$

$$x^2 = 49 + x^2 + 1 - 2x$$

$$2x = 50 \Rightarrow x = 25 \text{ cm}$$

$$\text{Now, } \sin C = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{7}{25}$$

Short Trick:-



Using pythagoras triplets : (7, 24, 25)

AB = 7 cm

And AC - BC = 25 - 24 = 1 cm (Satisfy the given condition)

$$\text{Then, } \sin C = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{7}{25}$$

Sol.514.(a) Formula :-

$$\frac{(1 - \cos \theta)}{\sin \theta} = \frac{\sin \theta}{(1 + \cos \theta)}$$

$$\frac{(1 - \cos \theta)}{\sin \theta} = \frac{1}{5} \text{ (given)}$$

$$\text{Then, } \frac{(1 + \cos \theta)}{\sin \theta} = 5$$

$$\text{Sol.515.(b)} \frac{\cos A}{1 + \tan A} - \frac{\sin A}{1 + \cot A}$$

$$\begin{aligned}
 &= \frac{\cos A}{1 + \frac{\sin A}{\cos A}} - \frac{\sin A}{1 + \frac{\cos A}{\sin A}} \\
 &= \frac{\cos^2 A}{\cos A + \sin A} - \frac{\sin^2 A}{\sin A + \cos A} \\
 &= \frac{\cos^2 A - \sin^2 A}{\cos A + \sin A} \\
 &= \frac{(\cos A + \sin A)(\cos A - \sin A)}{(\cos A + \sin A)} \\
 &= (\cos A - \sin A)
 \end{aligned}$$

Sol.516.(a)

$$\begin{aligned}
 &\frac{1}{1 + \cos(90^\circ - \theta)} + \frac{1}{1 - \cos(90^\circ - \theta)} \\
 &= \frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta} \\
 &= \frac{(1 - \sin \theta) + (1 + \sin \theta)}{1 - \sin^2 \theta} \\
 &= \frac{2}{\cos^2 \theta} \Rightarrow 2 \sec^2 \theta
 \end{aligned}$$

Sol.517.(d) We know that ,

$$\cos^2 \theta = (1 - \sin^2 \theta)$$

$$\sin^2 \theta + \cos^2 \theta = 1 \text{eq. (1)}$$

On squaring eq. (1), we get

$$(\sin^2 \theta + \cos^2 \theta)^2 = (1)^2$$

$$\sin^4 \theta + \cos^4 \theta + 2\sin^2 \theta \cos^2 \theta = 1$$

$$\sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta \cos^2 \theta$$

$$\sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta (1 - \sin^2 \theta)$$

$$\sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta + 2\sin^4 \theta$$

$$\text{Sol.518.(c)} \frac{\cos x - \sqrt{3} \sin x}{2}$$

$$\begin{aligned}
 &= \frac{1}{2} \times \cos x - \frac{\sqrt{3}}{2} \times \sin x \\
 &= \sin 30^\circ \cos x - \cos 30^\circ \sin x \\
 &= \sin(30^\circ - x) = \sin[90^\circ - (60^\circ + x)] \\
 &= \cos(60^\circ + x) = \cos\left(\frac{\pi}{3} + x\right)
 \end{aligned}$$

Sol.519.(b)

$$\begin{aligned}
 \sqrt{\frac{1 + \cos A}{1 - \cos A}} &= \sqrt{\frac{1 + \cos A}{1 - \cos A} \times \frac{1 + \cos A}{1 + \cos A}} \\
 &= \sqrt{\frac{(1 + \cos A)^2}{1 - \cos^2 A}} = \frac{1 + \cos A}{\sin A} \\
 &= \operatorname{cosec} A + \cot A
 \end{aligned}$$

Sol.520.(a)

$$\sin \theta + \cos \theta = \frac{\sqrt{3} - 1}{2\sqrt{2}} \text{eq. (1)}$$

Squaring eq. (1), we get

$$\sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta$$

$$= \left(\frac{\sqrt{3} - 1}{2\sqrt{2}}\right)^2 = \frac{4 - 2\sqrt{3}}{8}$$

$$2\sin \theta \cos \theta = \frac{4 - 2\sqrt{3}}{8} - 1$$

$$= \frac{(4 - 2\sqrt{3}) - 8}{8} = \frac{-(2 + \sqrt{3})}{4}$$

$$\text{Now, } \tan \theta + \cot \theta = \frac{2}{2\sin \theta \cos \theta}$$

$$= \frac{-8}{(2 + \sqrt{3})} \times \frac{(2 - \sqrt{3})}{(2 - \sqrt{3})} = 8(\sqrt{3} - 2)$$

Sol.521.(d)If $A + B = 90^\circ$, then $\sec A = \operatorname{cosec} B$

$$\sec 2\theta = \operatorname{cosec}(\theta - 36^\circ)$$

$$\text{Now, } 2\theta + (\theta - 36^\circ) = 90^\circ \Rightarrow \theta = 42^\circ$$

Sol.522.(d) Formula :-

$$\tan 2x = \frac{2 \tan x}{(1 - \tan^2 x)}$$

$$\cos x \cdot \operatorname{cosec} x - \sin x \cdot \sec x$$

$$= \cot x - \tan x = \frac{1}{\tan x} - \frac{\tan x}{\tan x}$$

$$= \frac{2 \times (1 - \tan^2 x)}{2 \times \tan x} = \frac{2 \times 1}{\tan 2x} = 2 \cot 2x$$

Sol.523.(b) $\cot^2 A - \cos^2 A$

$$= \frac{\cos^2 A}{\sin^2 A} - \cos^2 A = \cos^2 A (\operatorname{cosec}^2 A - 1)$$

$$= \cos^2 A \cdot \cot^2 A$$

Sol.524.(a) $b \cdot \sin \theta = a$

$$\Rightarrow \sin \theta = \frac{a}{b} = \frac{\text{Perpendicular}}{\text{hypotenuse}}$$

by, using pythagoras theorem, we get

$$\text{Base} = \sqrt{b^2 - a^2} \text{ . Now,}$$

$$\sec \theta + \tan \theta = \frac{b}{\sqrt{b^2 - a^2}} + \frac{a}{\sqrt{b^2 - a^2}}$$

$$= \frac{b + a}{\sqrt{b^2 - a^2}} = \frac{\sqrt{b+a} \times \sqrt{b+a}}{\sqrt{b+a} \times \sqrt{b-a}} = \sqrt{\frac{b+a}{b-a}}$$

Sol.525.(b) Formula :-

$$\sin 3\theta = 3\sin \theta - 4\sin^3 \theta$$

$$\sin 10^\circ - \frac{4}{3} \sin^3 10^\circ = \frac{3\sin 10^\circ - 4\sin^3 10^\circ}{3}$$

$$= \frac{\sin 30^\circ}{3} = \frac{1}{6}$$

Sol.526.(c) (i) $\cos(-\theta) = \cos \theta$

$$(ii) \cos(2n\pi - \theta) = \cos \theta$$

$$\text{Now, } \cos\left(\frac{-17\pi}{3}\right) = \cos\left(\frac{17\pi}{3}\right)$$

$$= \cos\left(6\pi - \frac{\pi}{3}\right) = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

Sol.527.(a) Formula :-

$$a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$$

$$(\sin A + \cos A)(1 - \sin A \cos A)$$

$$= (\sin A + \cos A)$$

$$(\sin^2 A + \cos^2 A - \sin A \cos A)$$

$$= \sin^3 A + \cos^3 A$$

$$\text{Sol.528.(d)} \sec \theta - 2\cos \theta = \frac{7}{2}$$

$$= \sec \theta - \frac{2}{\sec \theta} = \frac{7}{2}$$

$$= \sec^2 \theta - 2 = \frac{7}{2} \sec \theta$$

$$= 2\sec^2 \theta - 7\sec \theta - 4 = 0$$

$$= 2\sec^2 \theta - 8\sec \theta + \sec \theta - 4 = 0$$

$$= (\sec \theta - 4)(2\sec \theta + 1) = 0$$

$$= \sec \theta = 4$$

Sol.529.(d) $\sec 3\theta = \operatorname{cosec}(4\theta - 15^\circ)$

$$\Rightarrow \operatorname{cosec}(90^\circ - 3\theta) = \operatorname{cosec}(4\theta - 15^\circ)$$

$$\Rightarrow 90^\circ - 3\theta = 4\theta - 15^\circ$$

$$\Rightarrow 7\theta = 90^\circ + 15^\circ$$

$$\Rightarrow 7\theta = 105^\circ \Rightarrow \theta = 15^\circ$$

$$\text{Then, } \tan 3\theta = \tan 45^\circ = 1$$

Sol.530.(a) In a right angled triangle ,
right angled at A ,

$$= \angle A = \angle B + \angle C = 90^\circ$$

$$= \angle A - \angle B - \angle C = 0^\circ$$

$$\tan(A - B - C) = \tan 0^\circ = 0$$

$$\tan(A - B - C)$$

$$= \frac{\tan A - \tan B - \tan C - \tan A \tan B \tan C}{1 + \tan A \tan B - \tan B \tan C + \tan C \tan A}$$

$$\tan A - \tan B - \tan C - \tan A \tan B \tan C = 0$$

$$\tan A - \tan B - \tan C = \tan A \tan B \tan C$$

Sol.531.(d)

$$1 + \tan^2 \alpha = 4 + Q^2 \text{ (adding 1 both sides)}$$

$$\sec^2 \alpha = 4 + Q^2$$

$$\cos \alpha = (4 + Q^2)^{-\frac{1}{2}}$$

then , $\sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha$

$$= \frac{1}{\cos \alpha} + \frac{\sin^3 \alpha}{\cos^3 \alpha \times \sin \alpha}$$

$$= \frac{1}{\cos \alpha} + \frac{\sin^2 \alpha}{\cos^2 \alpha \times \cos \alpha}$$

$$= \frac{1 + \tan^2 \alpha}{\cos \alpha} = \frac{(4 + Q^2)}{(4 + Q^2)^{-1/2}} = (4 + Q^2)^{\frac{3}{2}}$$

Short Trick:- Put , $= 60^\circ$, then ,

$$\tan^2 60^\circ = 3 + Q^2 \Rightarrow Q = 0$$

$$\text{Now, } \sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha = 2 + 3\sqrt{3} \times$$

$$\frac{2}{\sqrt{3}} = 8 \text{ Clearly , option (d) satisfy the}$$

given equation

$$\Rightarrow (4 + Q^2)^{\frac{3}{2}} = (4 + 0^2)^{\frac{3}{2}}$$

$$= (2^2)^{\frac{3}{2}} = 2^3 = 8$$

Sol.532.(b) $1 + \sin \theta = m \cos \theta$

$$\frac{1 + \sin \theta}{\cos \theta} = m \Rightarrow \frac{(1 + \sin \theta)^2}{\cos^2 \theta} = m^2$$

$$\frac{(1 + \sin \theta)^2}{(1 + \sin \theta)(1 - \sin \theta)} = m^2$$

$$\frac{(1 + \sin \theta)}{(1 - \sin \theta)} = m^2$$

By using componendo and dividendo ,

$$\text{we get } \frac{(1 + \sin \theta + 1 - \sin \theta)}{(1 + \sin \theta - 1 + \sin \theta)} = \frac{m^2 + 1}{m^2 - 1}$$

$$\frac{(2)}{(2\sin \theta)} = \frac{m^2 + 1}{m^2 - 1} \Rightarrow \sin \theta = \frac{m^2 - 1}{m^2 + 1}$$

Sol.533.(a) Put $\theta = 90^\circ$

$$5(\sin^4 \theta + \cos^4 \theta) + 3(\sin^6 \theta + \cos^6 \theta) +$$

$$19\sin^2 \theta \cos^2 \theta$$

$$= 5 \times 1 + 3 \times 1 + 19 \times 0 = 8$$

Sol.534.(d) Formula: $\cos 2A = 1 - 2\sin^2 A$

$$\sin Y = x \text{ then, } \cos 2Y = 1 - 2x^2$$

Sol.535.(d) Formula:- $\tan(A + B + C)$

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

In a right angled triangle, right angled at C,

$$= \angle C = \angle A + \angle B = 90^\circ$$

$$= \angle A + \angle B - \angle C = 0^\circ$$

$$\tan(A + B - C) = \tan 0^\circ = 0$$

$$\Rightarrow \tan(A + B - C)$$

$$= \frac{\tan A + \tan B - \tan C + \tan A \tan B \tan C}{1 - \tan A \tan B + \tan B \tan C + \tan C \tan A}$$

$$\Rightarrow \tan A + \tan B - \tan C + \tan A \tan B \tan C = 0$$

$$\Rightarrow \tan A + \tan B - \tan C = -\tan A \tan B \tan C$$

$$\Rightarrow \tan C - \tan A - \tan B = \tan A \tan B \tan C$$

Sol.536.(d) $x = 8(\sin \theta + \cos \theta)$ and $y = 9(\sin \theta - \cos \theta)$,

$$\Rightarrow \frac{x^2}{8^2} + \frac{y^2}{9^2}$$

$$= \frac{64(\sin^2 \theta + \cos^2 \theta + 2\sin \theta \cos \theta)}{64}$$

$$+ \frac{81(\sin^2 \theta + \cos^2 \theta - 2\sin \theta \cos \theta)}{81}$$

$$= 1 + 2\sin \theta \cos \theta + 1 - 2\sin \theta \cos \theta = 2$$

Sol.537.(b) $2 \cos 135^\circ \sin 15^\circ$

$$= [\sin(135^\circ + 15^\circ) - \sin(135^\circ - 15^\circ)]$$

$$= \sin 150^\circ - \sin 120^\circ$$

$$= \cos 60^\circ - \cos 30^\circ = \frac{1}{2} - \frac{\sqrt{3}}{2} = \frac{1 - \sqrt{3}}{2}$$

Sol.538.(c) $\sec x + \cos x =$

$$\sqrt{(\sec x - \cos x)^2 + 4 \times \sec x \cos x}$$

$$= \sqrt{4^2 + 4} = \sqrt{20} = 2\sqrt{5}$$

$$\frac{(1 + \cos^2 x)}{\cos x} = \frac{1}{\cos x} + \frac{\cos^2 x}{\cos x}$$

$$= \sec x + \cos x = 2\sqrt{5}$$

Sol.539.(c) $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$

$$= \frac{\sin A \sin B}{\cos A \cos B} + \frac{\cos x}{\cos A \cos B} = 1$$

$$= \frac{\sin A \sin B + \cos x}{\cos A \cos B} = 1$$

$$= \sin A \sin B + \cos x = \cos A \cos B$$

$$= \cos x = \cos A \cos B - \sin A \sin B$$

$$= \cos x = \cos(A + B) \Rightarrow x = A + B$$

Sol.540.(a) $\cos^3 60^\circ - \cos^3 240^\circ - \cos^3 360^\circ$

$$= \cos^3 60^\circ - \cos^3(180^\circ + 60^\circ) - 1$$

$$= \cos^3 60^\circ + \cos^3 60^\circ - 1$$

$$= \left(\frac{1}{2}\right)^3 \times 2 - 1 = \frac{1}{4} - 1 = \frac{-3}{4}$$

Sol.541.(c) $5\sin^2 A + 3\cos^2 A = 4$

$$\Rightarrow 2\sin^2 A + 3\sin^2 A + 3\cos^2 A = 4$$

$$\Rightarrow 2\sin^2 A + 3(\sin^2 A + \cos^2 A) = 4$$

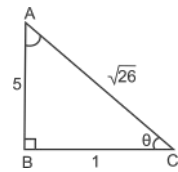
$$\Rightarrow 2\sin^2 A + 3 = 4 \Rightarrow 2\sin^2 A = 1$$

$$\sin^2 A = \frac{1}{2} \Rightarrow \sin A = \frac{1}{\sqrt{2}} \Rightarrow A = 45^\circ$$

$$\text{then, } \tan A = \tan 45^\circ = 1$$

Sol.542.(b) $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = \frac{3}{2}$

$$\frac{\sin \theta}{\cos \theta} = \frac{3+2}{3-2} = \frac{5}{1} \Rightarrow \tan \theta = \frac{5}{1}$$



$$\sin^4 \theta - \cos^4 \theta = \left(\frac{5}{\sqrt{26}}\right)^4 - \left(\frac{1}{\sqrt{26}}\right)^4$$

$$= \frac{625}{676} - \frac{1}{676} = \frac{624}{676} = \frac{12}{13}$$

Sol.543.(c)

$$\operatorname{cosec}^2 \theta + \sec^2 \theta = \frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta} = \frac{1}{\frac{2}{9}} = \frac{9}{2}$$

Sol.544.(c) Formula :- $(\cos 2x = 2\cos^2 x - 1)$

$$2 \frac{\cos^2 x - \sec^2 x}{\tan^2 x} = a + b \cos 2x$$

$$\Rightarrow 2 \frac{\cos^2 x - \frac{1}{\cos^2 x}}{\frac{\sin^2 x}{\cos^2 x}} = 2 \frac{\cos^4 x - 1}{\sin^2 x}$$

$$= 2 \frac{(\cos^2 x - 1)(\cos^2 x + 1)}{1 - \cos^2 x}$$

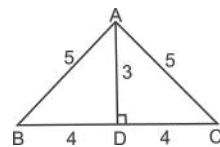
$$= -2(\cos^2 x + 1) = -2\cos^2 x - 2$$

$$= -(\cos 2x + 1) - 2 = a + b \cos 2x$$

$$-3 - \cos 2x = a + b \cos 2x$$

Comparing both side, we get,

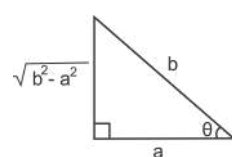
$$a = -3 \text{ and } b = -1$$

Sol.545.(b)Since, $\triangle ABC$ is an isosceles triangle then, AD(median) divides BC in the ratio of 1 : 1.

By the given figure,

$$\tan C = \frac{3}{4}, \cot B = \frac{4}{3}$$

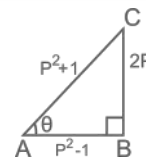
$$\text{So, } \tan C - \cot B = \frac{3}{4} - \frac{4}{3} = \frac{9-16}{12} = \frac{-7}{12}$$

Sol.546.(c) $b \cos \theta = a \Rightarrow \cos \theta = \frac{a}{b}$ 

$$p = \sqrt{b^2 - a^2} \text{ (using pythagoras theorem)}$$

$$\operatorname{cosec} \theta + \cot \theta = \frac{b}{\sqrt{b^2 - a^2}} + \frac{a}{\sqrt{b^2 - a^2}}$$

$$= \frac{b+a}{(\sqrt{b-a})(\sqrt{b+a})} = \sqrt{\frac{b+a}{b-a}}$$

Sol.547.(a)

$$\operatorname{cosec} \theta + \cot \theta = p \text{ ----- (1)}$$

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{p} \text{ ----- (2)}$$

$$(\text{As } \operatorname{cosec}^2 \theta - \cot^2 \theta = 1)$$

Adding eq. (1) and (2) we get :

$$\operatorname{cosec} \theta = \frac{p^2+1}{2p} \text{ . then, } \cot \theta = \frac{p^2-1}{2p}$$

$$\text{Now, } \frac{\frac{p^2-1}{2p}}{\frac{p^2+1}{2p}} = \frac{p^2-1}{p^2+1} = \frac{\cot \theta}{\operatorname{cosec} \theta} = \cos \theta$$

Sol.548.(d) $\sin^2 \theta - 3\sin \theta + 2 = 0$

$$\sin^2 \theta - 2\sin \theta - \sin \theta + 2 = 0$$

$$\sin \theta (\sin \theta - 2) - 1(\sin \theta - 2) = 0$$

$$(\sin \theta - 1)(\sin \theta - 2) = 0 \Rightarrow \sin \theta = 1 \text{ (} \sin \theta \neq 2 \text{)}$$

$$\therefore (-1 \leq \sin \theta \leq 1))$$

$$\text{So, } \sin \theta = 1 \text{ (As, } 0^\circ \leq \theta \leq 90^\circ) \Rightarrow \theta = 90^\circ$$

Sol.549.(b) $\tan 3\theta \cdot \tan 7\theta = 1$

$$\tan 3\theta = \frac{1}{\tan 7\theta} = \cot 7\theta$$

$$\tan 3\theta = \tan(90^\circ - 7\theta)$$

$$3\theta = 90^\circ - 7\theta \Rightarrow 10\theta = 90^\circ$$

$$\Rightarrow \theta = \frac{90}{10} = 9^\circ$$

$$\text{So, } \cot 15\theta = \cot 135^\circ = -1$$

Sol.550.(b) $2\sin \theta + 2\sin^2 \theta = 2$

$$\sin \theta + \sin^2 \theta = 1 \Rightarrow \sin \theta = 1 - \sin^2 \theta$$

$$\Rightarrow \sin \theta = \cos^2 \theta$$

$$\text{So, } 2\cos^4 \theta + 2\cos^2 \theta = 2(\cos^4 \theta + \cos^2 \theta)$$

$$= 2(\sin^2 \theta + \cos^2 \theta) = 2 \times 1 = 2$$

Sol.551.(a) $\cos \theta + \cos^2 \theta = 1$

$$\cos \theta = 1 - \cos^2 \theta \Rightarrow \cos \theta = \sin^2 \theta$$

$$\text{So, } \sqrt{\sin^4 \theta + \cos^2 \theta} = \sqrt{\cos^2 \theta + \cos^2 \theta}$$

$$= \sqrt{2\cos^2 \theta} = \sqrt{2}\cos \theta$$

Sol.552.(c) $\sec^2 A + \tan^2 A = 3$

$$1 + \tan^2 A + \tan^2 A = 3$$

$$1 + 2\tan^2 A = 3 \Rightarrow 2\tan^2 A = 2$$

$$\tan^2 A = 1 \Rightarrow \frac{1}{\cot^2 A} = 1$$

$$\cot^2 A = 1 \Rightarrow \cot A = 1$$

Sol.553.(b) $\operatorname{cosec} 15^\circ \sec 15^\circ$

$$= \frac{1}{\sin 15^\circ \cos 15^\circ} = \frac{2}{2\sin 15^\circ \cos 15^\circ}$$

$$= \frac{2}{\sin 30^\circ} = \frac{2}{\frac{1}{2}} = 4$$

Sol.554.(d) $\frac{\cos A}{1 - \tan A} + \frac{\sin A}{1 - \cot A}$

$$= \frac{\cos A}{1 - \frac{\sin A}{\cos A}} + \frac{\sin A}{1 - \frac{\cos A}{\sin A}}$$

$$\begin{aligned}
 &= \frac{\cos^2 A}{\cos A - \sin A} - \frac{\sin^2 A}{\cos A - \sin A} \\
 &= \frac{\cos^2 A - \sin^2 A}{\cos A - \sin A} \\
 &= \frac{(\cos A + \sin A)(\cos A - \sin A)}{\cos A - \sin A} \\
 &= \sin A + \cos A
 \end{aligned}$$

Sol.555.(c) $\tan 40^\circ = \cot 50^\circ = a$

$$\begin{aligned}
 \text{So, } &\frac{\tan 320^\circ - \tan 310^\circ}{1 + \tan 320^\circ \tan 310^\circ} \\
 &= \frac{\tan(360^\circ - 40^\circ) - \tan(360^\circ - 50^\circ)}{1 + \tan(360^\circ - 40^\circ) \tan(360^\circ - 50^\circ)} \\
 &= \frac{-\tan 40^\circ + \tan 50^\circ}{1 + \tan 40^\circ \tan 50^\circ} \\
 &= \frac{-a + \frac{1}{a}}{1 + a \times \frac{1}{a}} = \frac{1 - a^2}{2a}
 \end{aligned}$$

Sol.556.(a) Formula :-

$$\begin{aligned}
 \tan(A + B) &= \frac{\tan A + \tan B}{1 - \tan A \tan B} \\
 \tan 2\alpha &= \tan\{(\alpha + \beta) + (\alpha - \beta)\} \\
 &= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta) \tan(\alpha - \beta)} = \frac{a + b}{1 - ab}
 \end{aligned}$$

Sol.557.(b) Short trick :

Putting $\theta = 45^\circ$, we have ,

$$\begin{aligned}
 x &= \frac{2\sin 45^\circ}{1 + \cos 45^\circ + \sin 45^\circ} = \frac{\sqrt{2}}{1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}} \\
 &= \frac{2}{\sqrt{2} + 2} = \frac{\sqrt{2}}{1 + \sqrt{2}} \\
 \text{Now,} \\
 \frac{1 - \cos \theta + \sin \theta}{1 + \sin \theta} &= \frac{1 - \cos 45^\circ + \sin 45^\circ}{1 + \sin 45^\circ} \\
 &= \frac{1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}} = \frac{\sqrt{2} - 1 + 1}{\sqrt{2} + 1} \\
 &= \frac{\sqrt{2}}{\sqrt{2} + 1} = x
 \end{aligned}$$